Computer Animation Workshops

Ball Animation Using Maya

Anargyros Sarafopoulos

National Centre for Computer Animation
Bournemouth University

workshop notes, animation 01

Outline

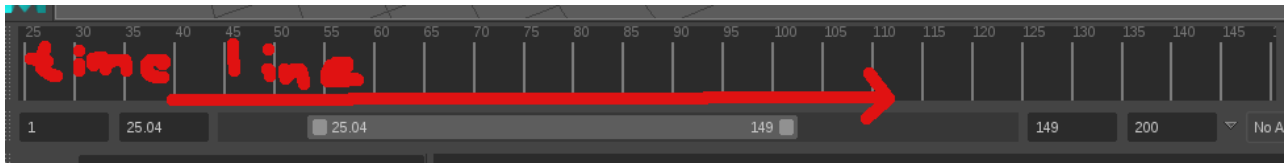
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   - The Animation Time Line
2. References

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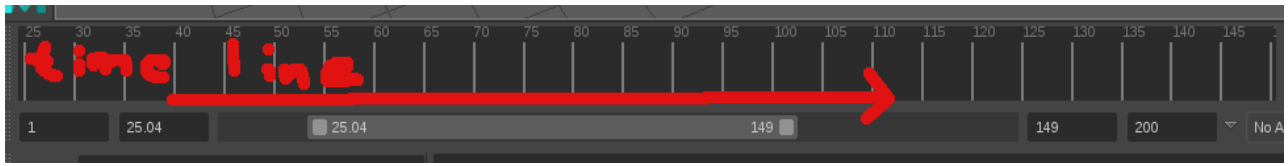
Maya's animation Straight Ahead and Pose to Pose tools

- The Maya interface is posed for "pose to pose" or key-frame animation.
- Maya's dynamics and simulation animation tools cater for "straight ahead" animation.
- Here we look at key-frame or pose to pose Maya techniques.
- The most important concept here is the Maya "Time Slider" or more affectionately "Time Line".



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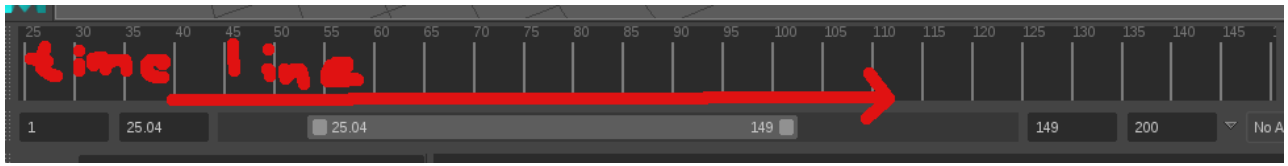
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Straight Ahead and Pose to Pose (slide 01)

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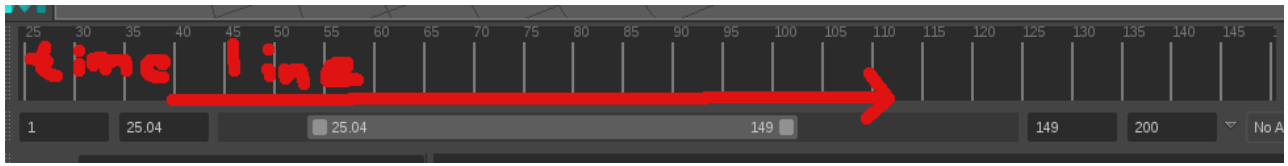
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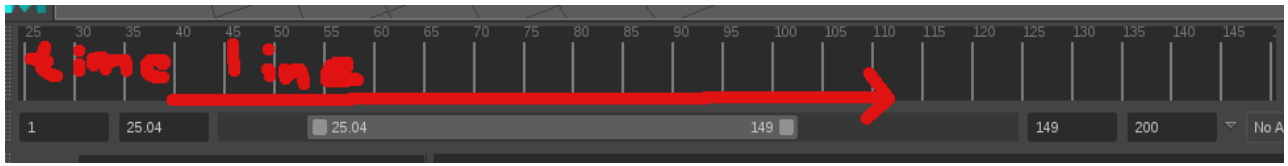
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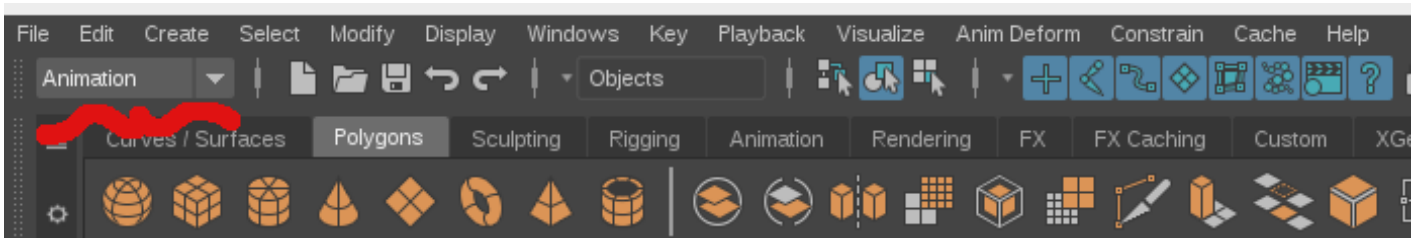
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Organising Playback and Time Slider Preferences

- In the top and leftmost drop-down menu select: Animation.
- Use the icon next to the time line to open the Animation Preferences.
- In the Animation Preferences select the "Settings" category and in the "Time" drop down menu select:(25 fps).
- In The Animation Preferences select the "Time Slider" category and in "Playback speed" drop-down menu select "[25 fps]"



Maya's Animation Preferences (slide 02)

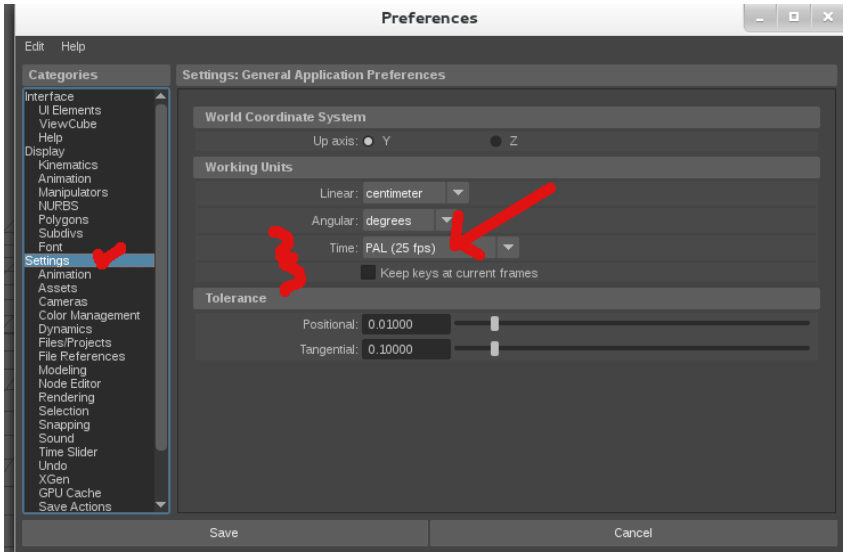
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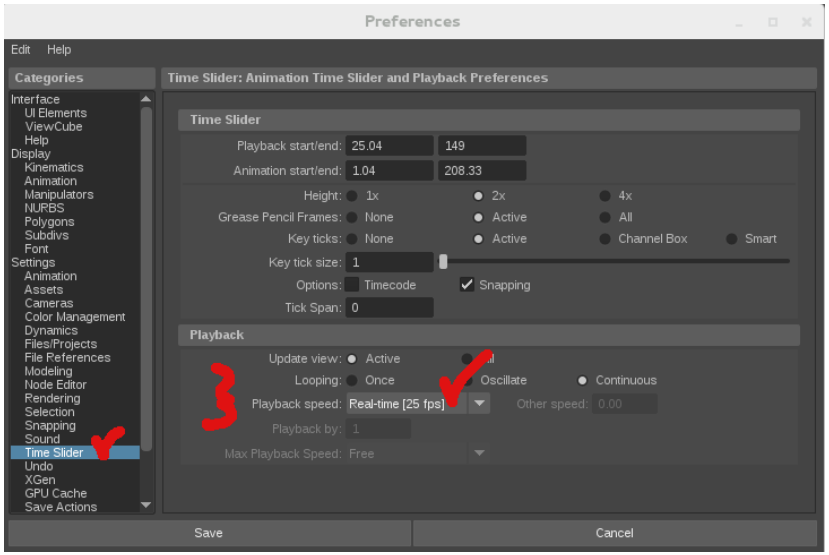
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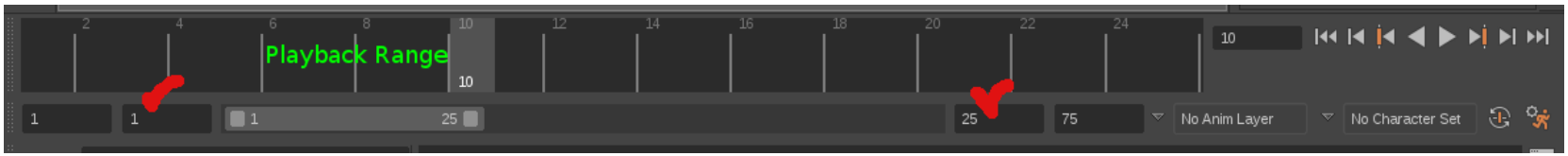
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Maya's Animation Playback Preferences (slide 03)

Organising Playback and Time Slider Preferences

- The (inner) slider below the time line controls the playback range.
- The values at the outer margins show the animation range.
- Let's set the animation start-end range to 1 to 75 (three frames).
- Let's set the playback start-end range to 1 to 25 (the first frame).
- Let's Set the current frame to '10'.



Maya's Animation Playback Preferences (slide 03)

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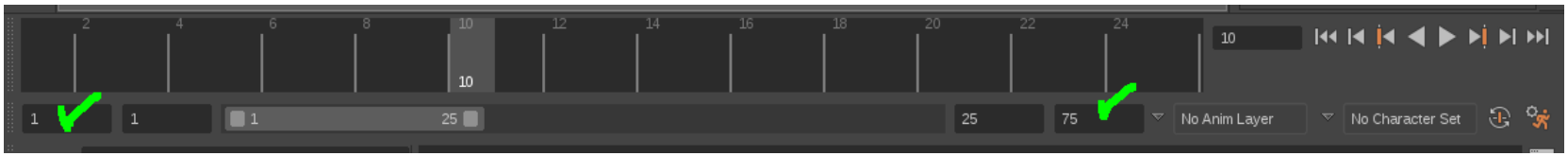
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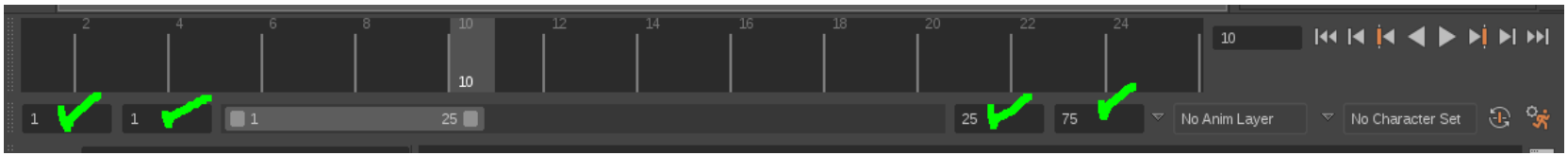
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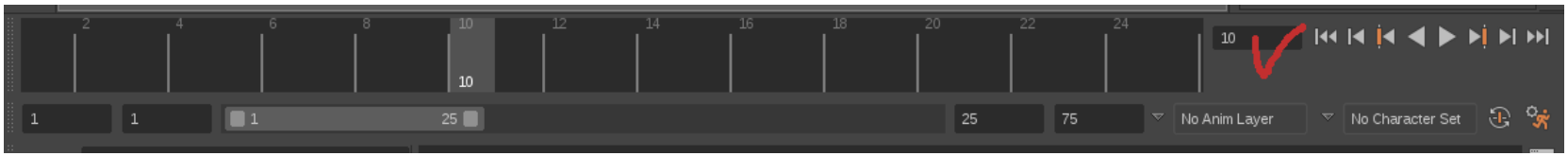
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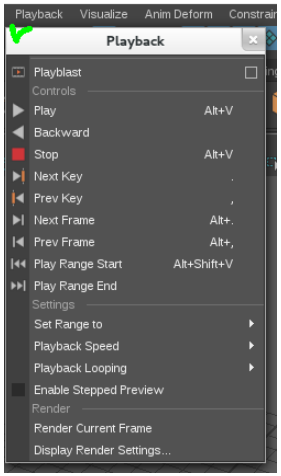
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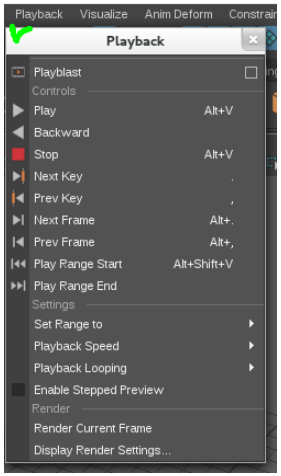
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- Opening the *Playback* menu shows the playback keyboard short-cuts.
- Pressing **Alt-v** set the animation to "play" and "stop".
- Pressing **Alt-.** and **Alt-,** increases and degreases the frame number by one.
- Pressing the full-stop "**.**" moves to the next key-frame.
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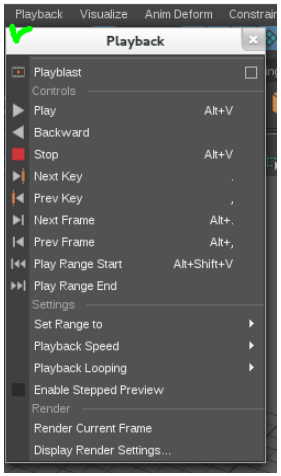
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Maya's Animation Playback Preferences (slide 04)

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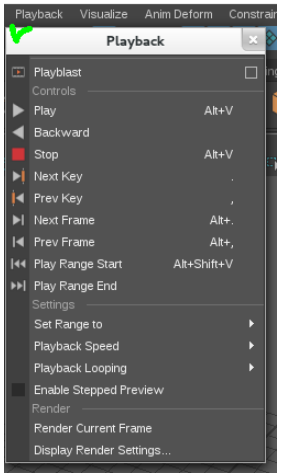
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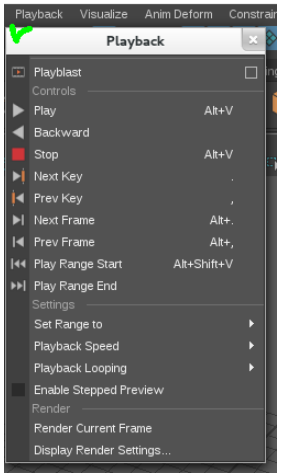
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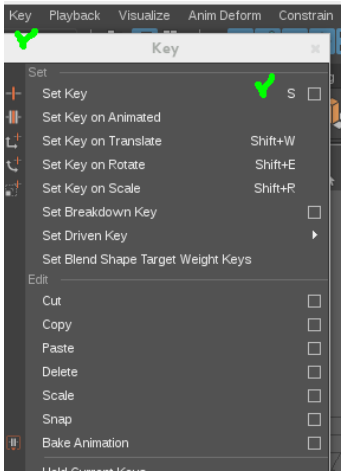
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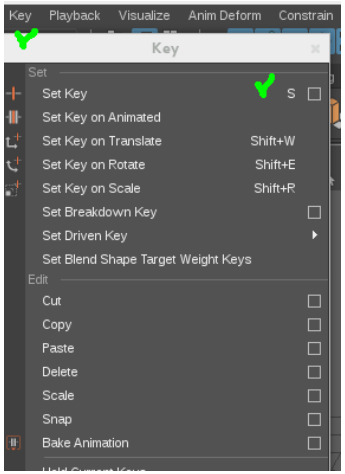
Setting and Editing Key Frames

- Opening the *Key* menu shows the key-frame related keyboard short-cuts and operations available.
- Pressing **S** sets a key frame to all transformation channels of the selected object.
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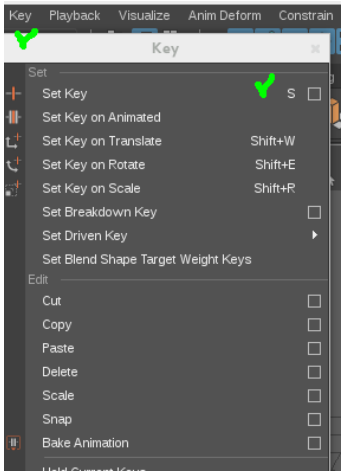
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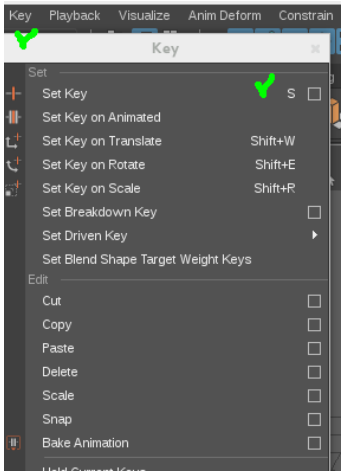
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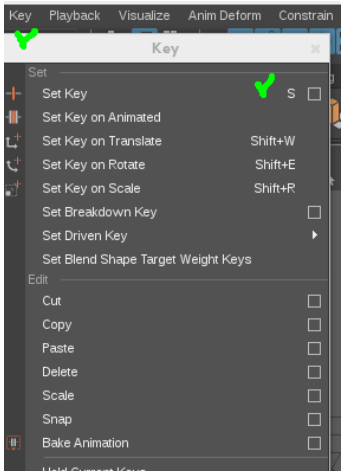
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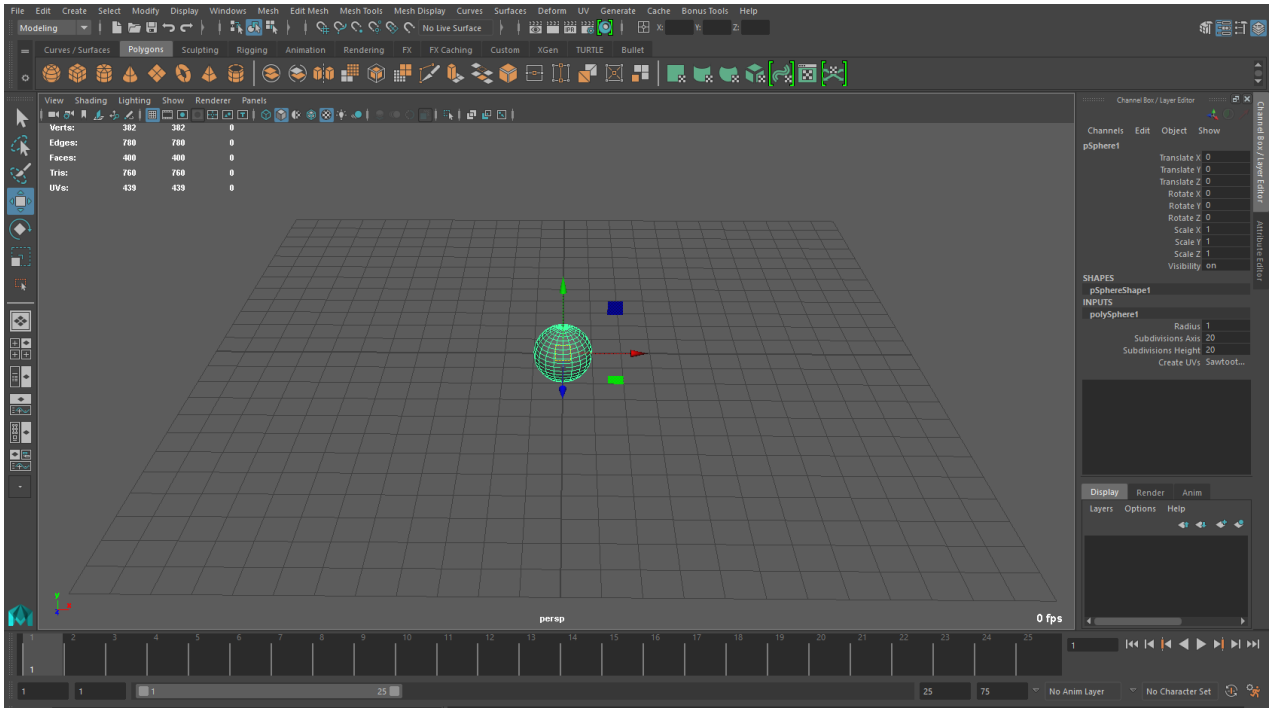
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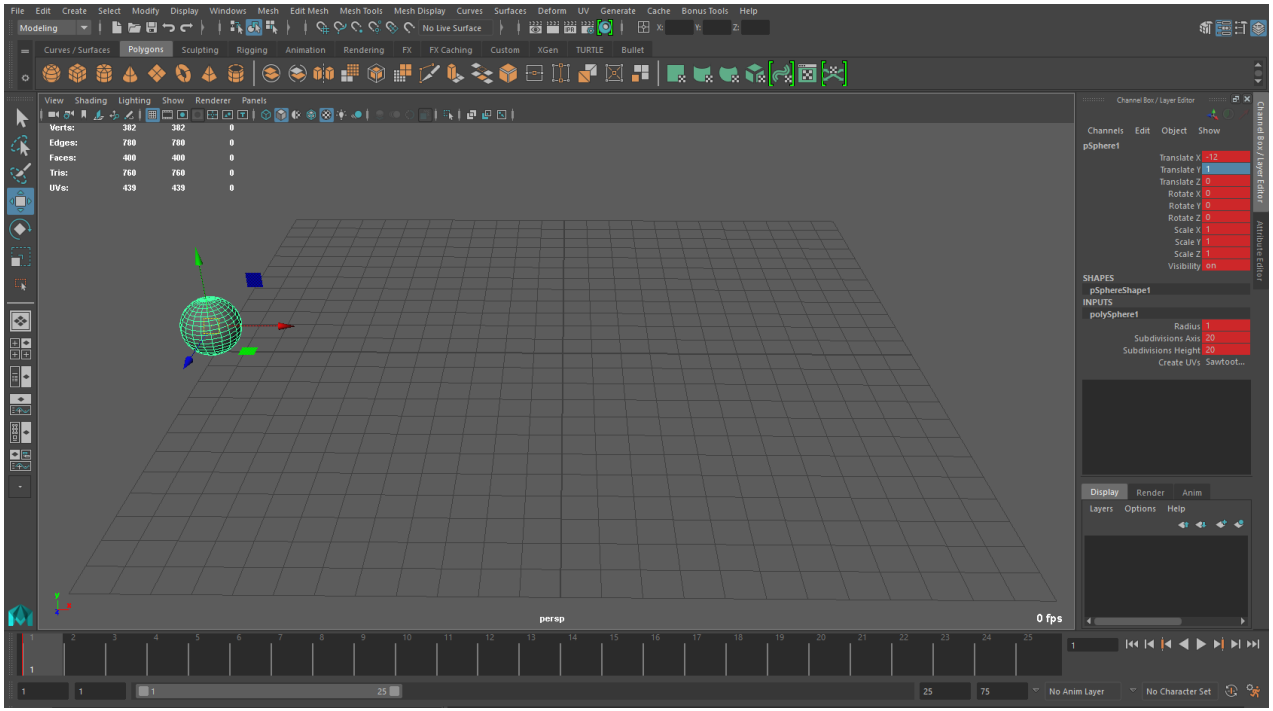
Setting a Key Frame

- Create a (polygonal) sphere and set the current frame to 1.
- Move the sphere at the global position (-12, 1, 0) and Press **S**.
- Set the current frame to 25.
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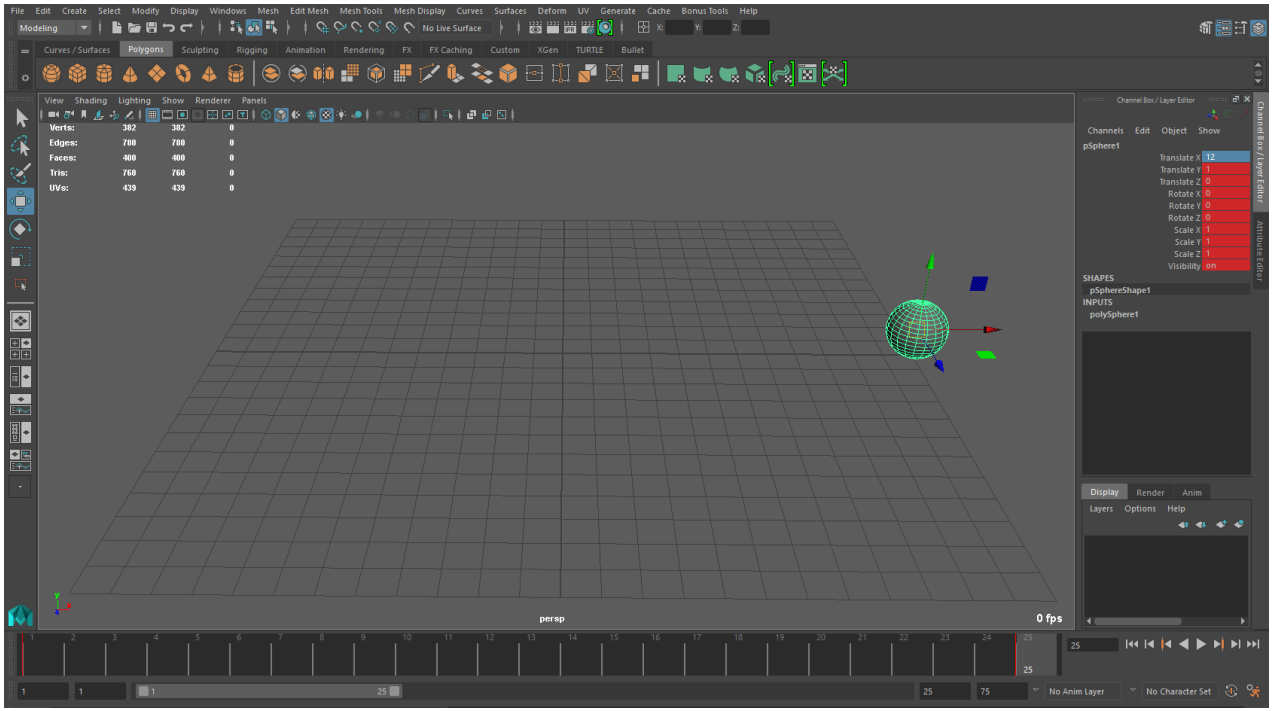
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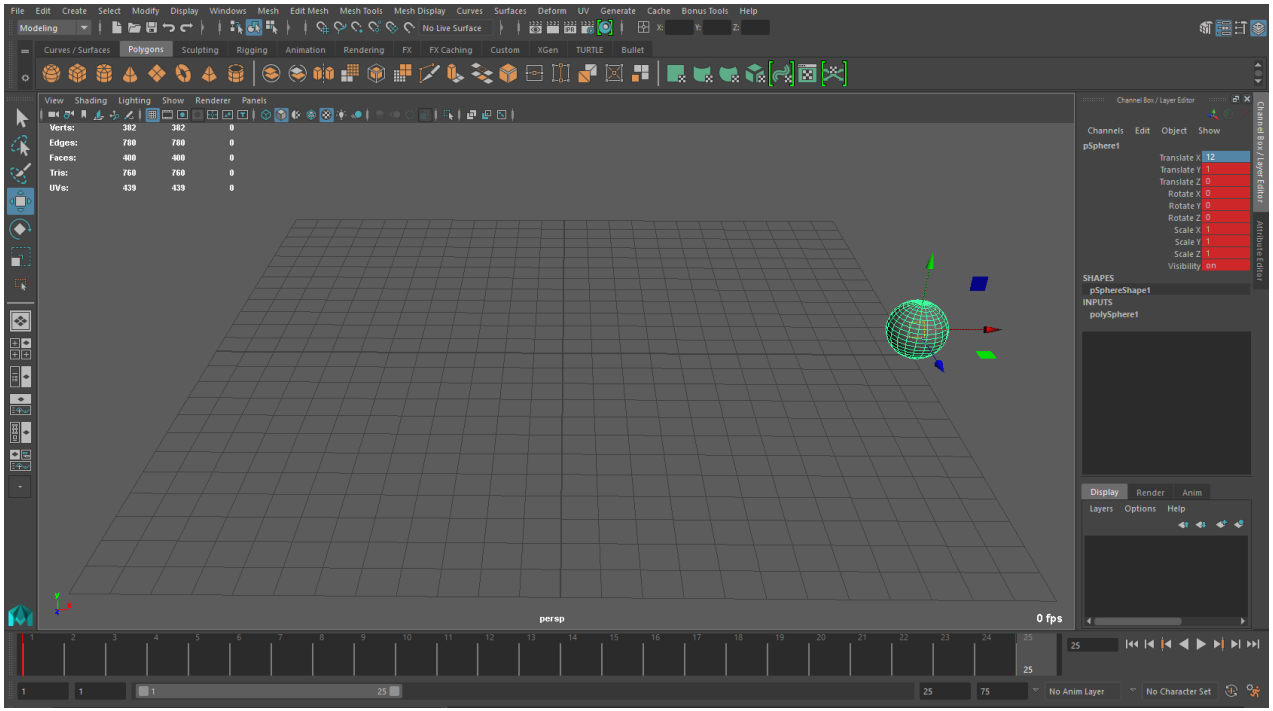
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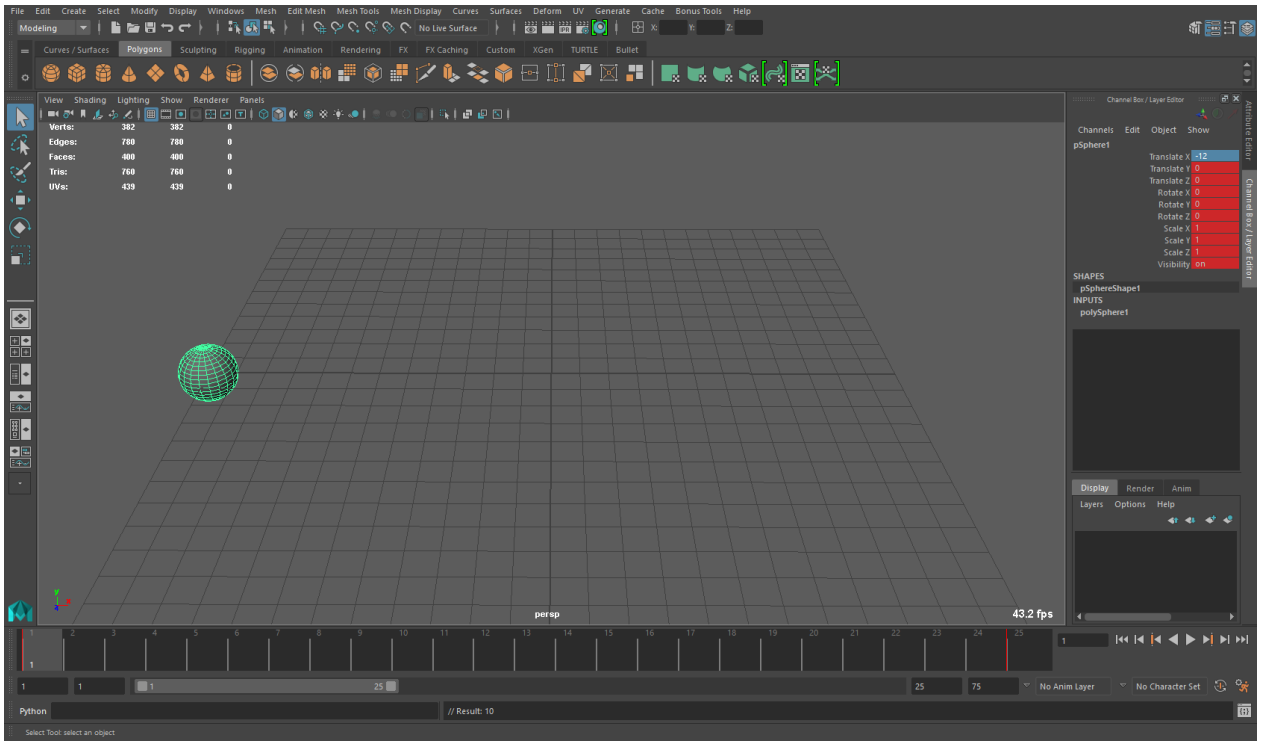
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Time Line Scrubbing (slide 07)

To play through the in-between frames:

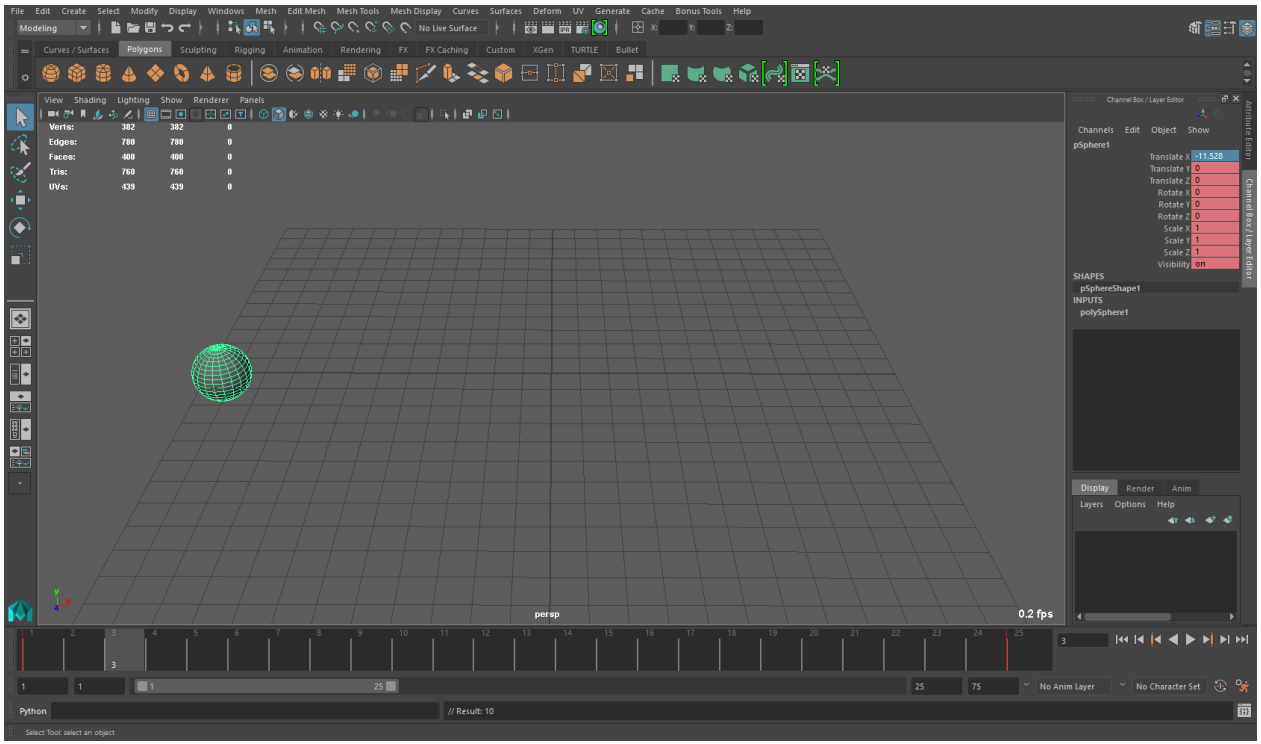
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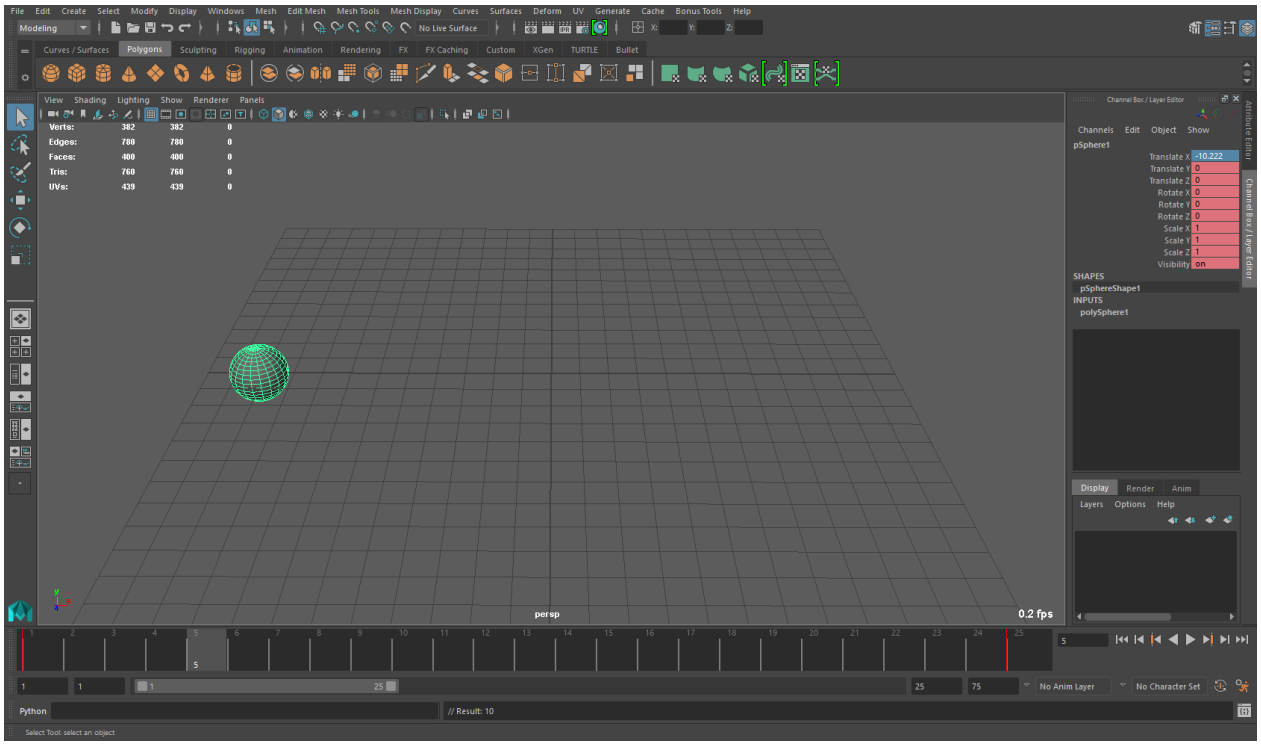
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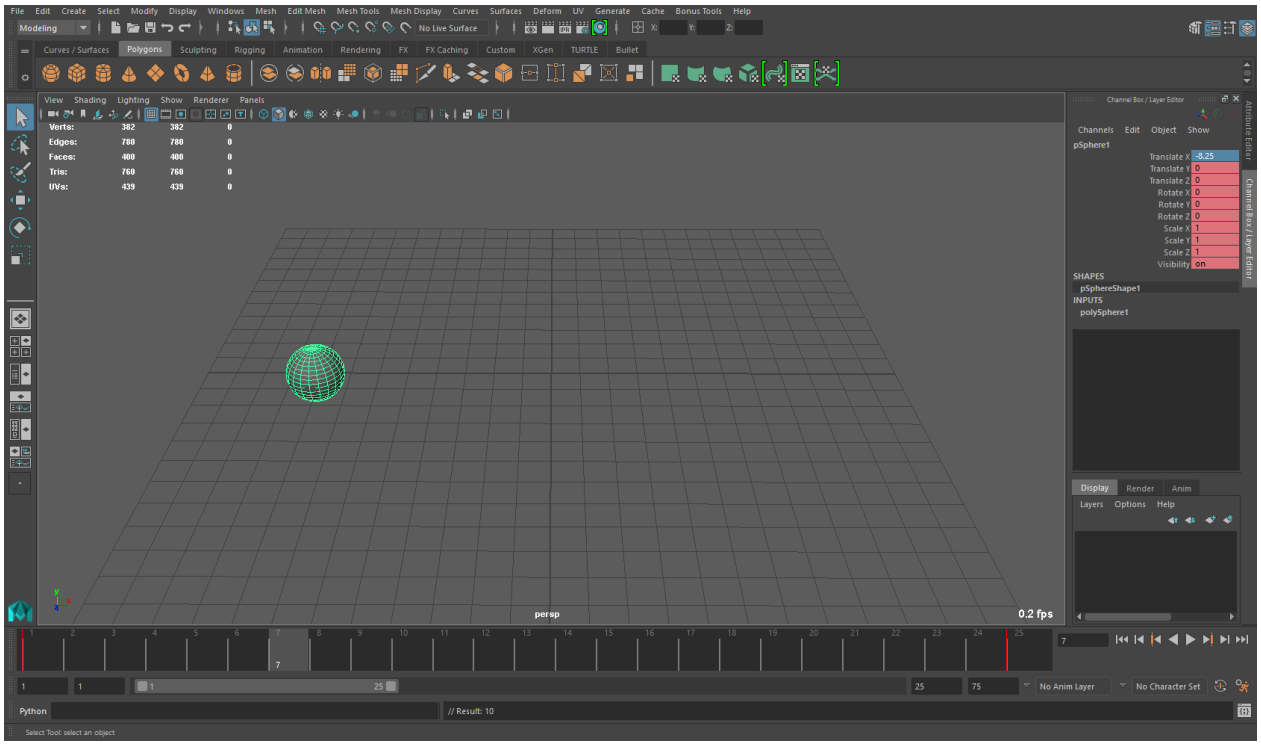
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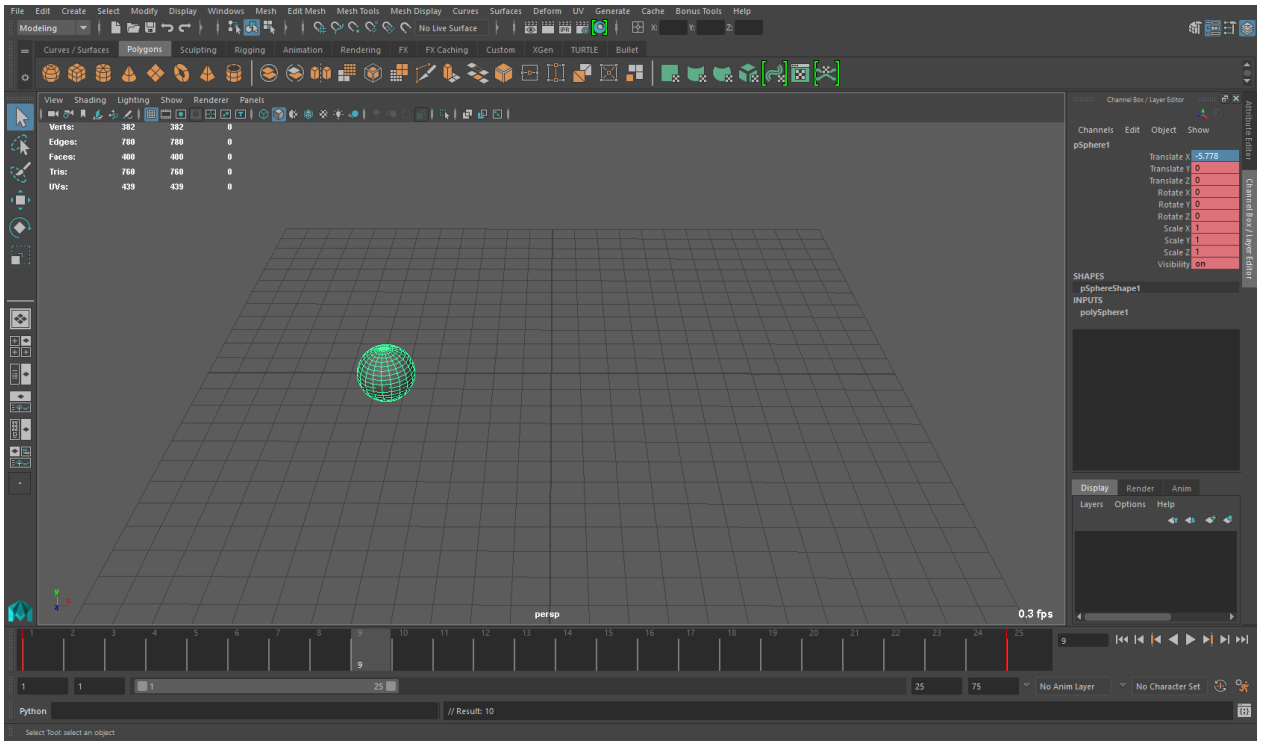
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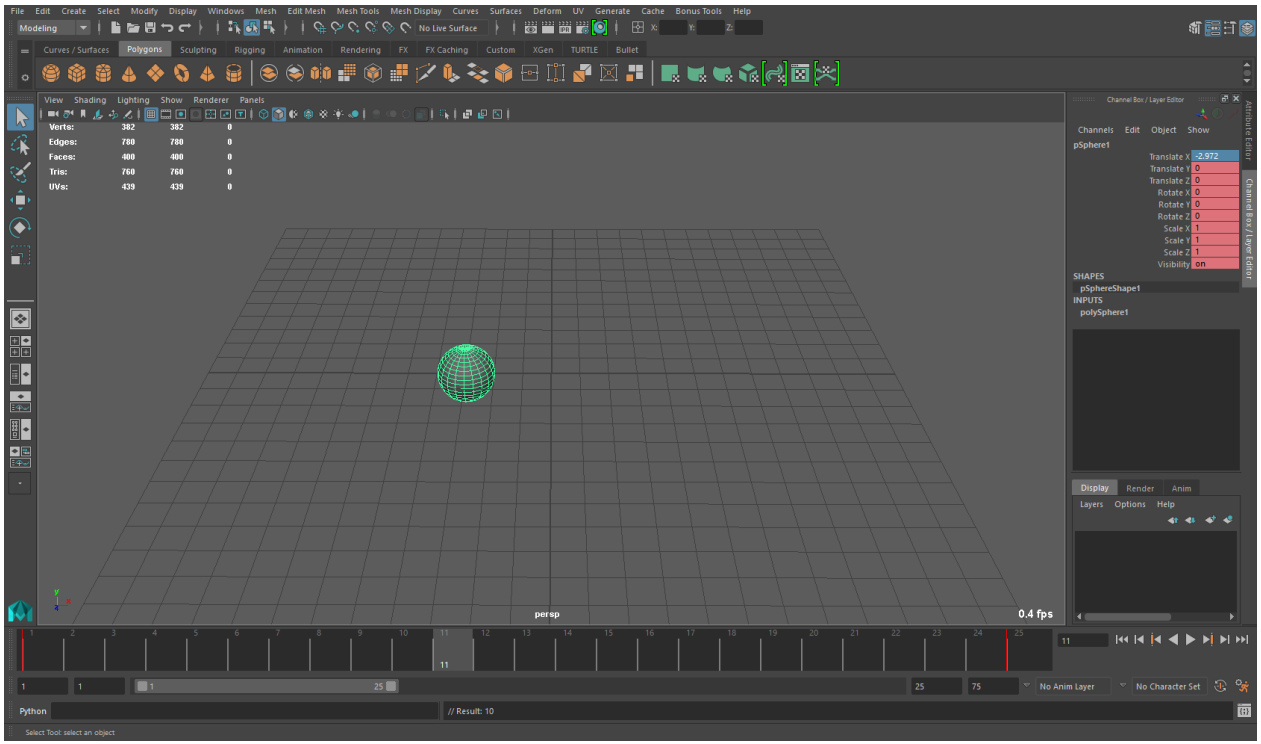
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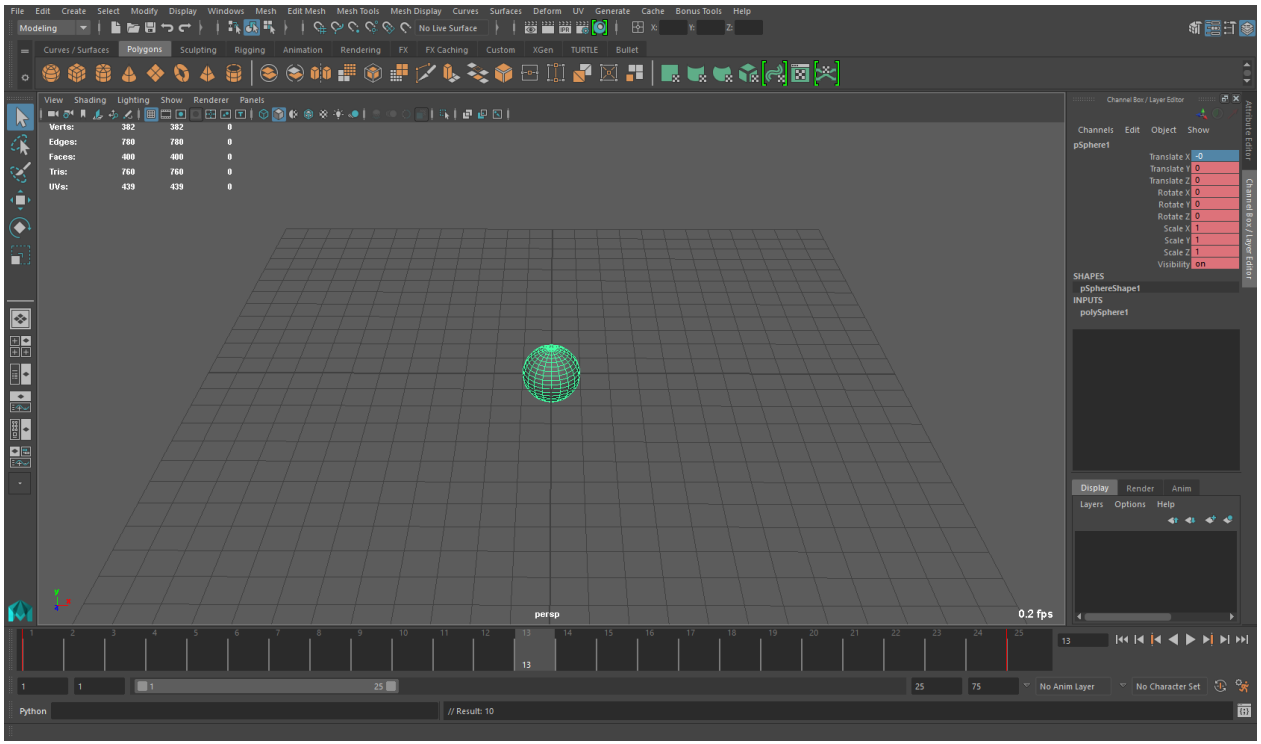
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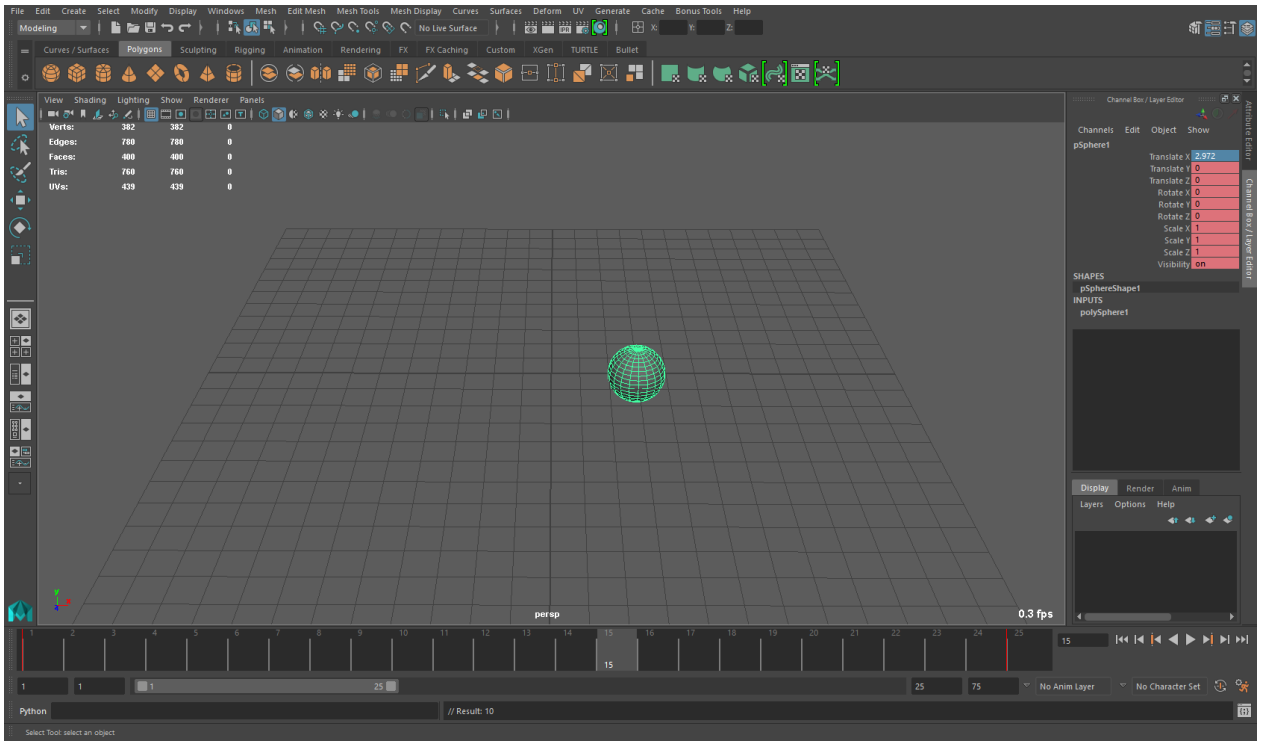
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Time Line Scrubbing (slide 07)

To play through the in-between frames:

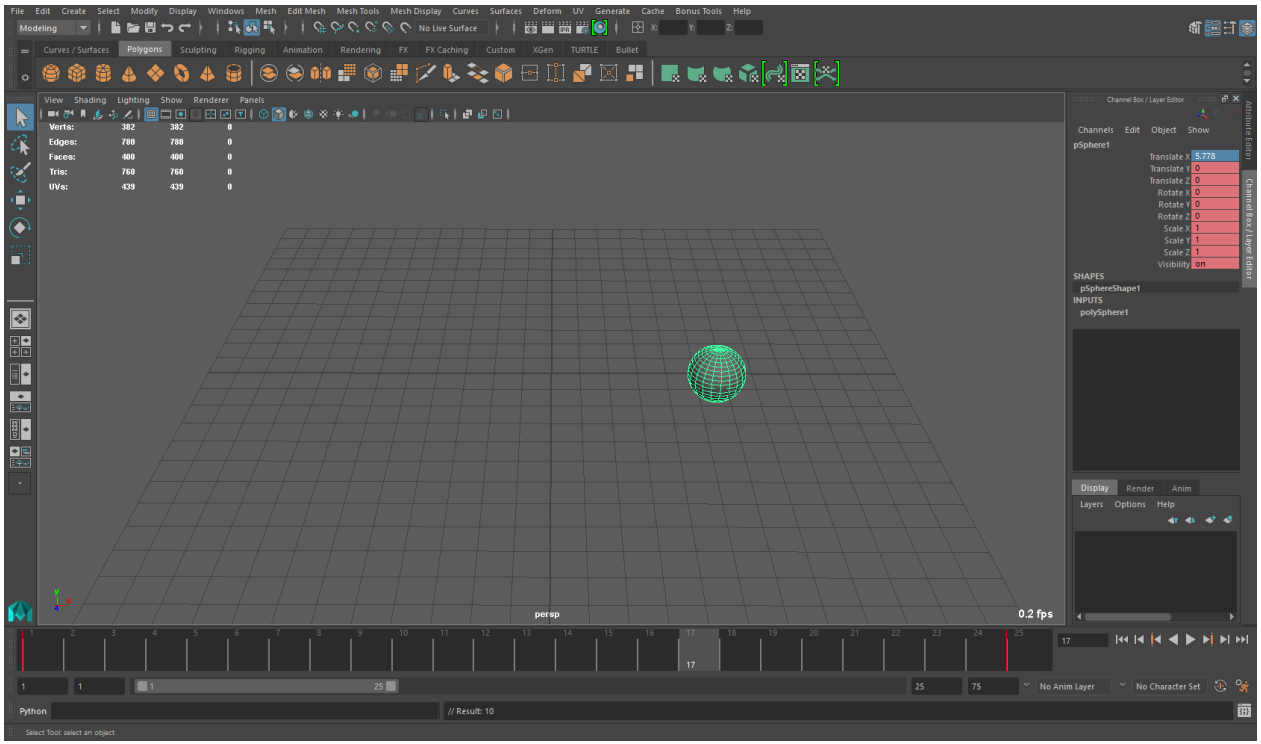
- Scrub across the time line while pressing the left mouse button (**LMB**).
- When away from the time line scrub by pressing **k** on the keyboard and the **LMB**.



Time Line Scrubbing (slide 07)

To play through the in-between frames:

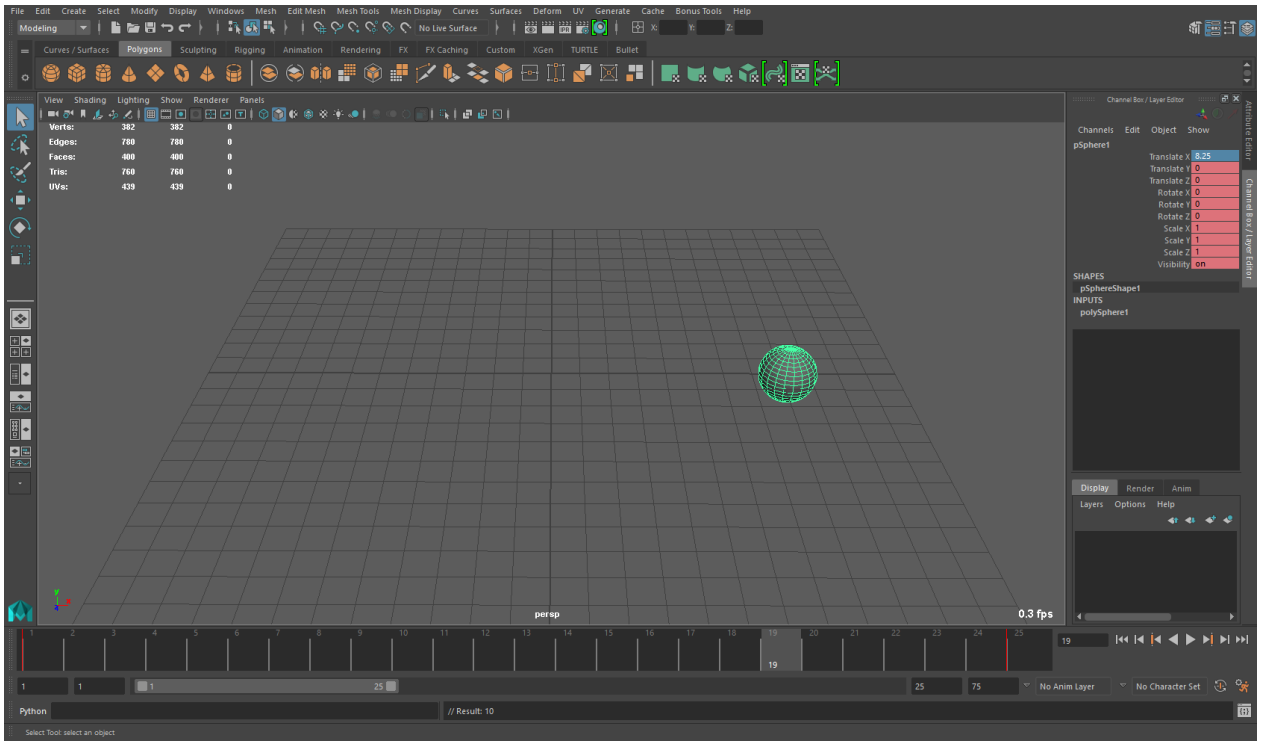
- Scrub across the time line while pressing the left mouse button (**LMB**).
- When away from the time line scrub by pressing **k** on the keyboard and the **LMB**.



Time Line Scrubbing (slide 07)

To play through the in-between frames:

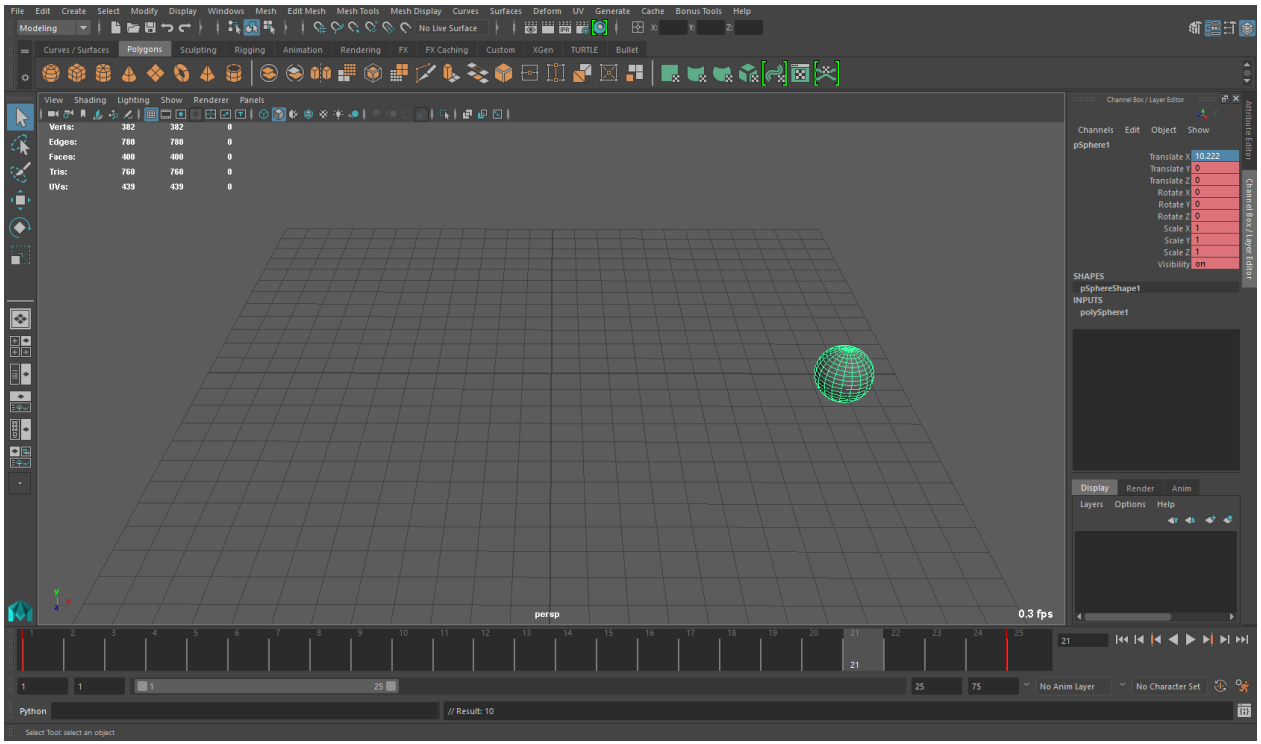
- Scrub across the time line while pressing the left mouse button (**LMB**).
- When away from the time line scrub by pressing **k** on the keyboard and the **LMB**.



Time Line Scrubbing (slide 07)

To play through the in-between frames:

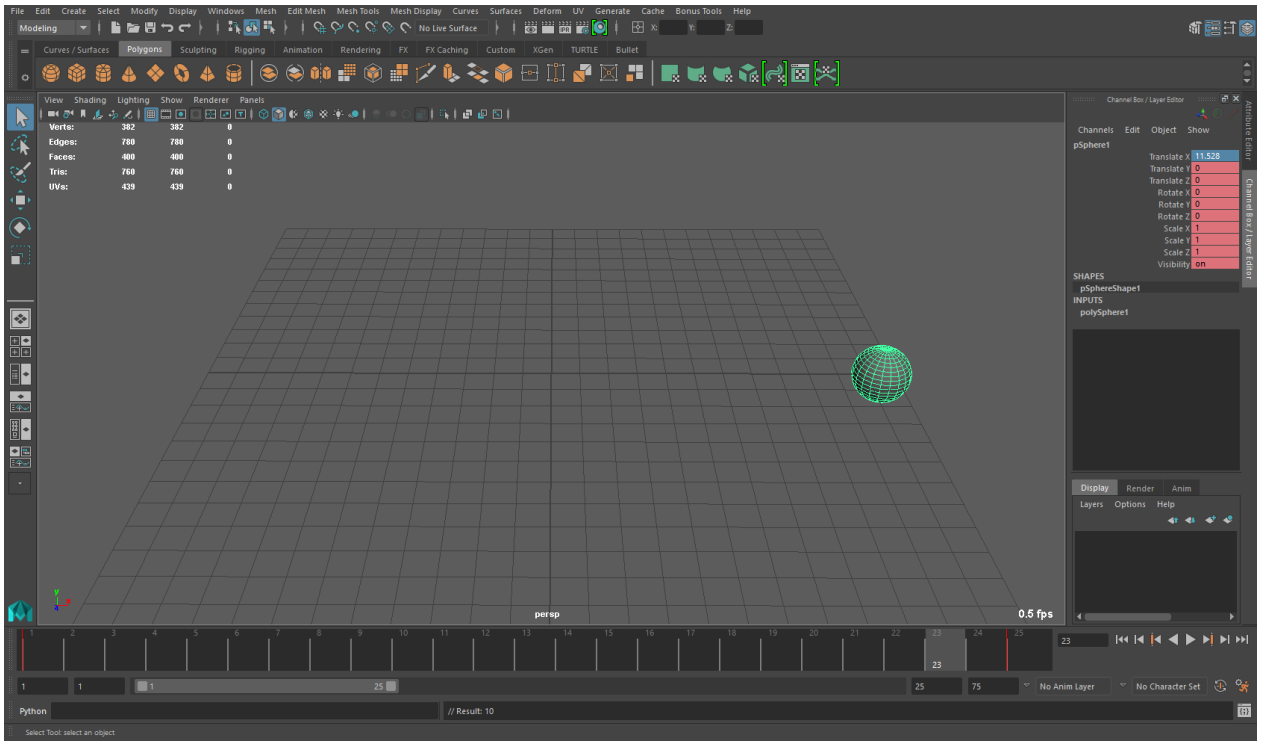
- Scrub across the time line while pressing the left mouse button (**LMB**).
- When away from the time line scrub by pressing **k** on the keyboard and the **LMB**.



Time Line Scrubbing (slide 07)

To play through the in-between frames:

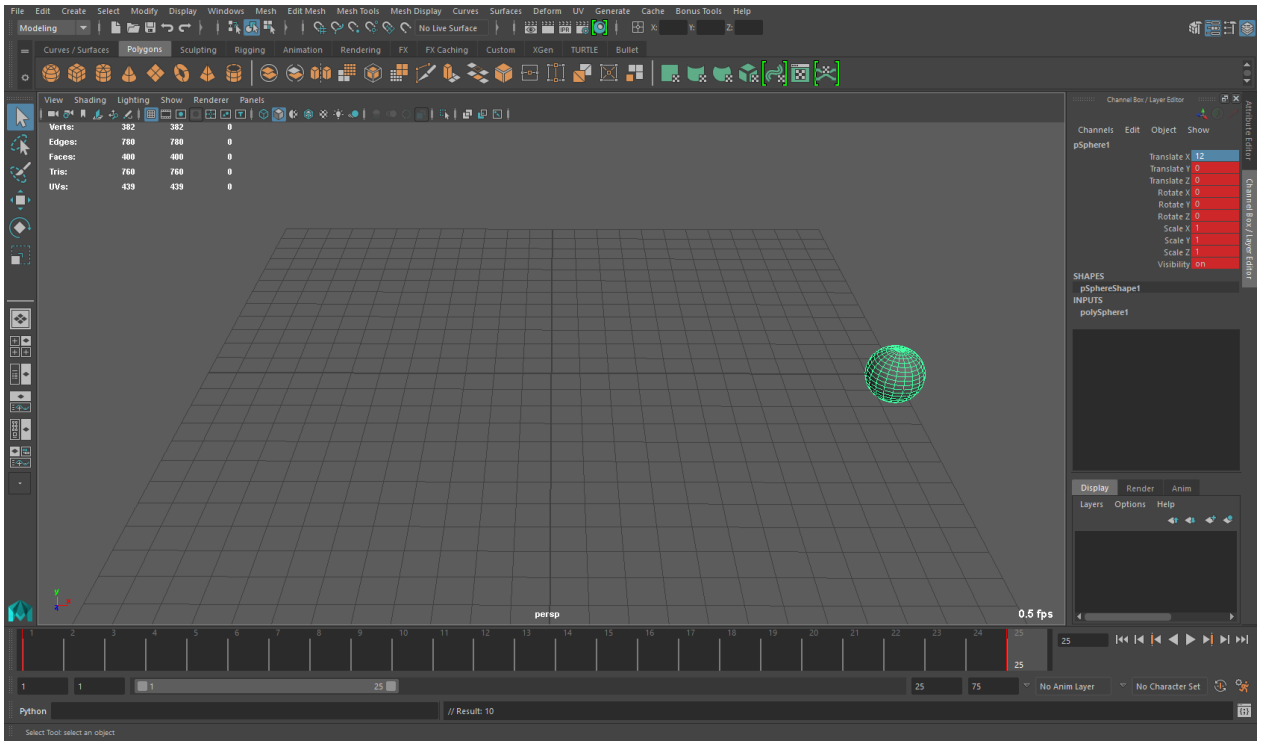
- Scrub across the time line while pressing the left mouse button (**LMB**).
- When away from the time line scrub by pressing **k** on the keyboard and the **LMB**.



Time Line Scrubbing (slide 07)

To play through the in-between frames:

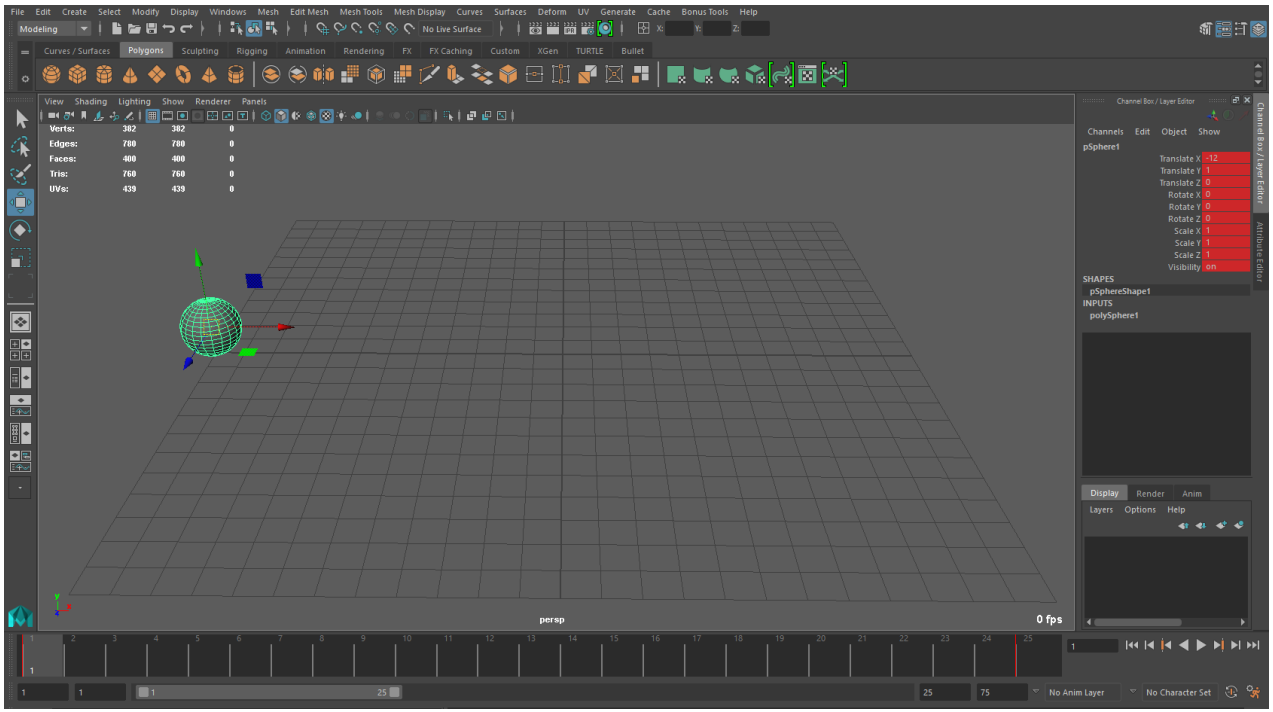
- Scrub across the time line while pressing the left mouse button (**LMB**).
- When away from the time line scrub by pressing **k** on the keyboard and the **LMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

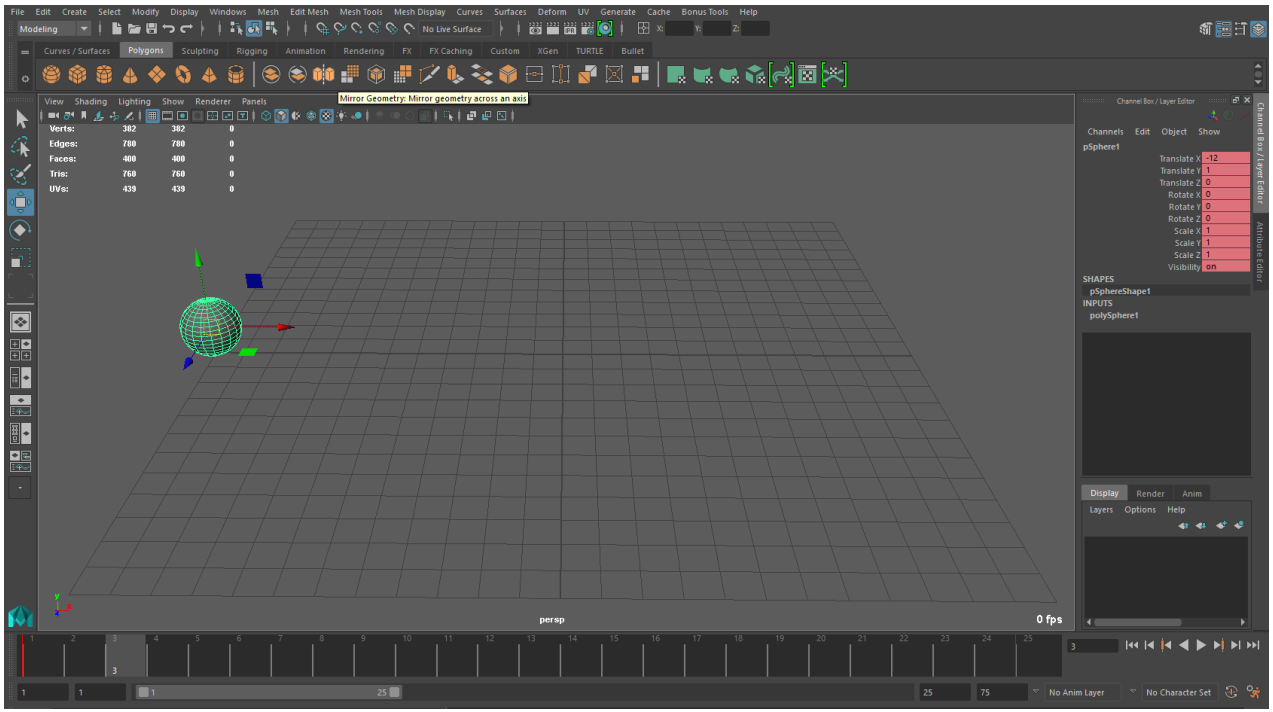
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

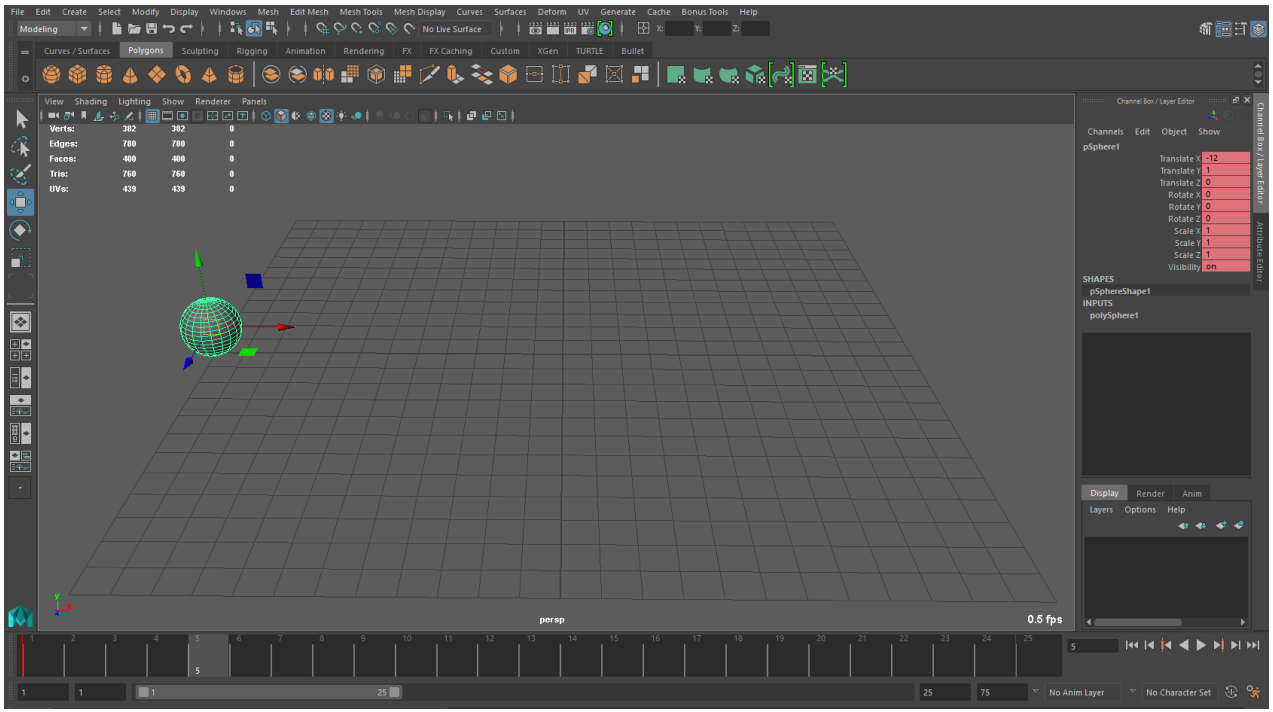
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

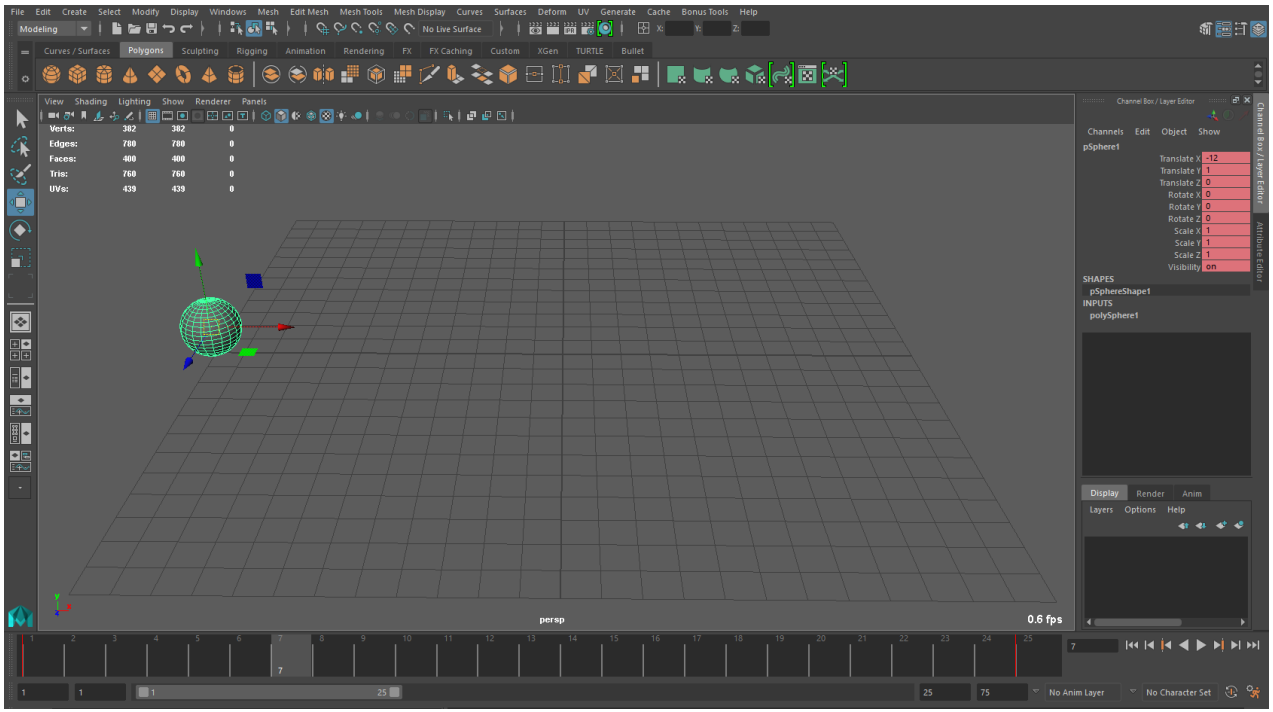
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

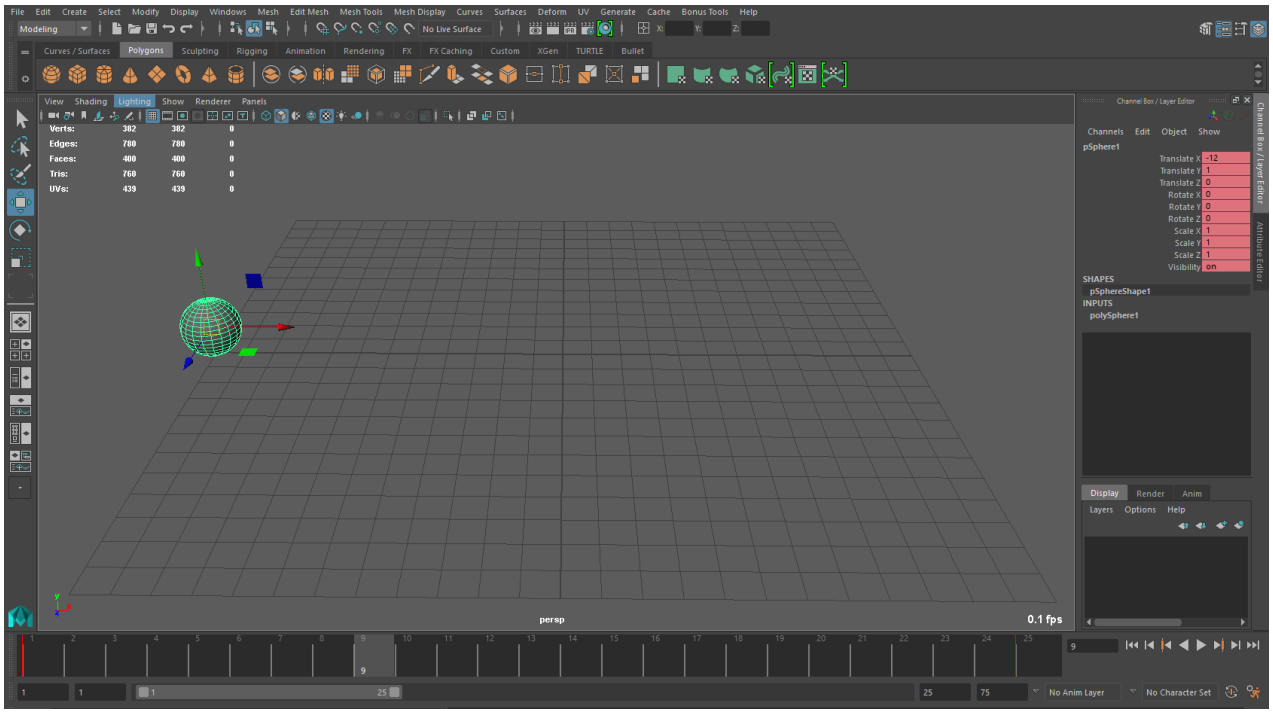
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

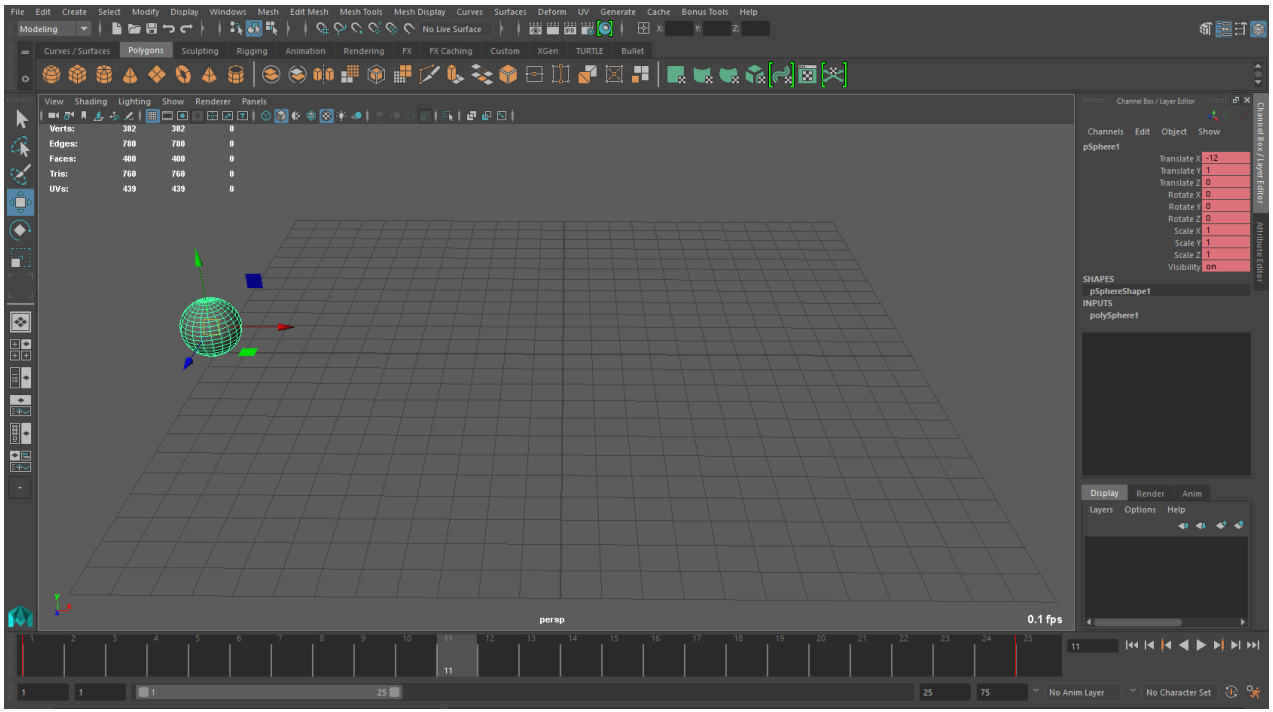
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

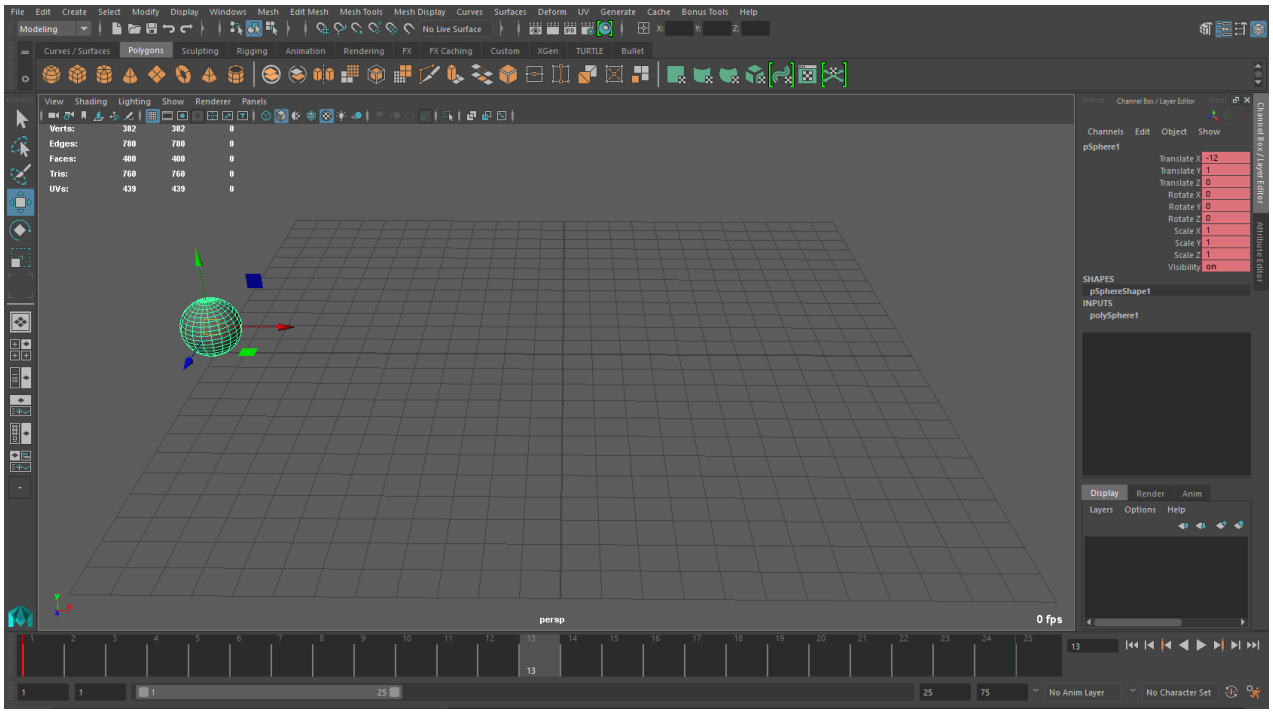
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

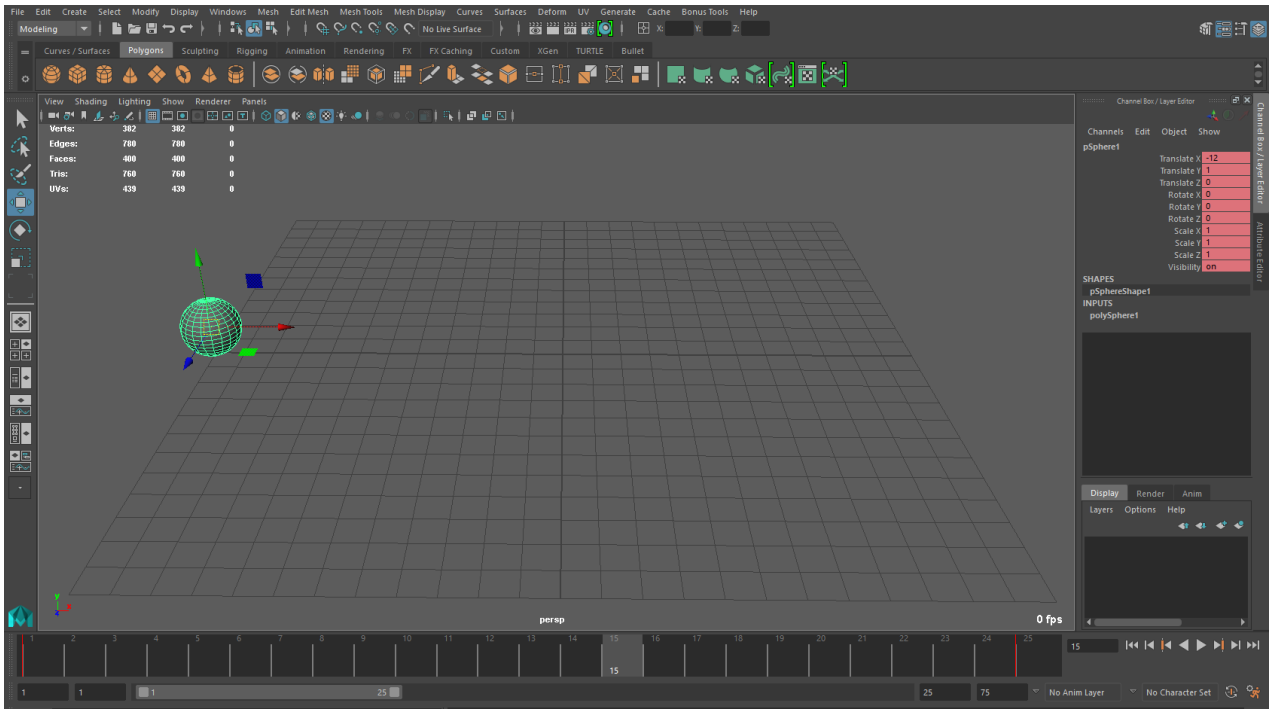
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

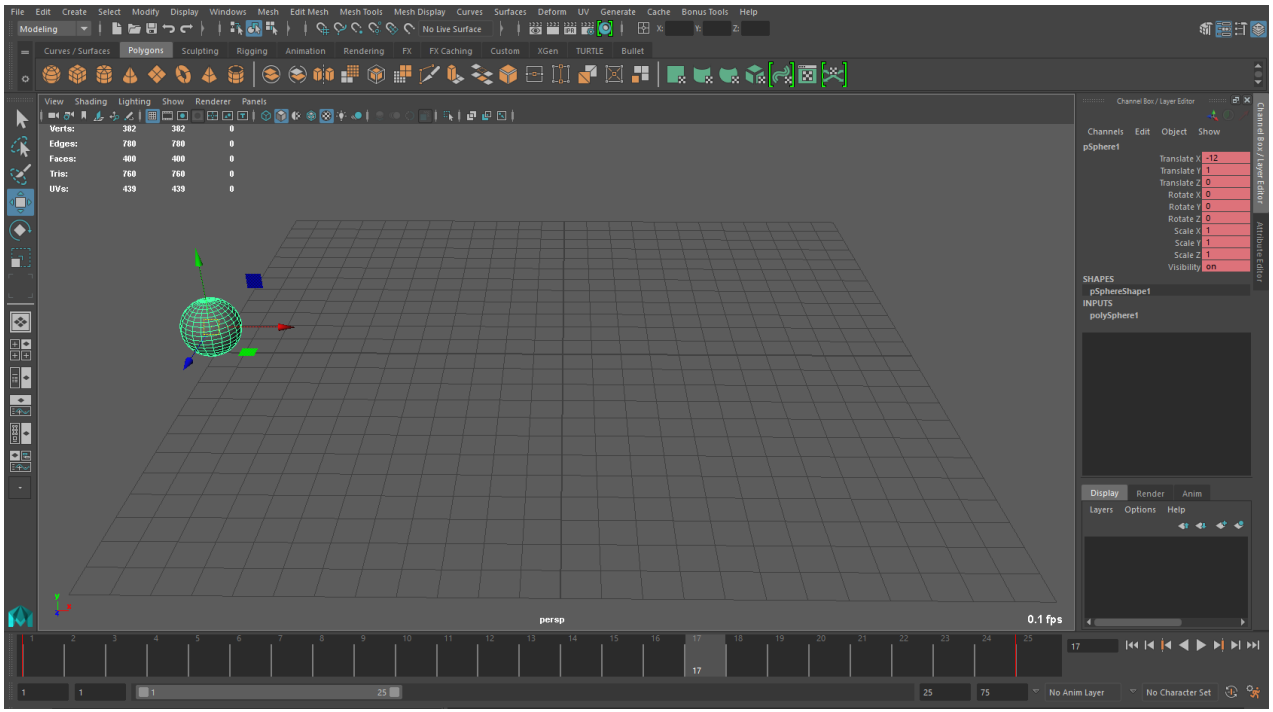
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

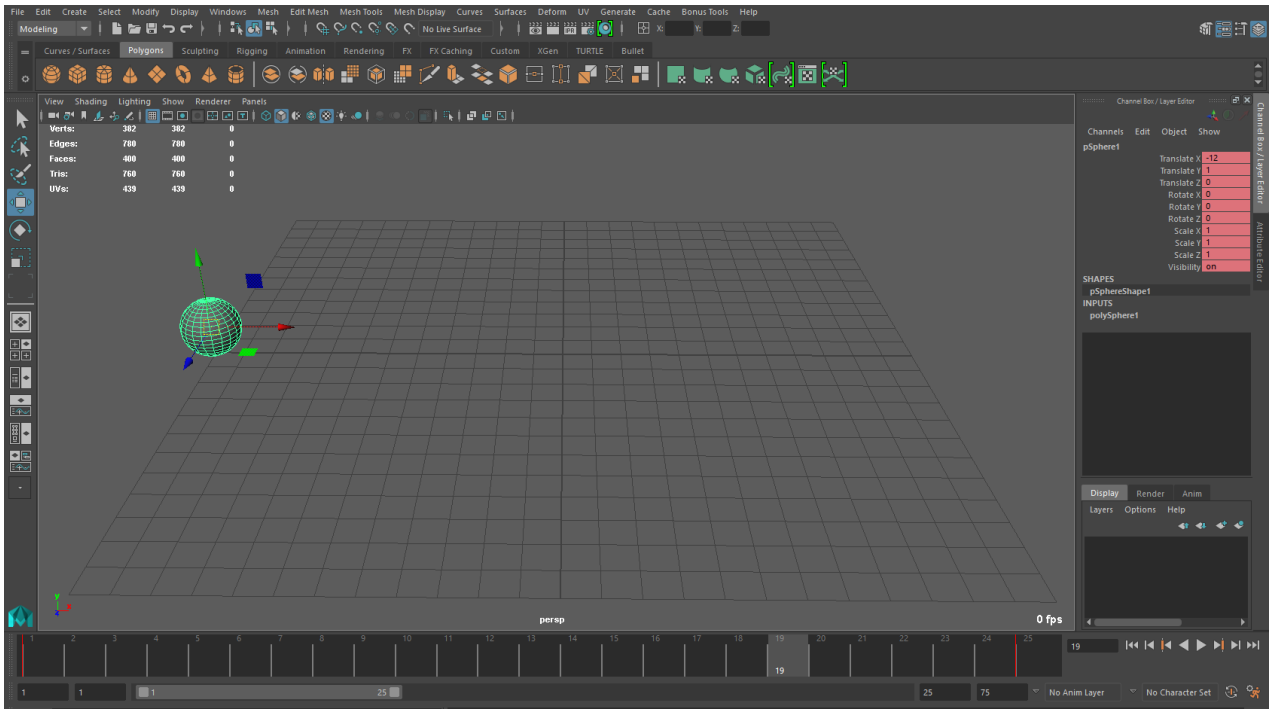
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

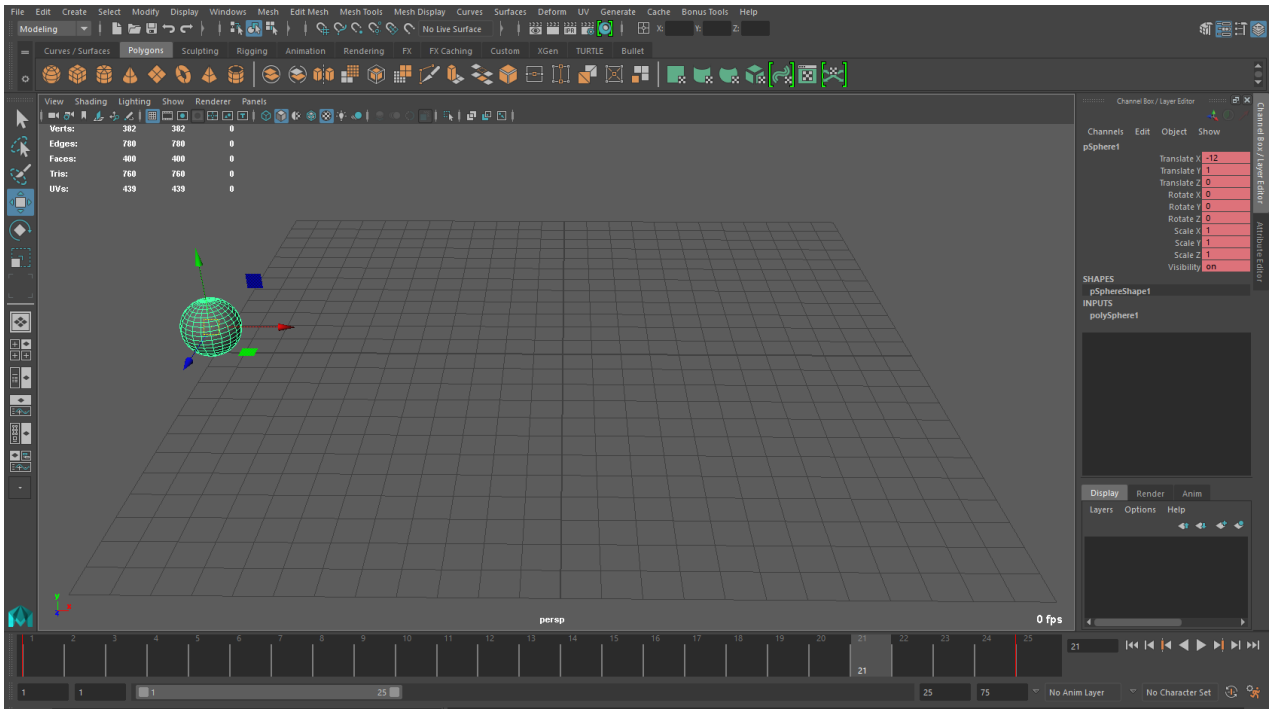
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

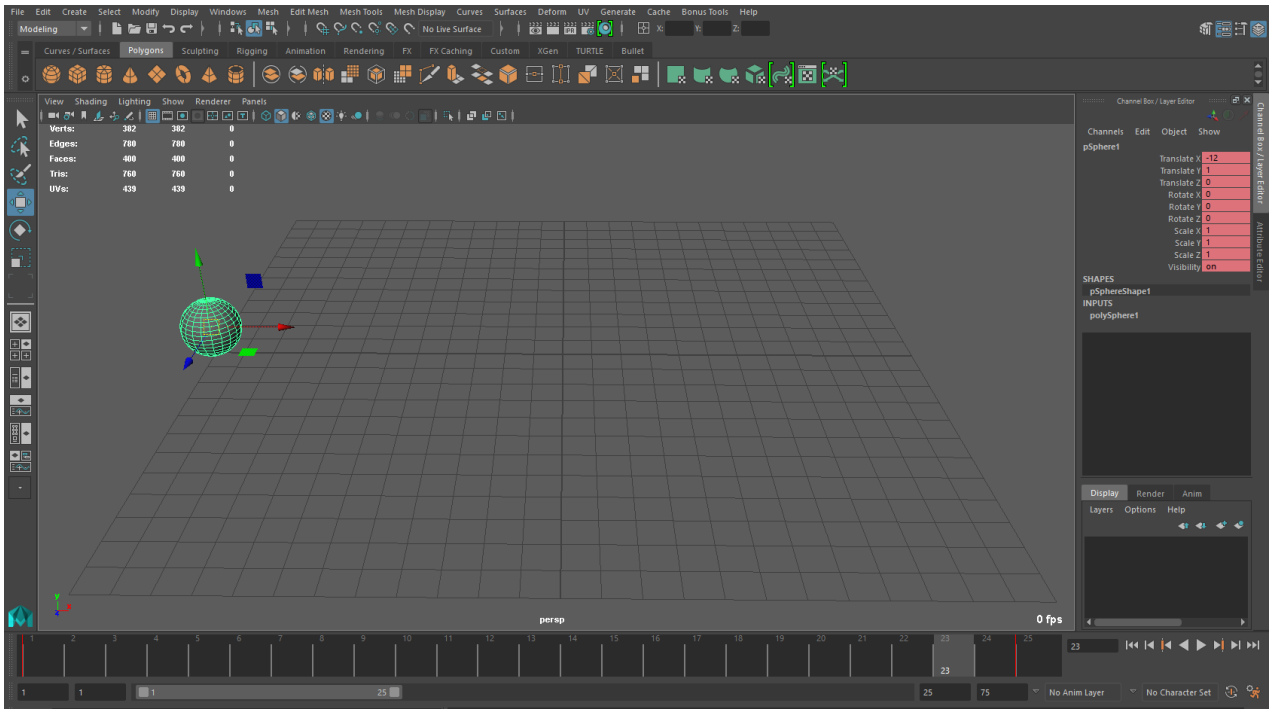
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

To change the current time:

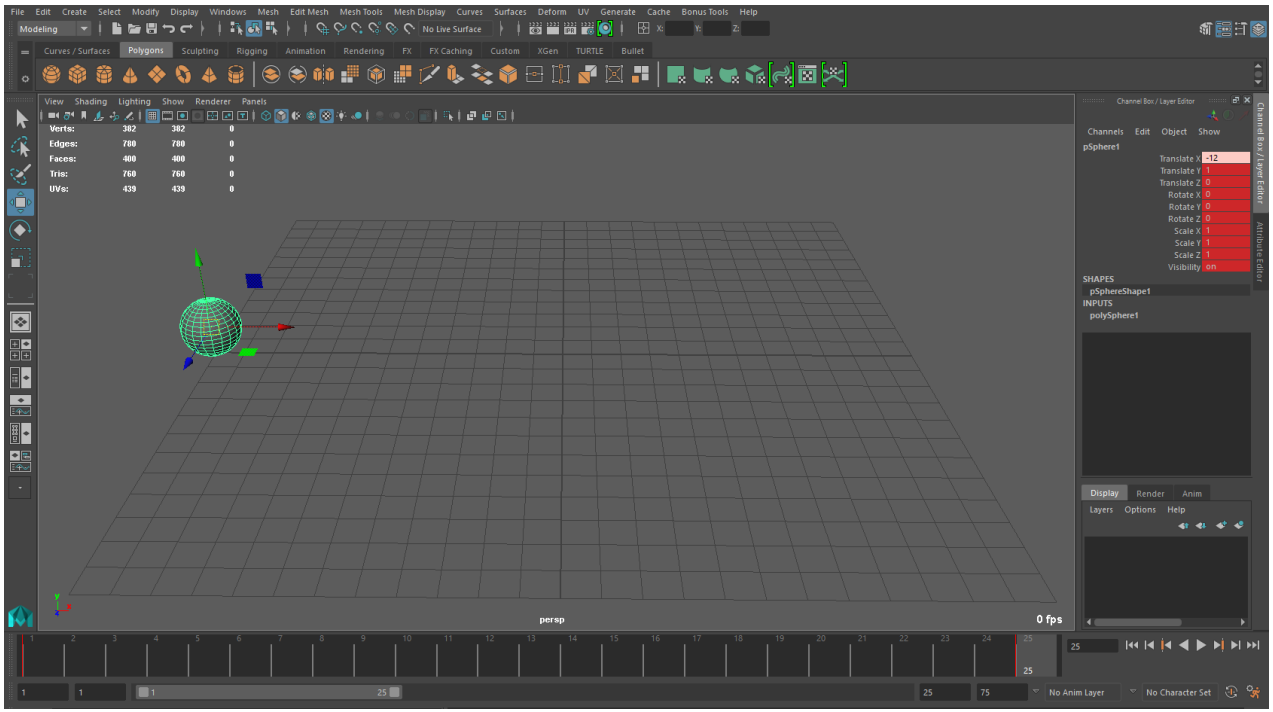
- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



Time Line Scrubbing Continued (slide 08)

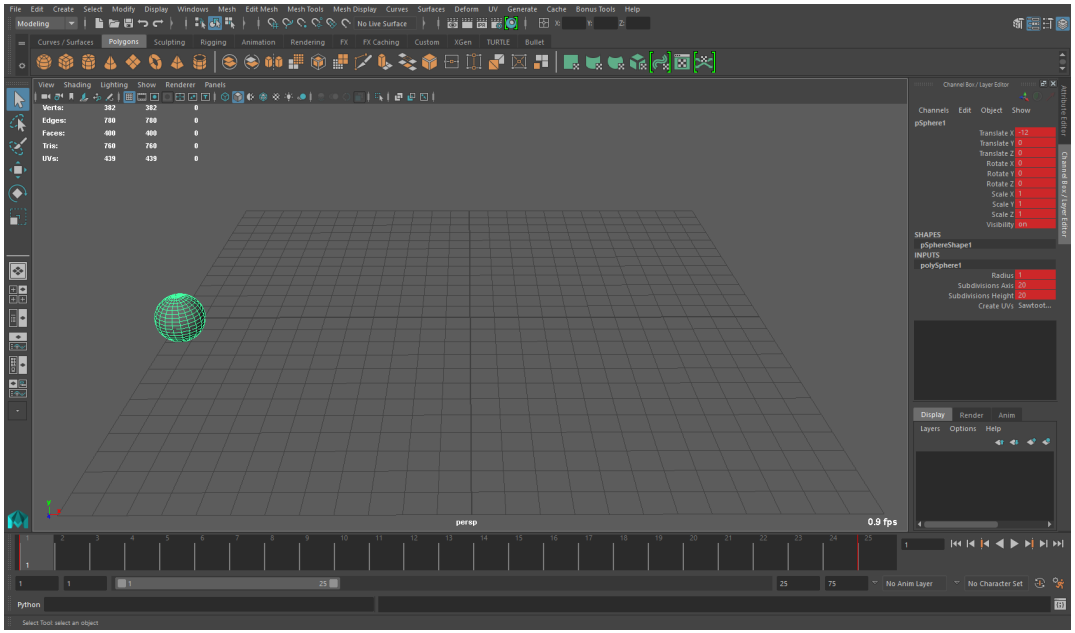
To change the current time:

- Change the current time (without updating objects' in-between positions) by scrubbing across the time line while pressing the middle mouse button (**MMB**).
- When away from the time change the current time (without updating objects' in-between positions) by scrubbing and pressing the key **k** and the **MMB**.



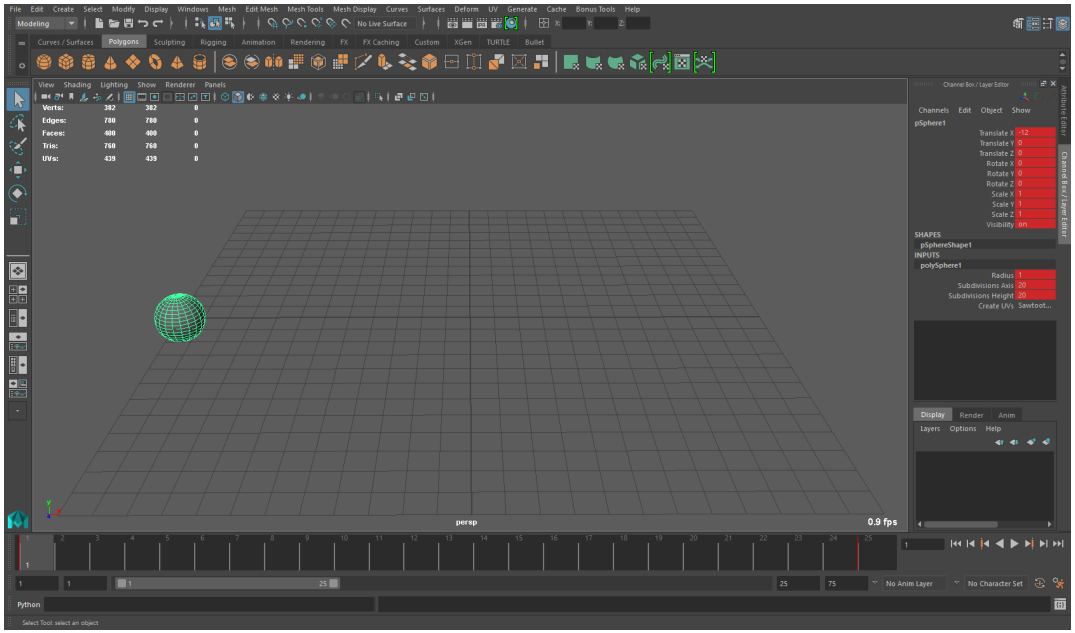
Setting Keys Using the Channel Editor

- We would like to animate only *Translate X/Y* as well as *Scale X/Y/Z*.
- Set the current time to 1.
- Select the *Translate Z* as well as the *Rotate X/Y/Z* and *Visibility* coefficients as well as all *Input* coefficients of the sphere in the Channel Editor;
- Right mouse click to pop-up the*Channels* menu and choose *Delete Selected*.
- The selected channels no longer hold keyed coefficients.



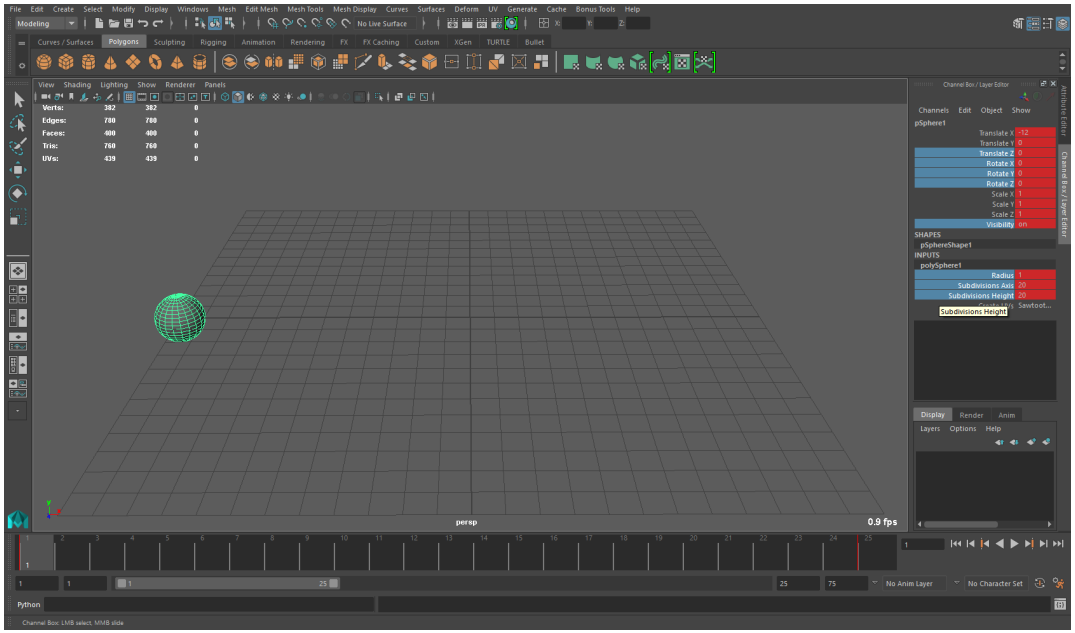
Setting Keys Using the Channel Editor

- We would like to animate only *Translate X/Y* as well as *Scale X/Y/Z*.
- Set the current time to 1.
- Select the *Translate Z* as well as the *Rotate X/Y/Z* and *Visibility* coefficients as well as all *Input* coefficients of the sphere in the Channel Editor;
- Right mouse click to pop-up the*Channels* menu and choose *Delete Selected*.
- The selected channels no longer hold keyed coefficients.



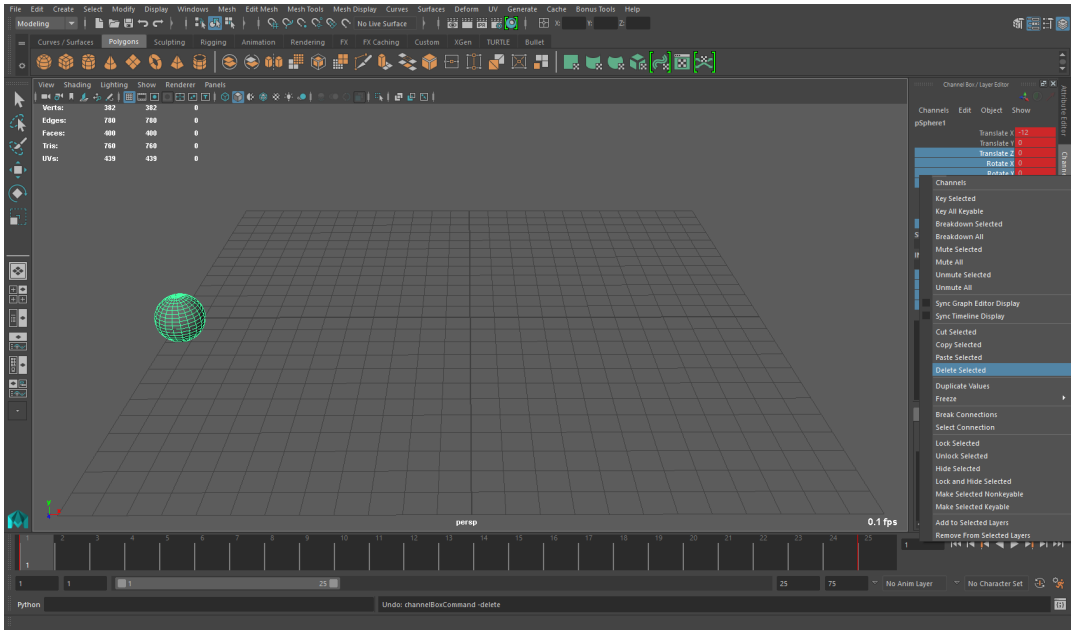
Setting Keys Using the Channel Editor

- We would like to animate only *Translate X/Y* as well as *Scale X/Y/Z*.
- Set the current time to 1.
- Select the *Translate Z* as well as the *Rotate X/Y/Z* and *Visibility* coefficients as well as all *Input* coefficients of the sphere in the Channel Editor;
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- The selected channels no longer hold keyed coefficients.



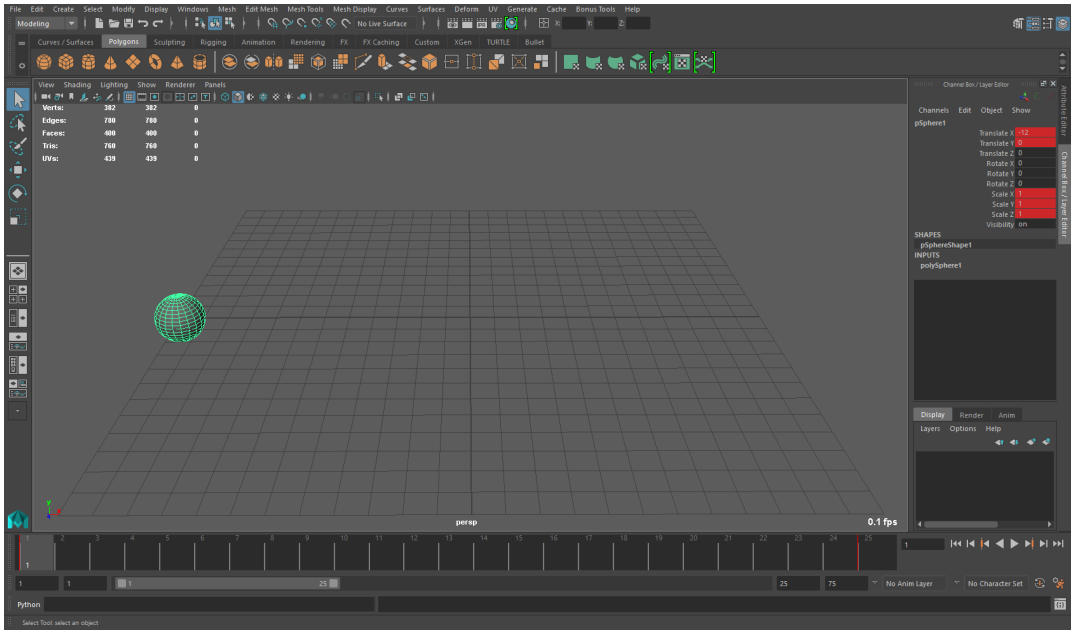
Setting Keys Using the Channel Editor

- We would like to animate only *Translate X/Y* as well as *Scale X/Y/Z*.
- Set the current time to 1.
- Select the *Translate Z* as well as the *Rotate X/Y/Z* and *Visibility* coefficients as well as all *Input* coefficients of the sphere in the Channel Editor;
- Right mouse click to pop-up the*Channels* menu and choose *Delete Selected*.
- The selected channels no longer hold keyed coefficients.



Setting Keys Using the Channel Editor

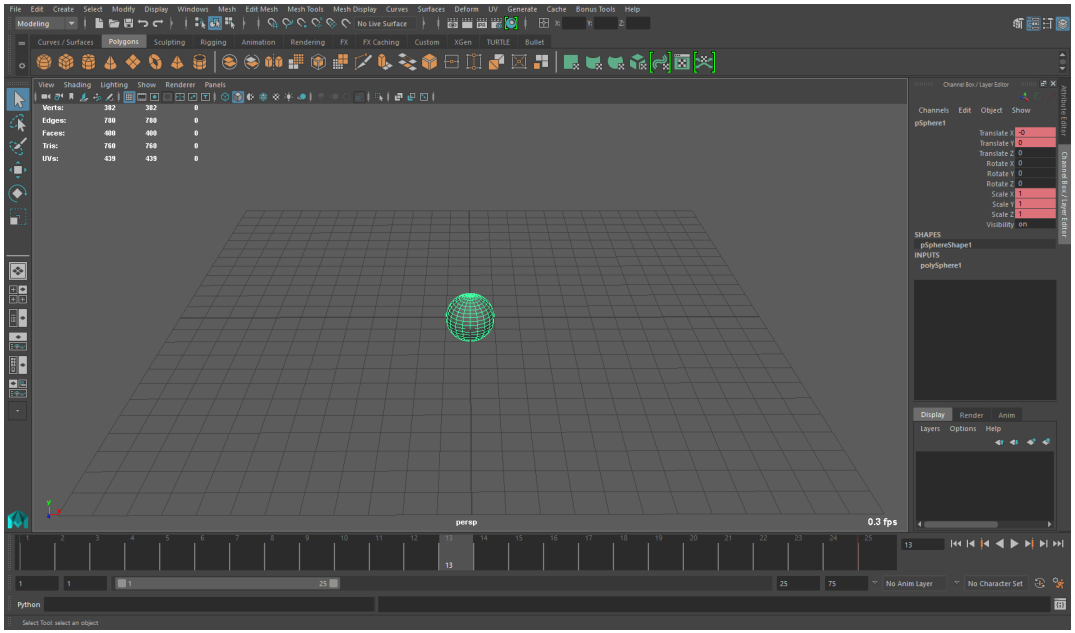
- We would like to animate only *Translate X/Y* as well as *Scale X/Y/Z*.
- Set the current time to 1.
- Select the *Translate Z* as well as the *Rotate X/Y/Z* and *Visibility* coefficients as well as all *Input* coefficients of the sphere in the Channel Editor;
- Right mouse click to pop-up the*Channels* menu and choose *Delete Selected*.
- The selected channels no longer hold keyed coefficients.



Adding More Key Frames (slide 09 β)

Setting Keys Using the Channel Editor

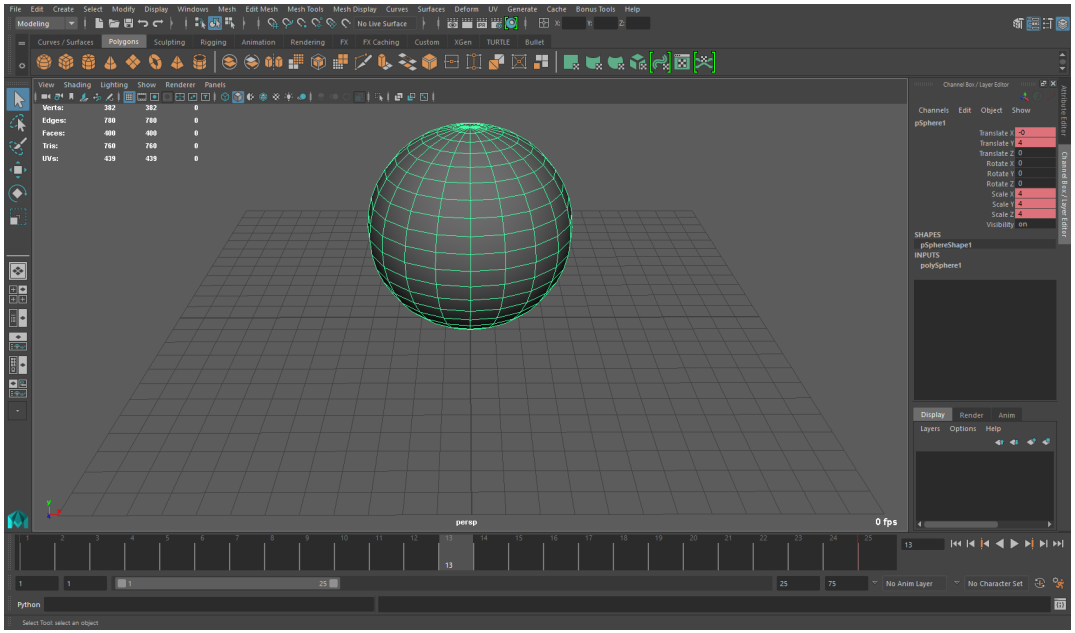
- Set the current time to 13.
- At frame 13 modify the scale coefficients of the sphere to (4,4,4) and move it to position (0, 4, 0).
- Select the *Translate X/Y* as well as the *Scale X/Y/Z* coefficients of the sphere in the Channel Editor; right mouse click to pop-up the*Channels* menu and choose *Key Selected*.
- You have added a new key frame (at frame 13).



Adding More Key Frames (slide 09 β)

Setting Keys Using the Channel Editor

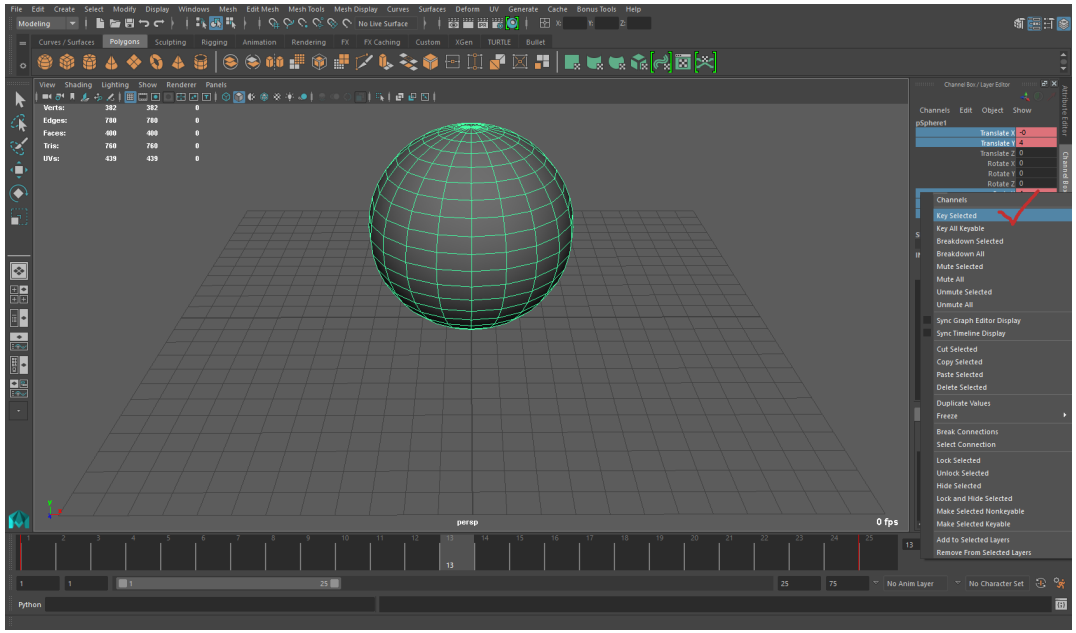
- Set the current time to 13.
- At frame 13 modify the scale coefficients of the sphere to (4,4,4) and move it to position (0, 4, 0).
- Select the *Translate X/Y* as well as the *Scale X/Y/Z* coefficients of the sphere in the Channel Editor; right mouse click to pop-up the*Channels* menu and choose *Key Selected*.
- You have added a new key frame (at frame 13).



Adding More Key Frames (slide 09 β)

Setting Keys Using the Channel Editor

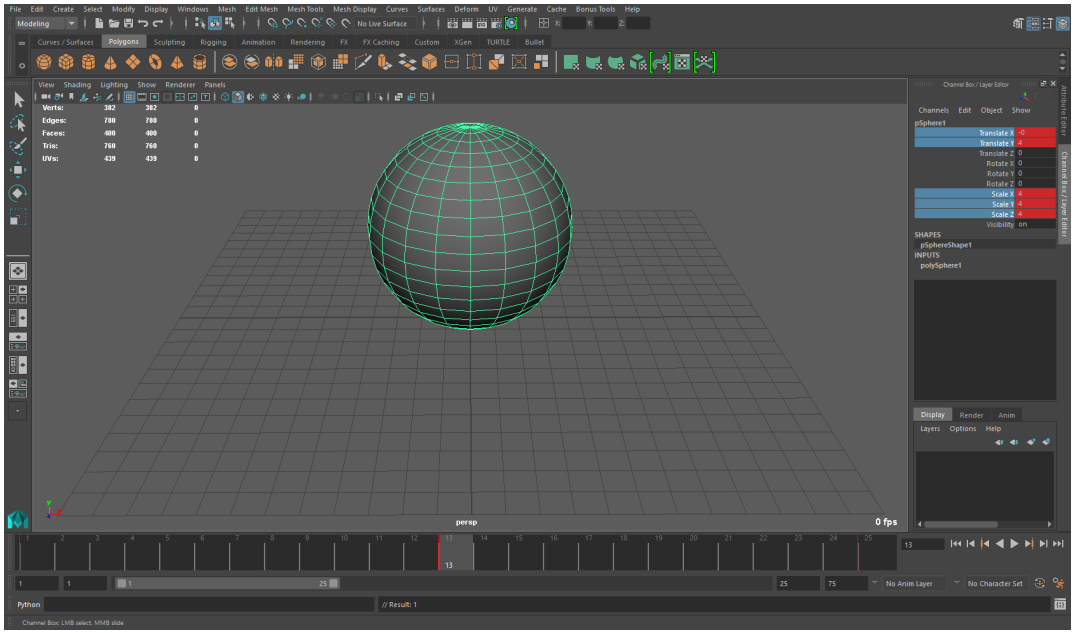
- Set the current time to 13.
- At frame 13 modify the scale coefficients of the sphere to (4,4,4) and move it to position (0, 4, 0).
- Select the *Translate X/Y* as well as the *Scale X/Y/Z* coefficients of the sphere in the Channel Editor; right mouse click to pop-up the*Channels* menu and choose *Key Selected*.
- You have added a new key frame (at frame 13).



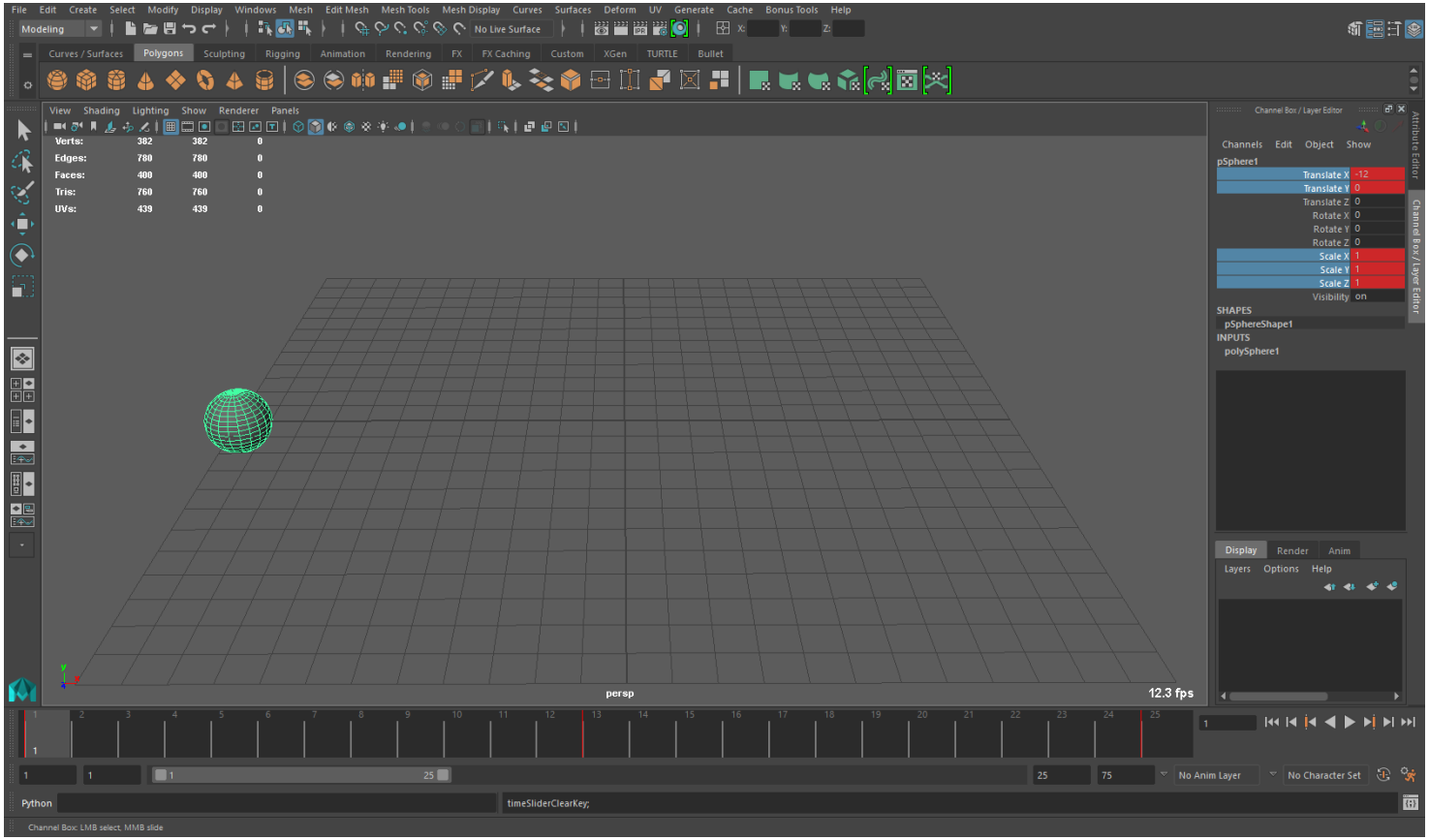
Adding More Key Frames (slide 09 β)

Setting Keys Using the Channel Editor

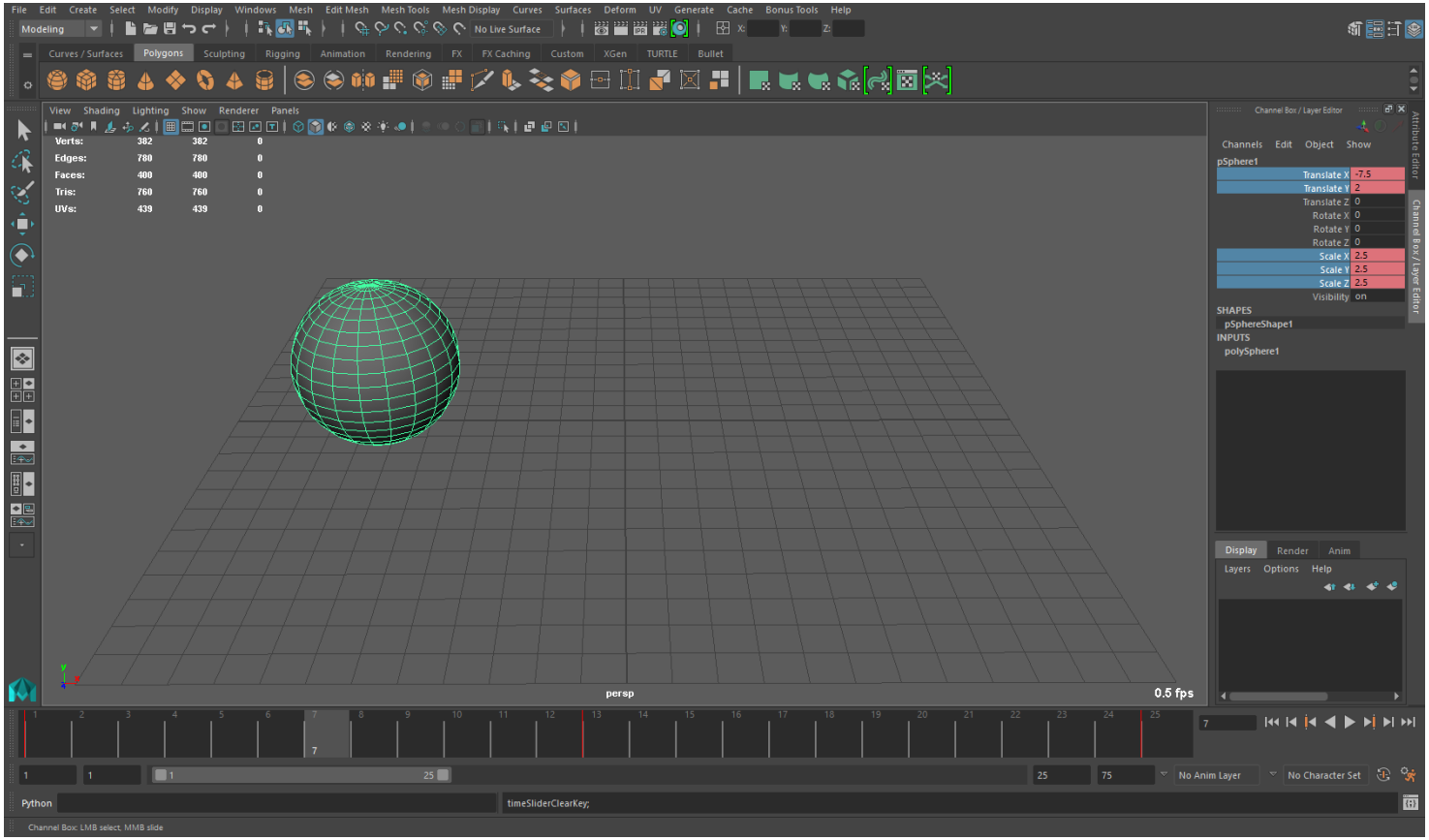
- Set the current time to 13.
- At frame 13 modify the scale coefficients of the sphere to (4,4,4) and move it to position (0, 4, 0).
- Select the *Translate X/Y* as well as the *Scale X/Y/Z* coefficients of the sphere in the Channel Editor; right mouse click to pop-up the*Channels* menu and choose *Key Selected*.
- You have added a new key frame (at frame 13).



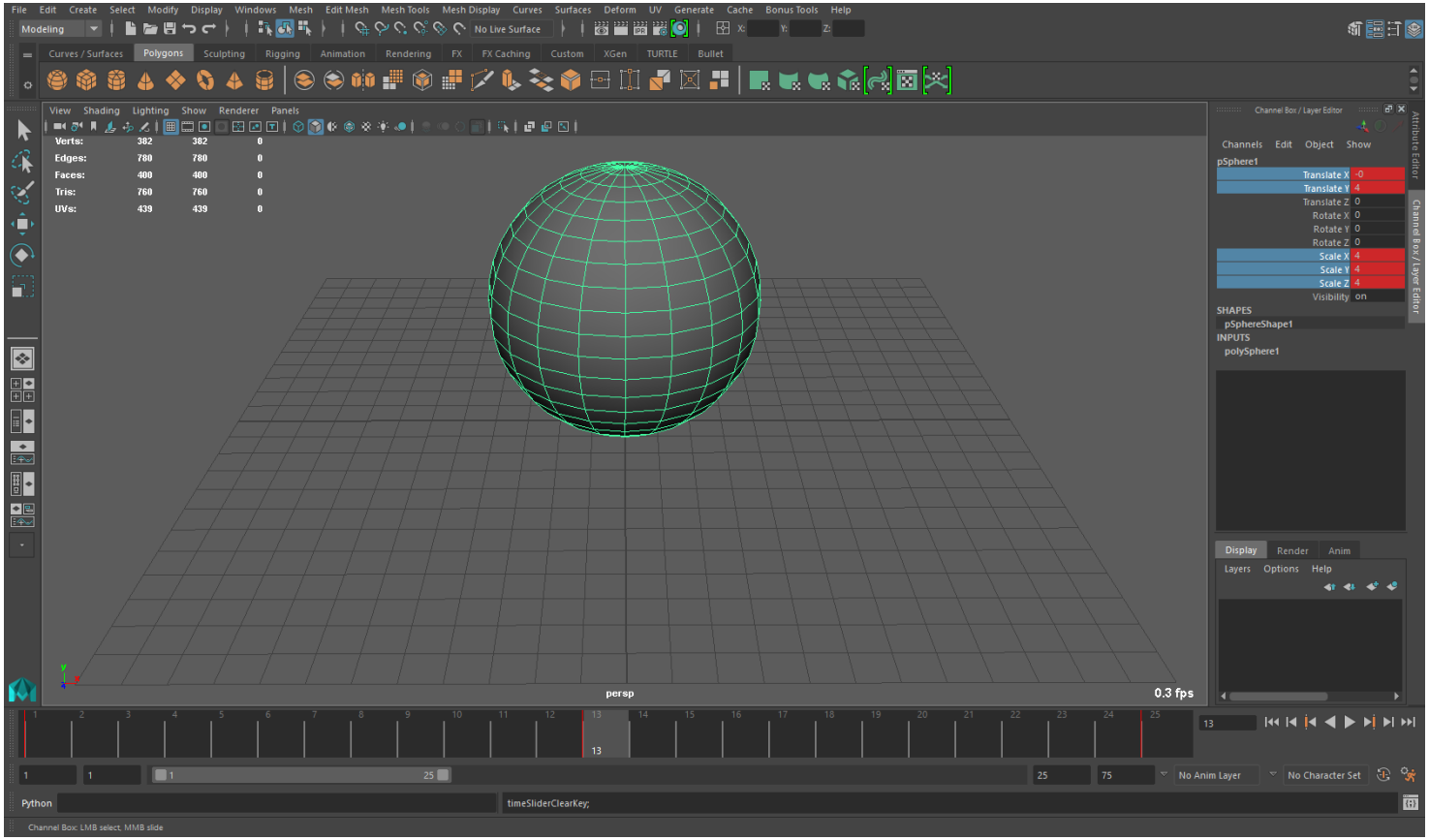
Adding More Key Frames Continued (slide 10)



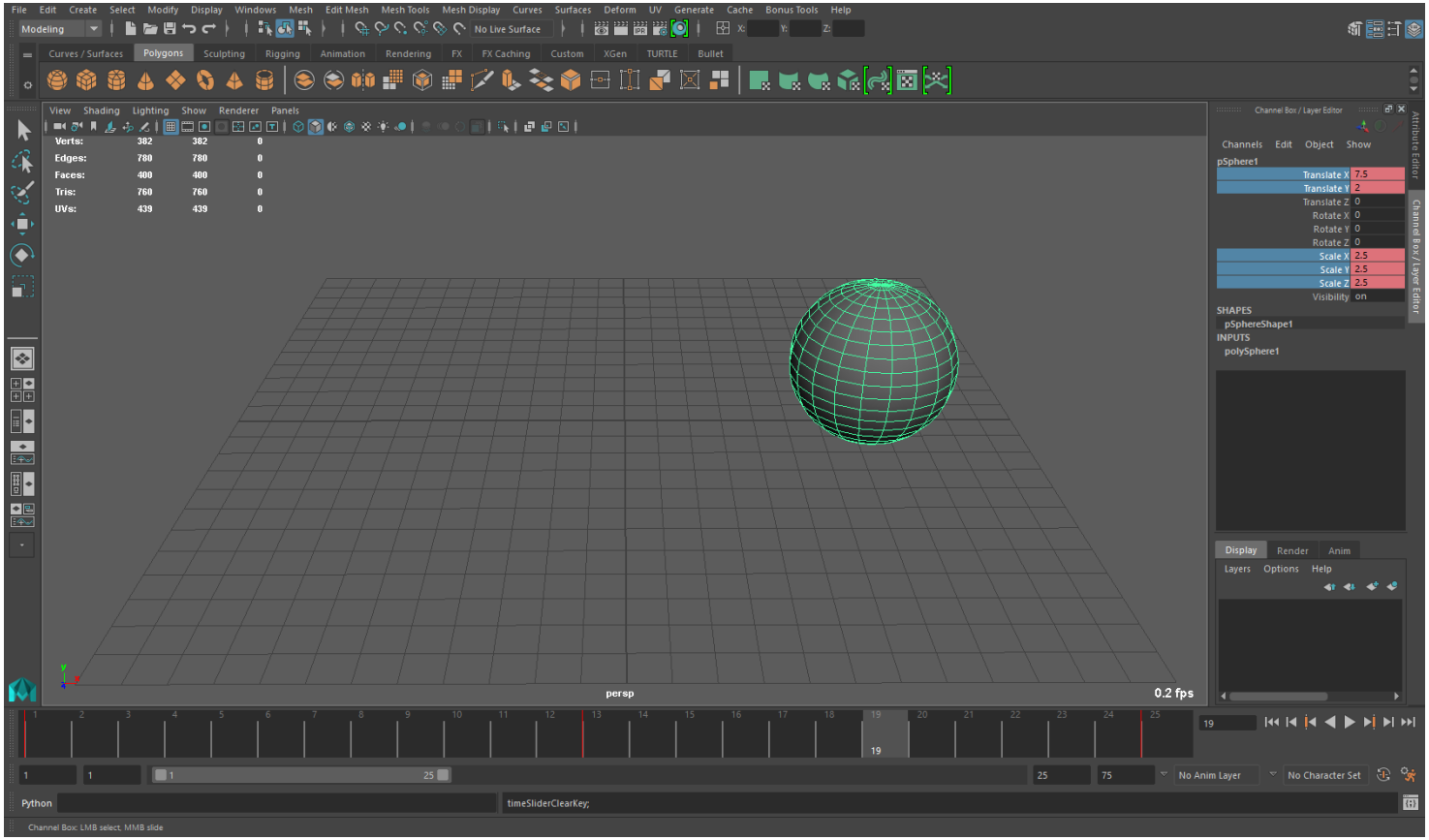
Adding More Key Frames Continued (slide 10)



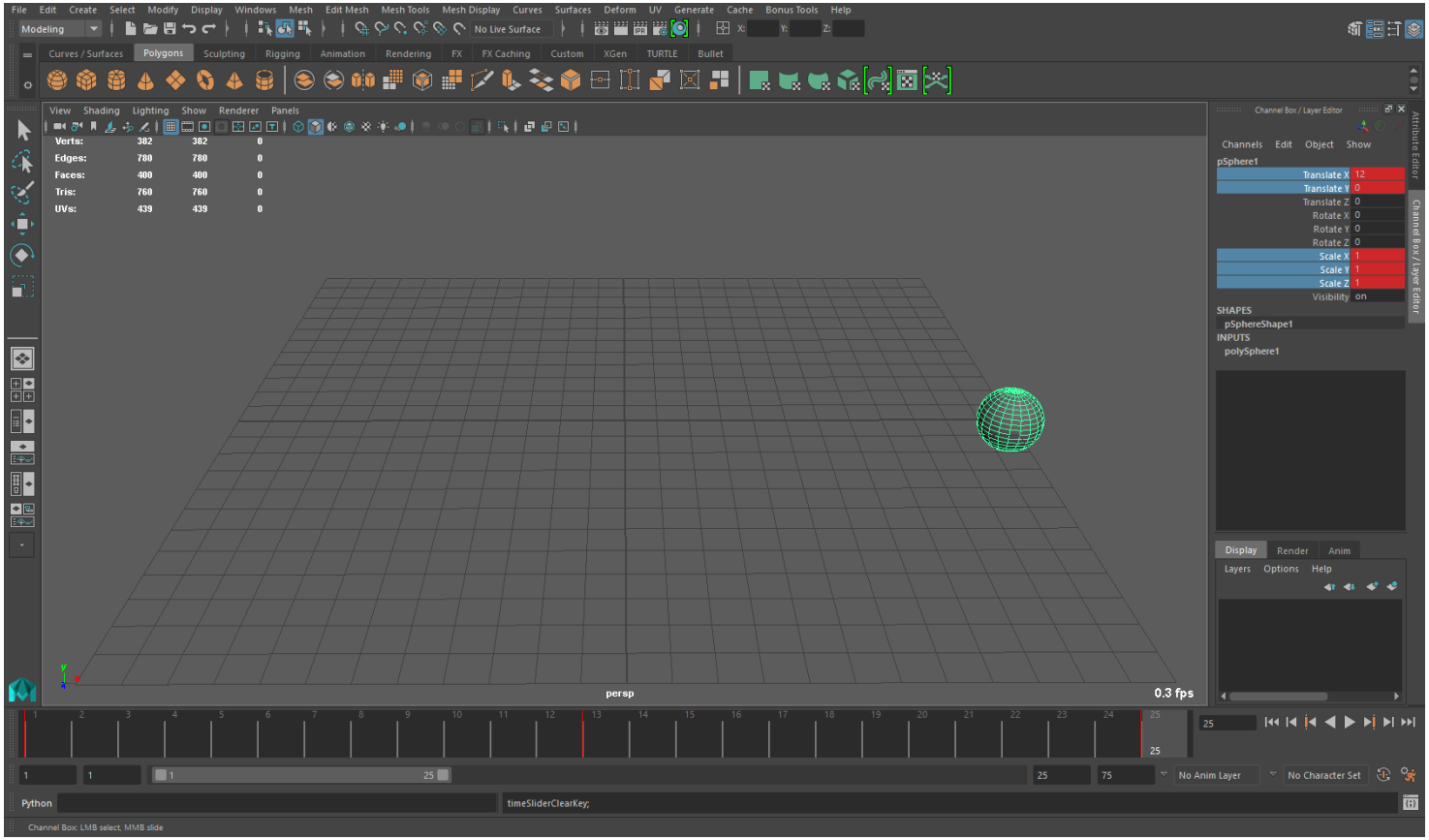
Adding More Key Frames Continued (slide 10)



Adding More Key Frames Continued (slide 10)



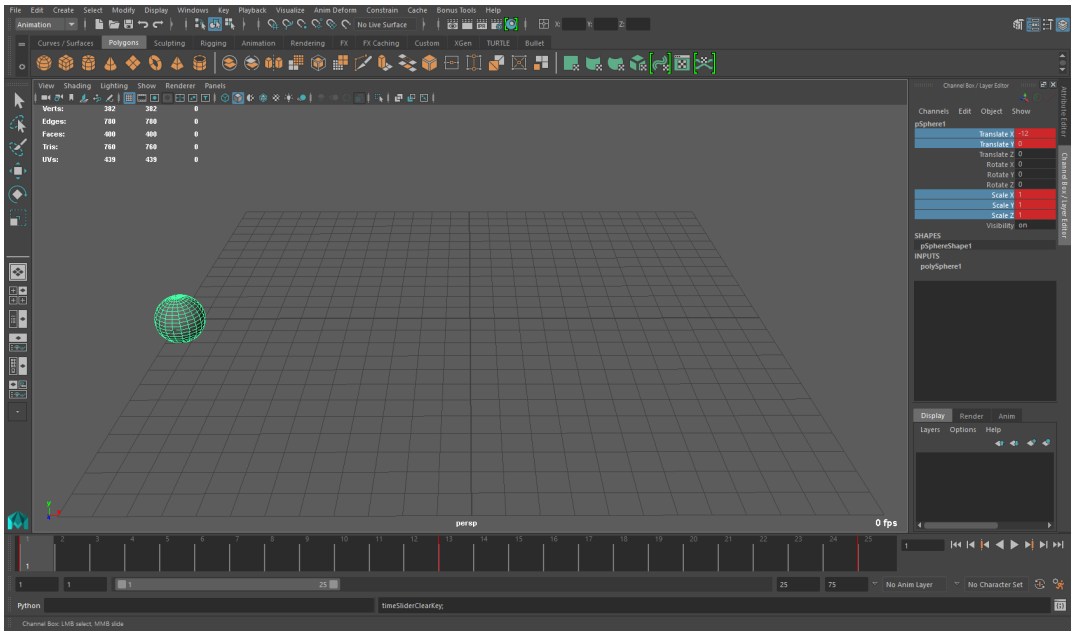
Adding More Key Frames Continued (slide 10)



Copy and Paste (slide 11)

Creating key frames using the Copy and Paste operations

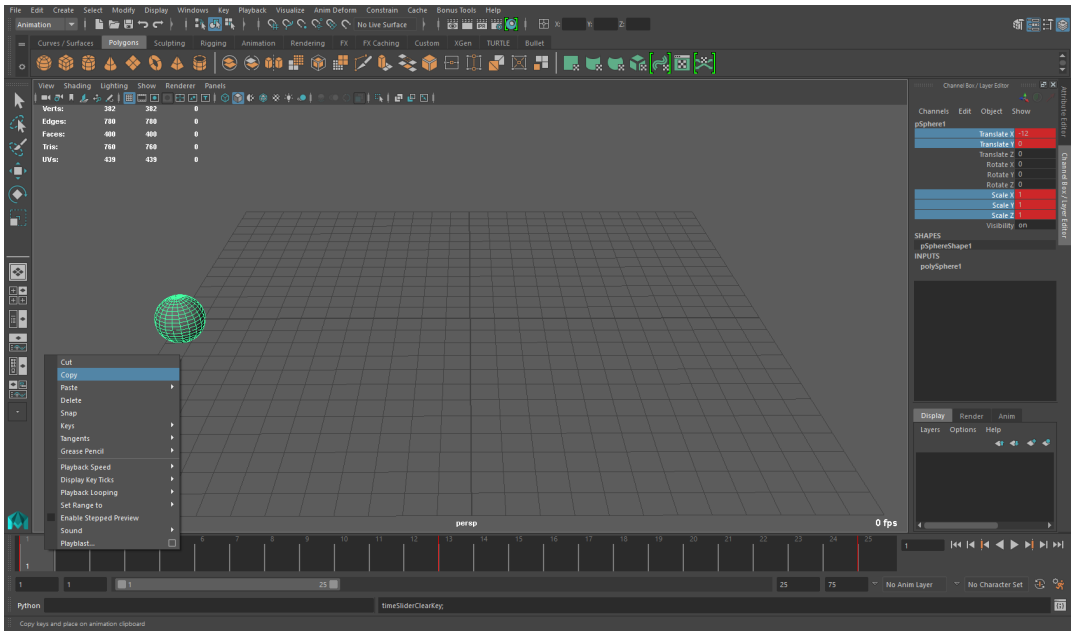
- Let's set current time to 1.
- On the time line right-click on frame 1 and in the pop-up menu select *Copy*.
- With the middle mouse button scrub to frame 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Paste*.
- A new key-frame is set in frame 17 with the same coefficients as frame 1.



Copy and Paste (slide 11)

Creating key frames using the Copy and Paste operations

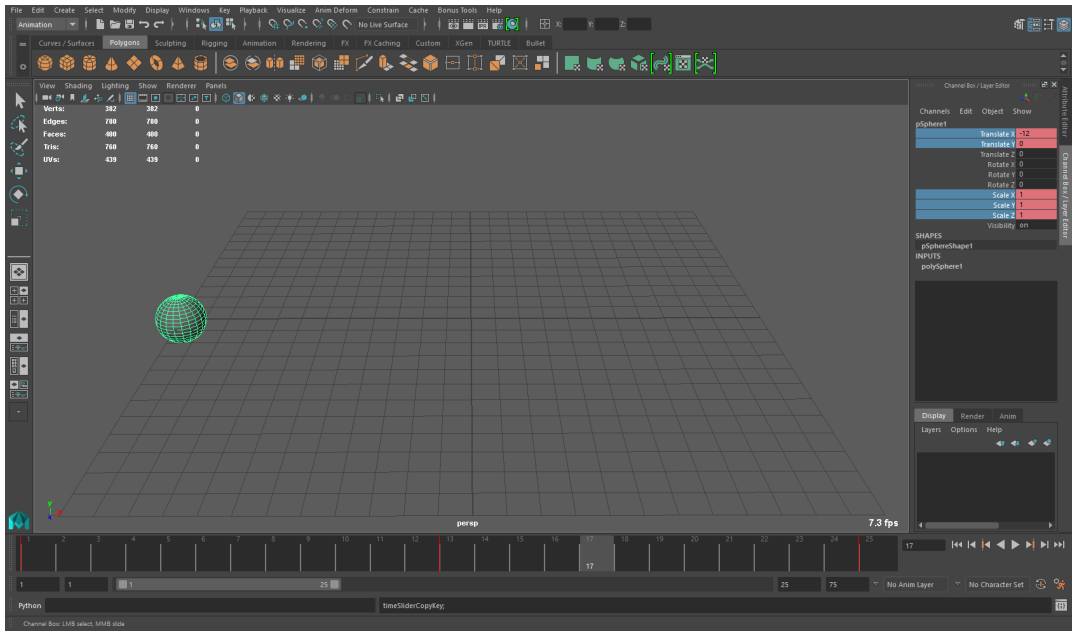
- Let's set current time to 1.
- On the time line right-click on frame 1 and in the pop-up menu select *Copy*.
- With the middle mouse button scrub to frame 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Paste*.
- A new key-frame is set in frame 17 with the same coefficients as frame 1.



Copy and Paste (slide 11)

Creating key frames using the Copy and Paste operations

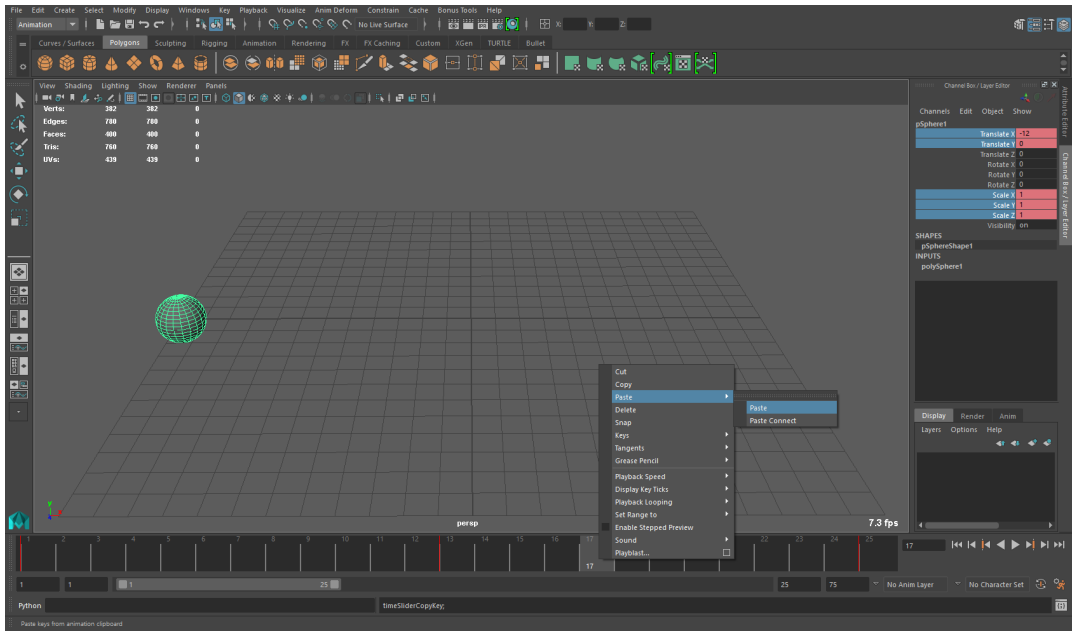
- Let's set current time to 1.
- On the time line right-click on frame 1 and in the pop-up menu select *Copy*.
- With the middle mouse button scrub to frame 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Paste*.
- A new key-frame is set in frame 17 with the same coefficients as frame 1.



Copy and Paste (slide 11)

Creating key frames using the Copy and Paste operations

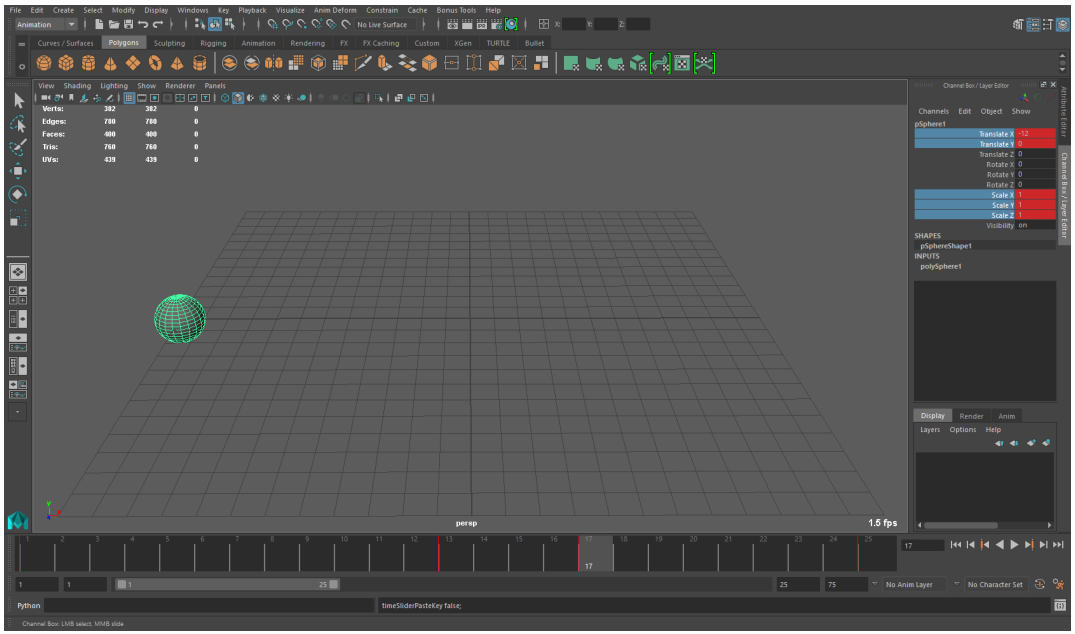
- Let's set current time to 1.
- On the time line right-click on frame 1 and in the pop-up menu select *Copy*.
- With the middle mouse button scrub to frame 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Paste*.
- A new key-frame is set in frame 17 with the same coefficients as frame 1.



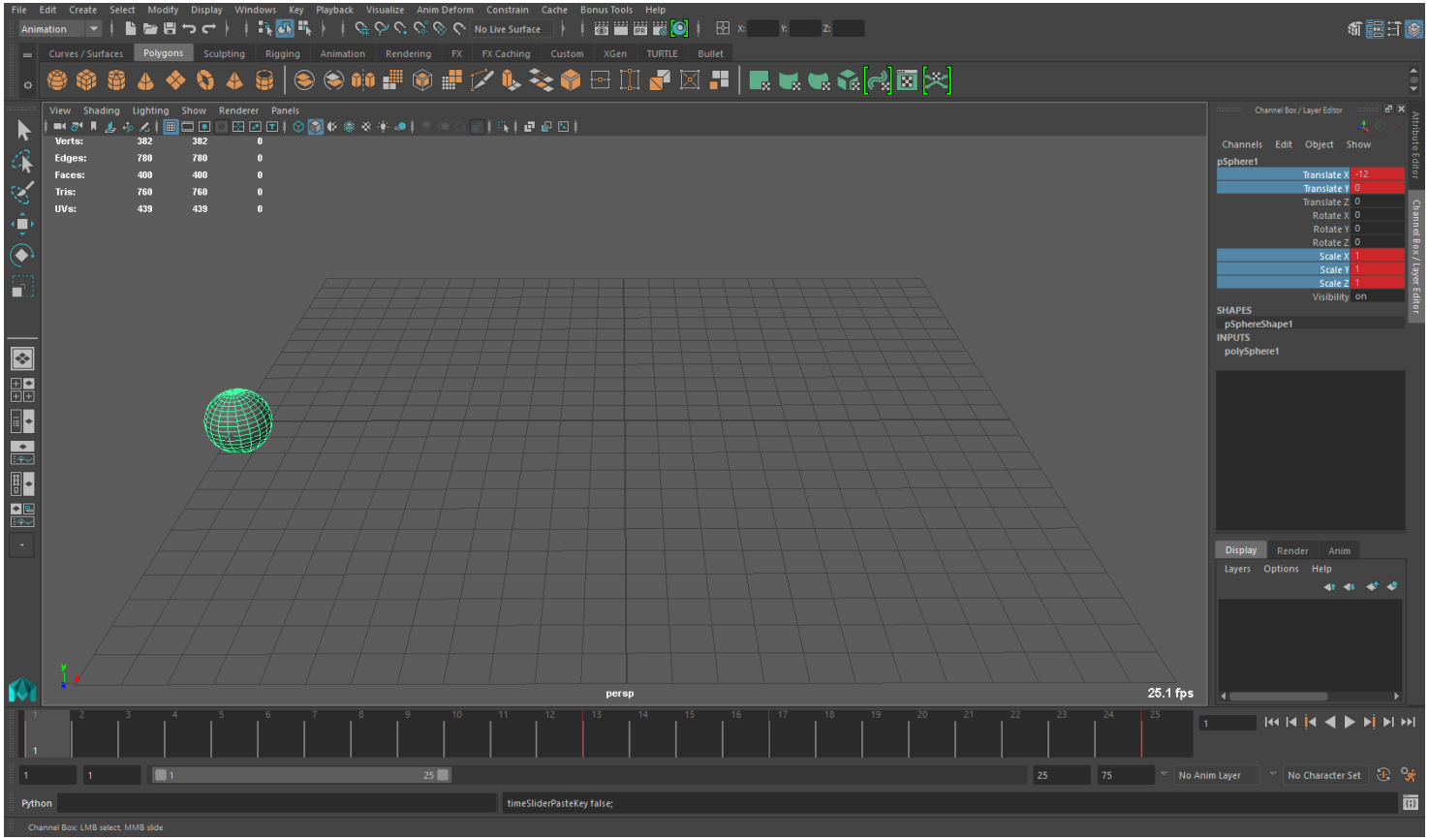
Copy and Paste (slide 11)

Creating key frames using the Copy and Paste operations

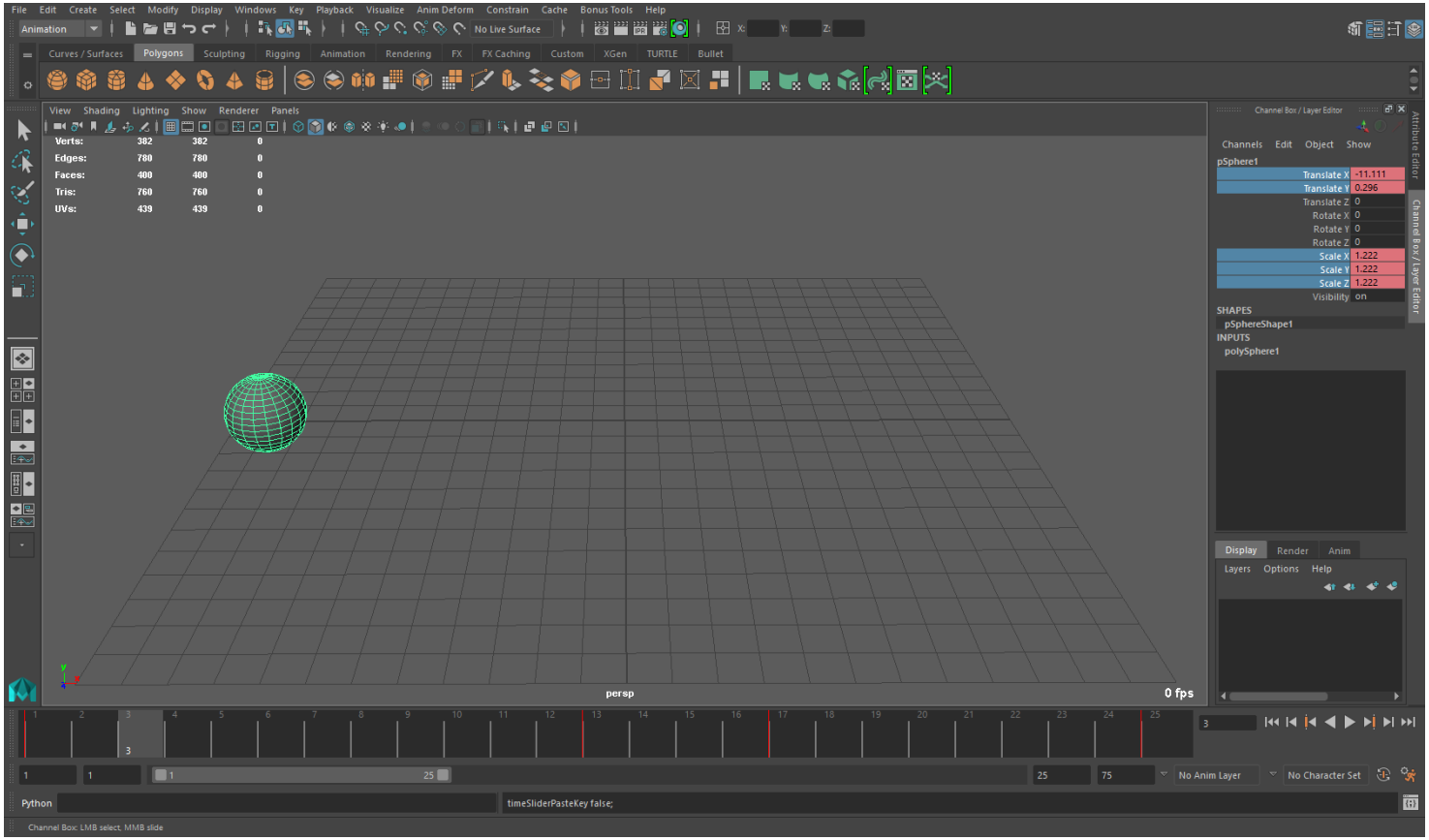
- Let's set current time to 1.
- On the time line right-click on frame 1 and in the pop-up menu select *Copy*.
- With the middle mouse button scrub to frame 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Paste*.
- A new key-frame is set in frame 17 with the same coefficients as frame 1.



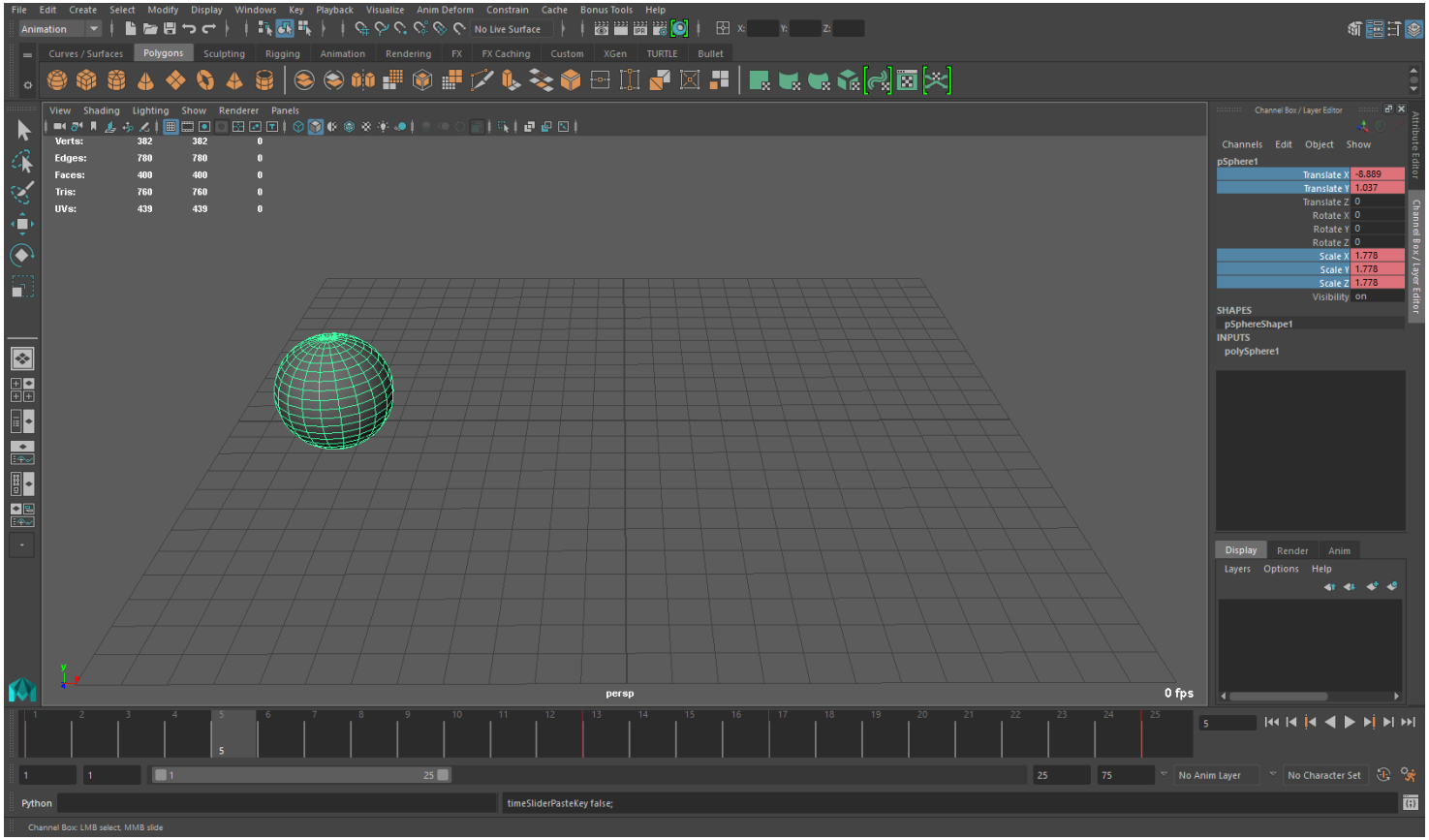
Copy Paste Continued (slide 12)



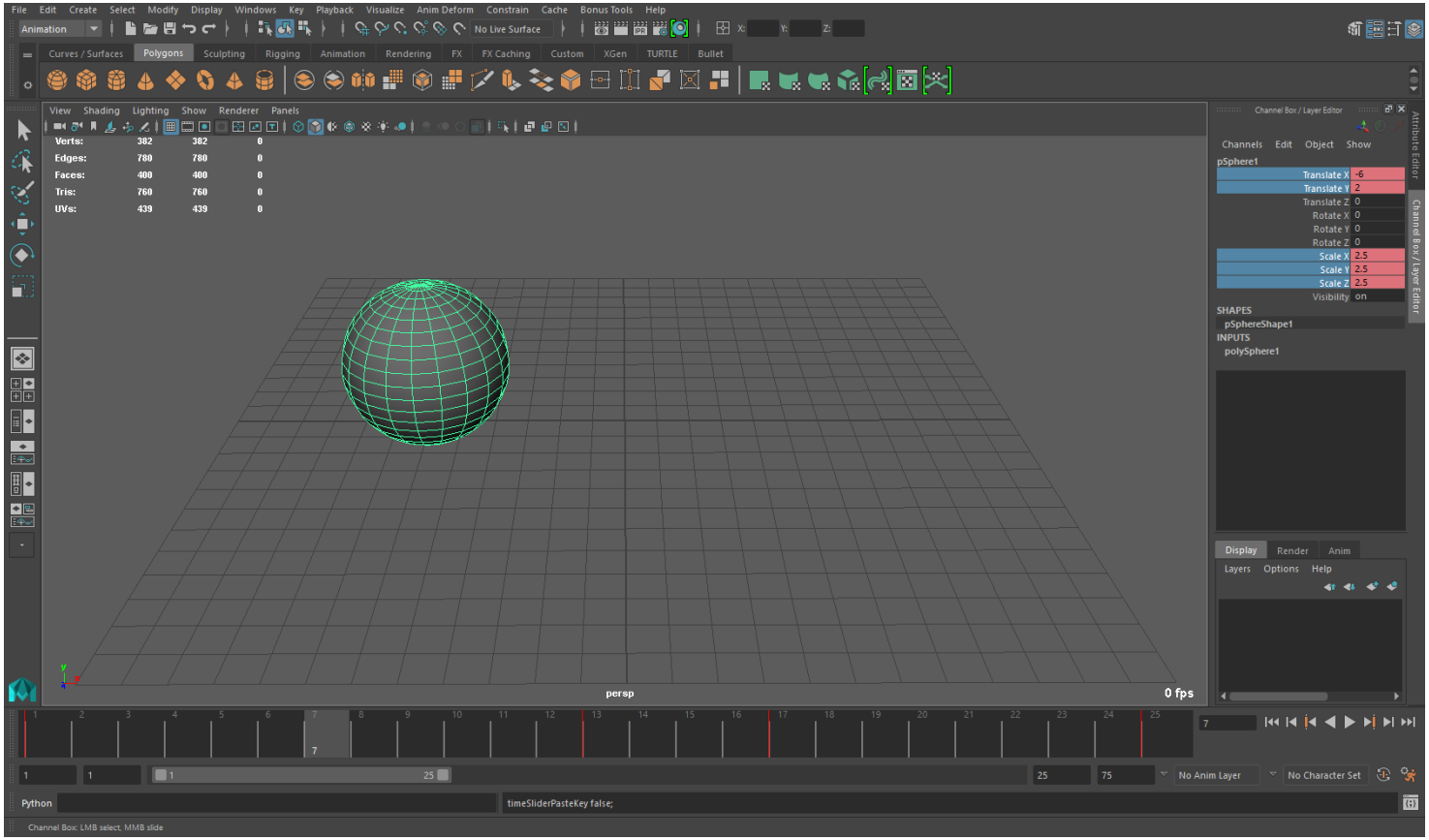
Copy Paste Continued (slide 12)



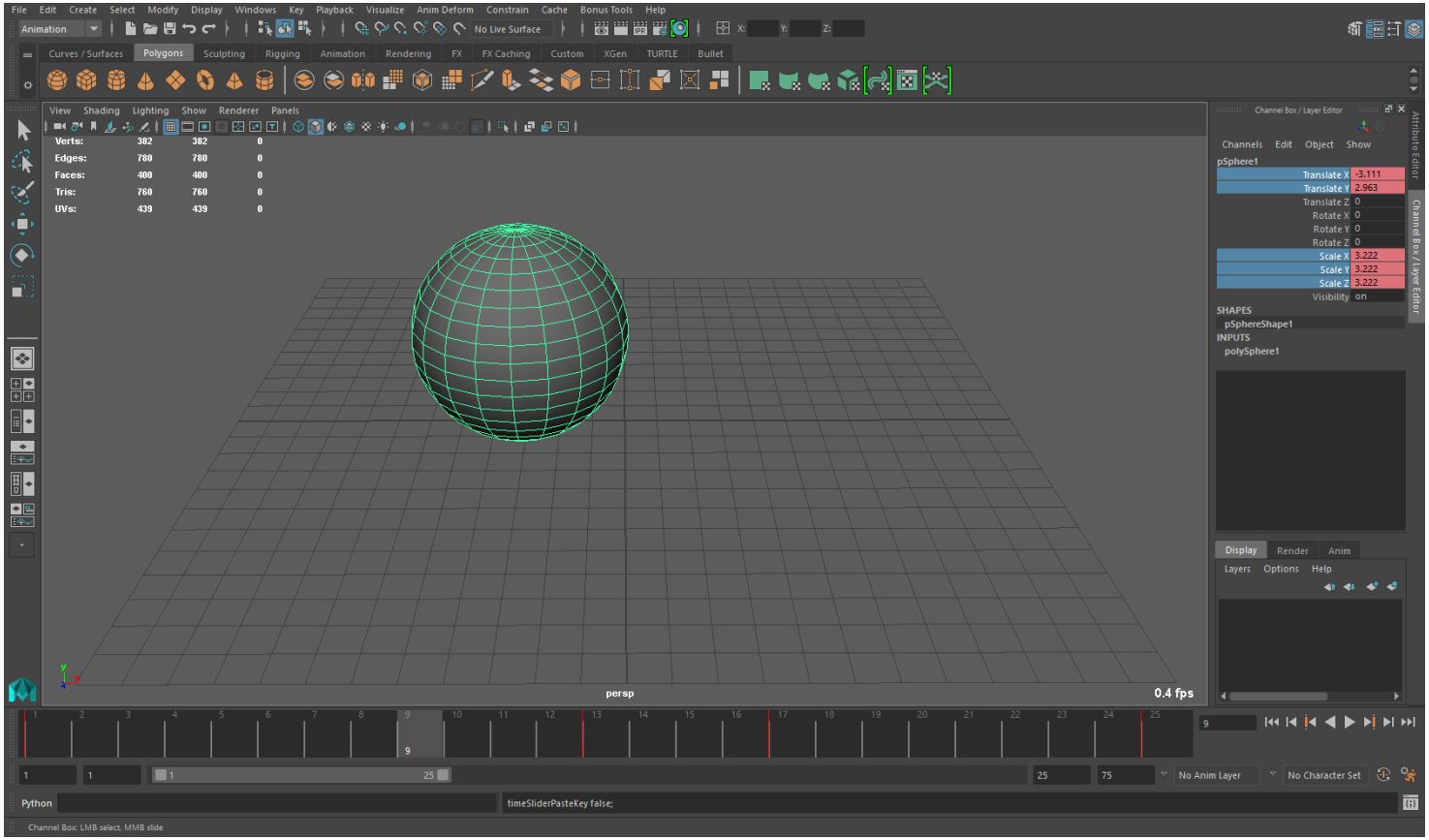
Copy Paste Continued (slide 12)



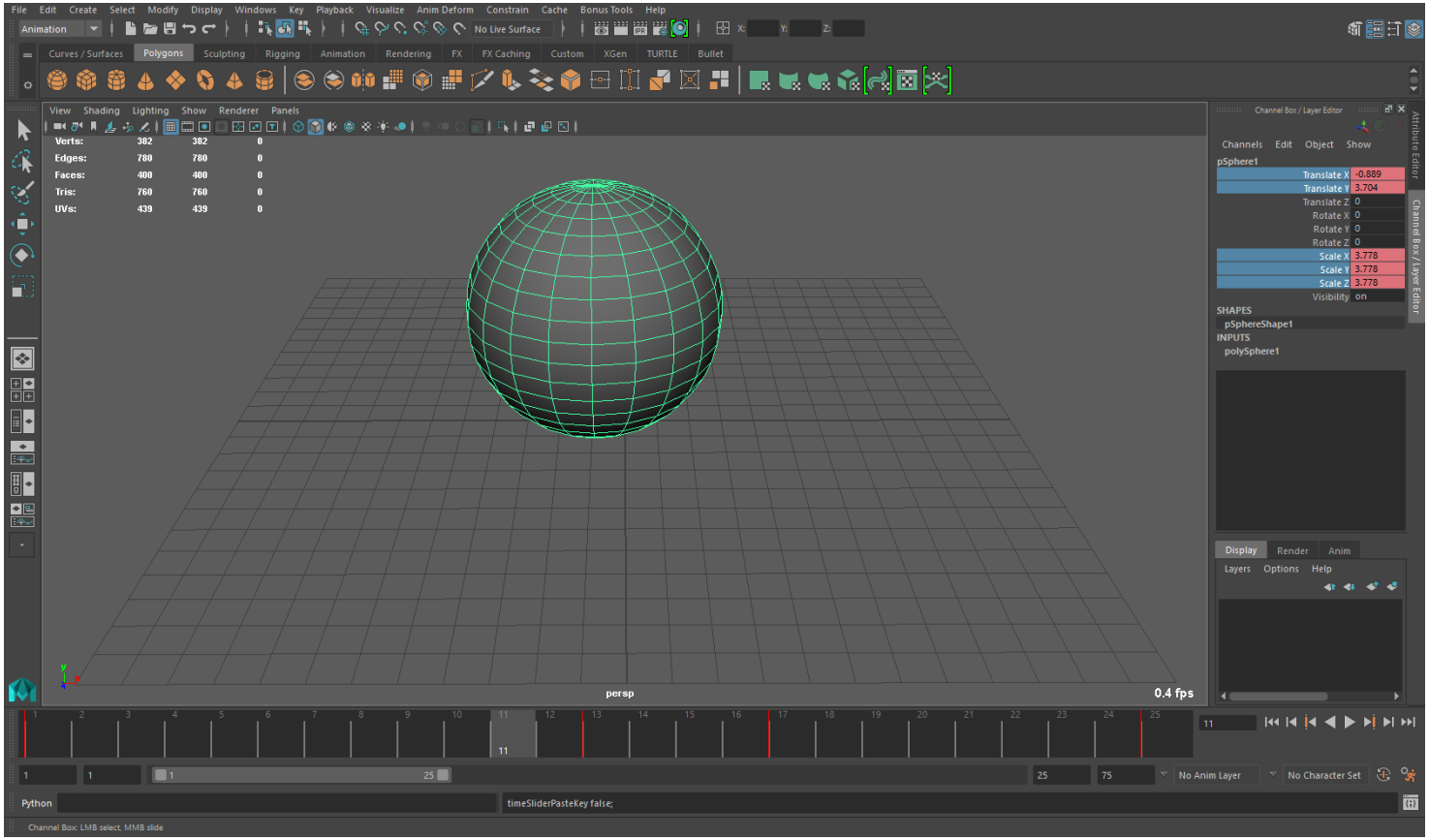
Copy Paste Continued (slide 12)



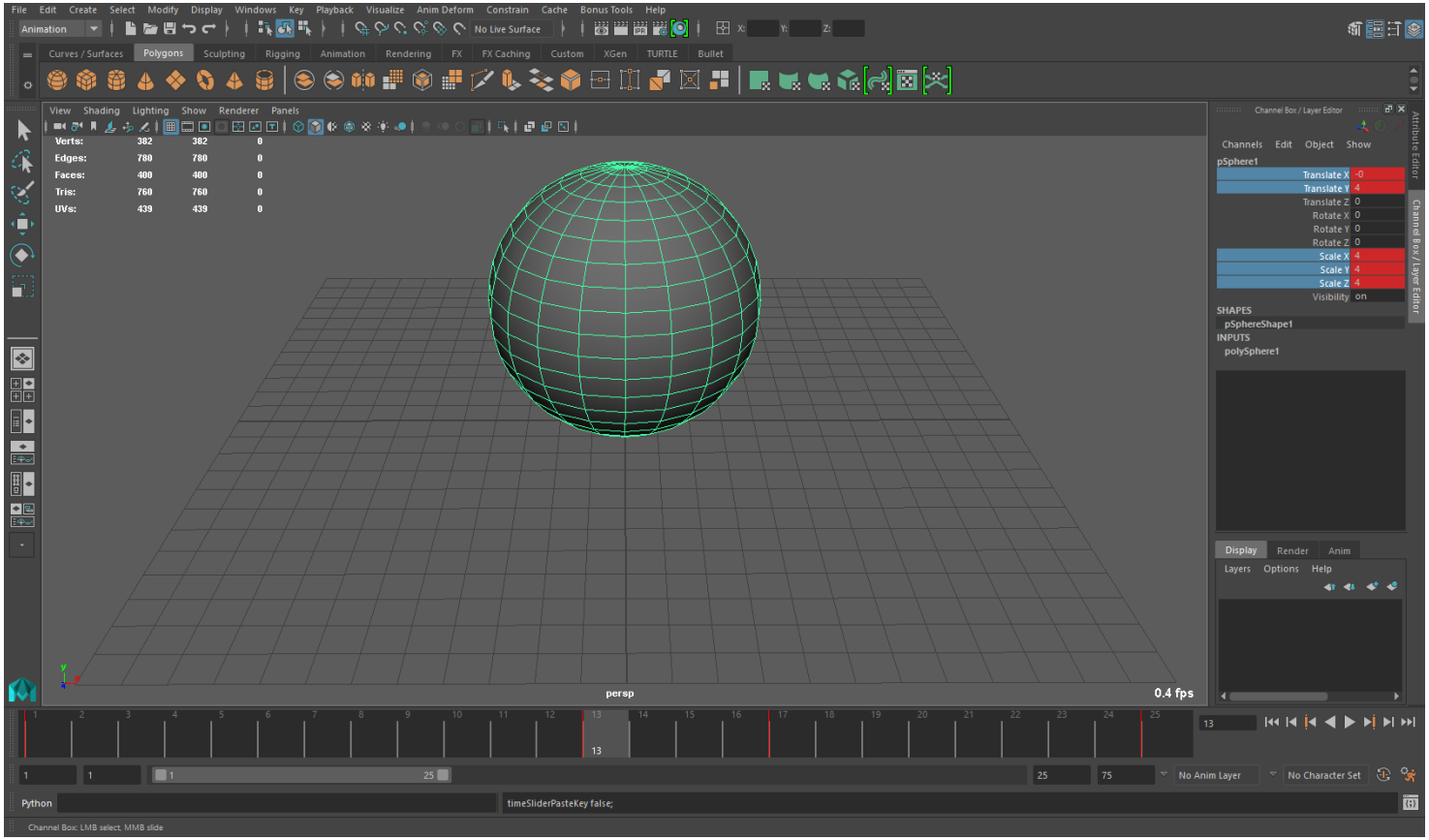
Copy Paste Continued (slide 12)



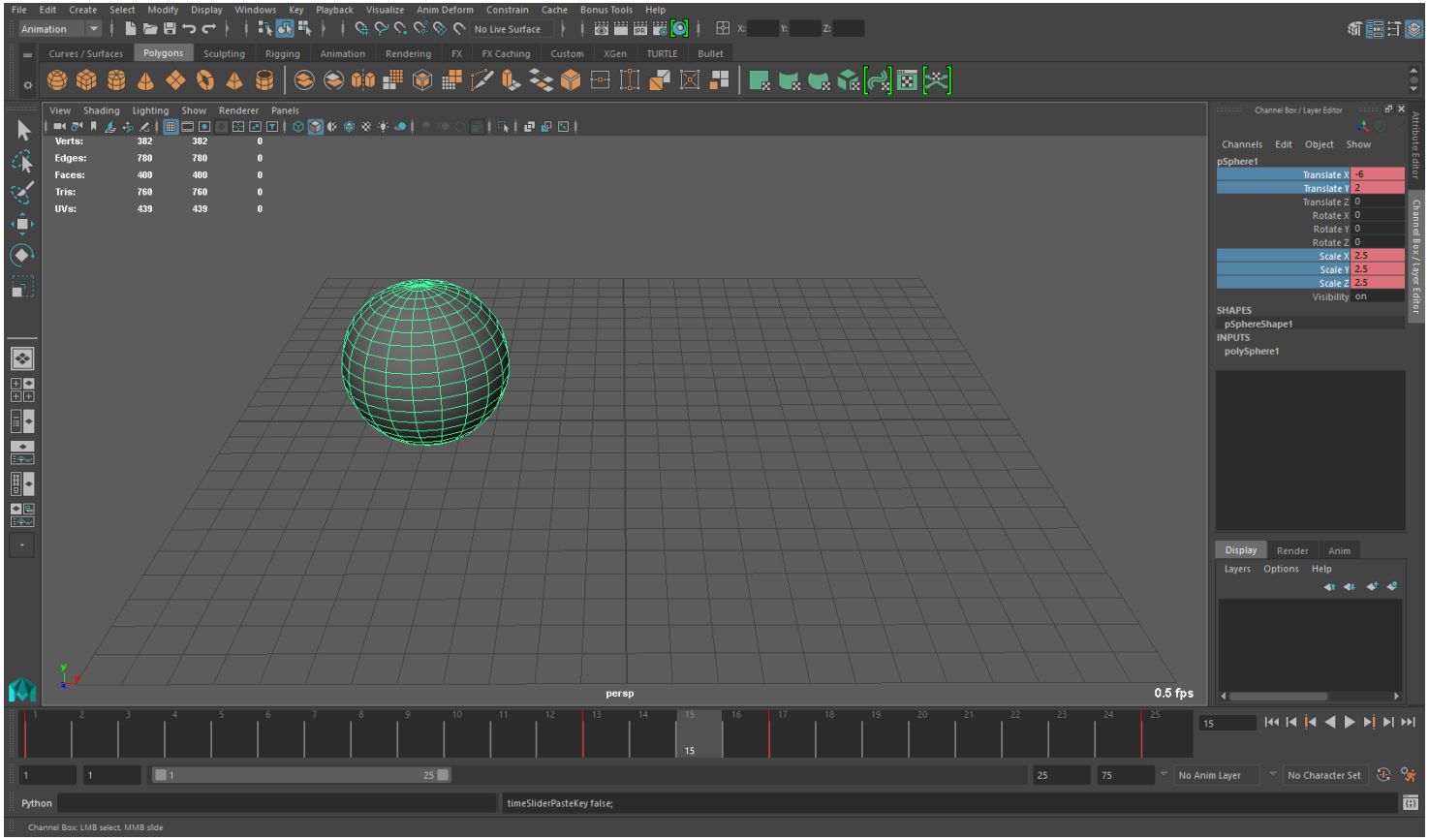
Copy Paste Continued (slide 12)



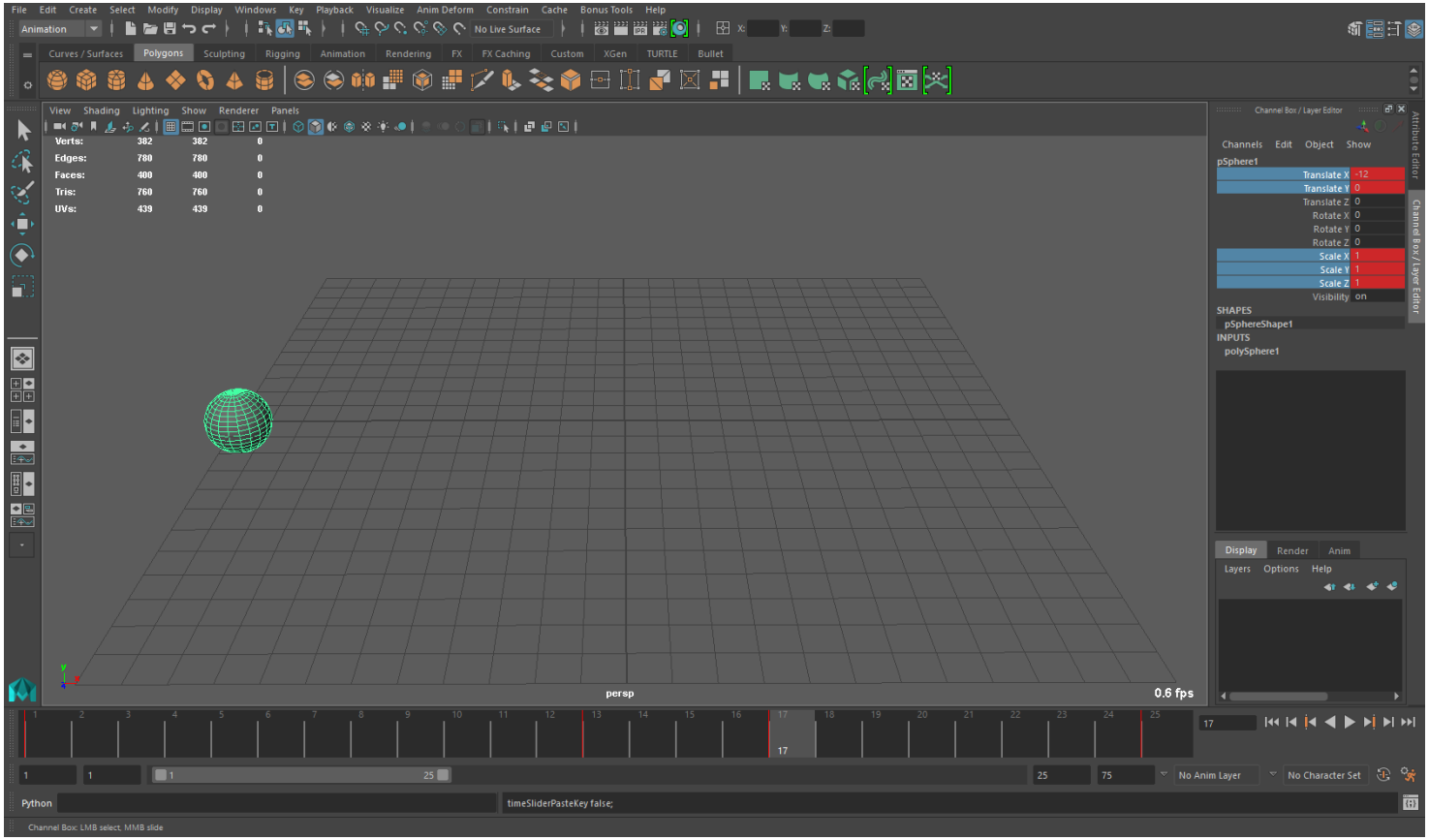
Copy Paste Continued (slide 12)



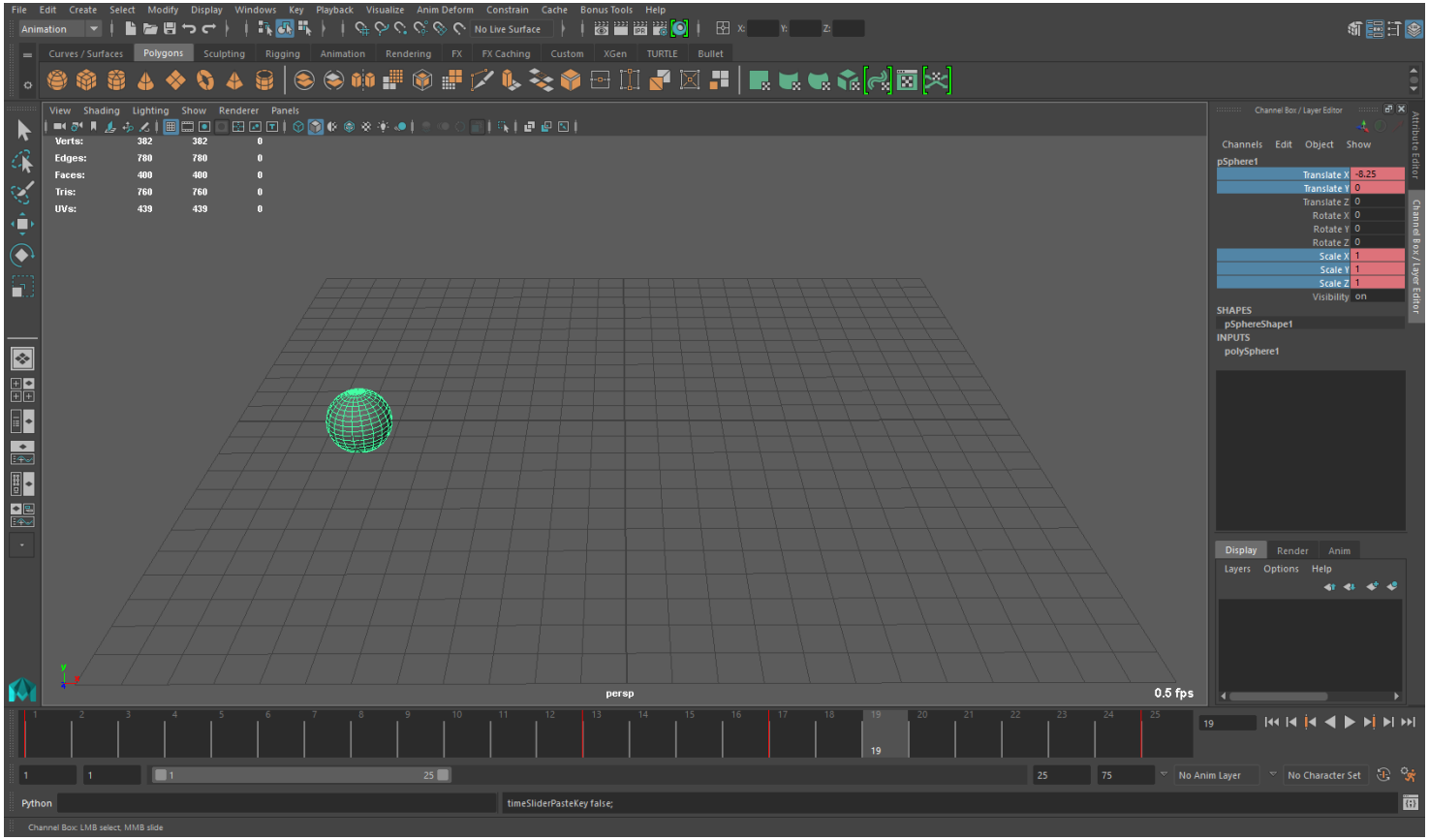
Copy Paste Continued (slide 12)



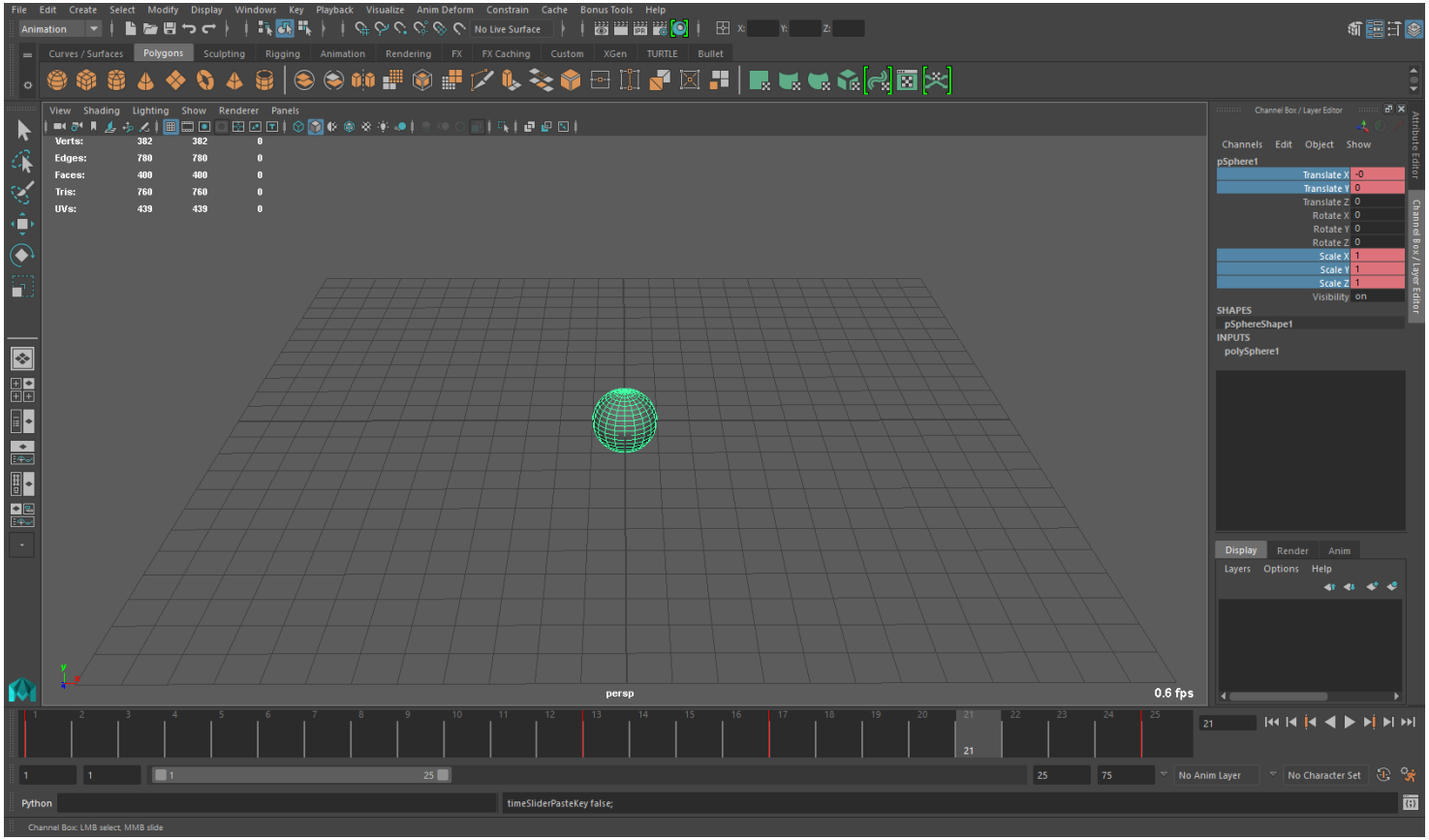
Copy Paste Continued (slide 12)



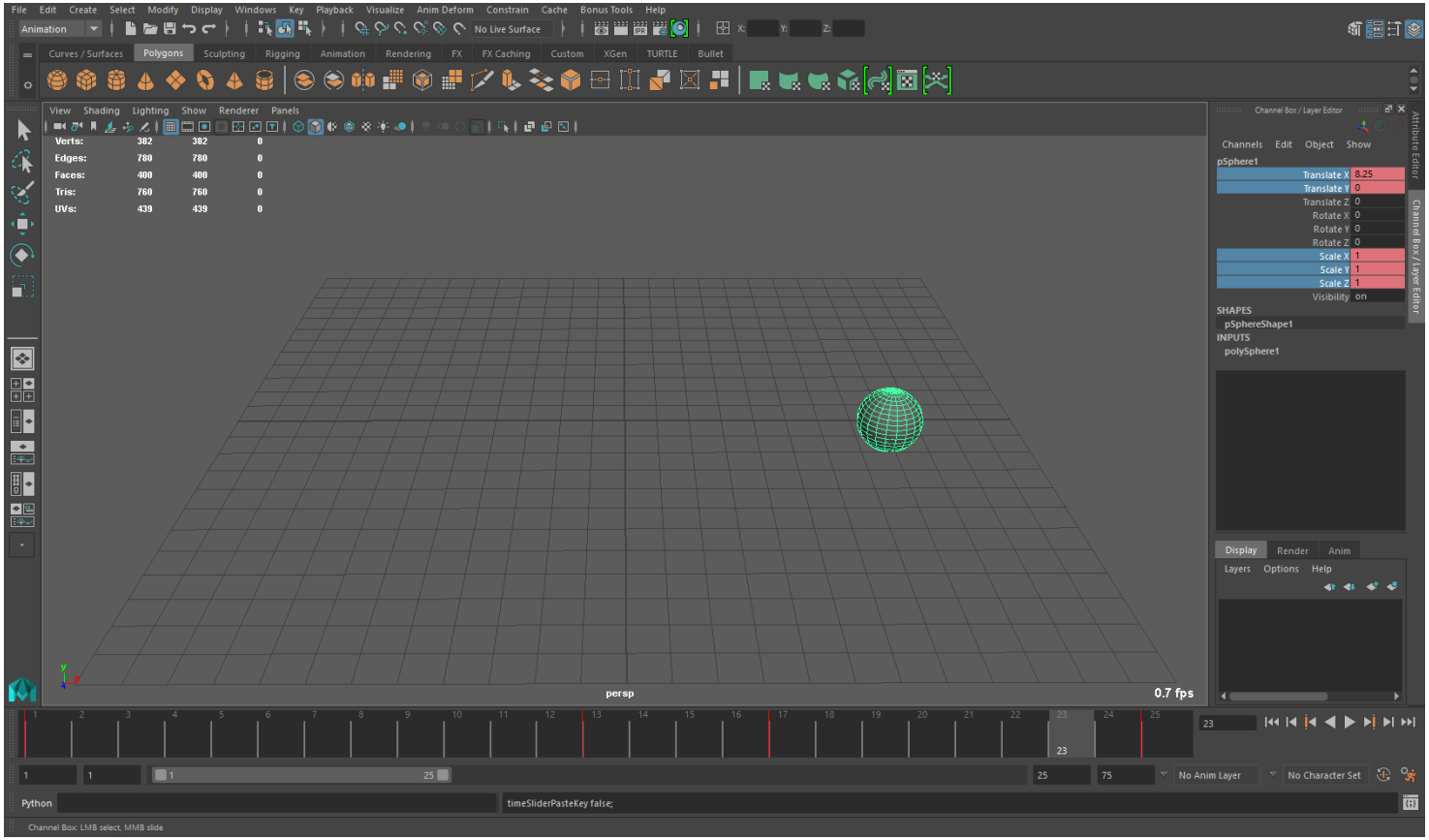
Copy Paste Continued (slide 12)



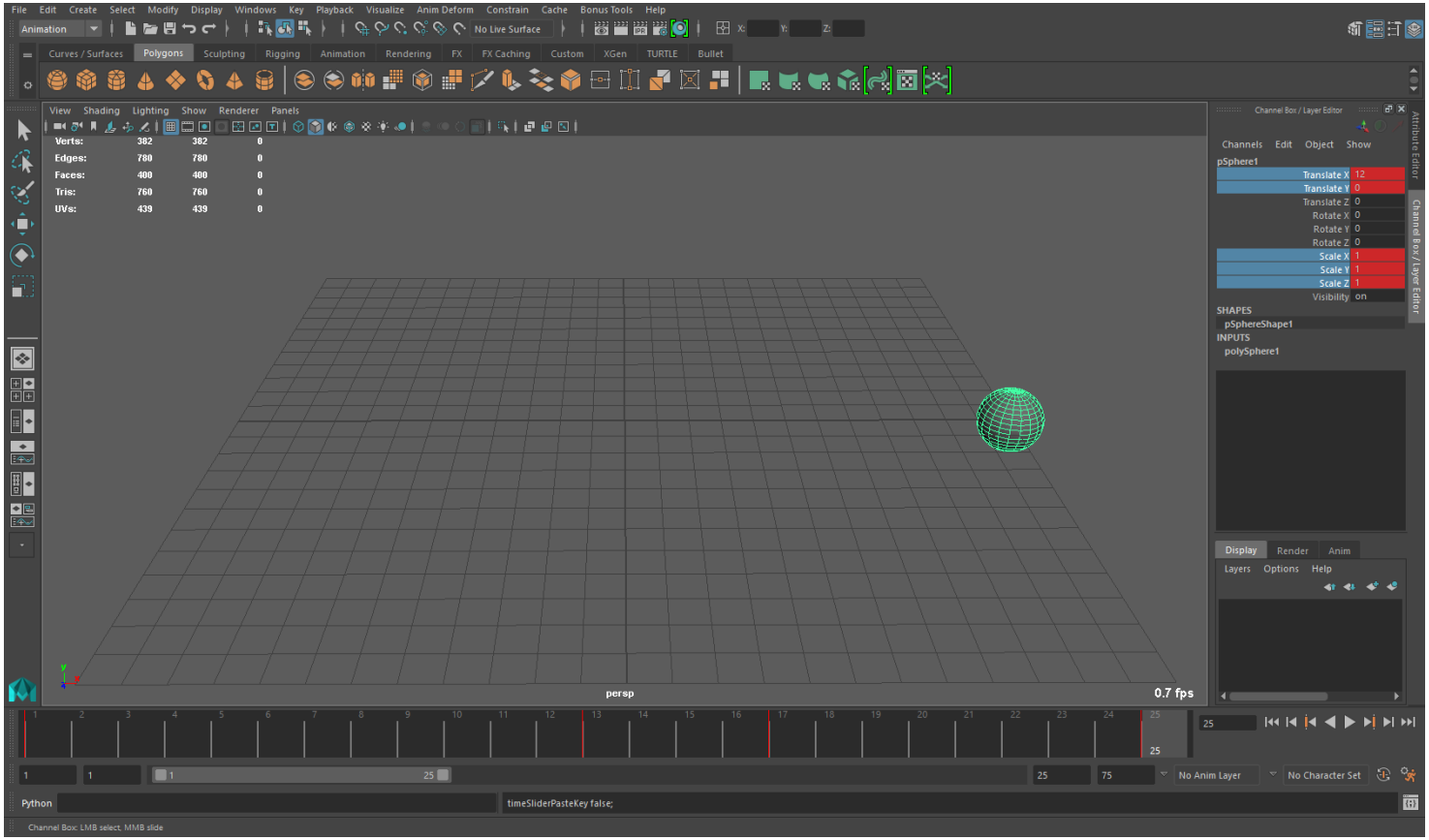
Copy Paste Continued (slide 12)



Copy Paste Continued (slide 12)



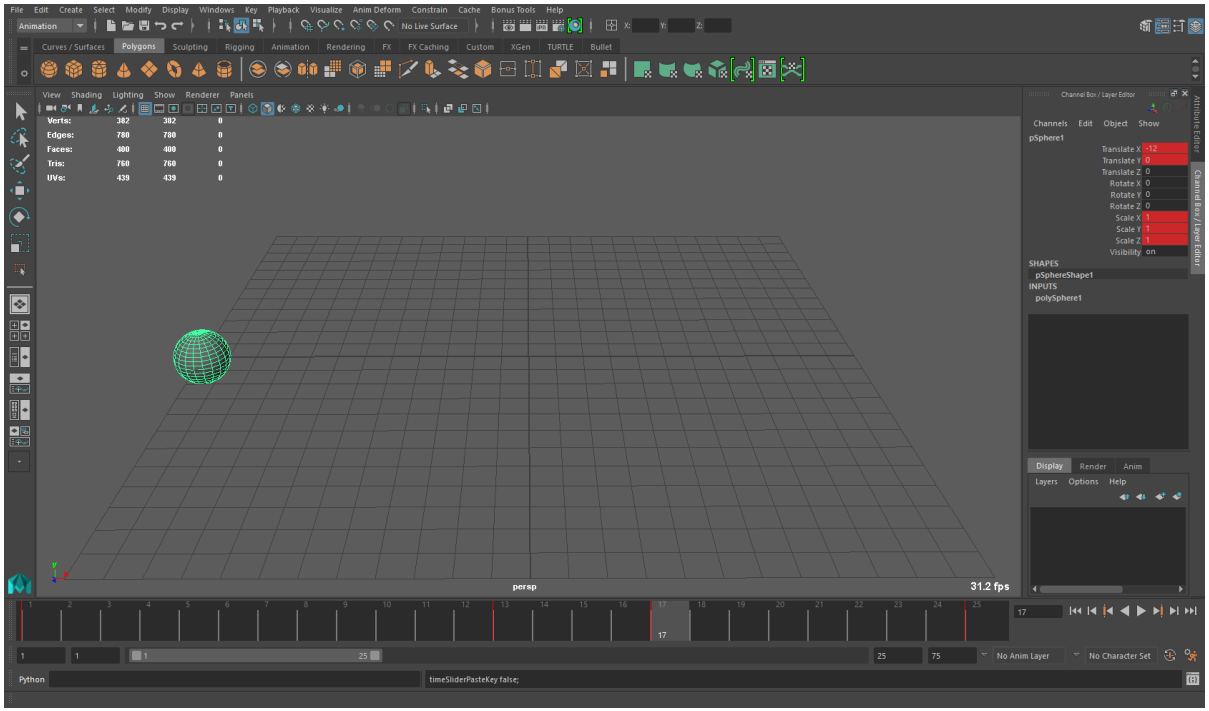
Copy Paste Continued (slide 12)



Cut and Delete (slide 13)

Deleting and Cutting Key Frames

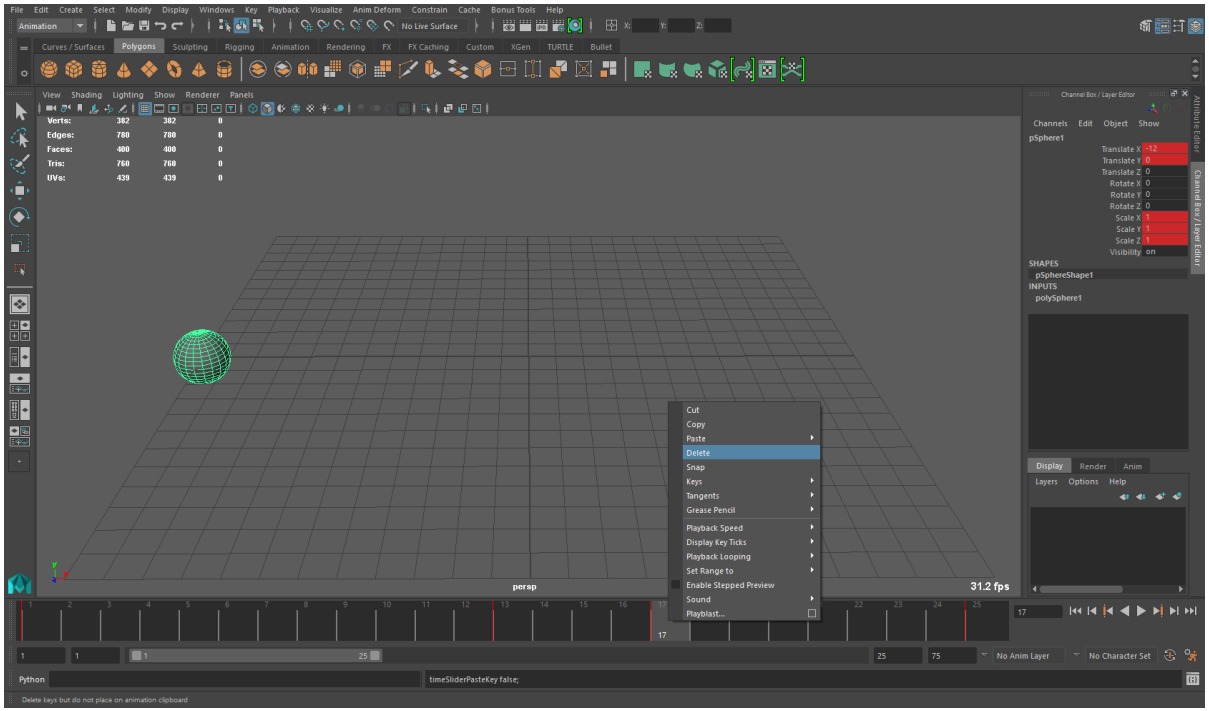
- Set the current time to 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Delete*.
- Alternatively select *Key*→*Delete*
- This deletes key-frame 17.
- The key frame *Cut* tool operates in a similar fashion.



Cut and Delete (slide 13)

Deleting and Cutting Key Frames

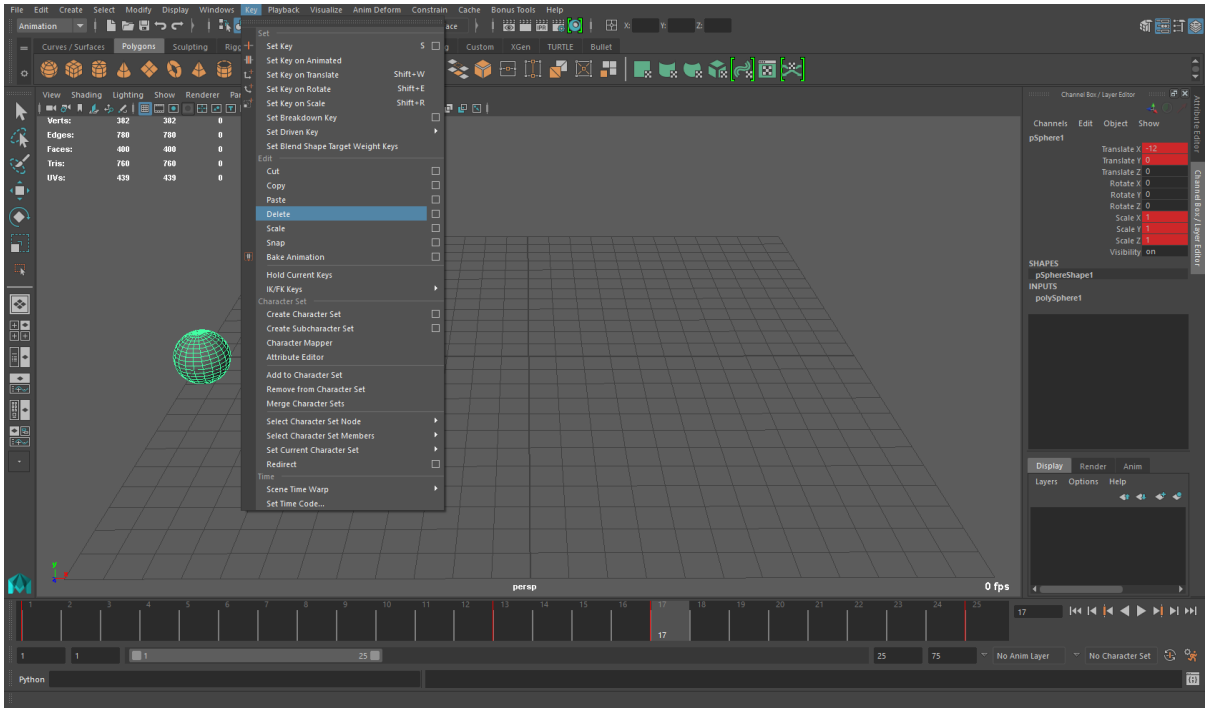
- Set the current time to 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Delete*.
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- This deletes key-frame 17.
- The key frame *Cut* tool operates in a similar fashion.



Cut and Delete (slide 13)

Deleting and Cutting Key Frames

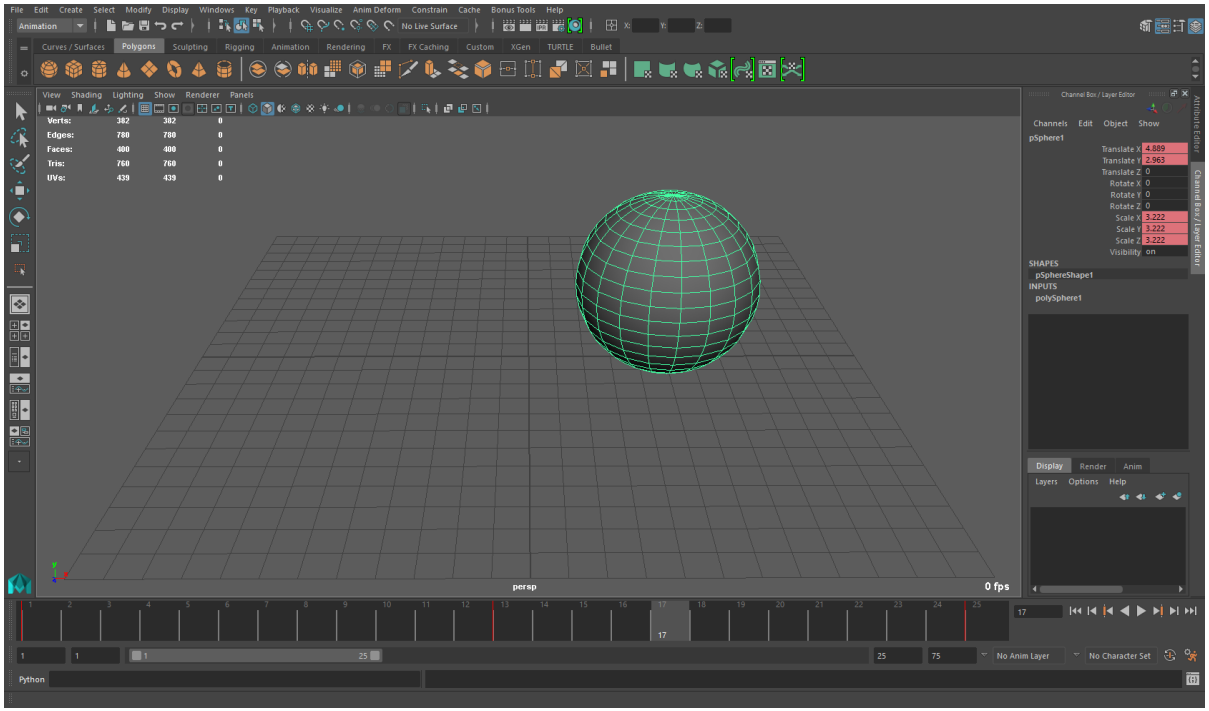
- Set the current time to 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Delete*.
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- The key frame *Cut* tool operates in a similar fashion.



Cut and Delete (slide 13)

Deleting and Cutting Key Frames

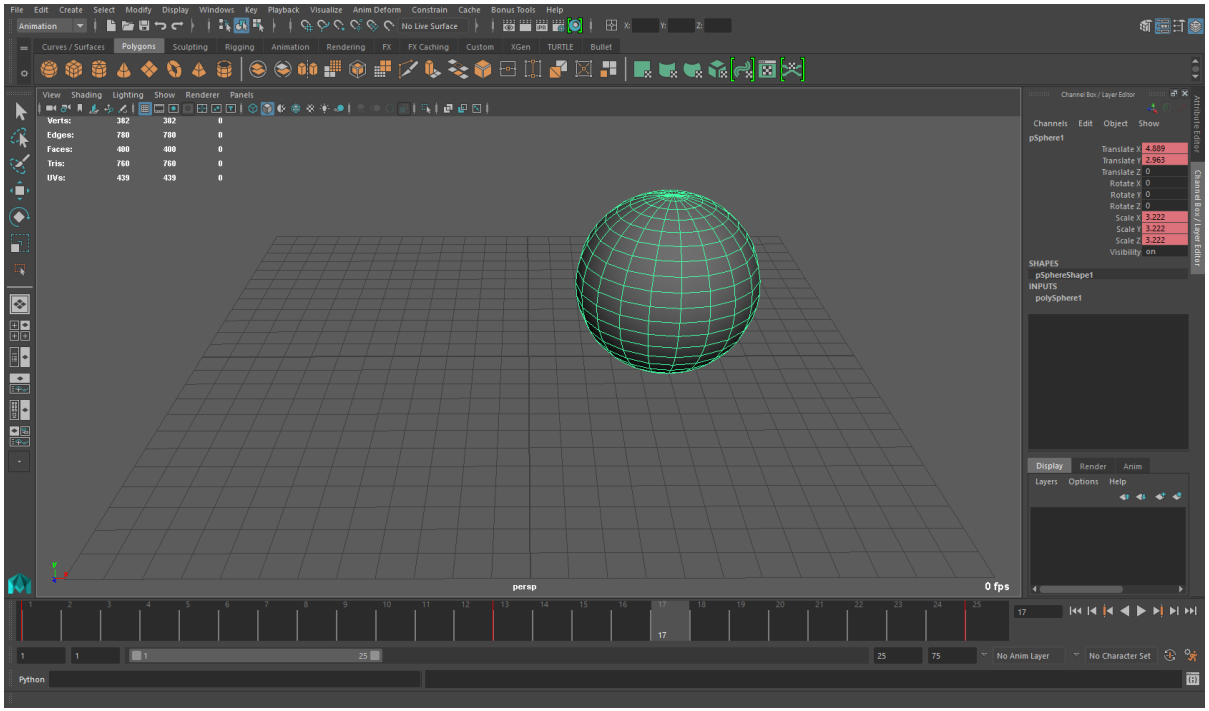
- Set the current time to 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Delete*.
- Alternatively select *Key*$\rightarrow$*Delete*
- This deletes key-frame 17.
- The key frame *Cut* tool operates in a similar fashion.



Cut and Delete (slide 13)

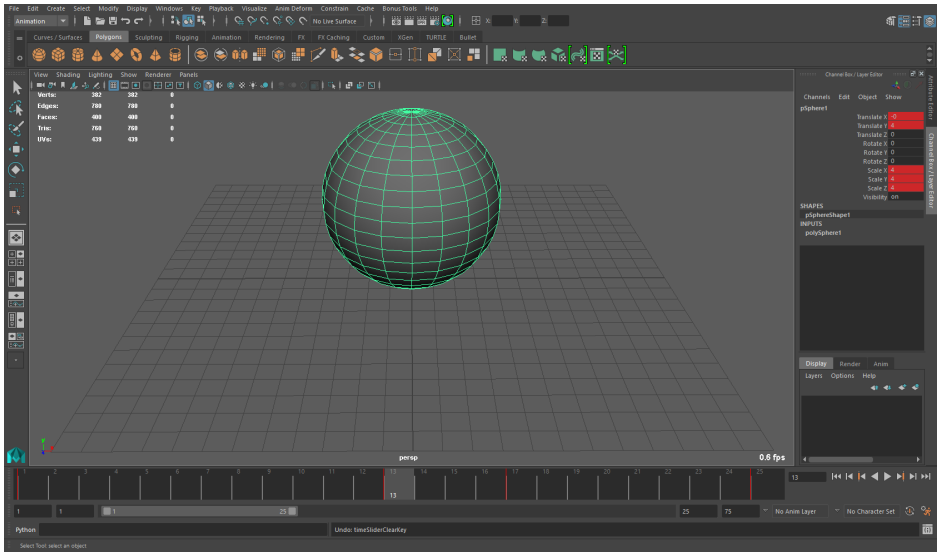
Deleting and Cutting Key Frames

- Set the current time to 17.
- On the time line right-click on frame 17 and in the pop-up menu select *Delete*.
- Alternatively select *Key*$\rightarrow$*Delete*
- This deletes key-frame 17.
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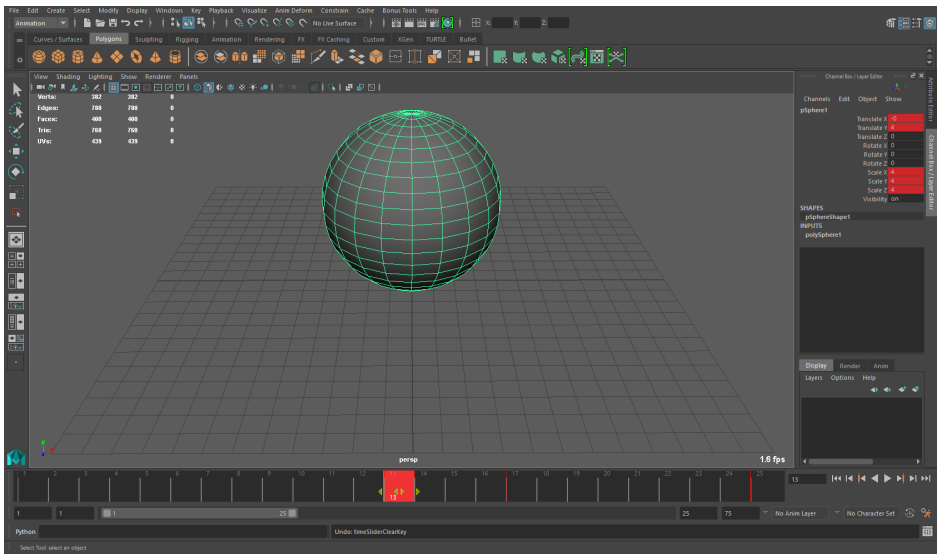
Operating on Groups of Key Frames

- Consider the key-frames set in slides 11/12 and set the current time to 13.
- On the time line press Shift and left-mouse click on frame 13.
- With the Shift key pressed, scrub (pressing the left-mouse button) to frame 17.
- This operation selects a sequence of key-frames which highlighted in red.
- With the left-mouse button pressed, at the centre of the highlighted frames,
- move the highlighted sequence of key-frames 4 frames to the left.
- This, indeed, moves the selected key-frames 4-frames to the left.



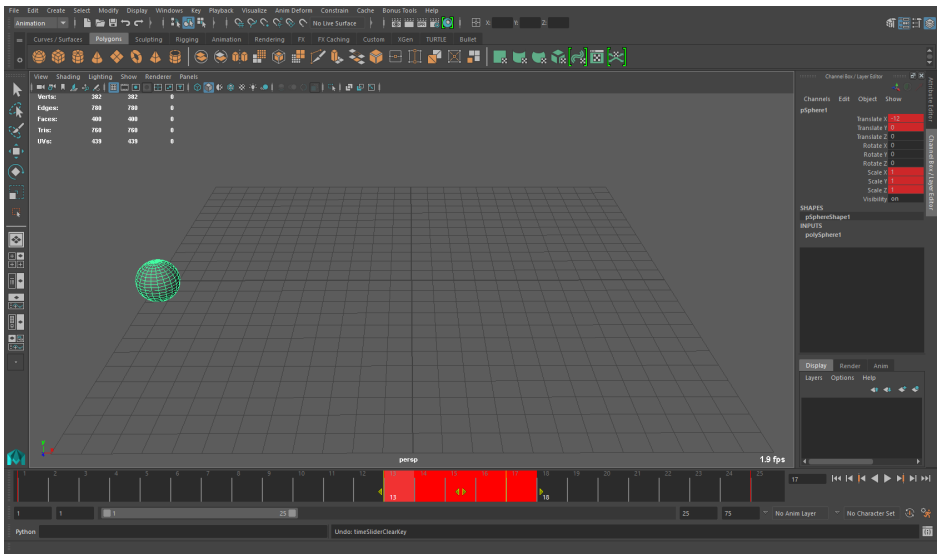
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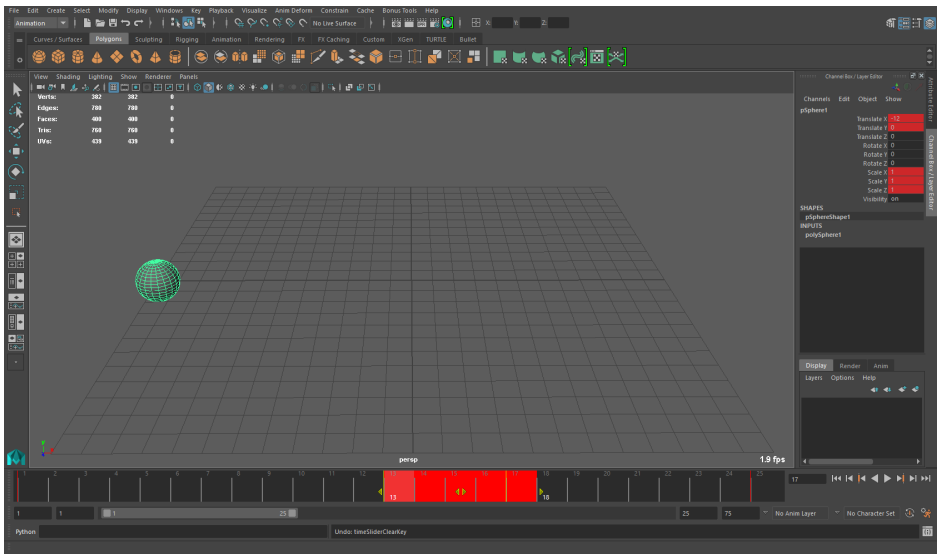
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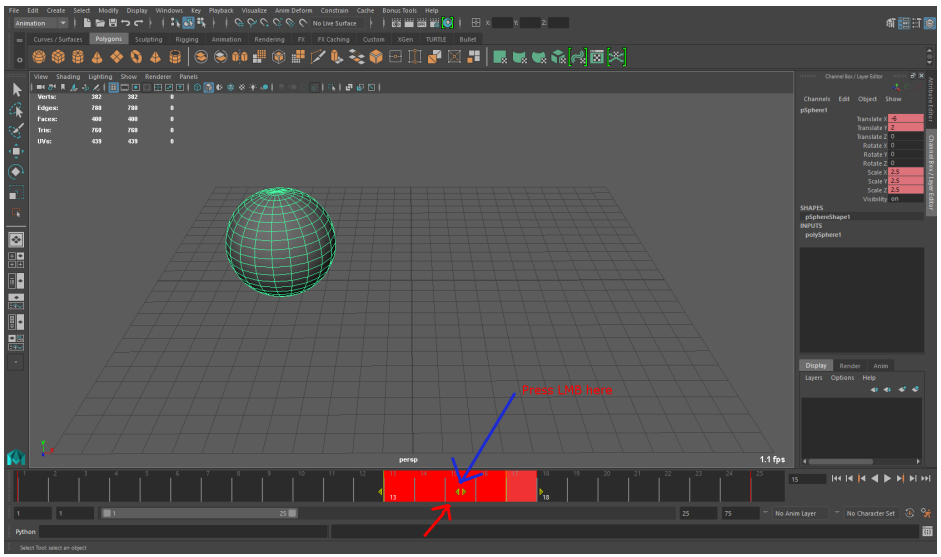
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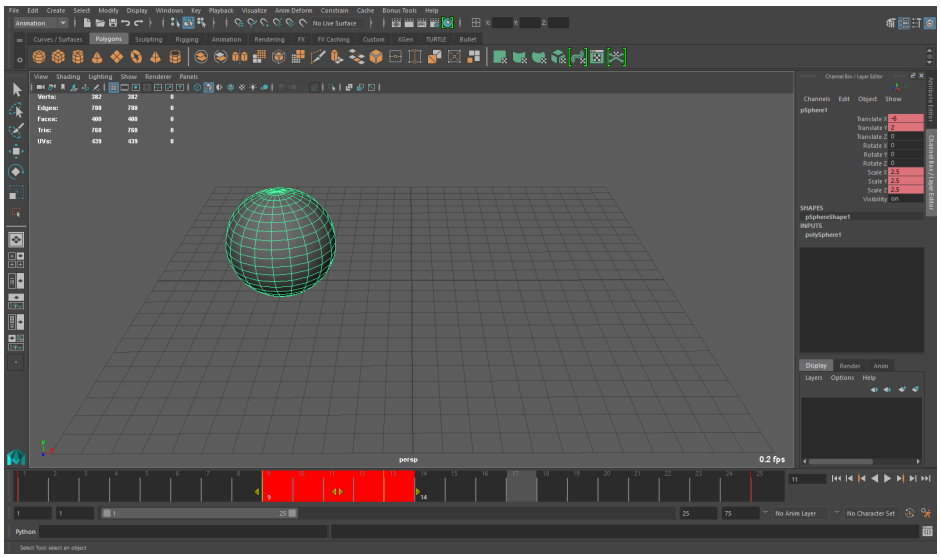
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- This, indeed, moves the selected key-frames 4-frames to the left.



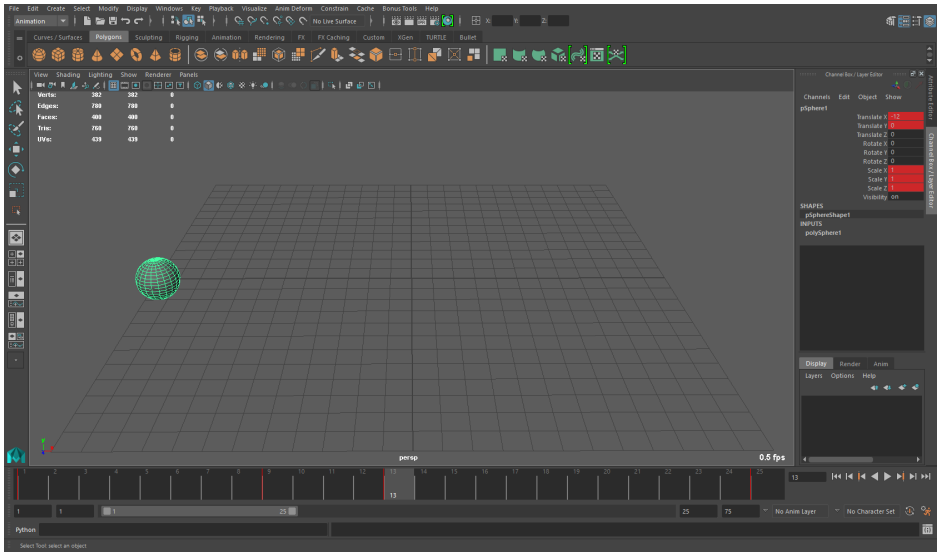
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Operating on Groups of Key Frames

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- With the Shift key pressed, scrub (pressing the left-mouse button) to frame 17.
- This operation selects a sequence of key-frames which highlighted in red.
- With the left-mouse button pressed, at the centre of the highlighted frames,
- move the highlighted sequence of key-frames 4 frames to the left.
- This, indeed, moves the selected key-frames 4-frames to the left.



Move Scale Snap Continued (slide 15)

Operating on Groups of Key Frames

- Press Shift and left-mouse click on frame 9.
- With the Shift key pressed, scrub (pressing the left-mouse button) to frame 13.
- Click the left-mouse button to the **right** edge/arrow of the highlighted frames; with left-mouse button pressed scale/stretch the highlighted sequence 6 frames to the left. This, indeed, scales the selected key-frames by approximately 6-frames to the left.
- Notice the last key-frame is set at a fractional time value (18.6).
- To fix this, right click at the key-frame and in the pop-up menu select *Snap*. The scaled key-frame is now set at time 19.

Move Scale Snap Continued (slide 15)

Operating on Groups of Key Frames

- Press Shift and left-mouse click on frame 9.
- With the Shift key pressed, scrub (pressing the left-mouse button) to frame 13.
- Click the left-mouse button to the **right** edge/arrow of the highlighted frames; with left-mouse button pressed scale/stretch the highlighted sequence 6 frames to the left. This, indeed, scales the selected key-frames by approximately 6-frames to the left.
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Move Scale Snap Continued (slide 15)

Operating on Groups of Key Frames

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Move Scale Snap Continued (slide 15)

Operating on Groups of Key Frames

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Move Scale Snap Continued (slide 15)

Operating on Groups of Key Frames

- Press Shift and left-mouse click on frame 9.
- With the Shift key pressed, scrub (pressing the left-mouse button) to frame 13.
- Click the left-mouse button to the **right** edge/arrow of the highlighted frames; with left-mouse button pressed scale/stretch the highlighted sequence 6 frames to the left. This, indeed, scales the selected key-frames by approximately 6-frames to the left.
- Notice the last key-frame is set at a fractional time value (18.6).
- To fix this, right click at the key-frame and in the pop-up menu select *Snap*. The scaled key-frame is now set at time 19.

Move Scale Snap Continued (slide 15 α)

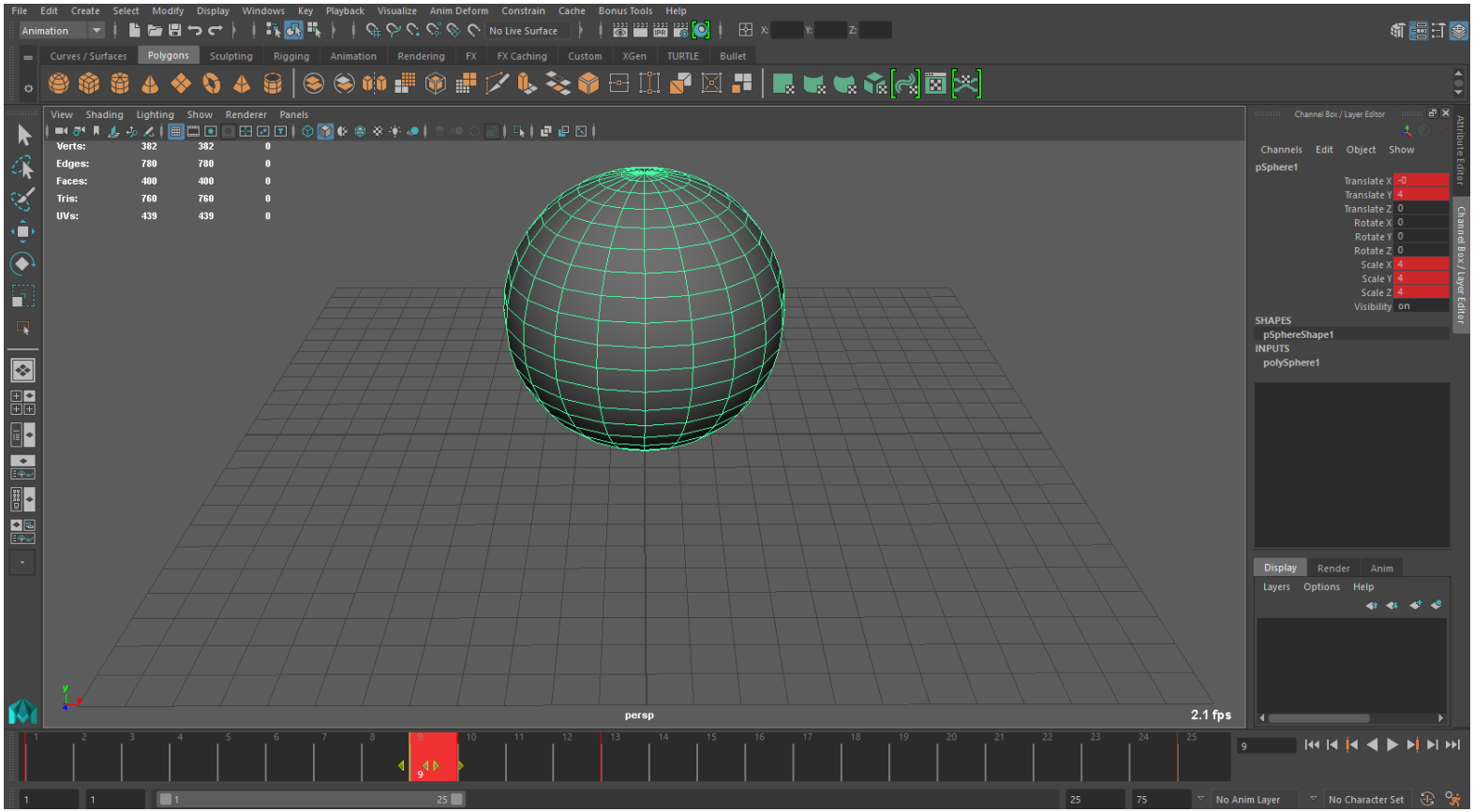


Figure: Press Shift and left-mouse click on frame 9.

Move Scale Snap Continued (slide 15 β)

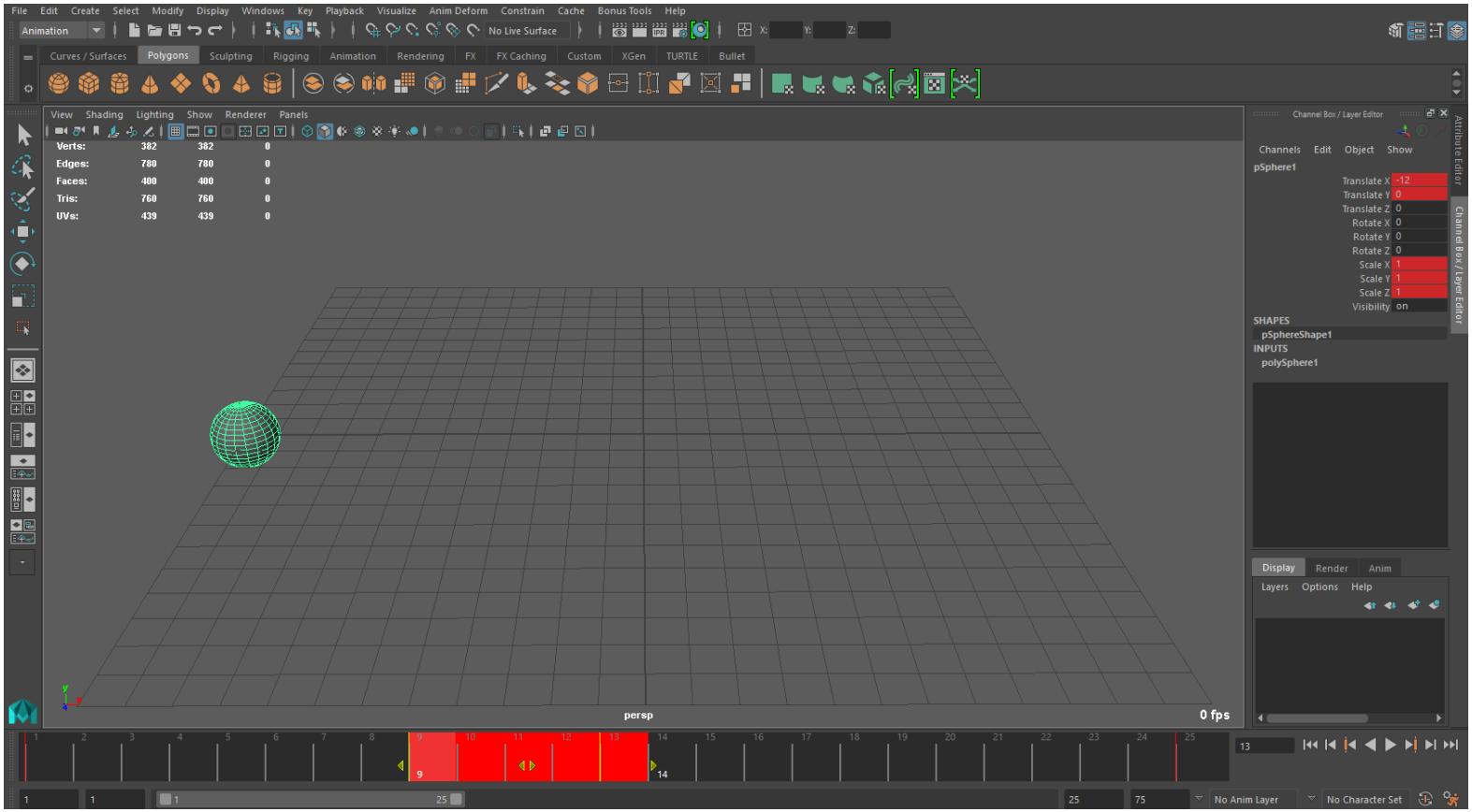


Figure: With the Shift key pressed, scrub (pressing the left-mouse button) to frame 13.

Move Scale Snap Continued (slide 15 γ)

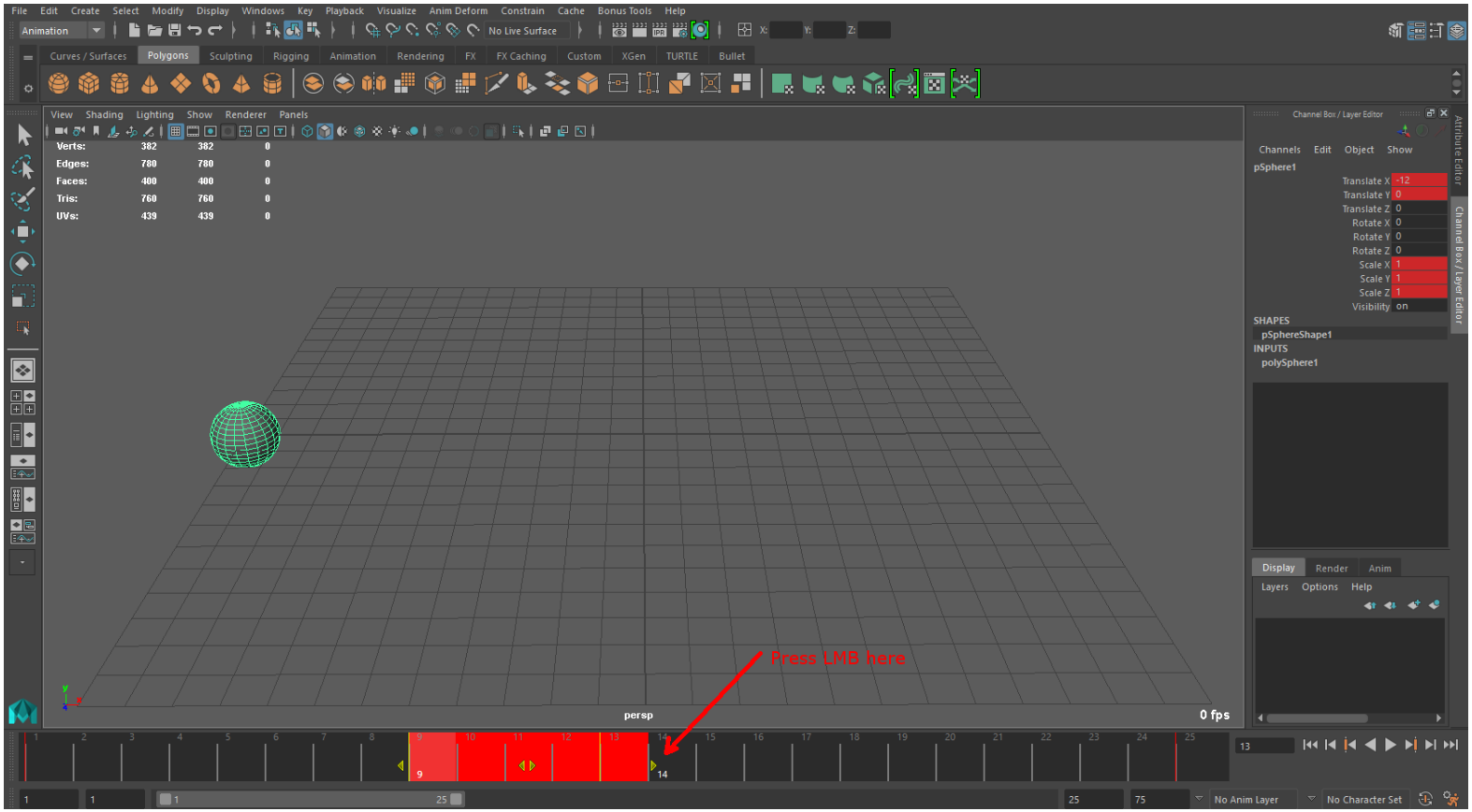


Figure: Click the left-mouse button to the **right** edge/arrow of the highlighted frames

Move Scale Snap Continued (slide 15 δ)

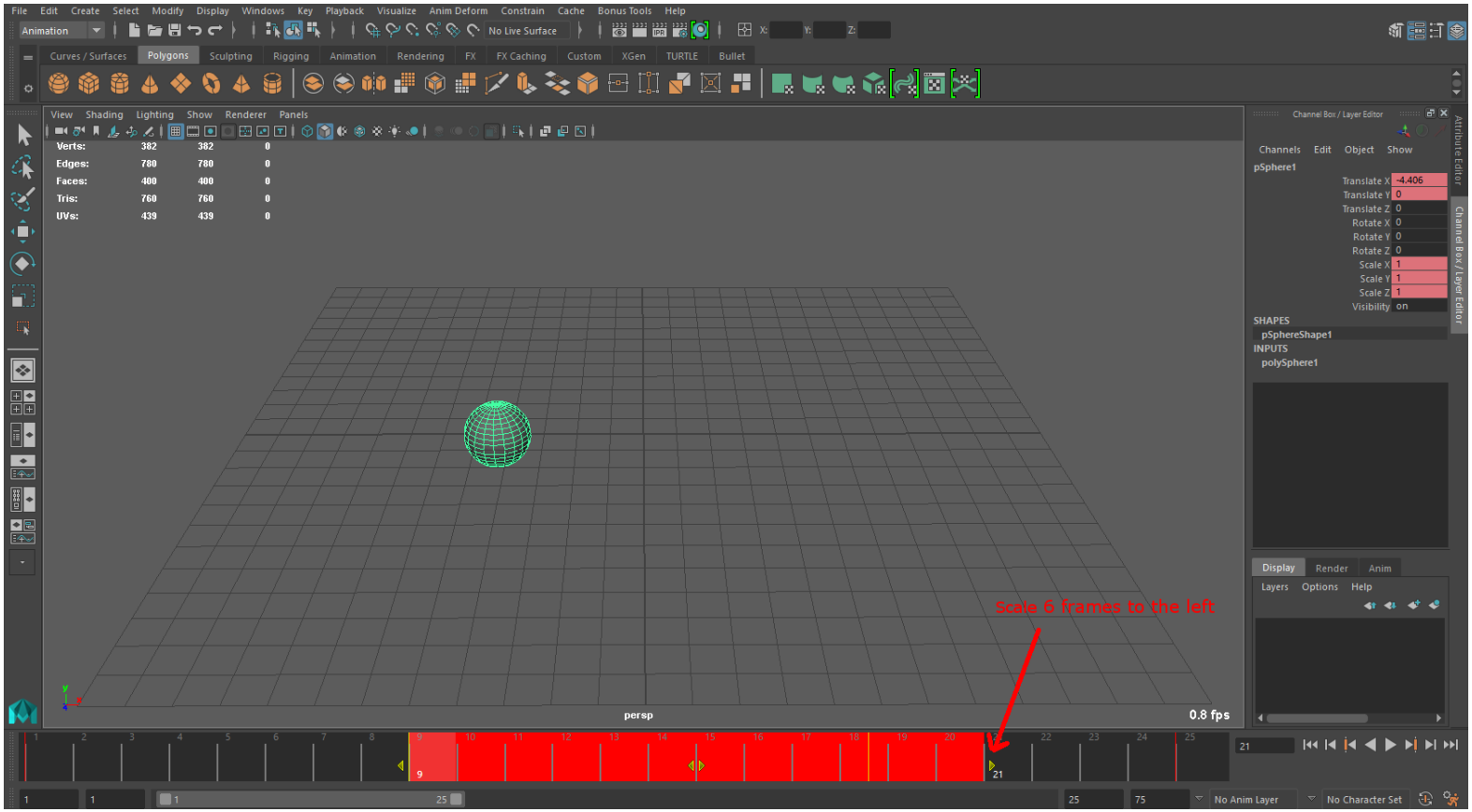


Figure: with left-mouse button pressed scale/stretch the highlighted sequence 6 frames to the left.

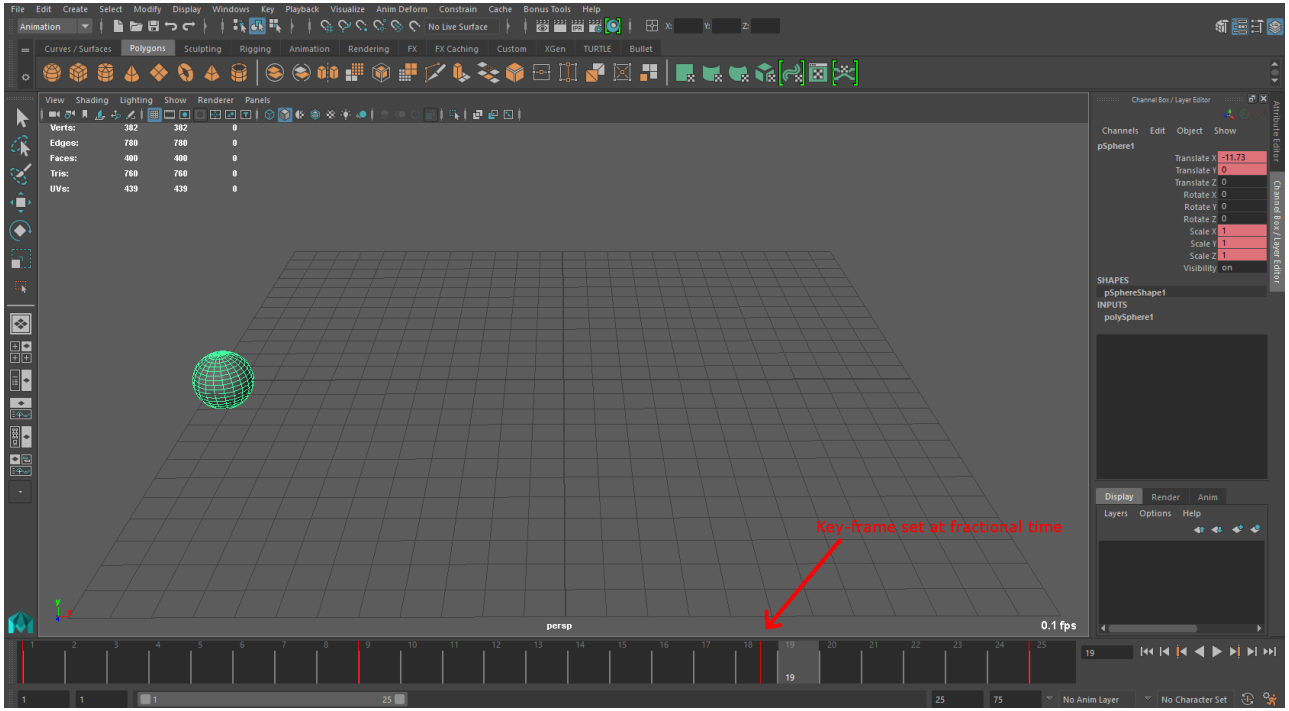


Figure: Notice the last key-frame is set at a fractional time value (18.6).

```
1 # to see the time a frame is set use
2 cmds.keyframe( 'ari', time=(18,19), query=True, timeChange=True)
```

Move Scale Snap Continued (slide 15 ζ)

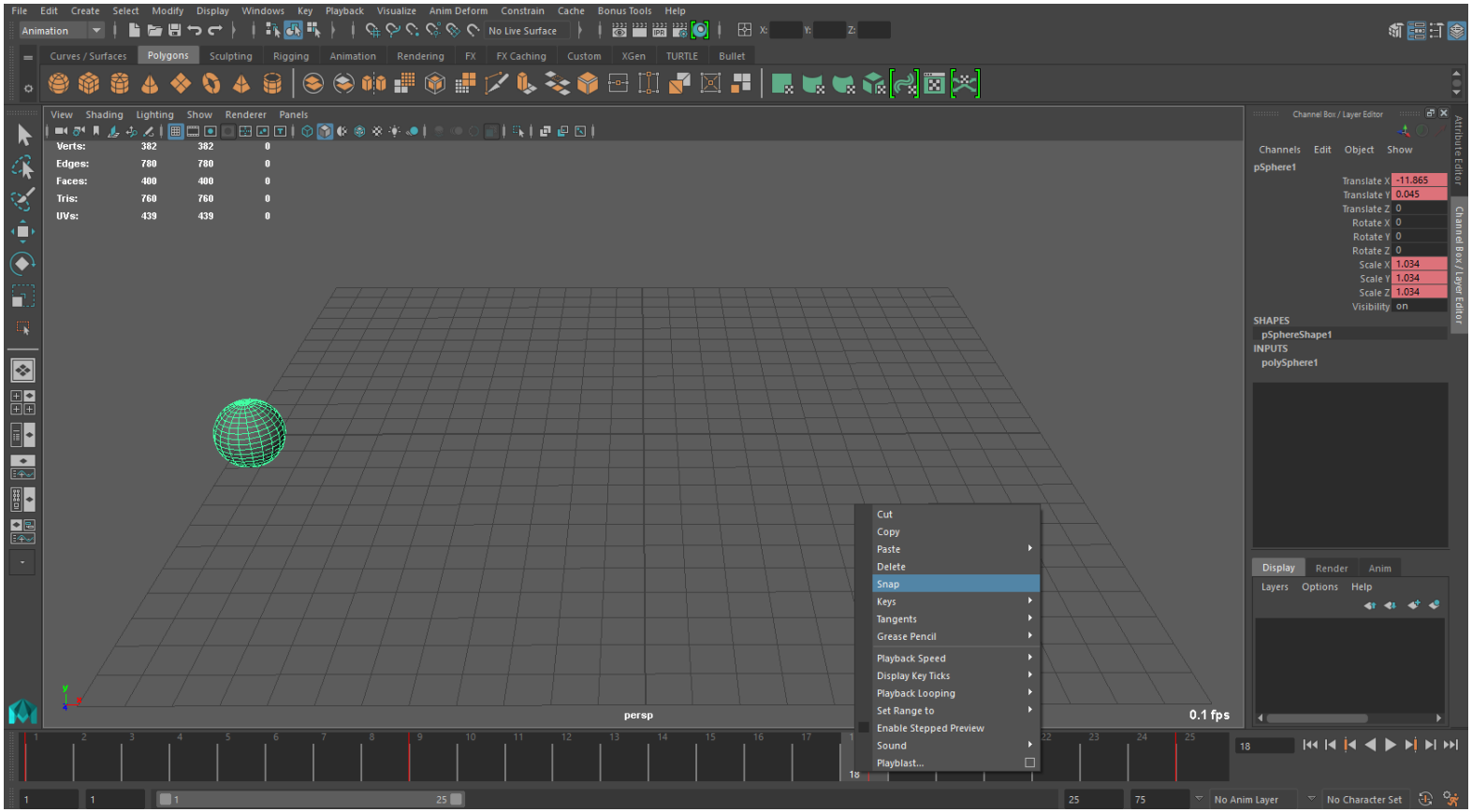


Figure: To fix this, right click at the key-frame and in the pop-up menu select *Snap*.

Move Scale Snap Continued (slide 15 η)

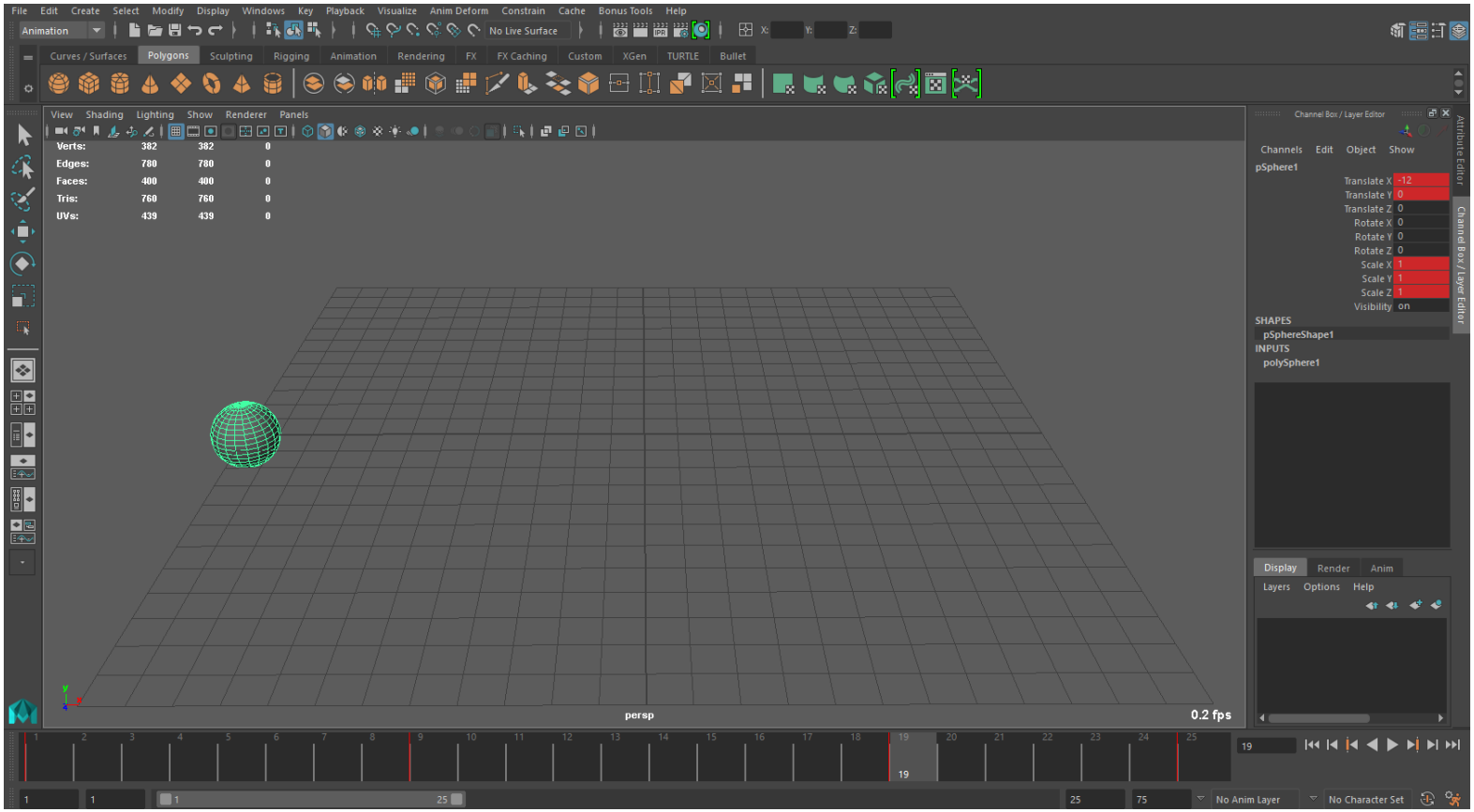
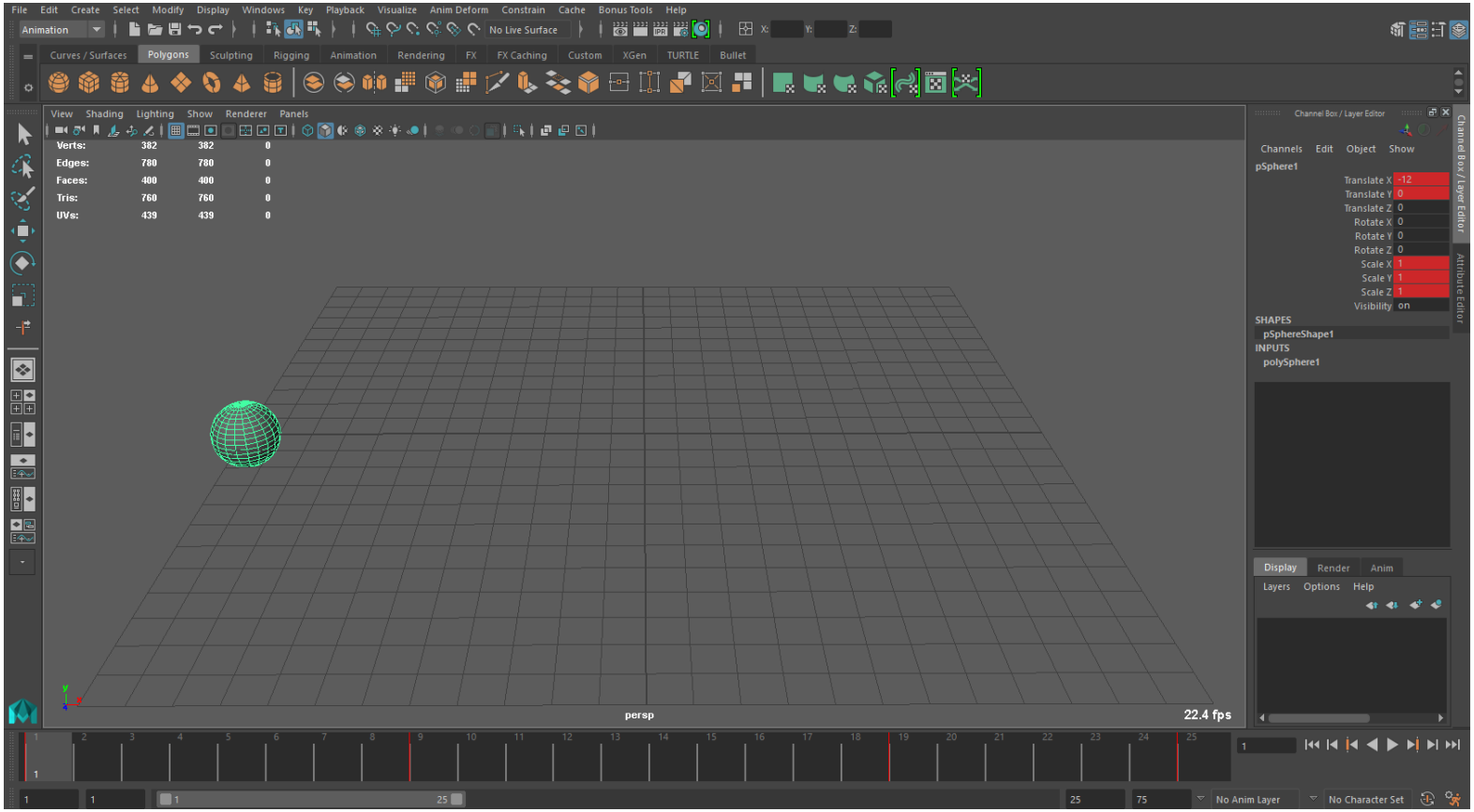
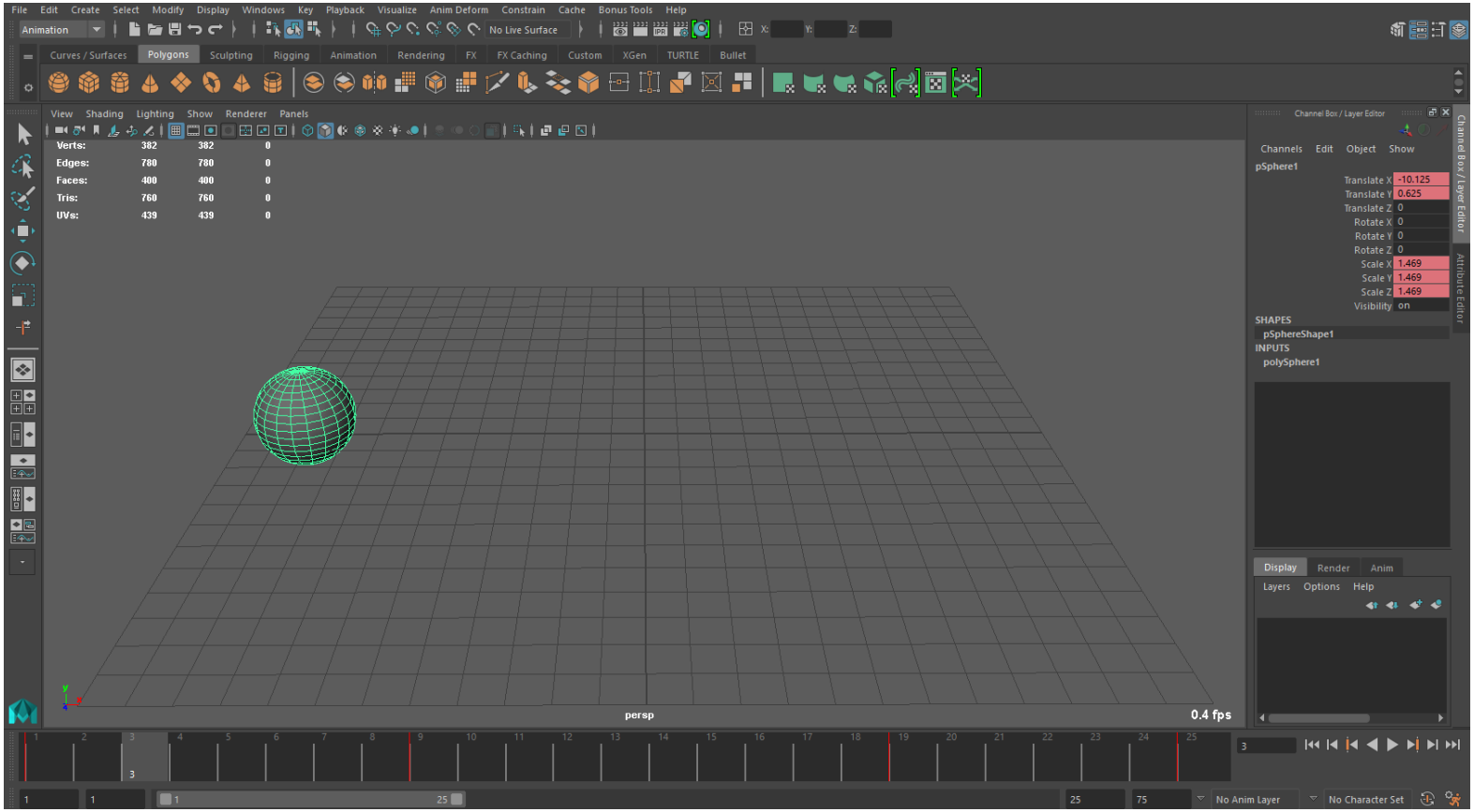


Figure: The scaled key-frame is now set at time 19.

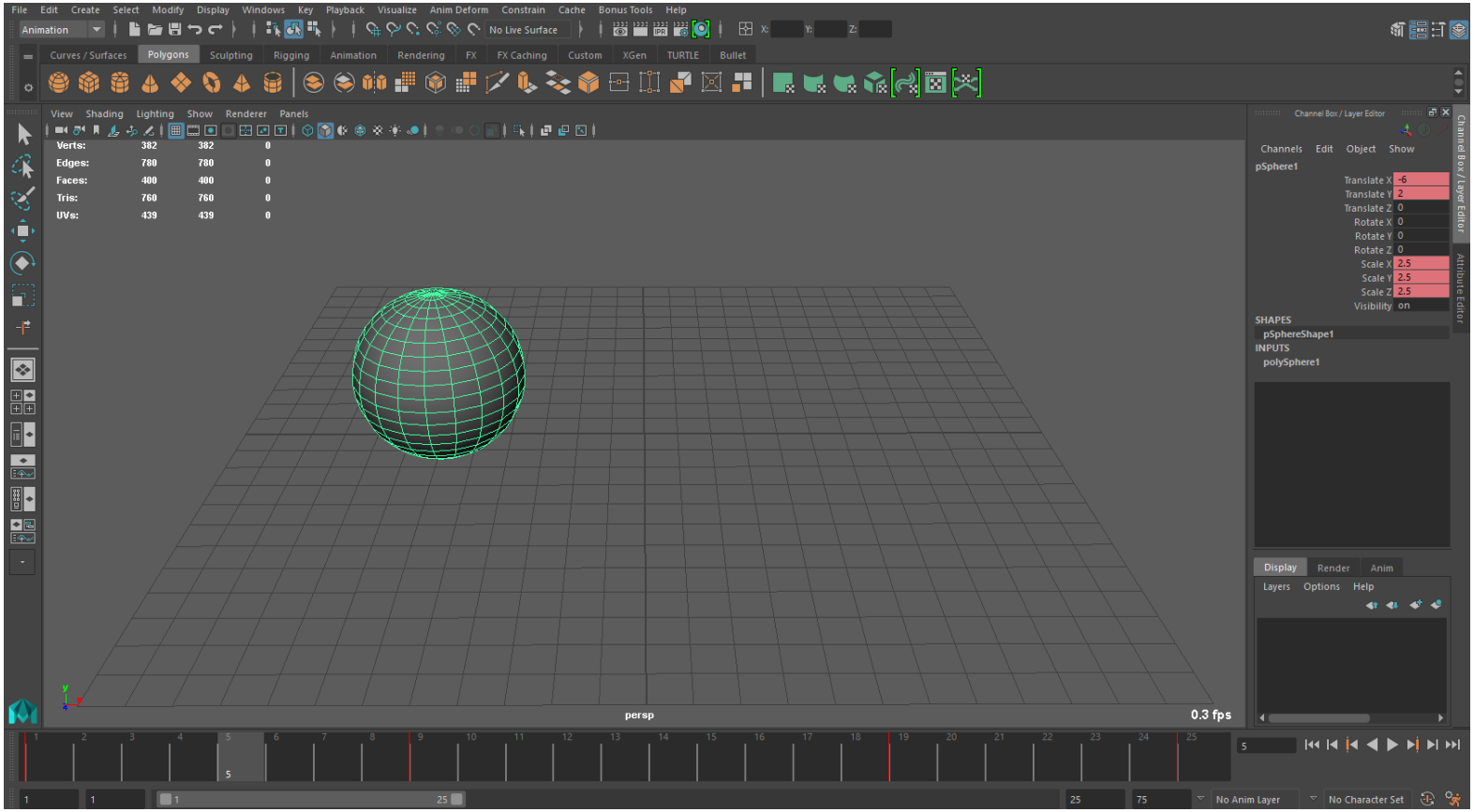
Move Scale Snap Last (slide 16)



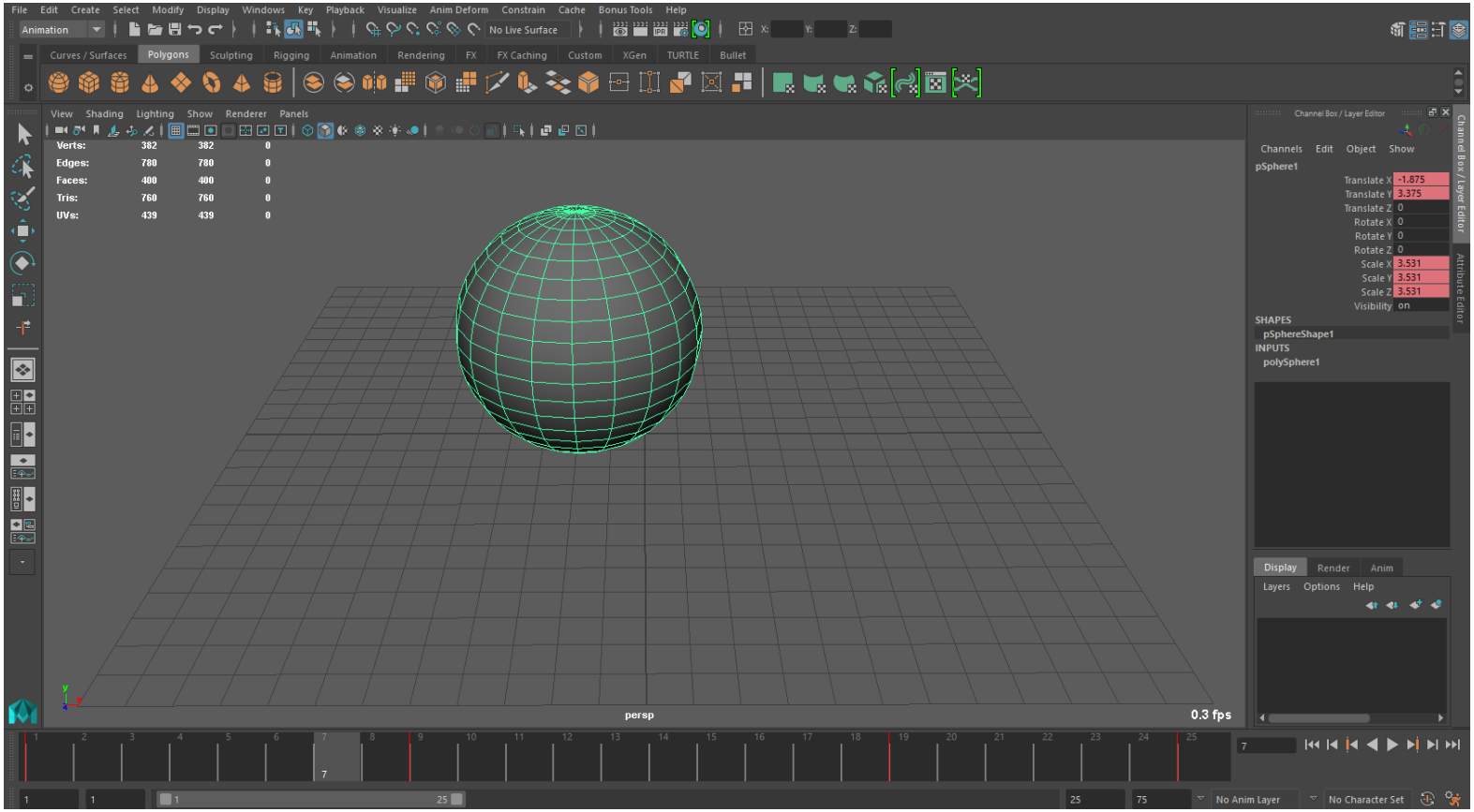
Move Scale Snap Last (slide 16)



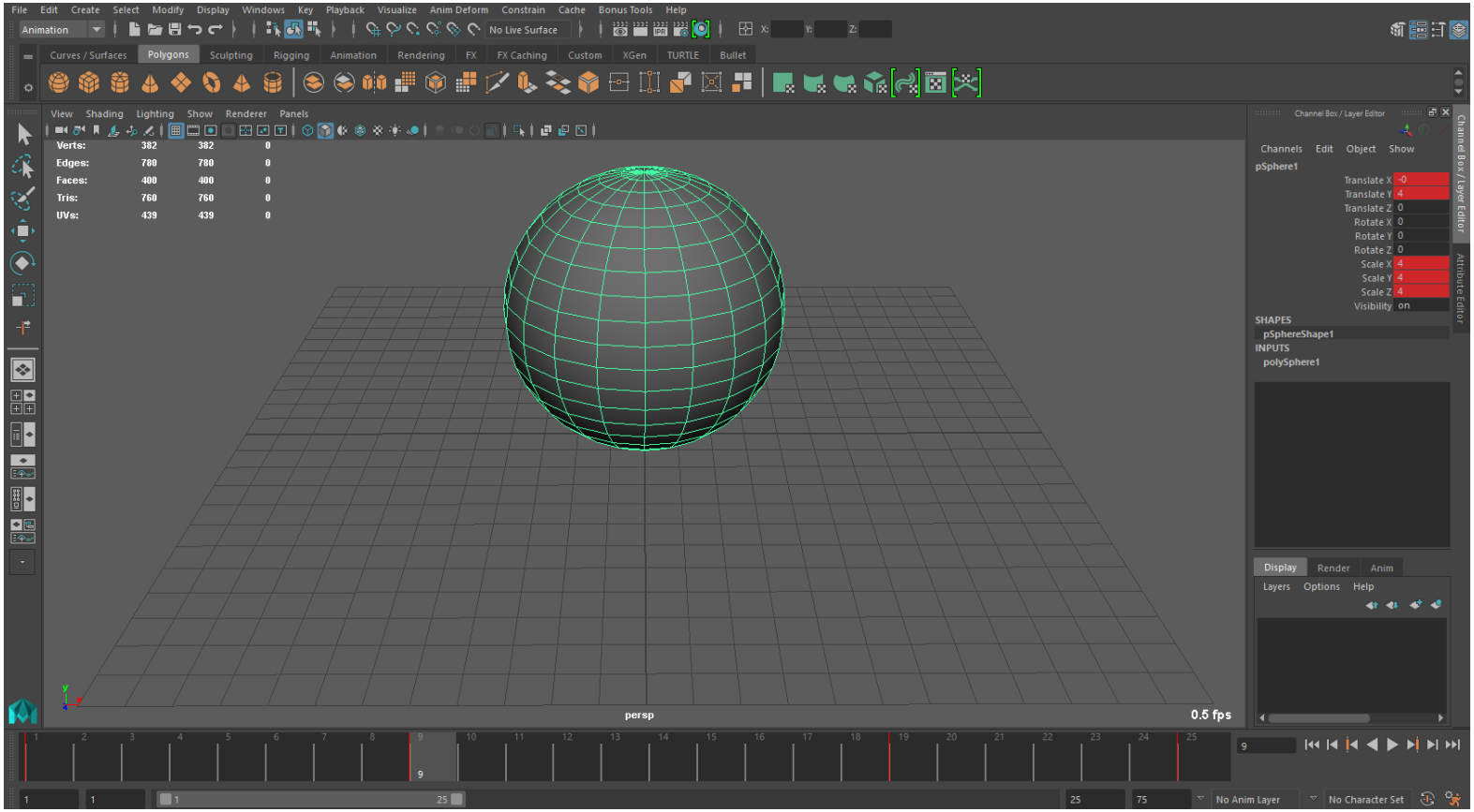
Move Scale Snap Last (slide 16)



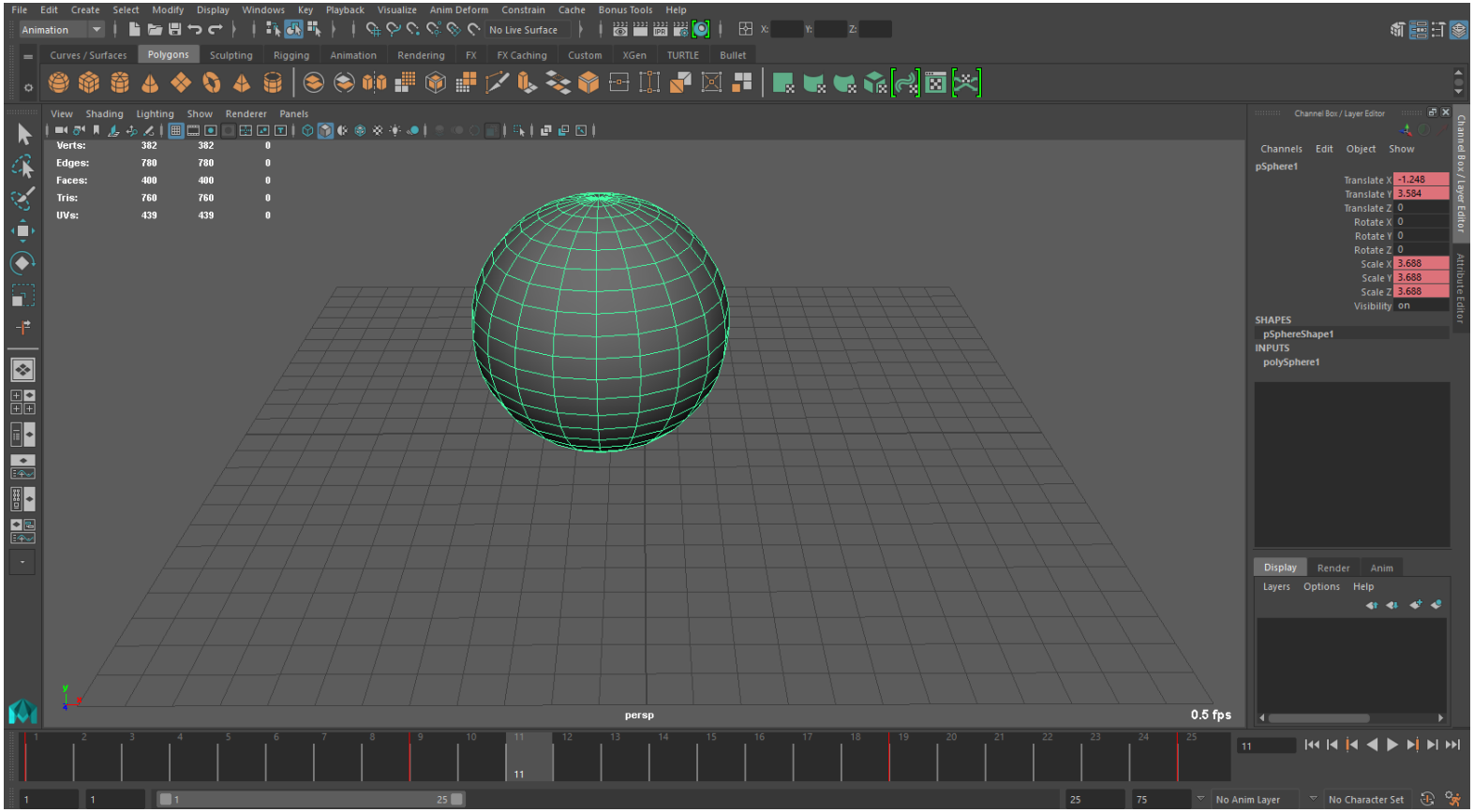
Move Scale Snap Last (slide 16)



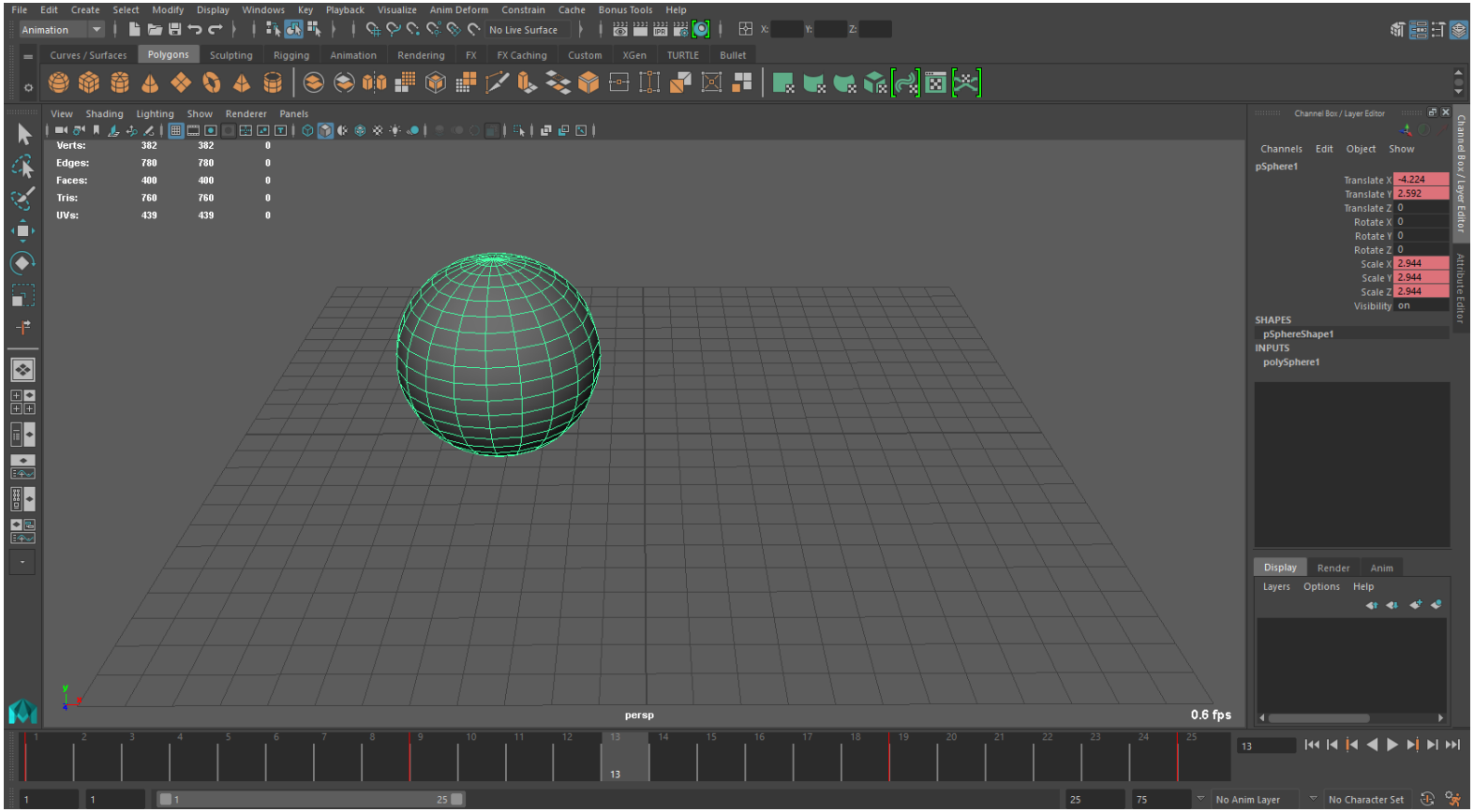
Move Scale Snap Last (slide 16)



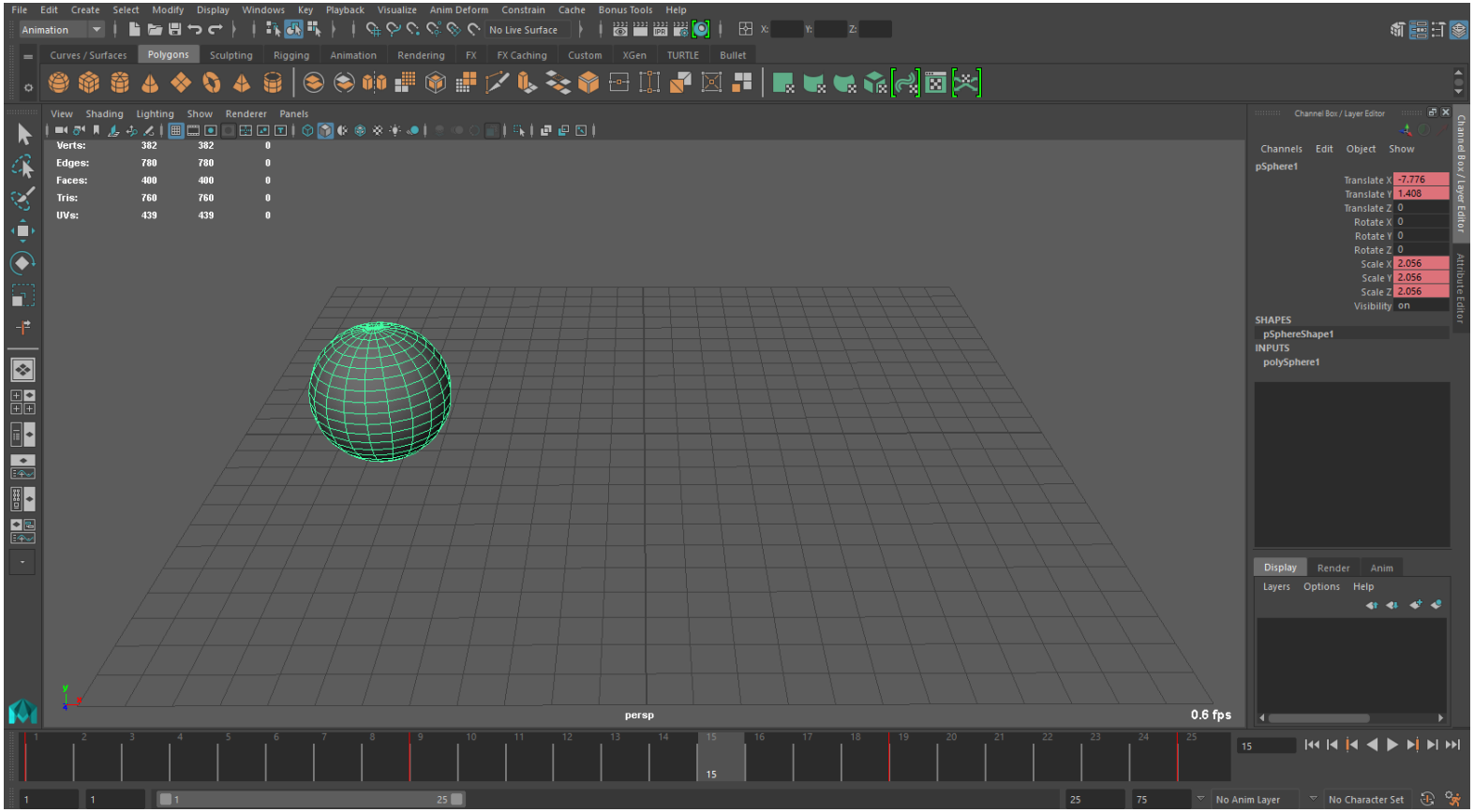
Move Scale Snap Last (slide 16)



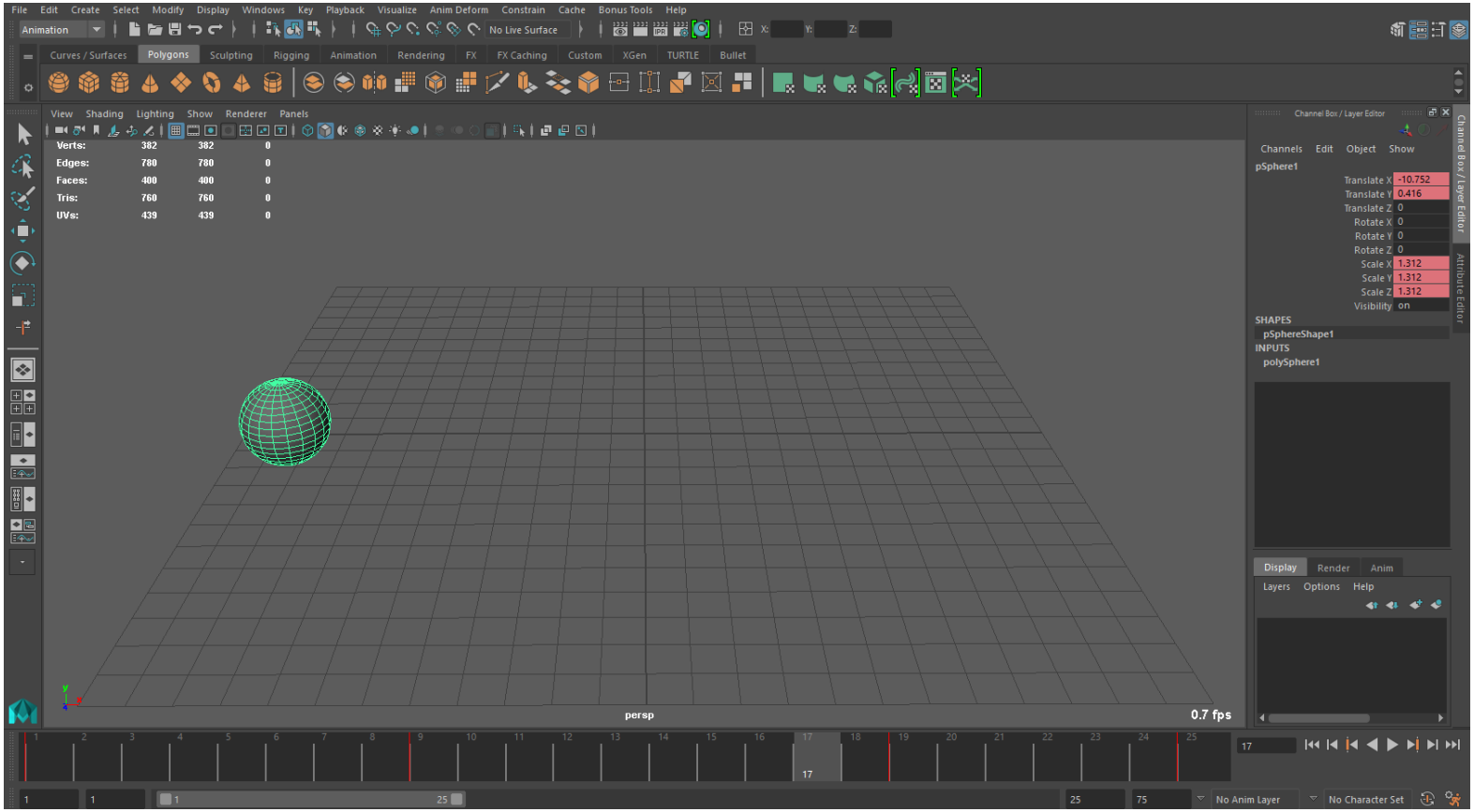
Move Scale Snap Last (slide 16)



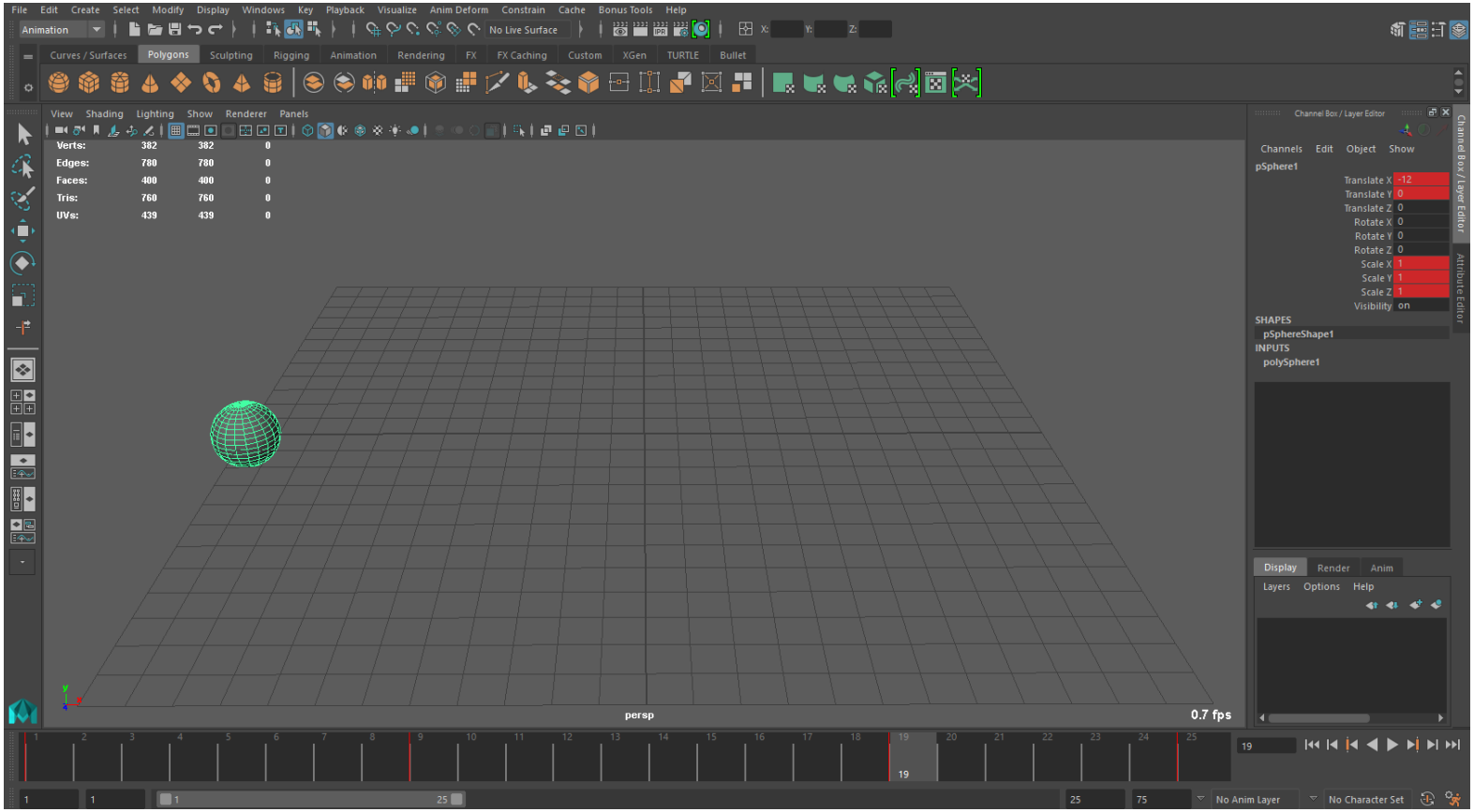
Move Scale Snap Last (slide 16)



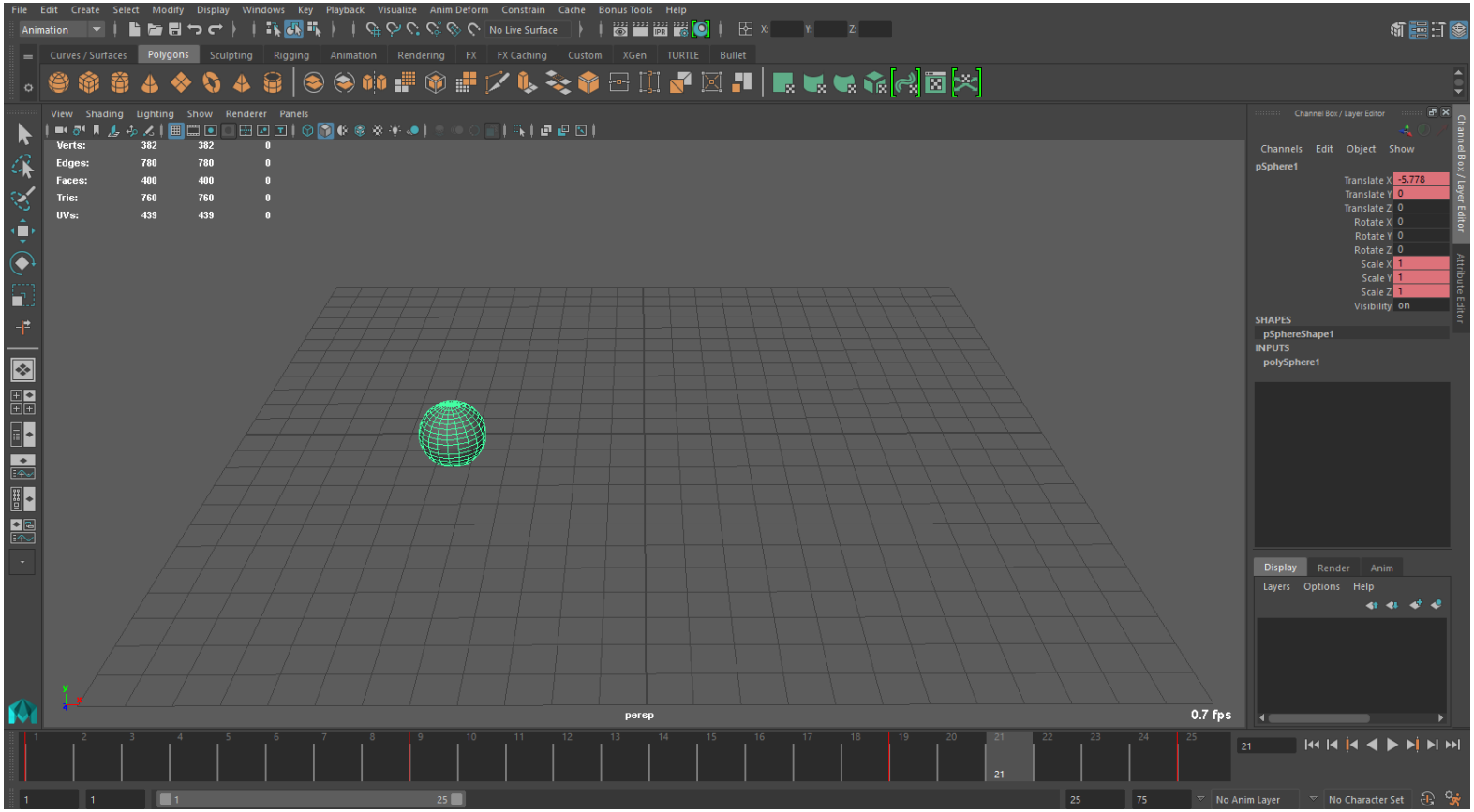
Move Scale Snap Last (slide 16)



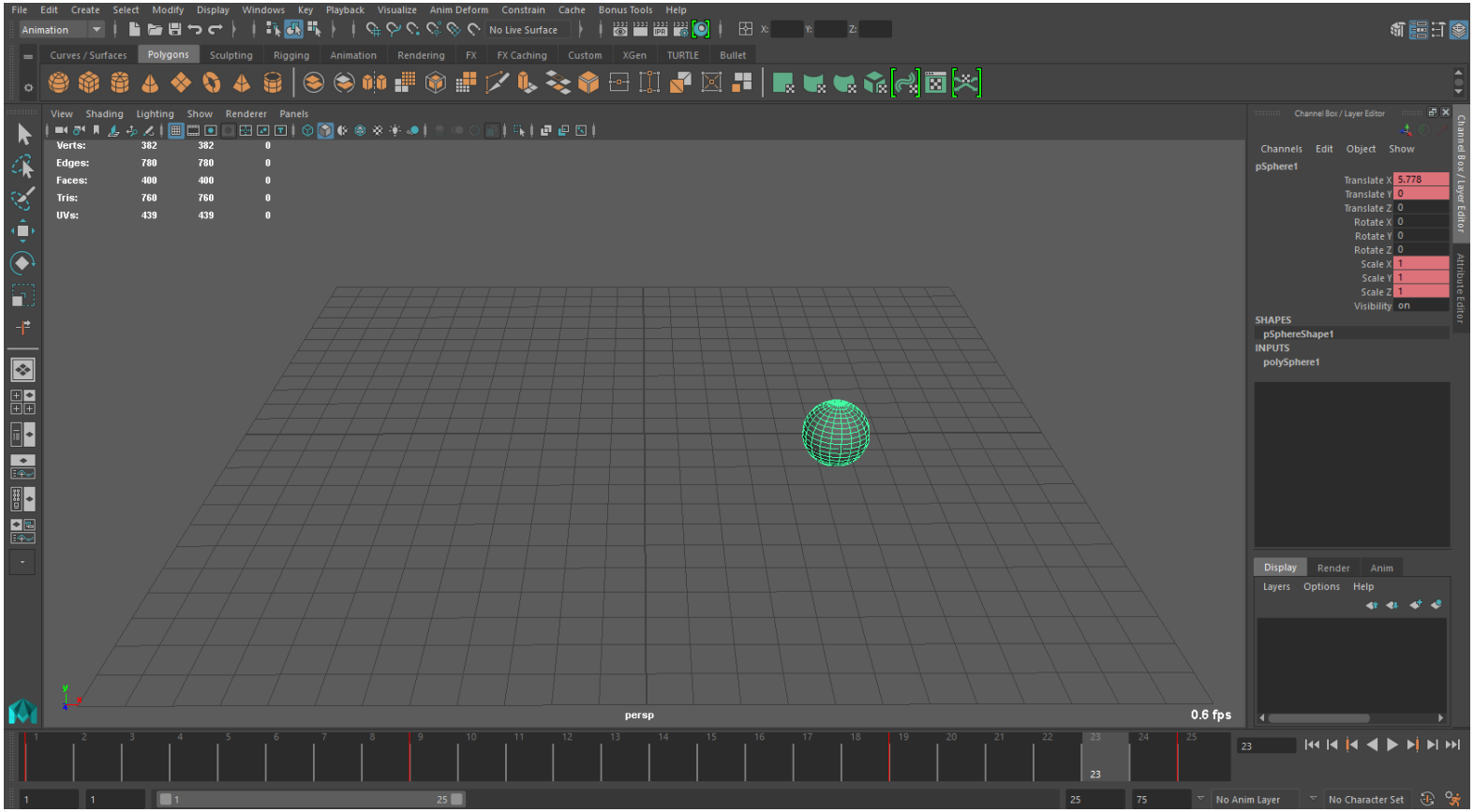
Move Scale Snap Last (slide 16)



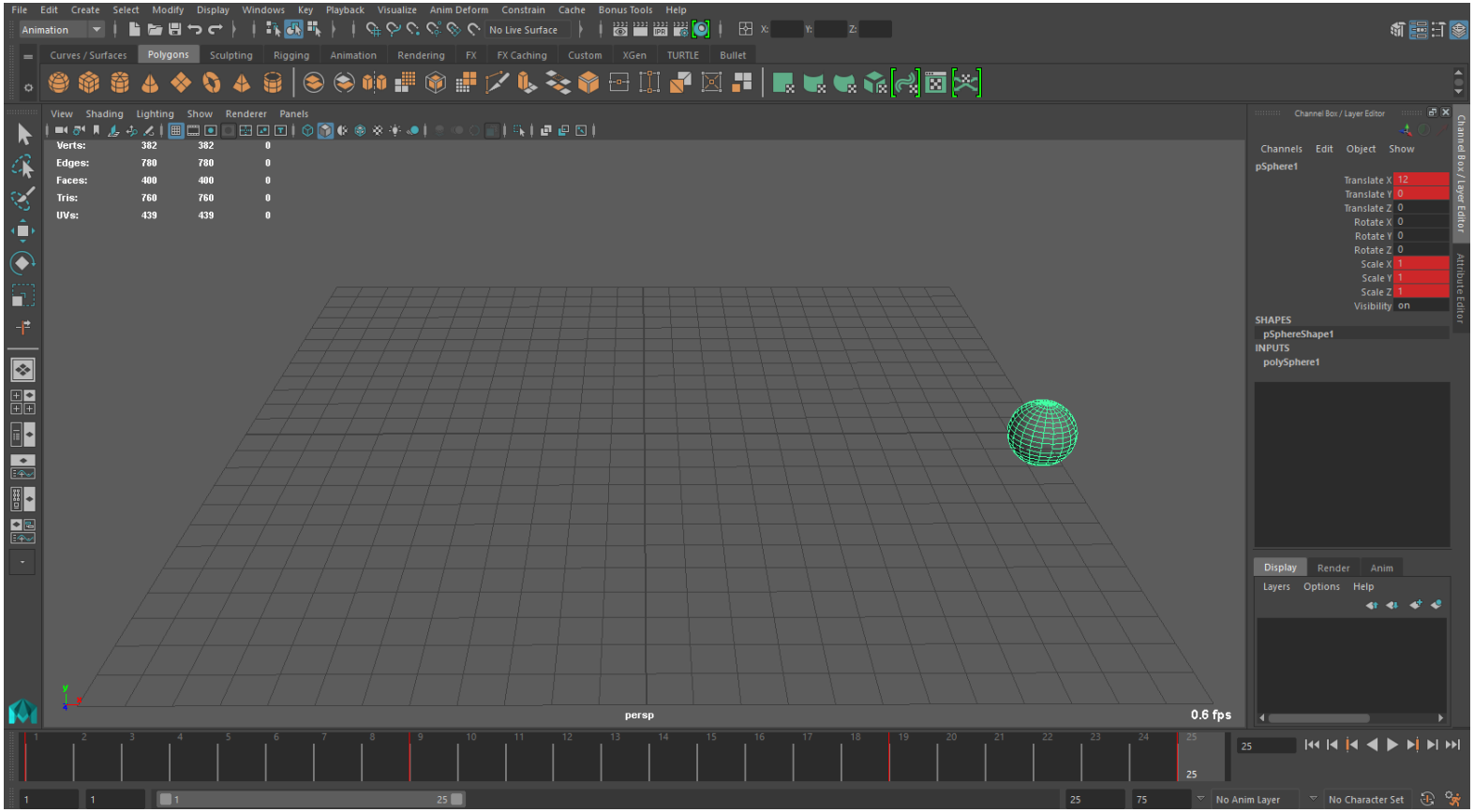
Move Scale Snap Last (slide 16)



Move Scale Snap Last (slide 16)

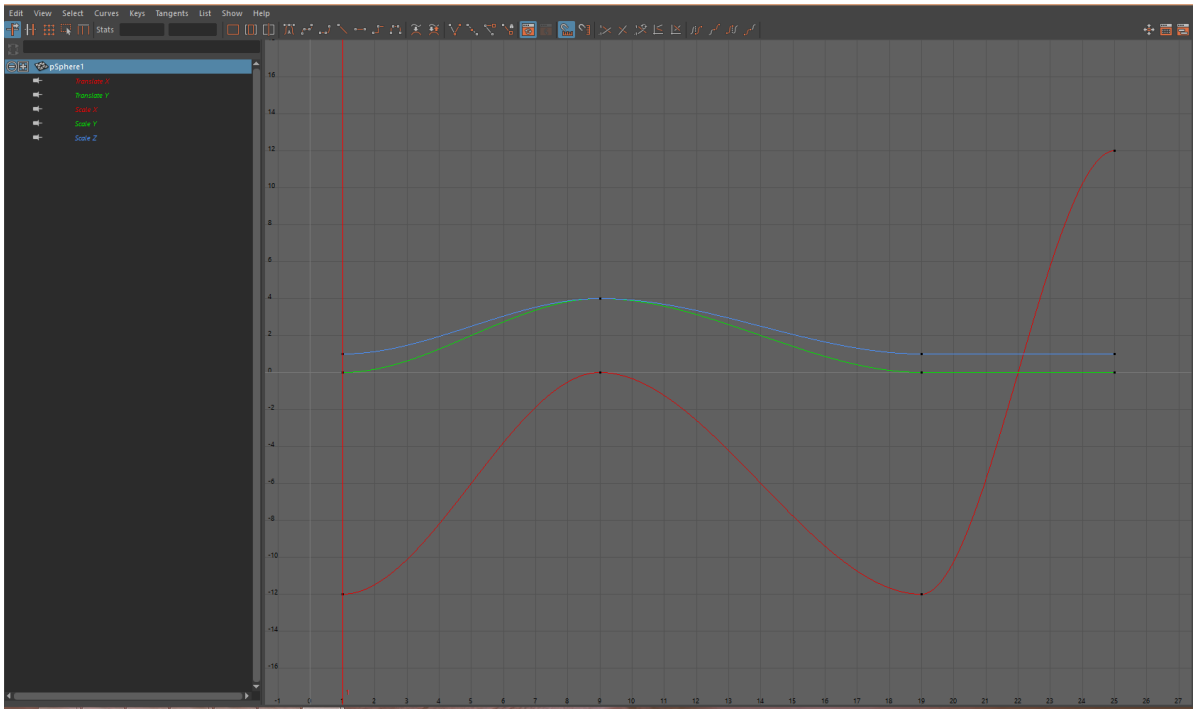


Move Scale Snap Last (slide 16)



The Graph Editor

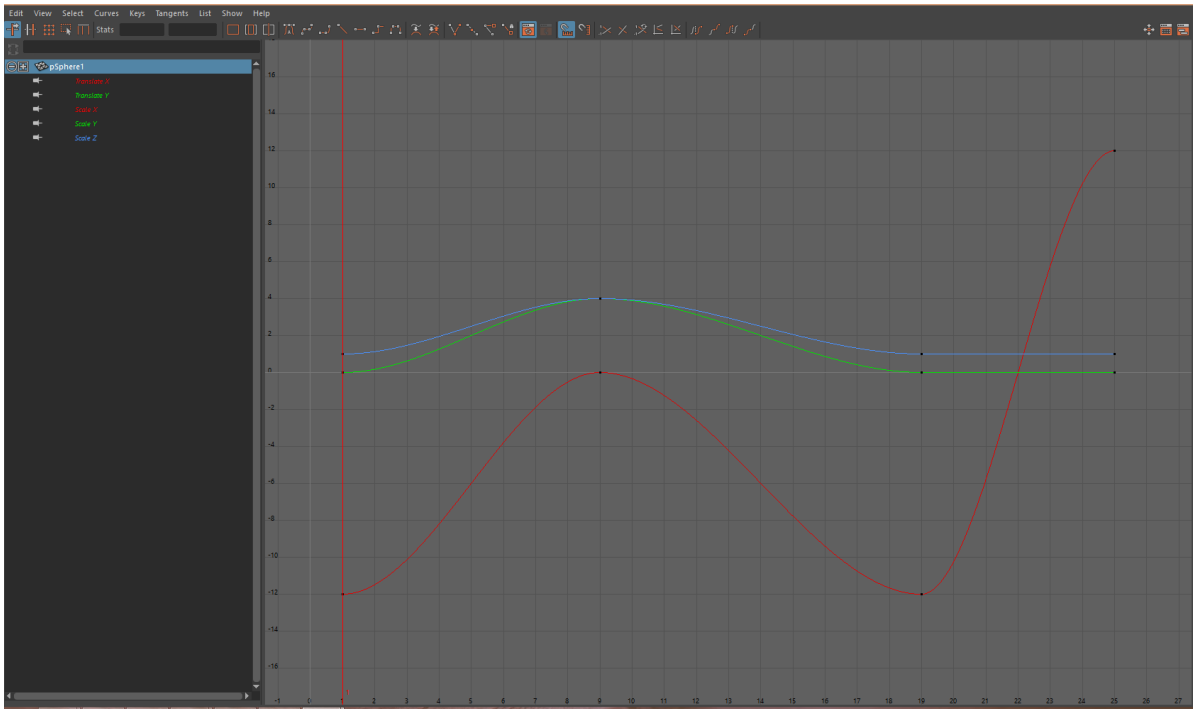
- Allows to edit move, scale, delete, cut and past key-frames.
- Allows to vary and change the way in-between frames are generated.
- Provides a convenient method of controlling timing.
- The figure shows the graph editor with curves which correspond to slide 16.



Animation Graph Editor (slide 17)

The Graph Editor

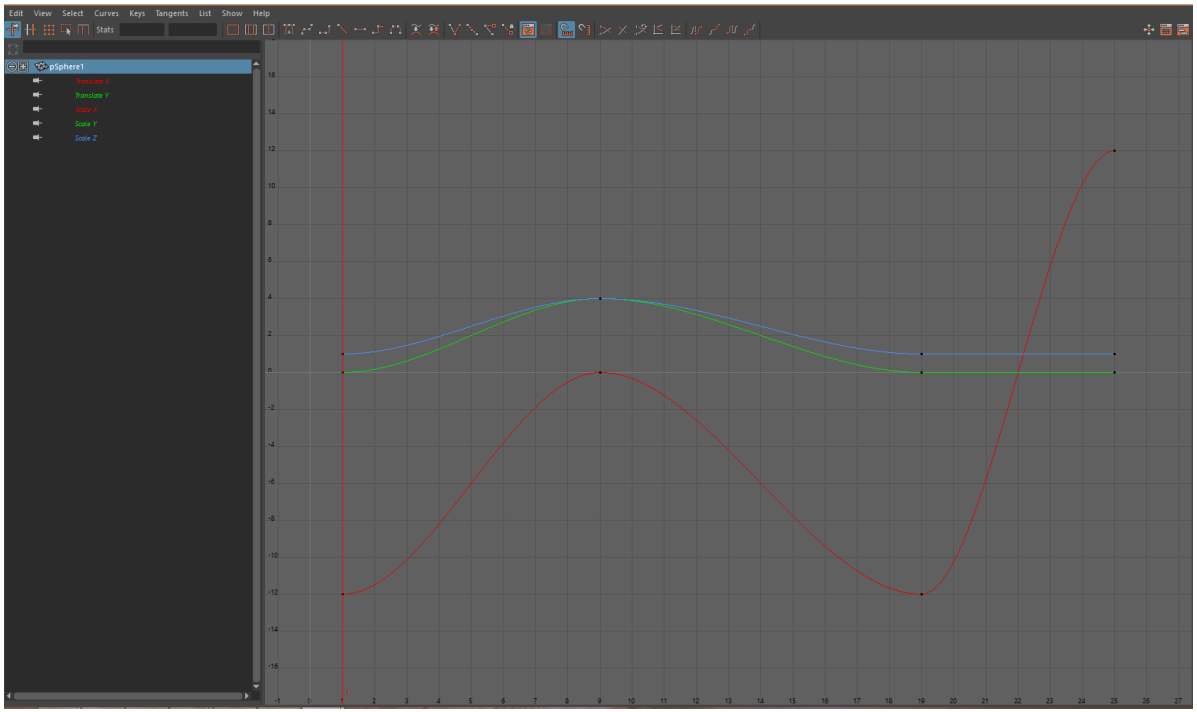
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Animation Graph Editor (slide 17)

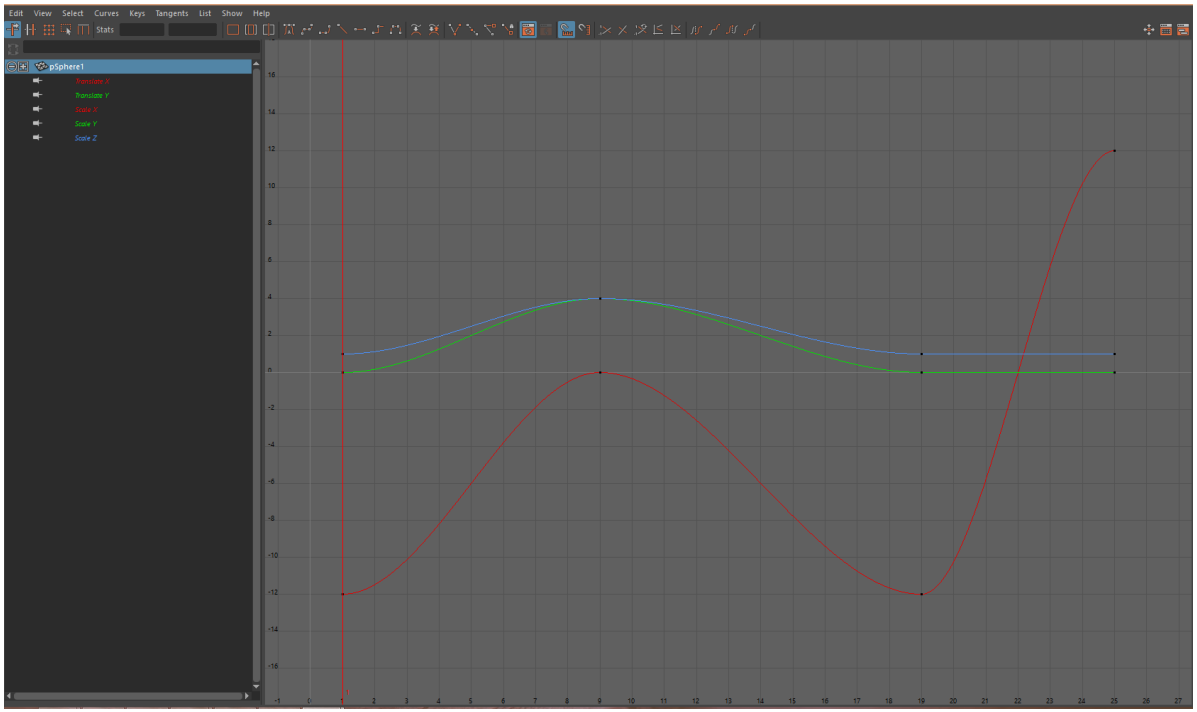
The Graph Editor

- Allows to edit move, scale, delete, cut and past key-frames.
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The Graph Editor

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Animation Graph Editor Tangents (slide 18)

The Graph Editor Tangents Options

- “Linear” tangents, interpolation is linear.
- “Spline” Tangents, interpolation is cubic polynomial using splines.
- “Clamped” Specifying a clamped tangent creates an animation curve that has the characteristics of linear and spline curves. The keys’ tangents will be splines unless the value of two adjacent keys are very close. In this case, the out tangent of the first key and the in tangent of the second key will be interpolated as linear.
- “Stepped” Specifying a stepped tangent creates an animation curve whose out tangent is a “flat” vertical or horizontal curve.
- “Plateau” Plateau tangents ease animation curves in and out of their key-frames like spline tangents, but also flatten curve segments that occur between equal-valued key-frames like clamped tangents. The plateau tangent normally behaves like a spline tangent, but ensures that the minimum and maximum values along a curve are all at key-frames.
- “Flat” Uses spline interpolation but sets the in and out spline tangents of the keys to be horizontal (with a slope of 0 degrees).
- “Auto” Specifying Auto tangents creates an animation curve where the first and last keyframes have flat tangents, and key-frames in between do not overshoot the neighboring key-frame value. This type of curve prevents issues with inter-penetration of closely animated objects that can occur with other tangent types, such as Spline.

Animation Graph Editor Tangents (slide 18)

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Animation Graph Editor Tangents (slide 18)

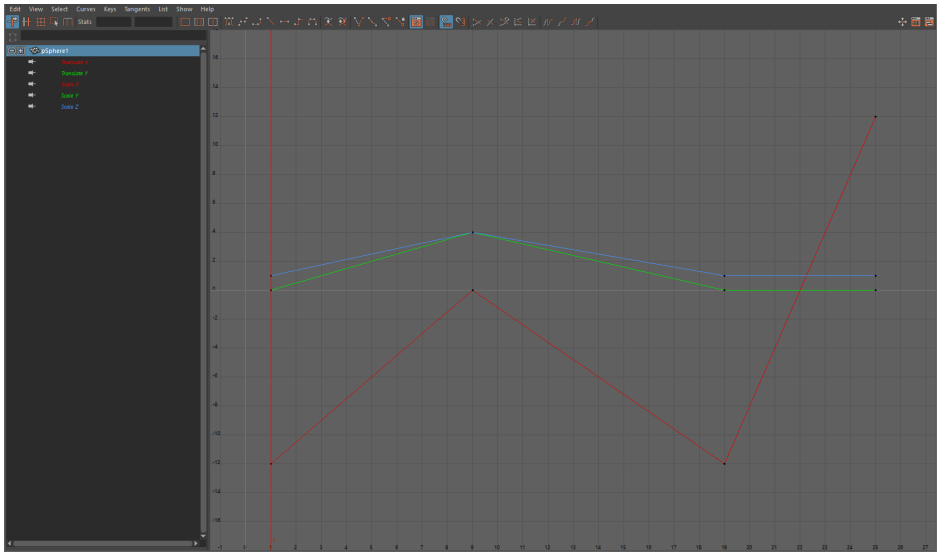
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Animation Graph Editor Tangents (slide 19)

The Graph Editor Tangents Options Continued

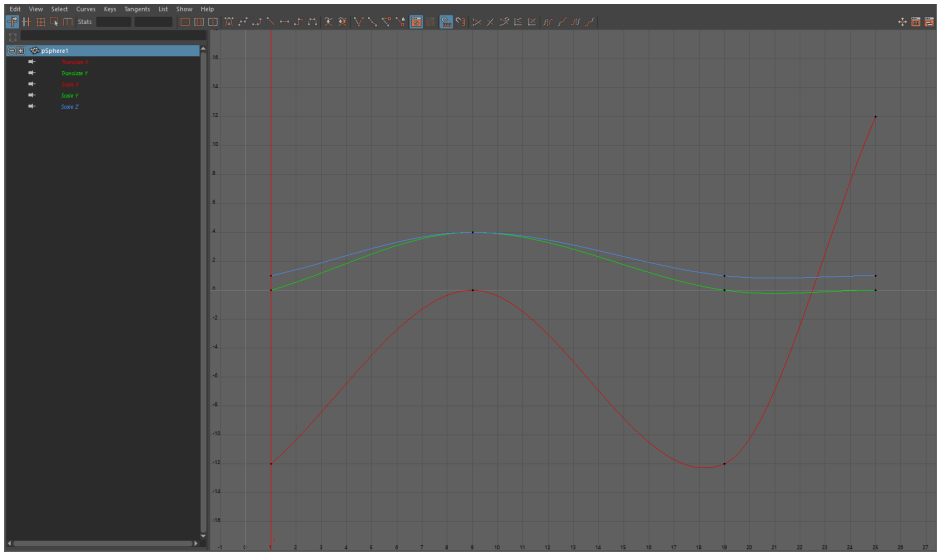
- “Linear”
- “Spline”
- “Clamped”
- “Stepped”
- “Plateau”
- “Flat”
- “Auto”



Animation Graph Editor Tangents (slide 19)

The Graph Editor Tangents Options Continued

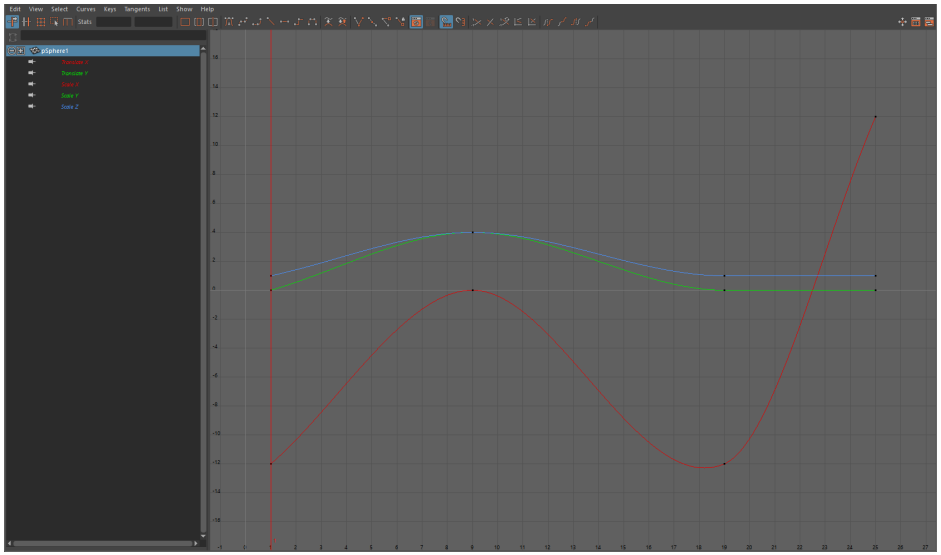
- “Linear”
- “Spline”
- “Clamped”
- “Stepped”
- “Plateau”
- “Flat”
- “Auto”



Animation Graph Editor Tangents (slide 19)

The Graph Editor Tangents Options Continued

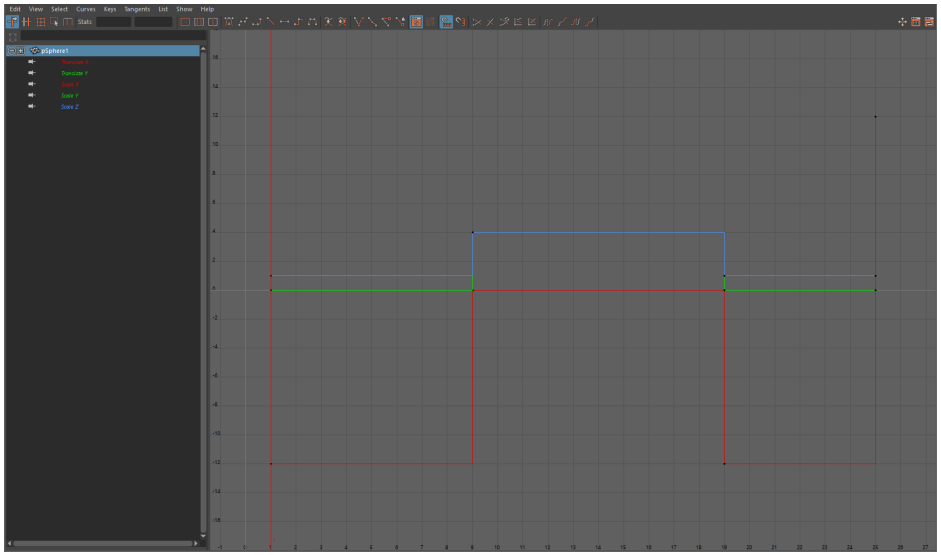
- “Linear”
- “Spline”
- “Clamped”
- “Stepped”
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Animation Graph Editor Tangents (slide 19)

The Graph Editor Tangents Options Continued

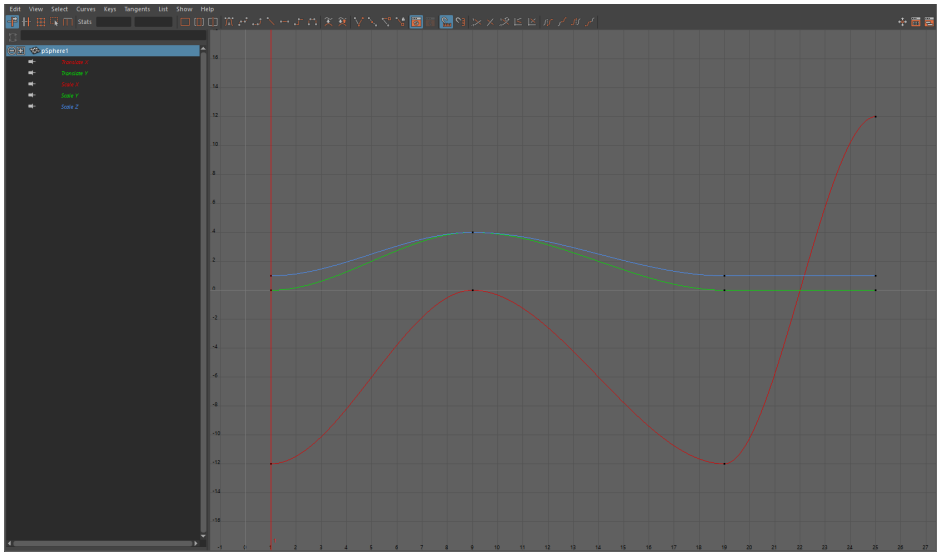
- “Linear”
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Animation Graph Editor Tangents (slide 19)

The Graph Editor Tangents Options Continued

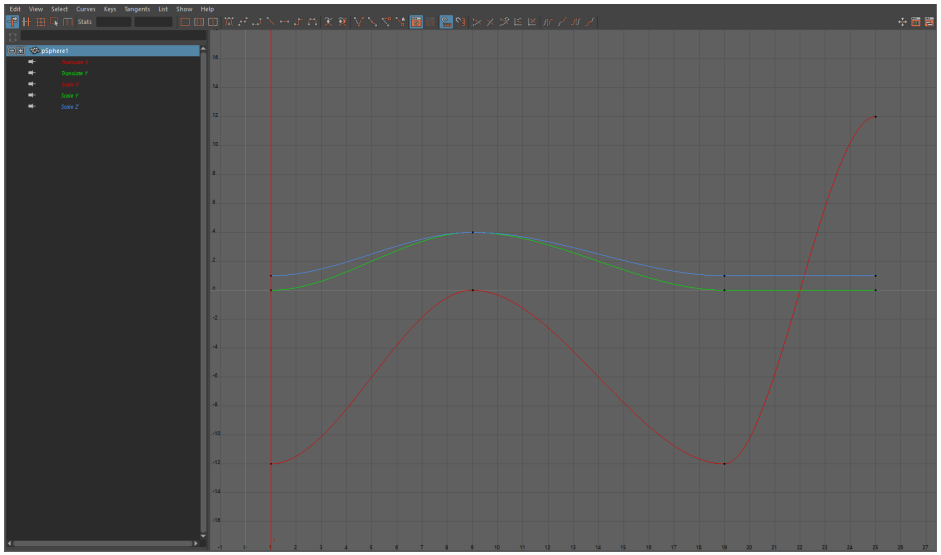
- “Linear”
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Animation Graph Editor Tangents (slide 19)

The Graph Editor Tangents Options Continued

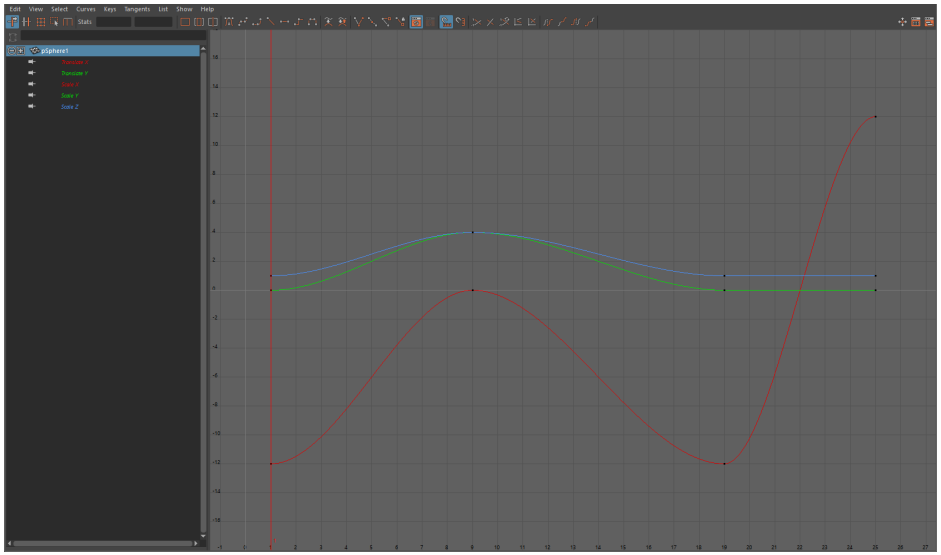
- “Linear”
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Animation Graph Editor Tangents (slide 19)

The Graph Editor Tangents Options Continued

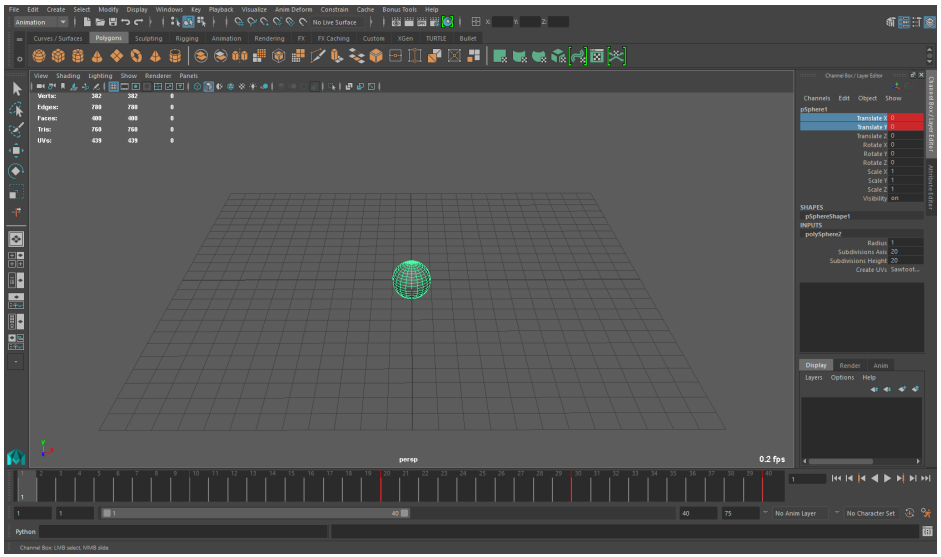
- “Linear”
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Bouncing Ball Attempt 01 (slide 20)

Bouncing ball at one spot with no friction.

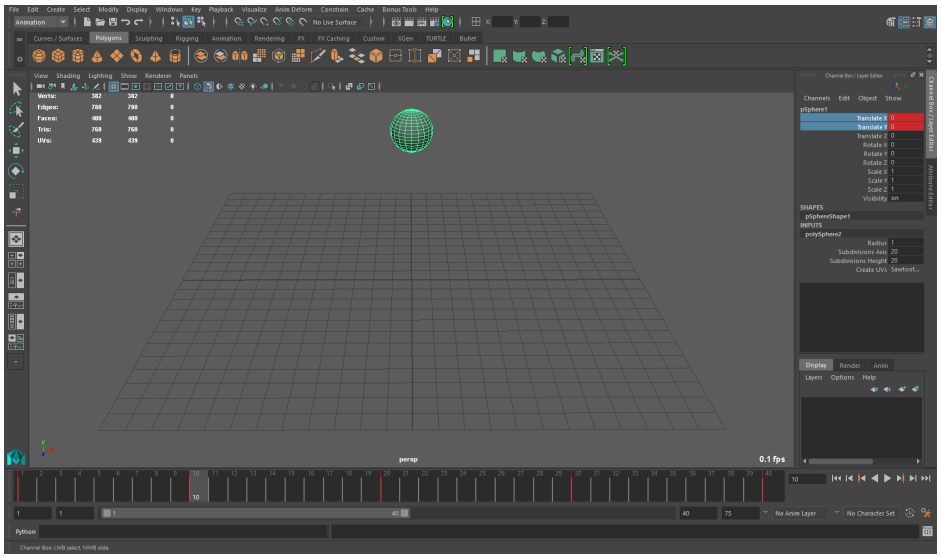
- Place a sphere at the origin (0, 0, 0) at (current) time 1. From the Channel Box key its selected translate Y/X coefficients.
- Set current time to 10 and translate its Y coefficient to (0, 8, 0). From the Channel Box key its selected translate Y/X coefficients.
- Set current time to 20 and translate its Y coefficient to (0, 0, 0). From the Channel Box key its selected translate Y/X coefficients.
- Repeat after 10 frames so the ball bounces to (0, 8, 0).
- Repeat after 10 frames so the ball lands at (0, 0, 0).



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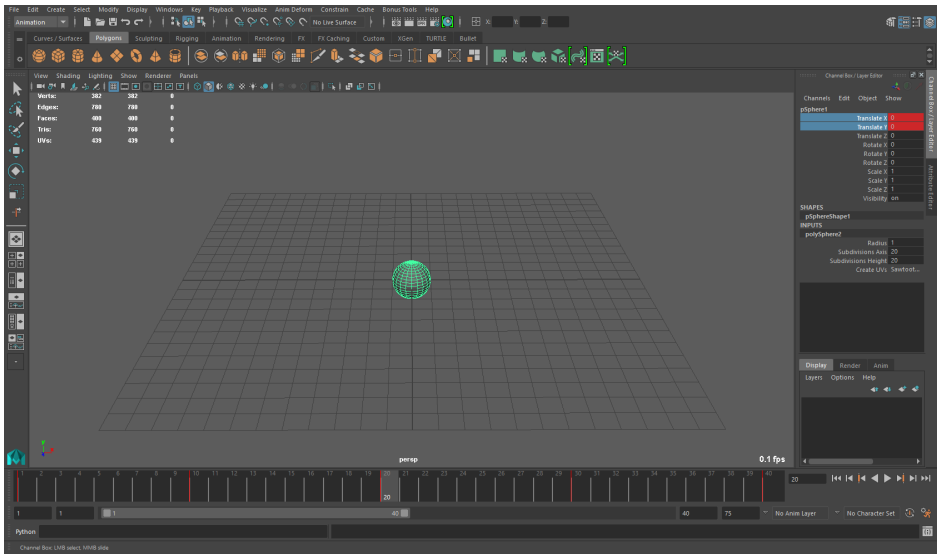
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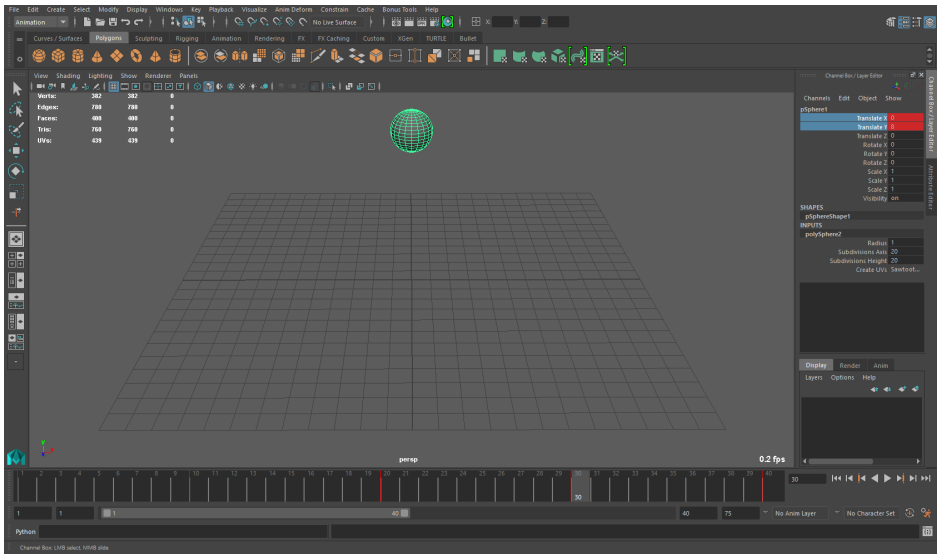
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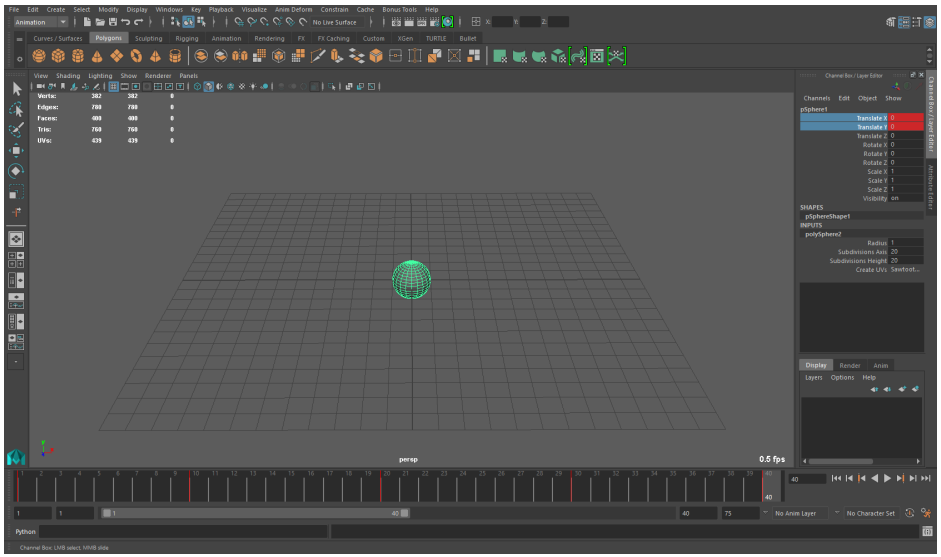
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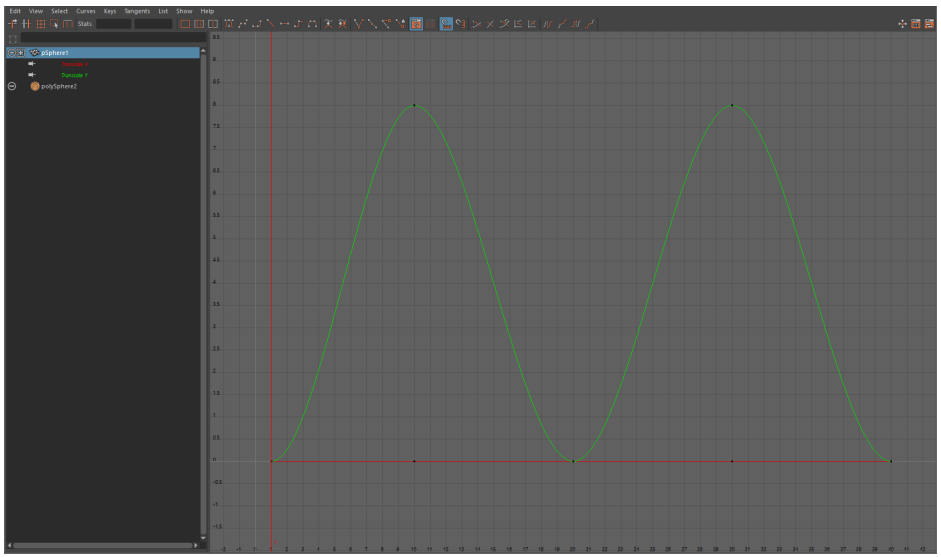
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Bouncing Ball Attempt 01 (slide 21)

Bouncing ball's graph editor and timing.

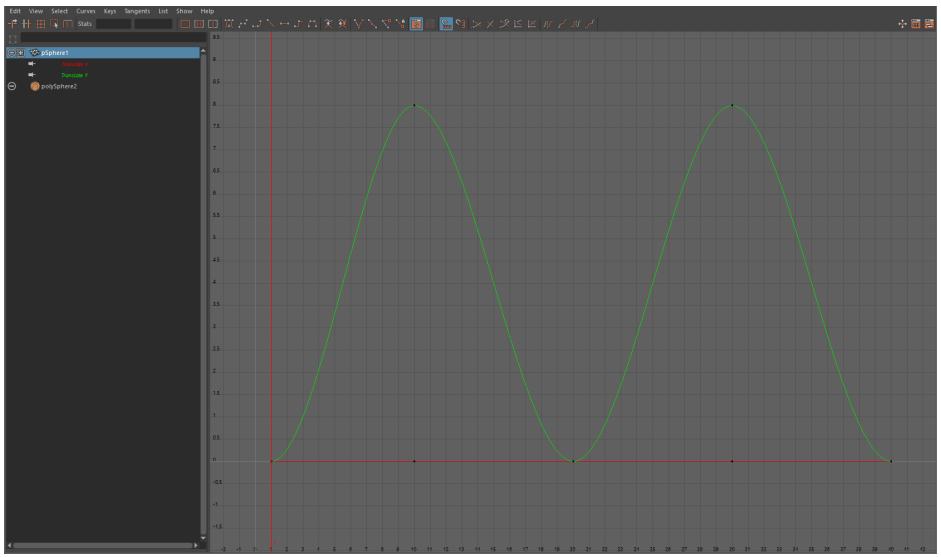
- Look at the animation curve of the translate Y coefficient in the graph editor.
- It is splined (by default) in a way which allows generation of slow in and slow out in-between frames.
- However, this is not appropriate for a bouncing ball movement. This because we know the ball needs to accelerate while falling, hitting the floor at its highest speed and decelerate while assenting, coming to a plateau when at its highest.
- We can improve the timing by editing the animation curve tangents manually.



Bouncing Ball Attempt 01 (slide 21)

Bouncing ball's graph editor and timing.

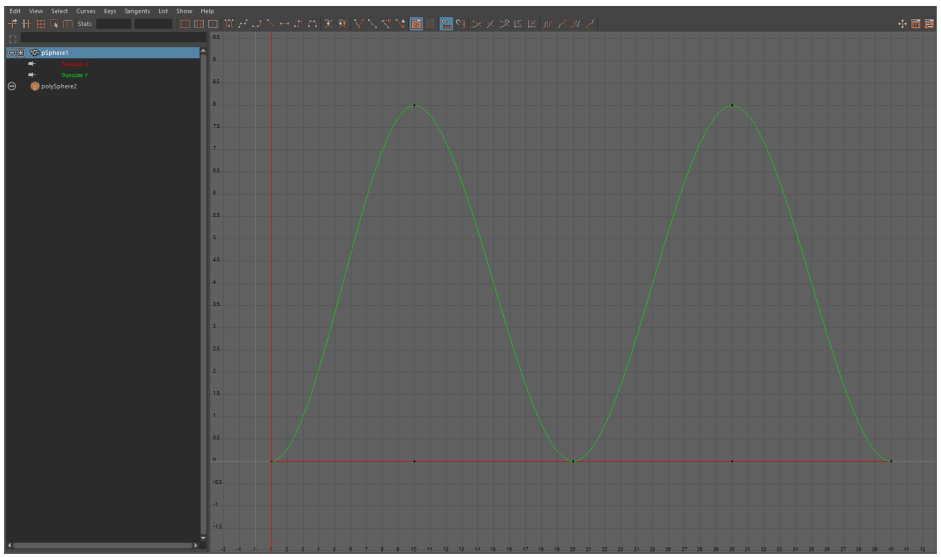
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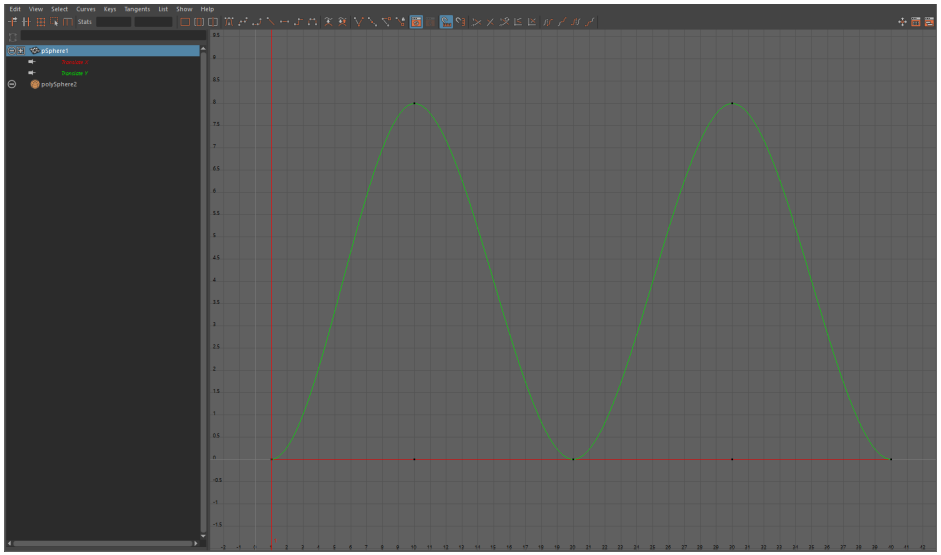
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- However, this is not appropriate for a bouncing ball movement. This because we know the ball needs to accelerate while falling, hitting the floor at its highest speed and decelerate while assenting, coming to a plateau when at its highest.
- We can improve the timing by editing the animation curve tangents manually.



Bouncing Ball Attempt 01 (slide 21)

Bouncing ball's graph editor and timing.

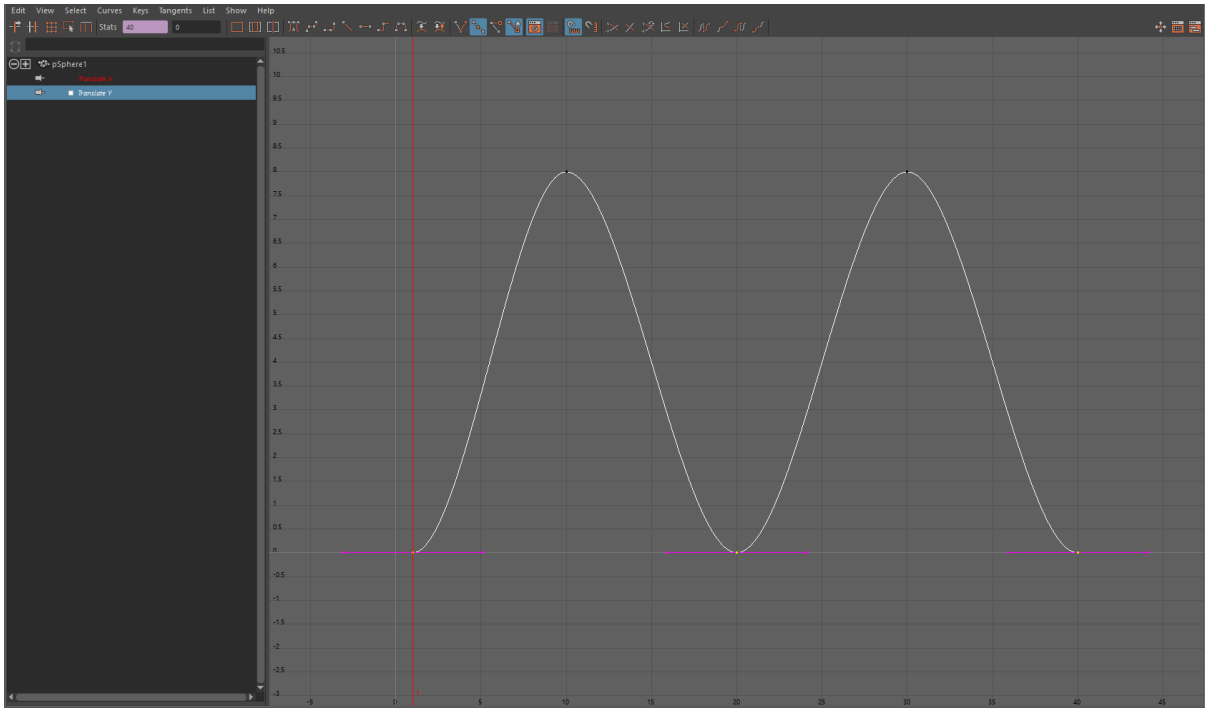
- Look at the animation curve of the translate Y coefficient in the graph editor.
- It is splined (by default) in a way which allows generation of slow in and slow out in-between frames.
- However, this is not appropriate for a bouncing ball movement. This because we know the ball needs to accelerate while falling, hitting the floor at its highest speed and decelerate while assenting, coming to a plateau when at its highest.
- We can improve the timing by editing the animation curve tangents manually.



Bouncing Ball Attempt 02 (slide 22)

Bouncing ball with improved timing.

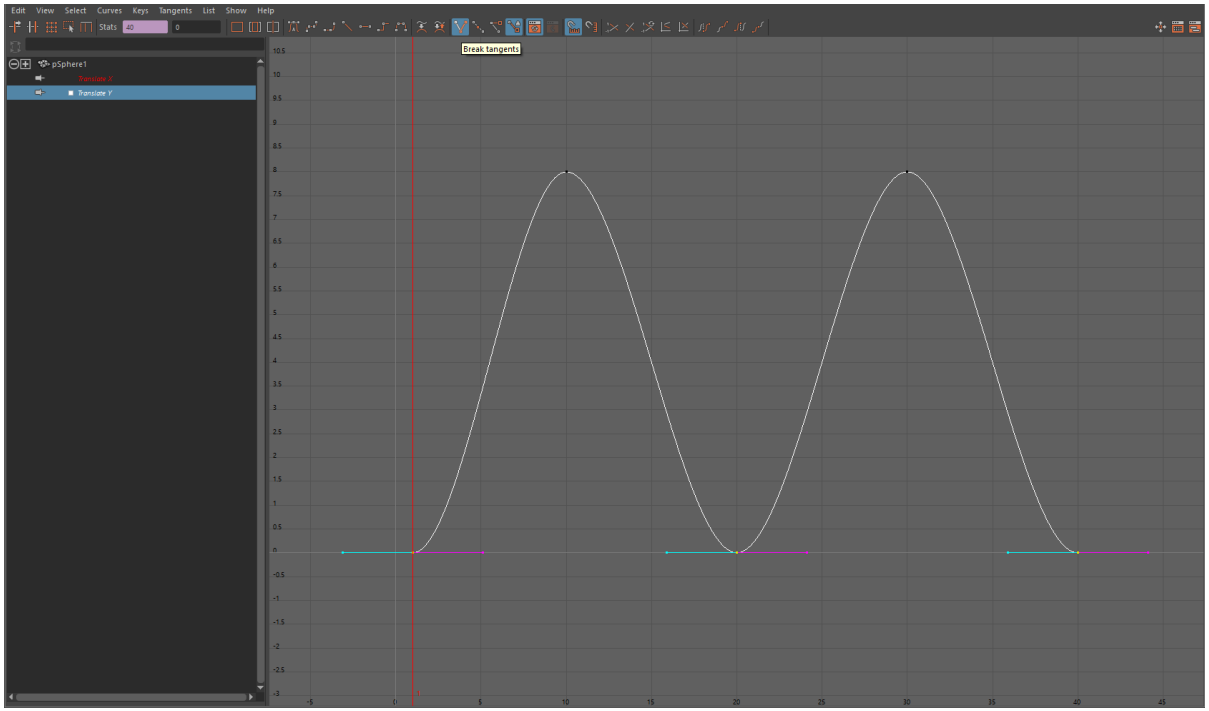
- We select the tangents of the key-frames when the ball hits the ground in the graph editor so we can edit them.
- We “break” the selected key-frame tangents so we can manipulate them.
- We change the tangents so they bend down more sharply.



Bouncing Ball Attempt 02 (slide 22)

Bouncing ball with improved timing.

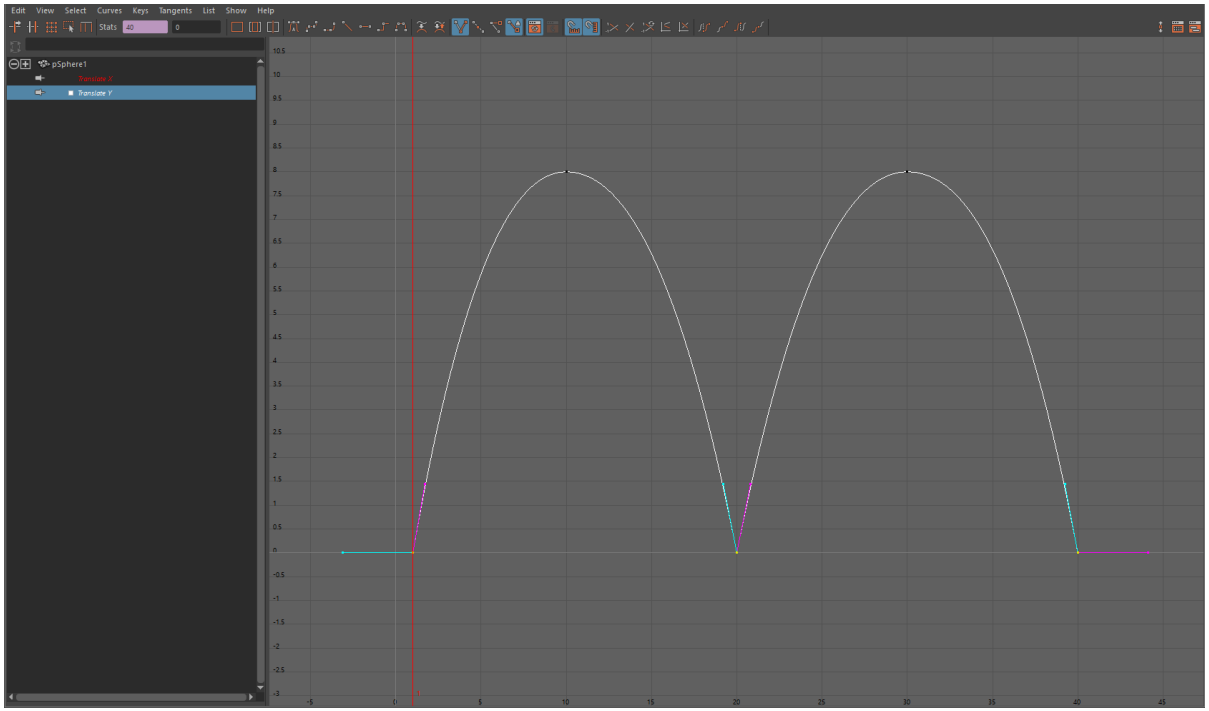
- We select the tangents of the key-frames when the ball hits the ground in the graph editor so we can edit them.
- We “break” the selected key-frame tangents so we can manipulate them.
- We change the tangents so they bend down more sharply.



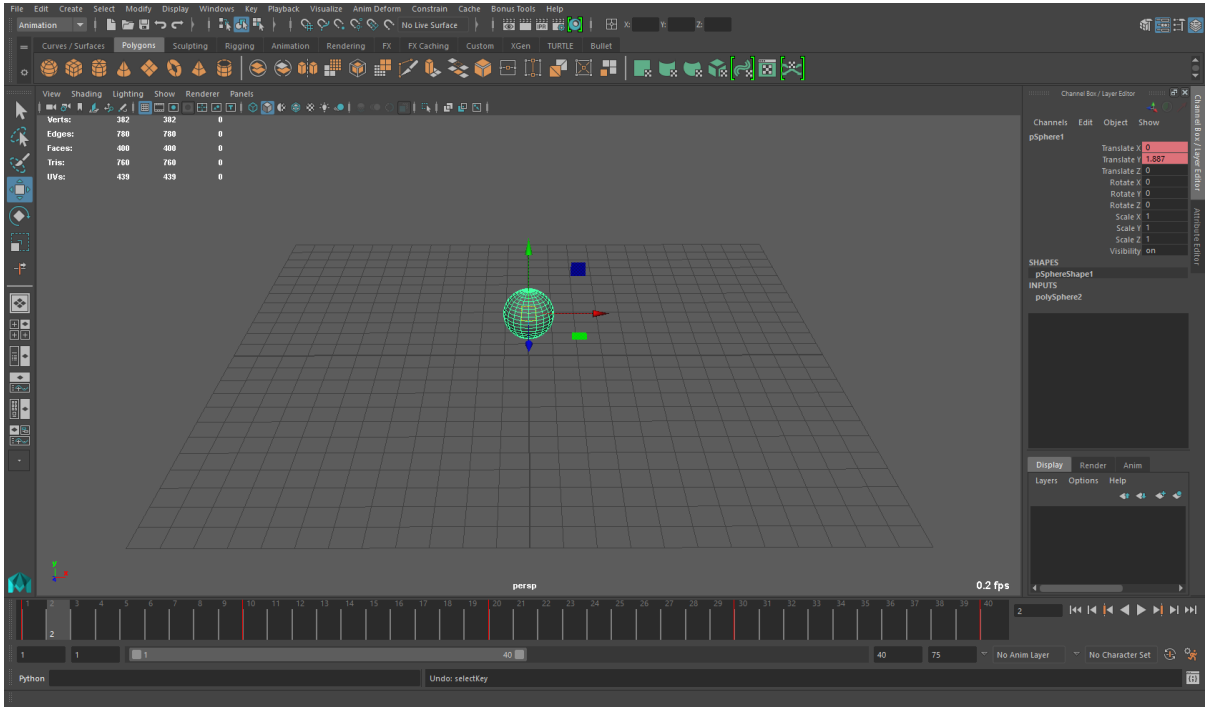
Bouncing Ball Attempt 02 (slide 22)

Bouncing ball with improved timing.

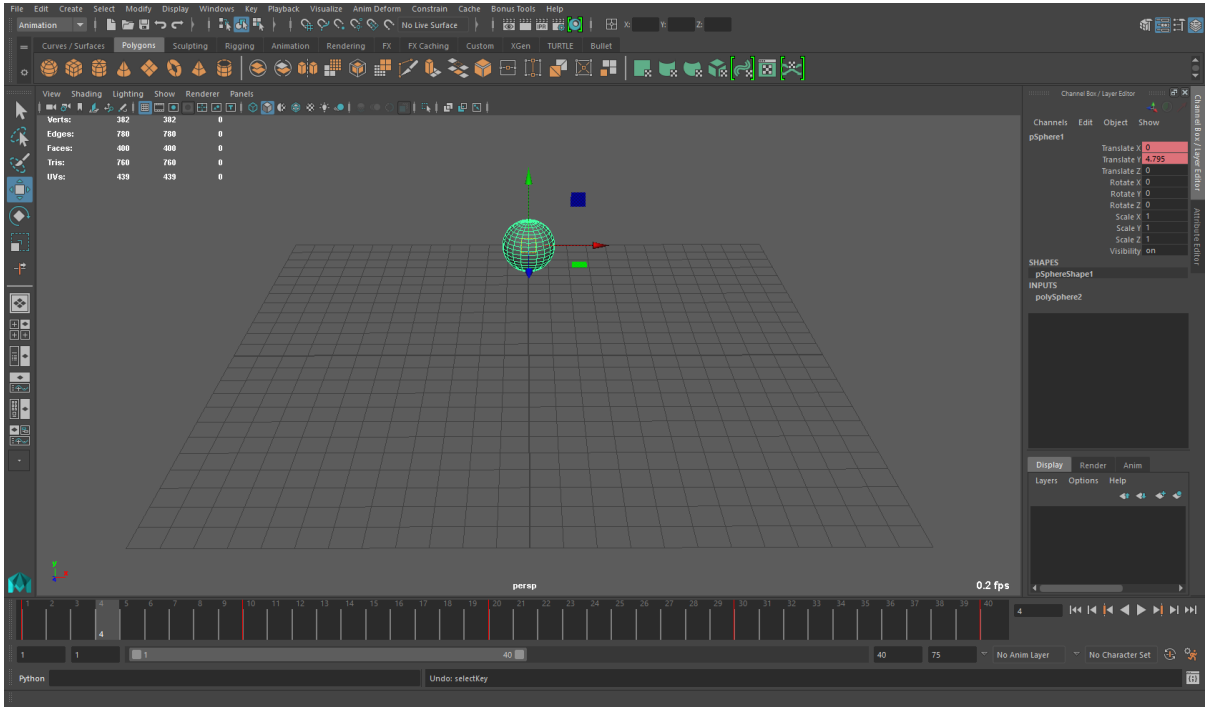
- We select the tangents of the key-frames when the ball hits the ground in the graph editor so we can edit them.
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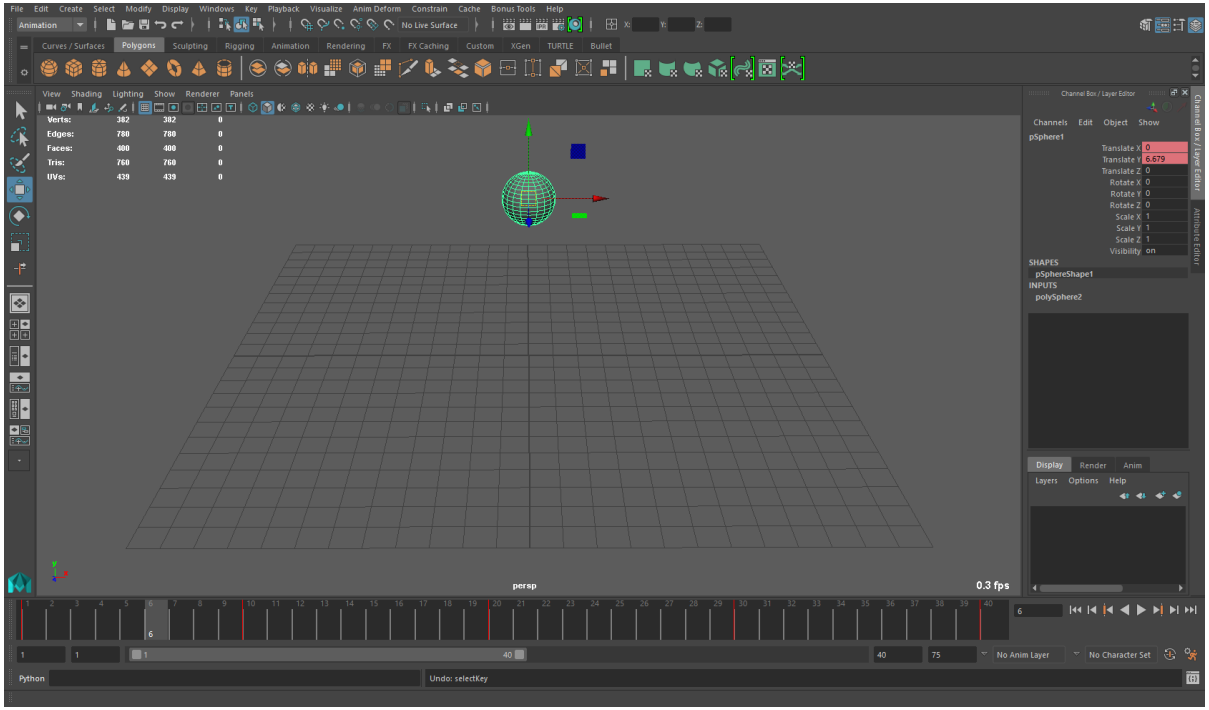
Bouncing Ball Attempt 02 (slide 23)



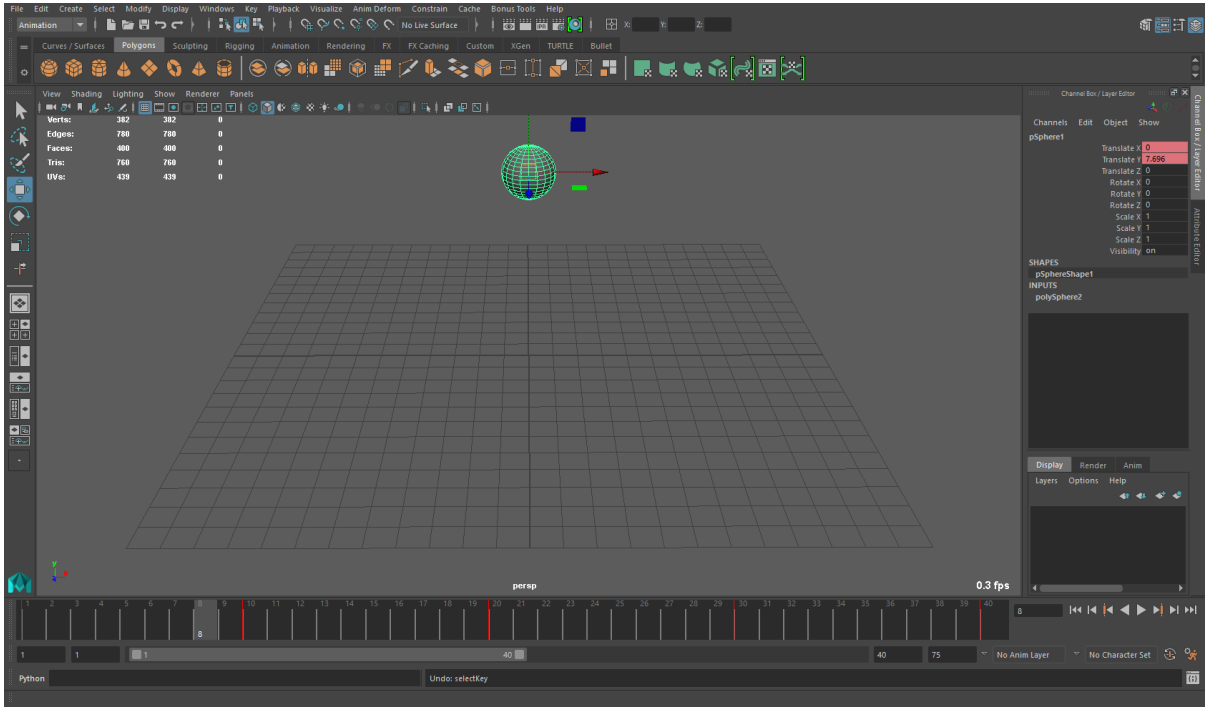
Bouncing Ball Attempt 02 (slide 23)



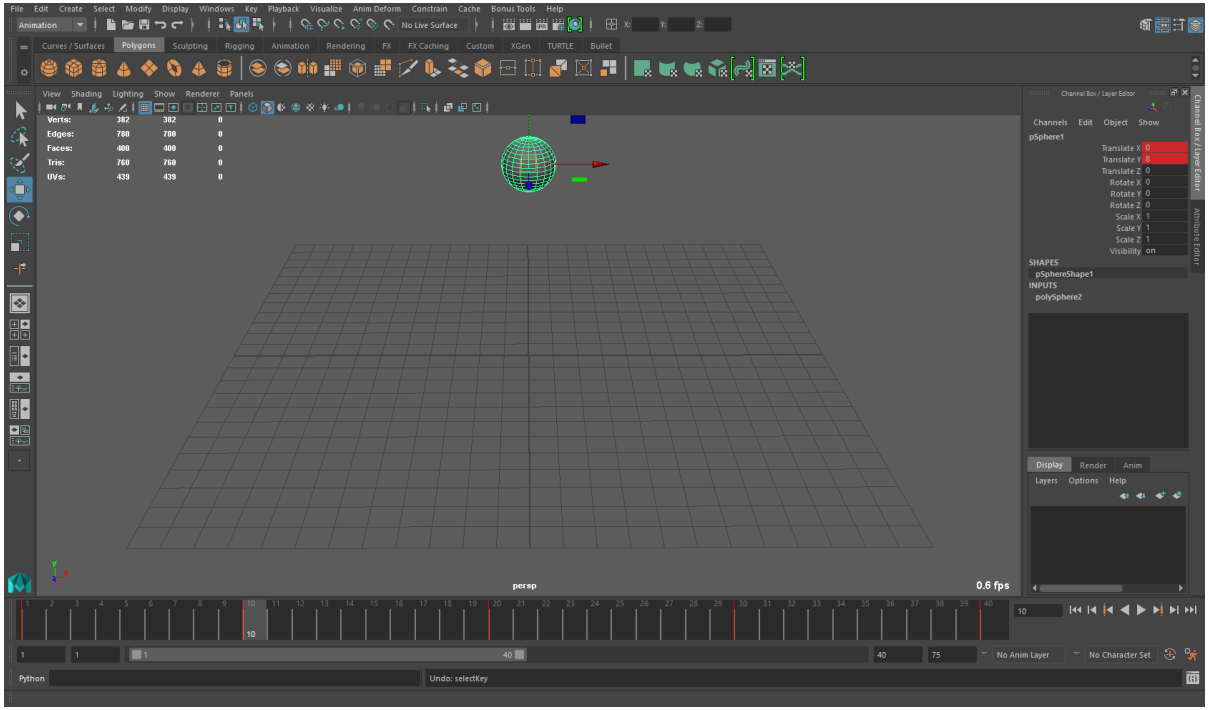
Bouncing Ball Attempt 02 (slide 23)



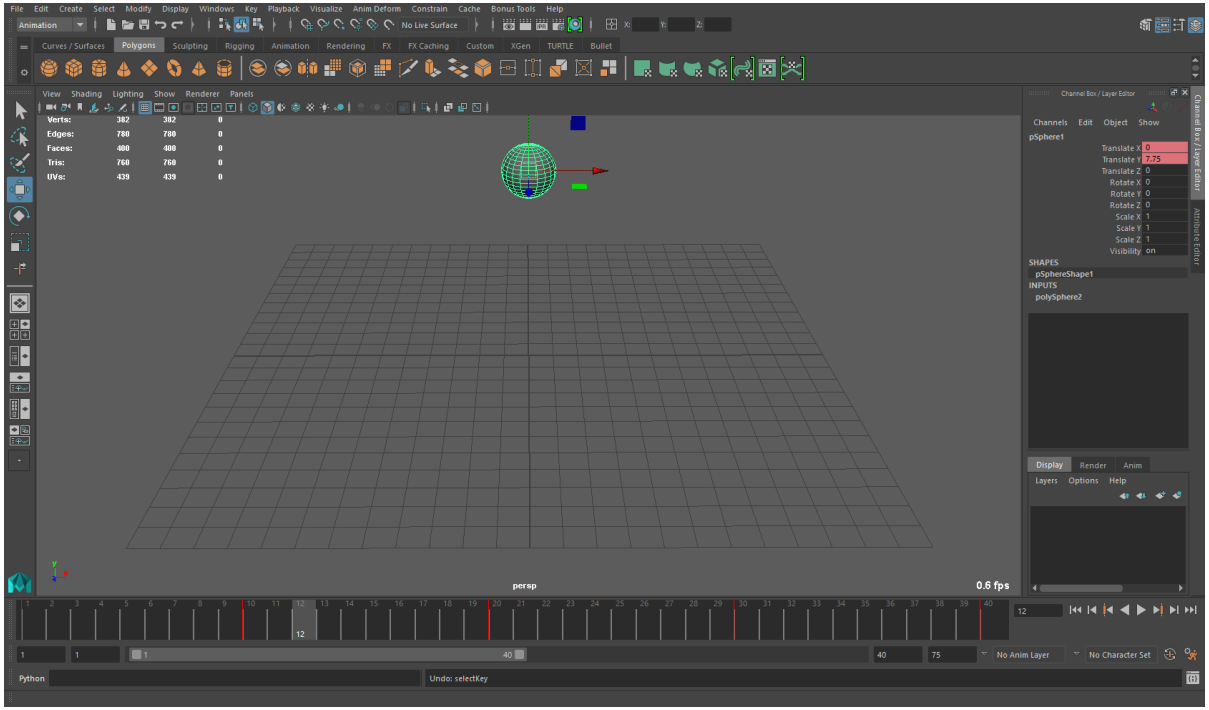
Bouncing Ball Attempt 02 (slide 23)



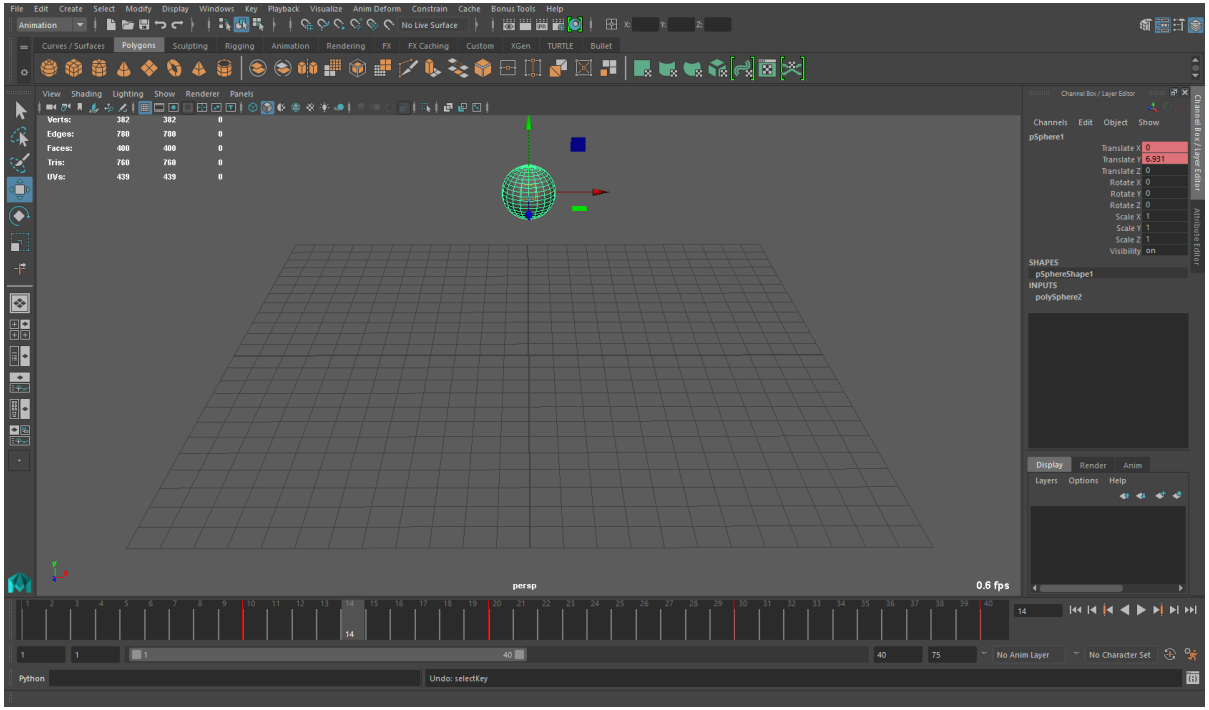
Bouncing Ball Attempt 02 (slide 23)



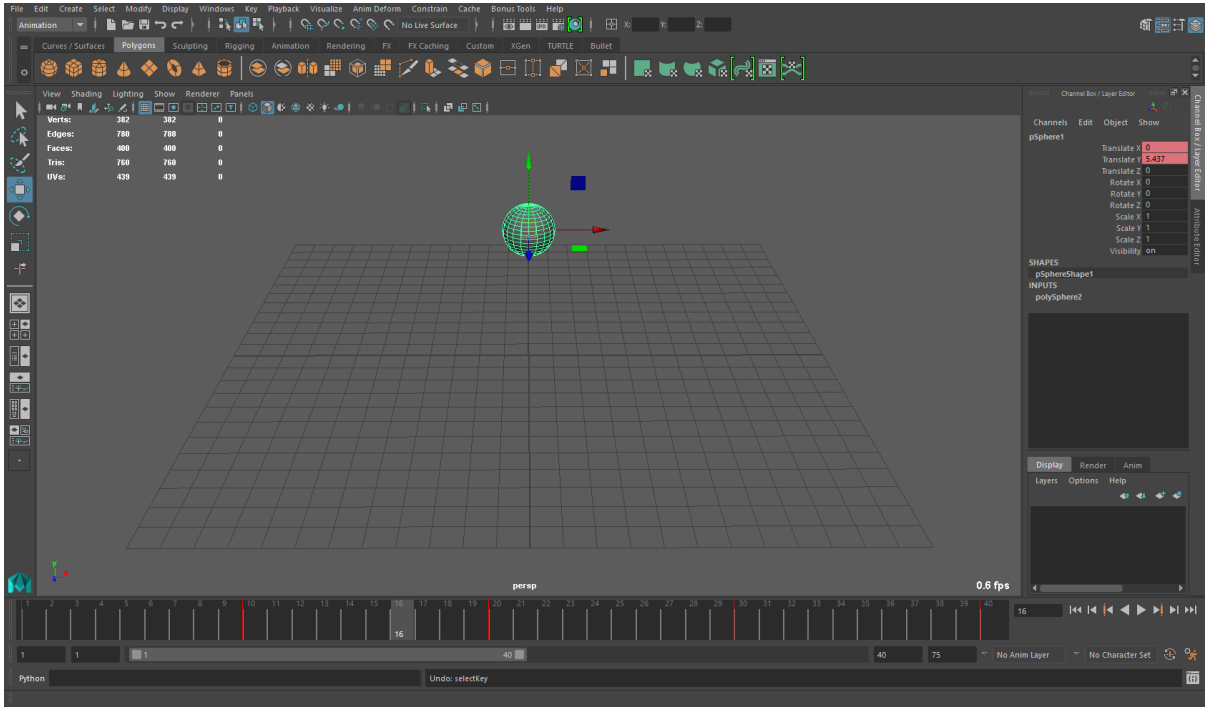
Bouncing Ball Attempt 02 (slide 23)



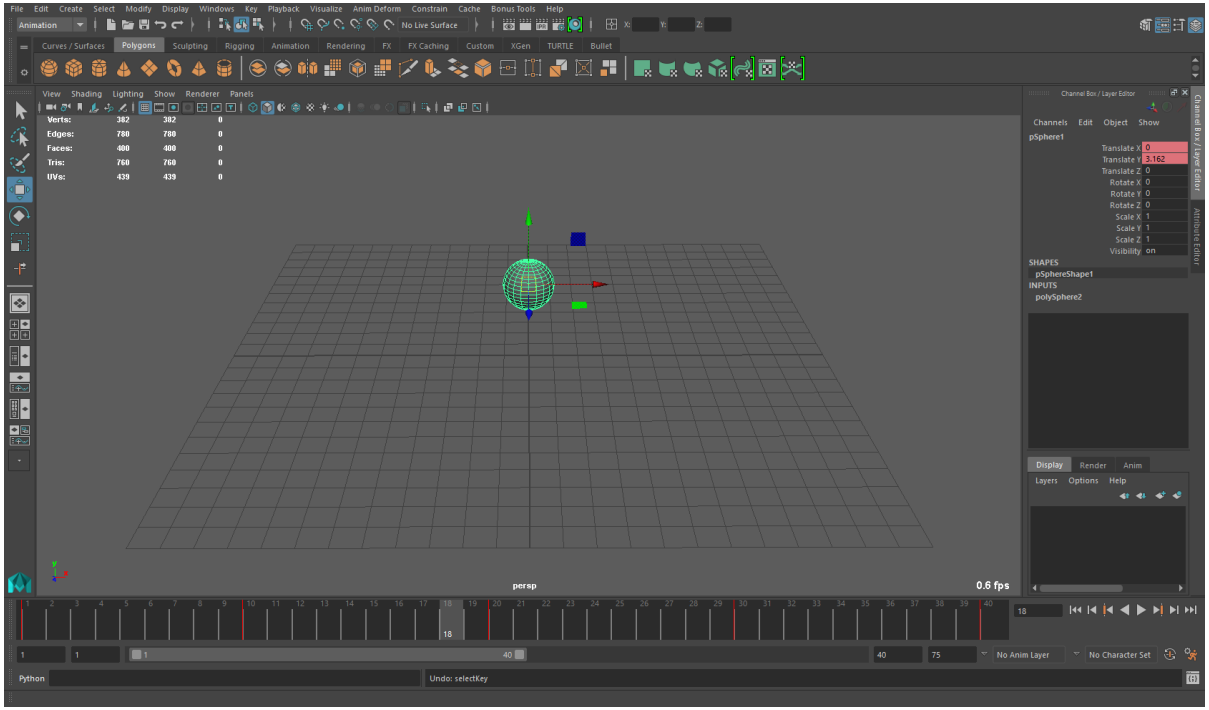
Bouncing Ball Attempt 02 (slide 23)



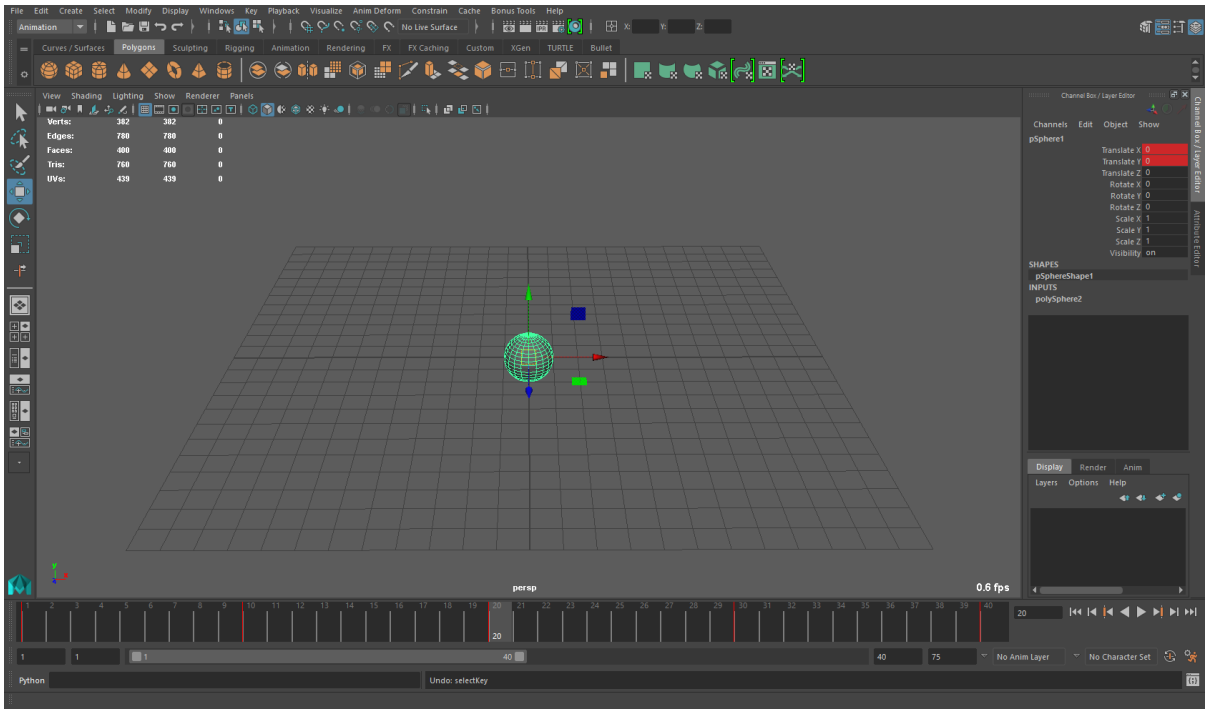
Bouncing Ball Attempt 02 (slide 23)



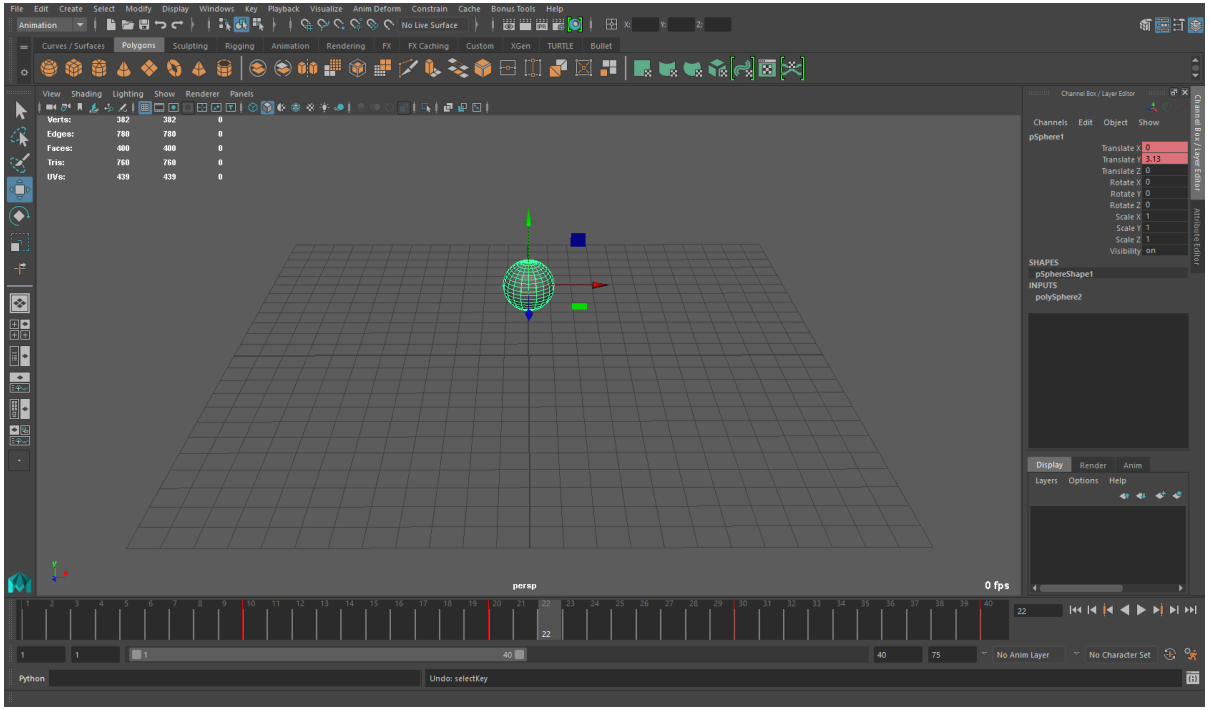
Bouncing Ball Attempt 02 (slide 23)



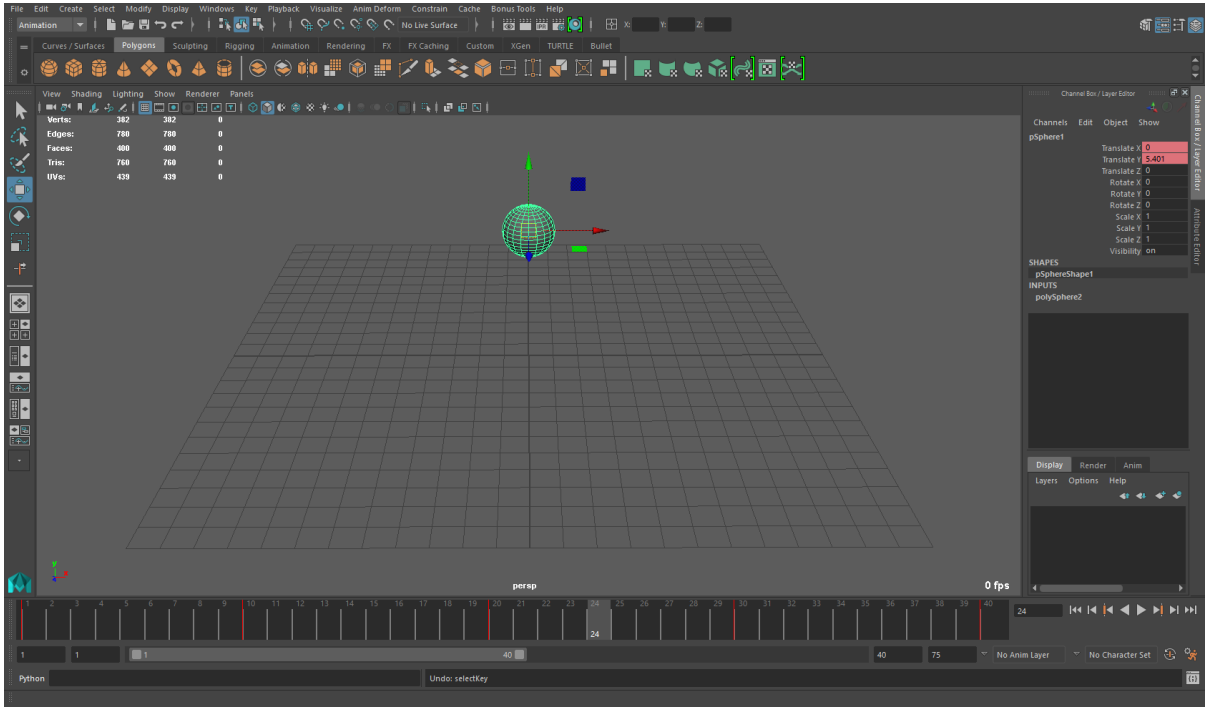
Bouncing Ball Attempt 02 (slide 23)



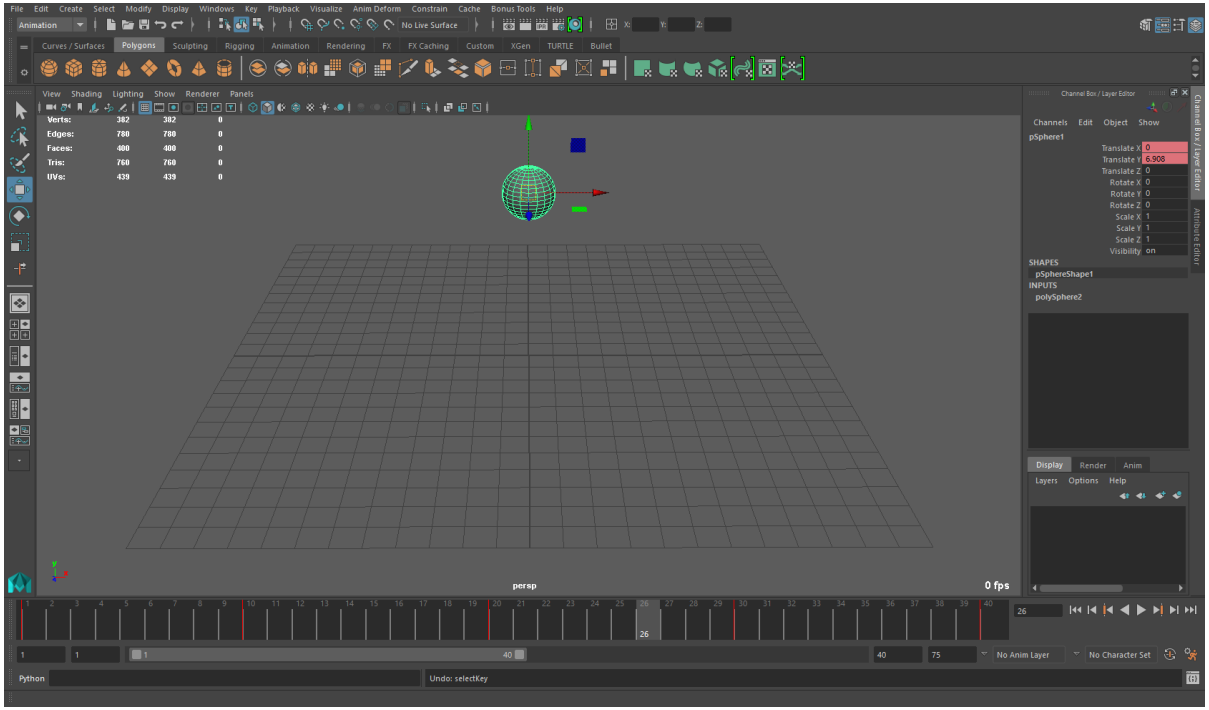
Bouncing Ball Attempt 02 (slide 23)



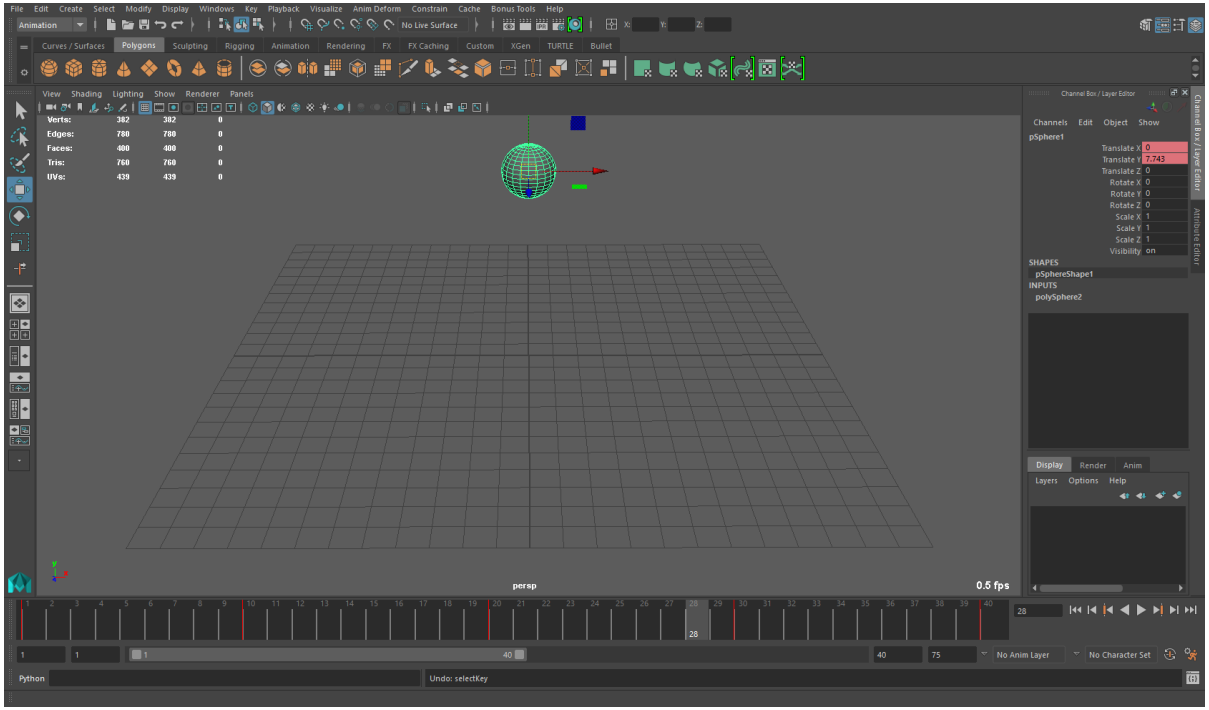
Bouncing Ball Attempt 02 (slide 23)



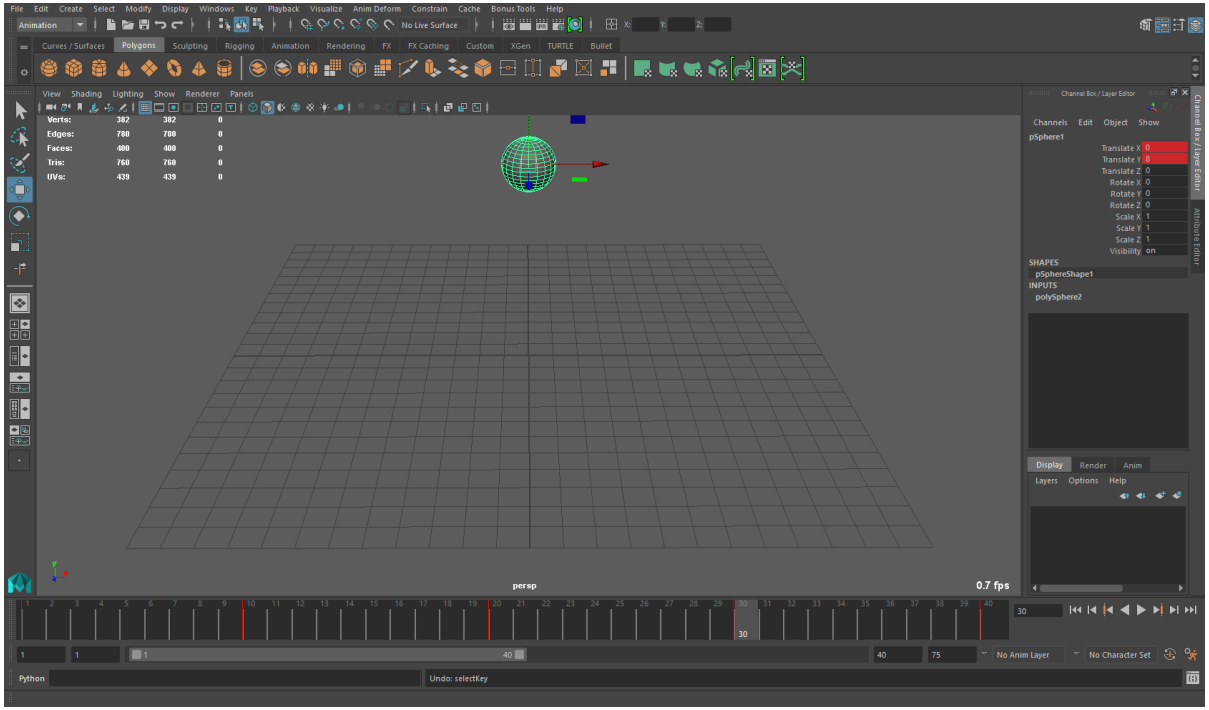
Bouncing Ball Attempt 02 (slide 23)



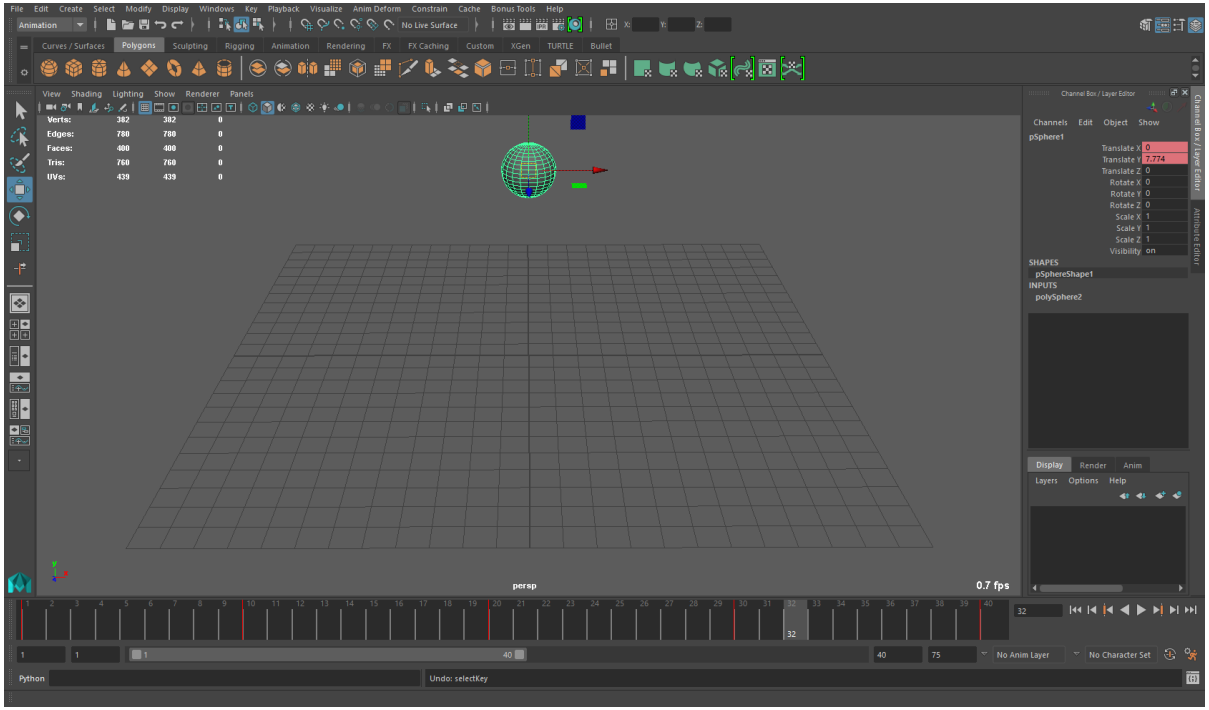
Bouncing Ball Attempt 02 (slide 23)



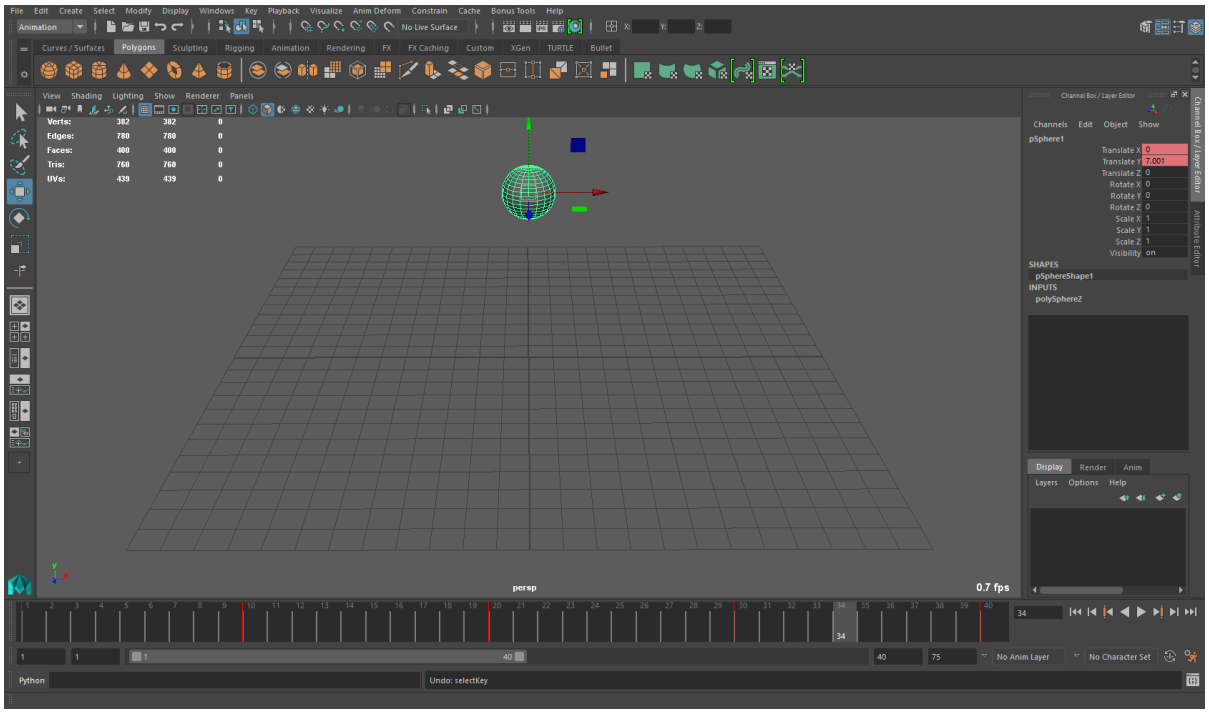
Bouncing Ball Attempt 02 (slide 23)



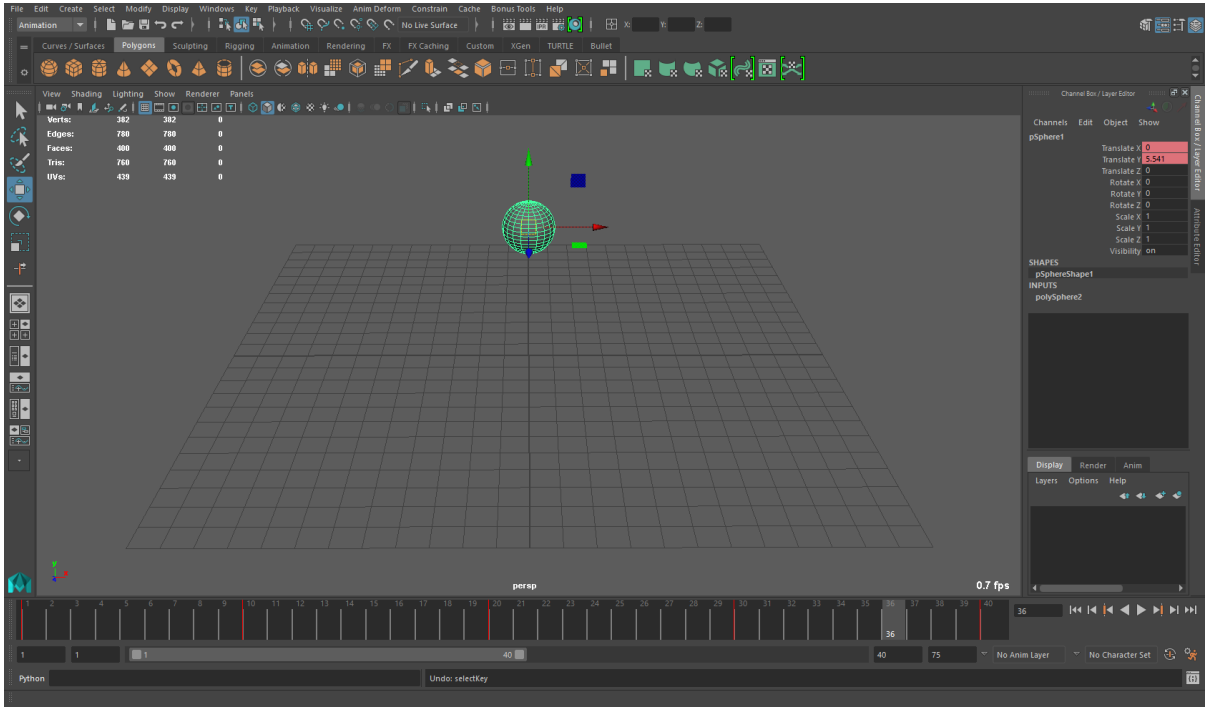
Bouncing Ball Attempt 02 (slide 23)



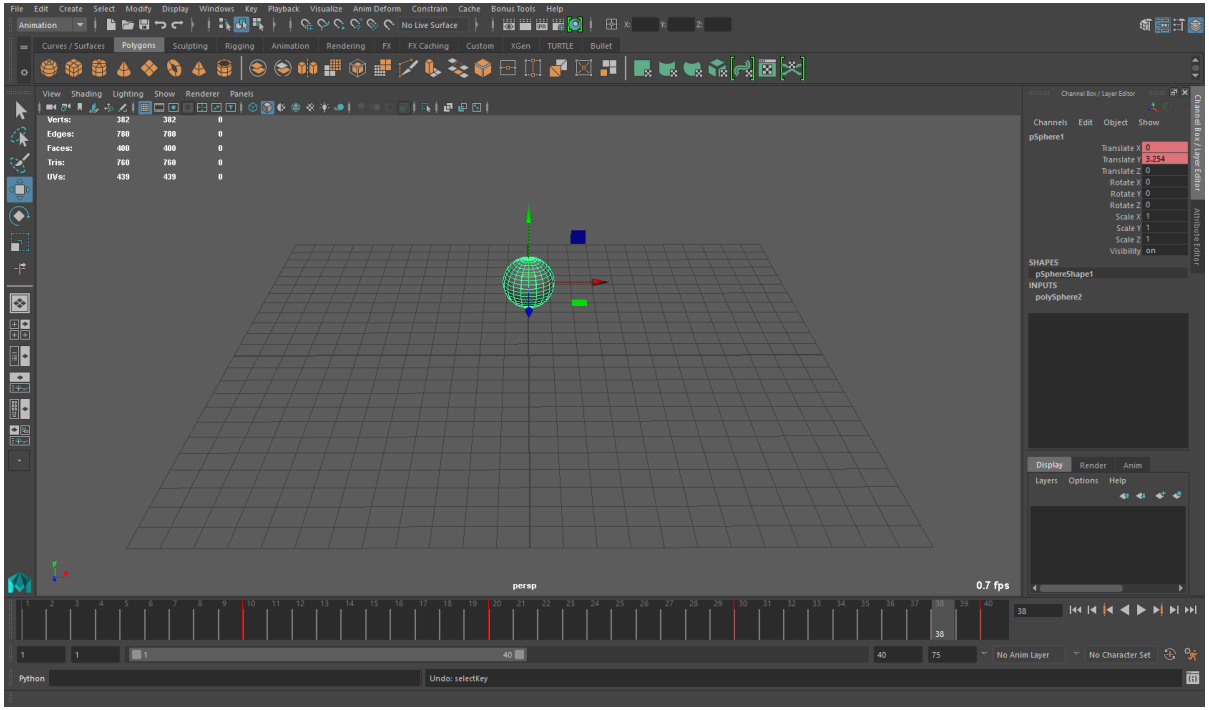
Bouncing Ball Attempt 02 (slide 23)



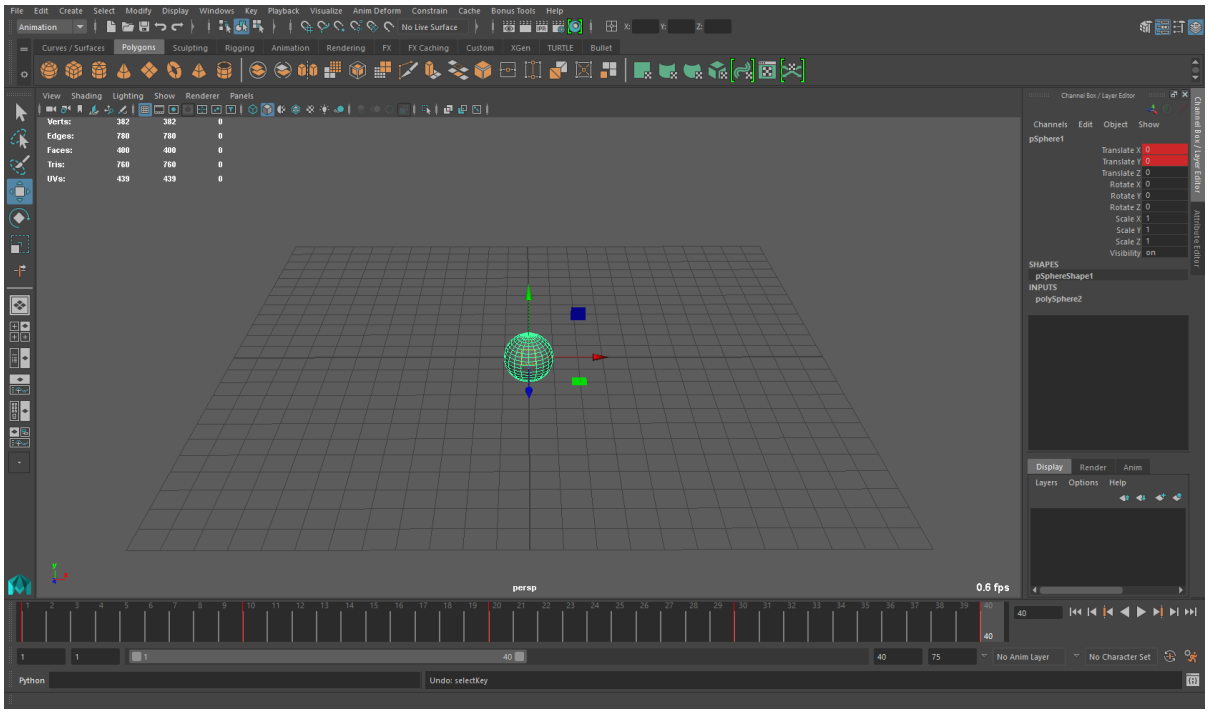
Bouncing Ball Attempt 02 (slide 23)



Bouncing Ball Attempt 02 (slide 23)



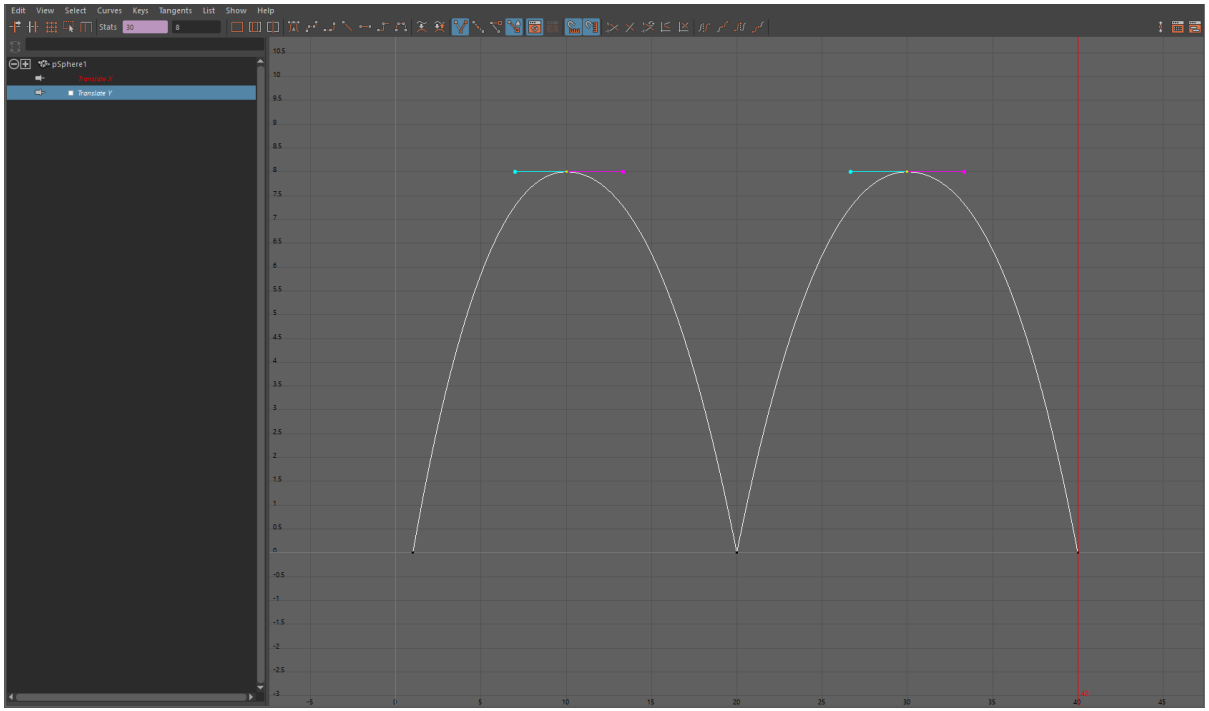
Bouncing Ball Attempt 02 (slide 23)



Bouncing Ball Attempt 03 (slide 24)

Bouncing ball with improved timing at plateau points.

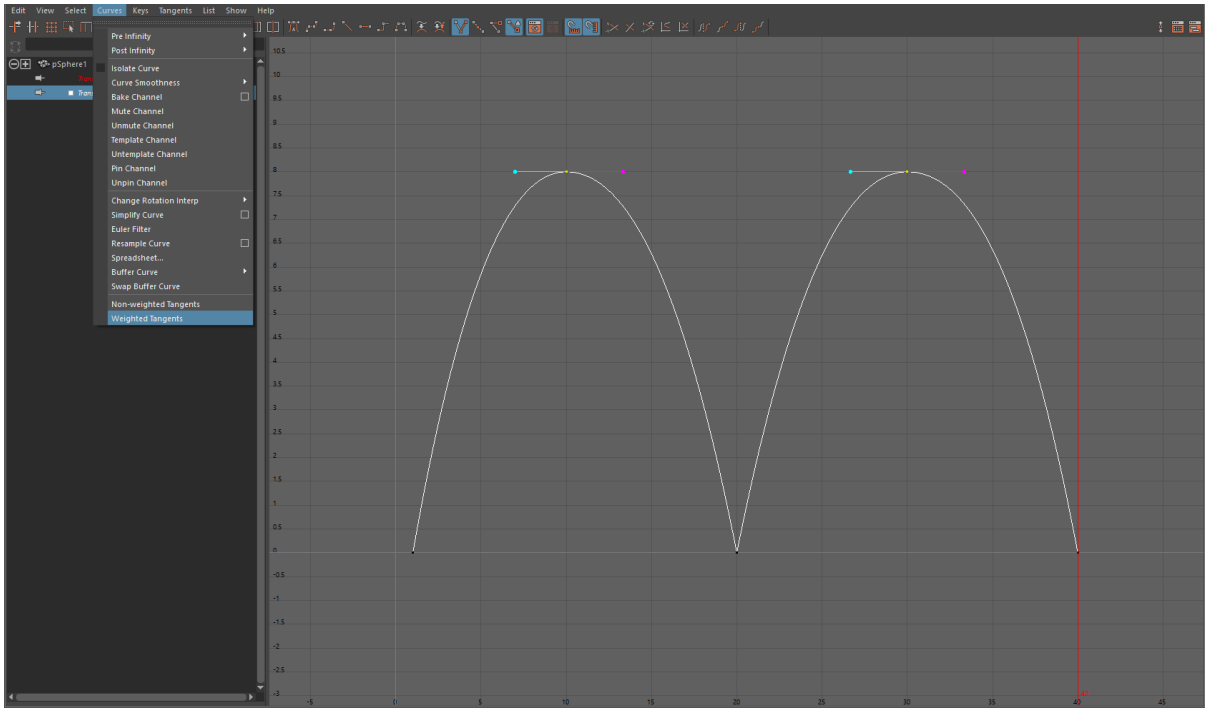
- We select the tangents of the key-frames when the ball is at its highest from the ground in the graph editor so we can edit them.
- In the *Curves* menu we select *Weighted Tangents*.
- We select the *Free Tangent Weight* option.
- We stretch the tangents to exaggerate the plateau effect. That is; the ball slows down sharply and remains slow when it reaches (0, 8, 0), until it starts falling again.



Bouncing Ball Attempt 03 (slide 24)

Bouncing ball with improved timing at plateau points.

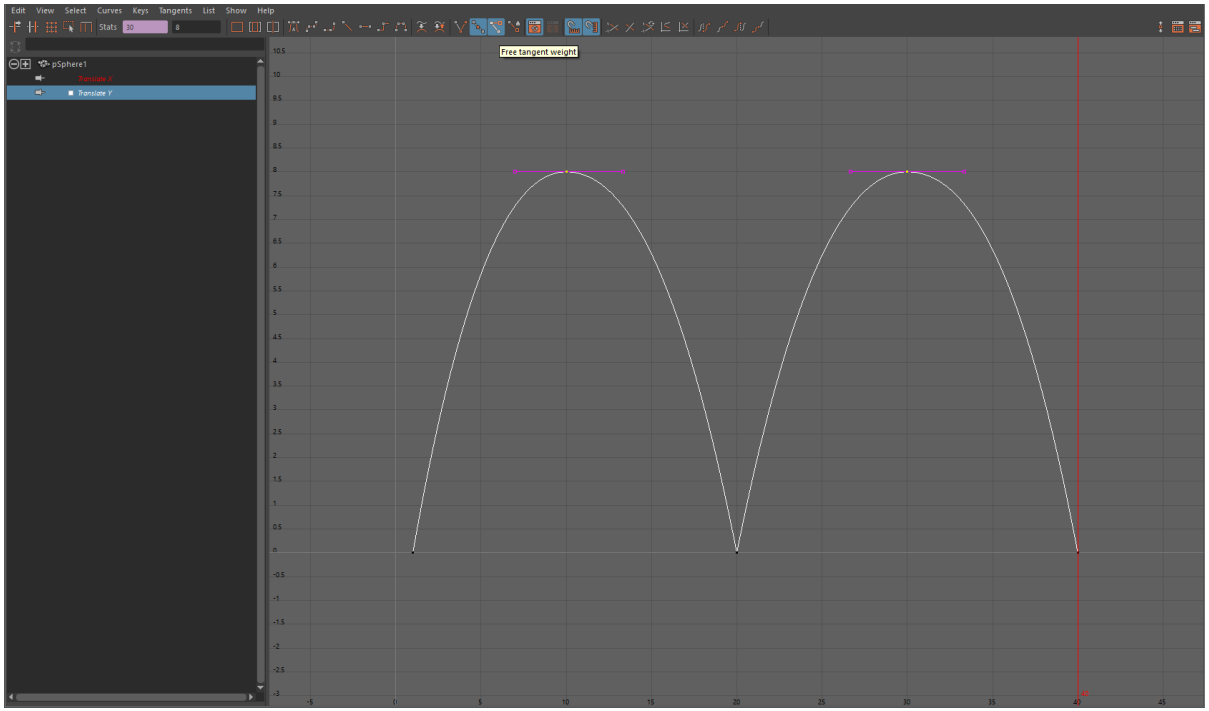
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Bouncing Ball Attempt 03 (slide 24)

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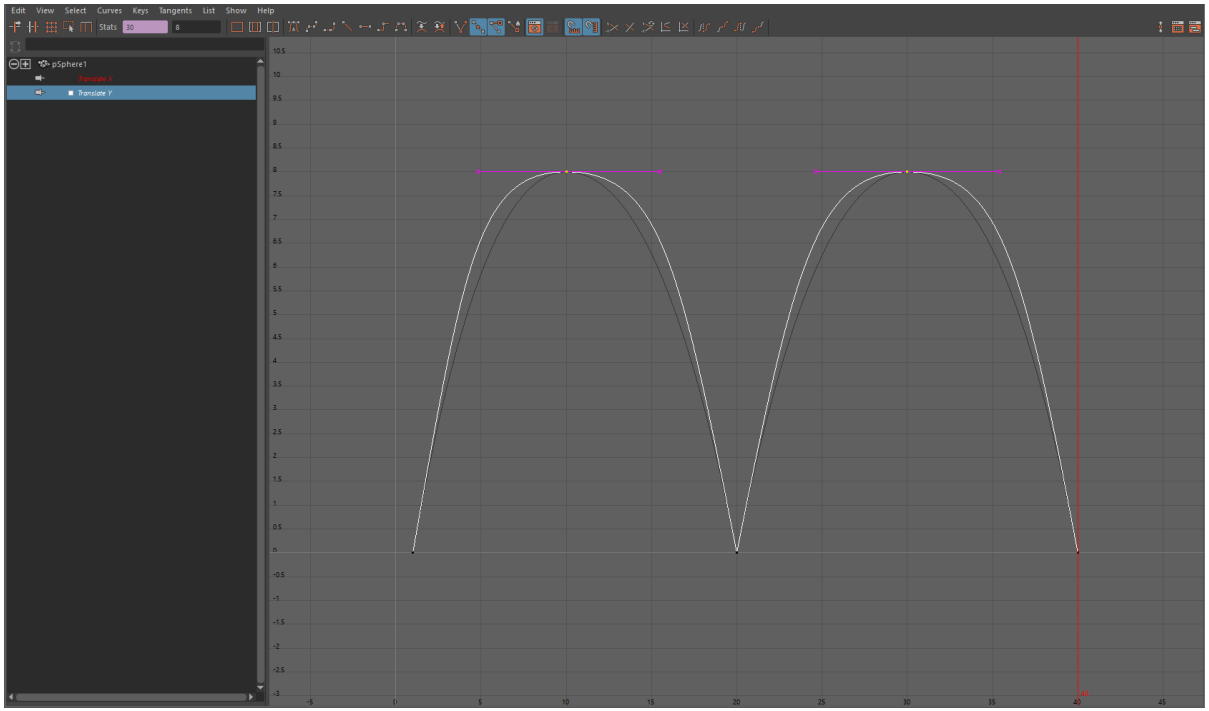
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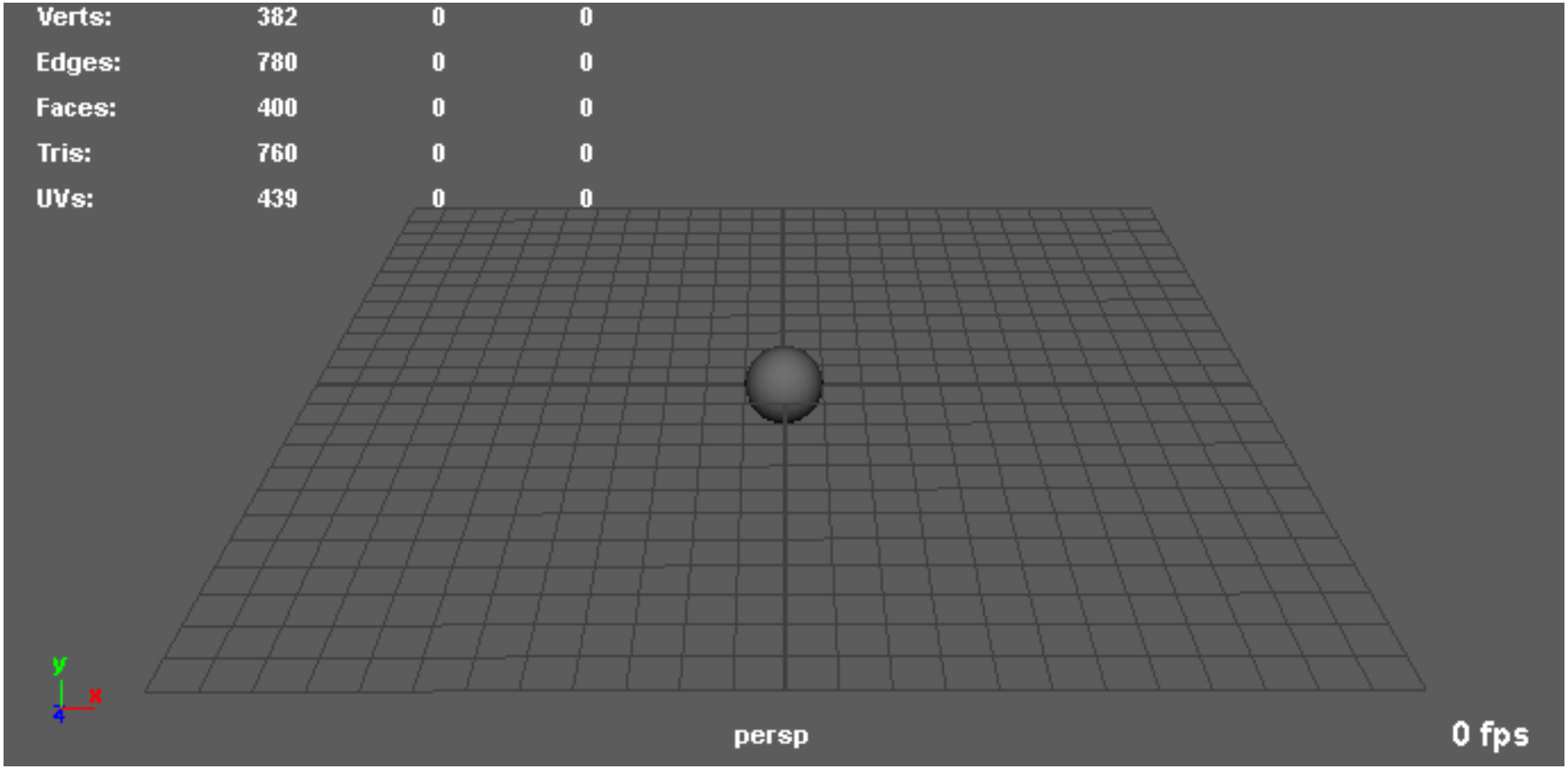
Bouncing Ball Attempt 03 (slide 24)

Bouncing ball with improved timing at plateau points.

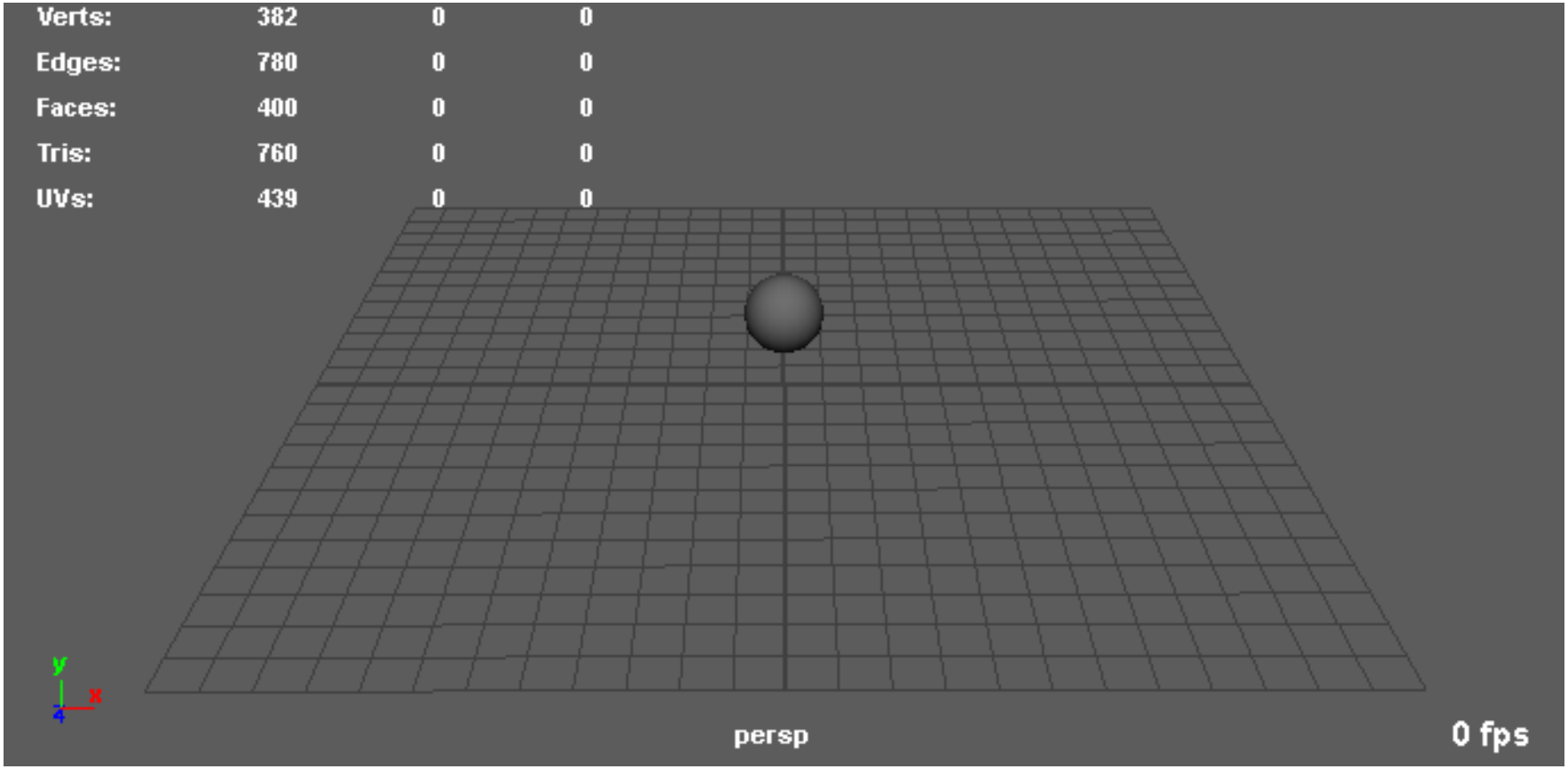
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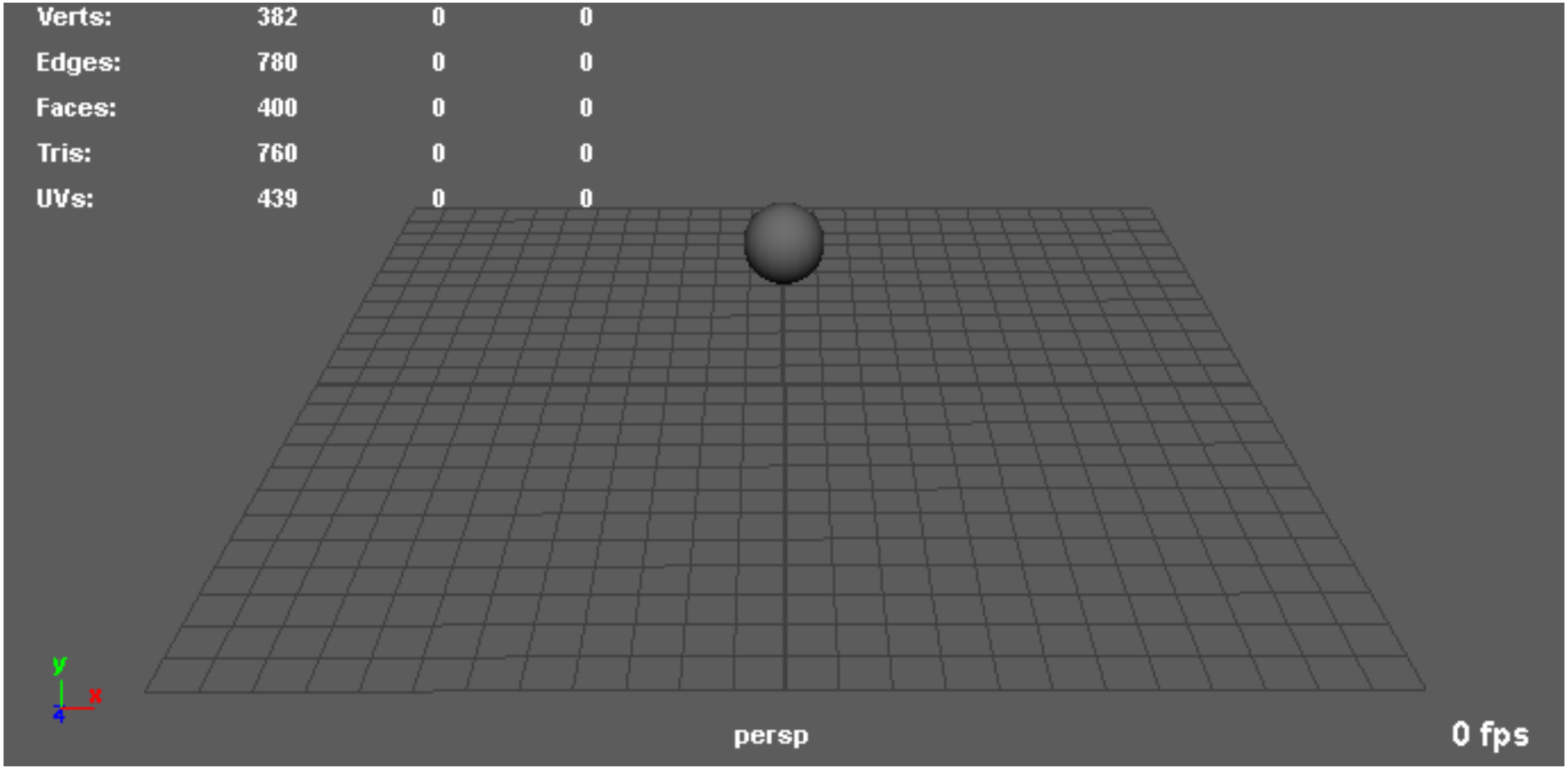
Bouncing Ball Animation Attempt 03 (slide 25)



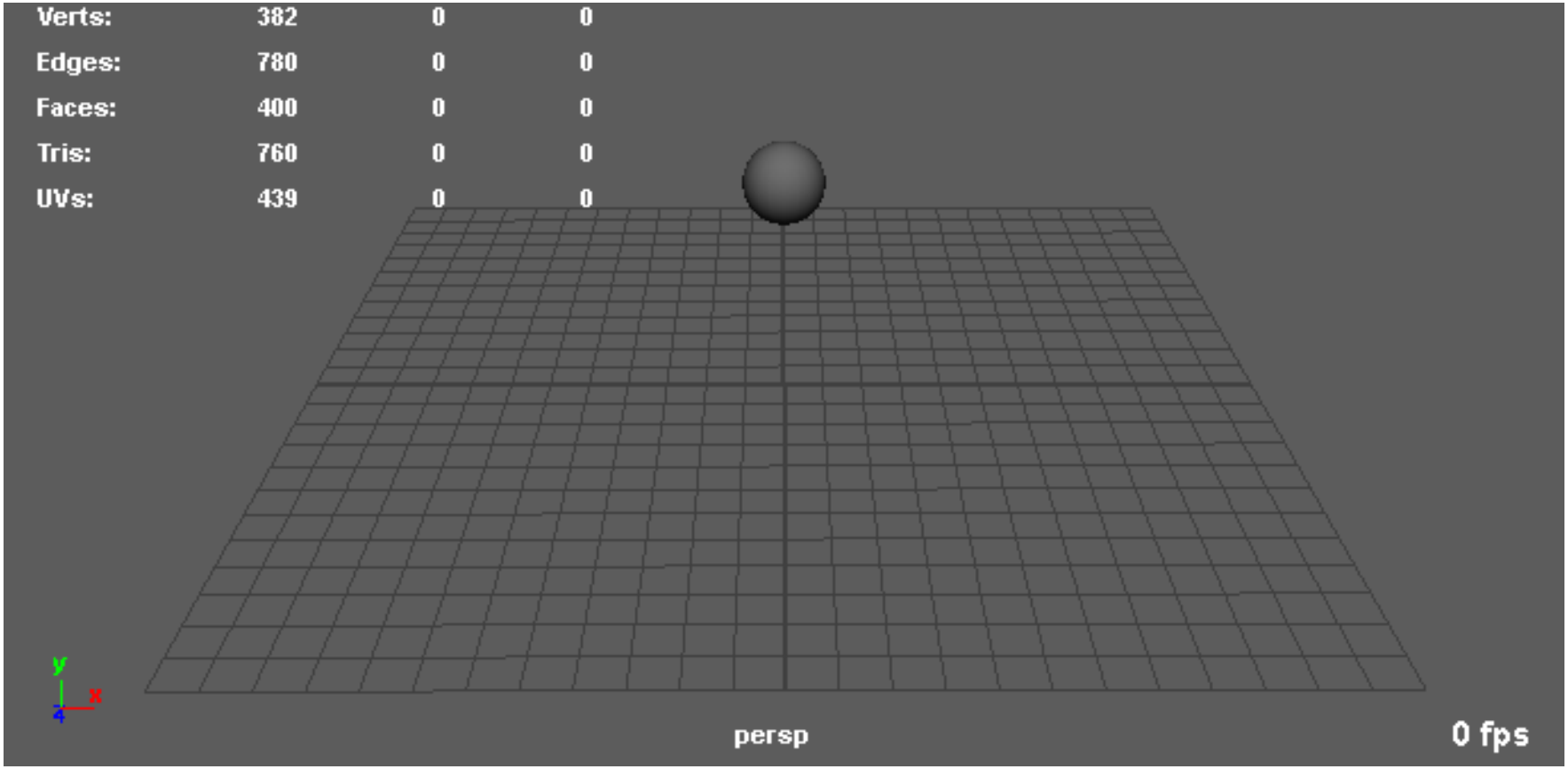
Bouncing Ball Animation Attempt 03 (slide 25)



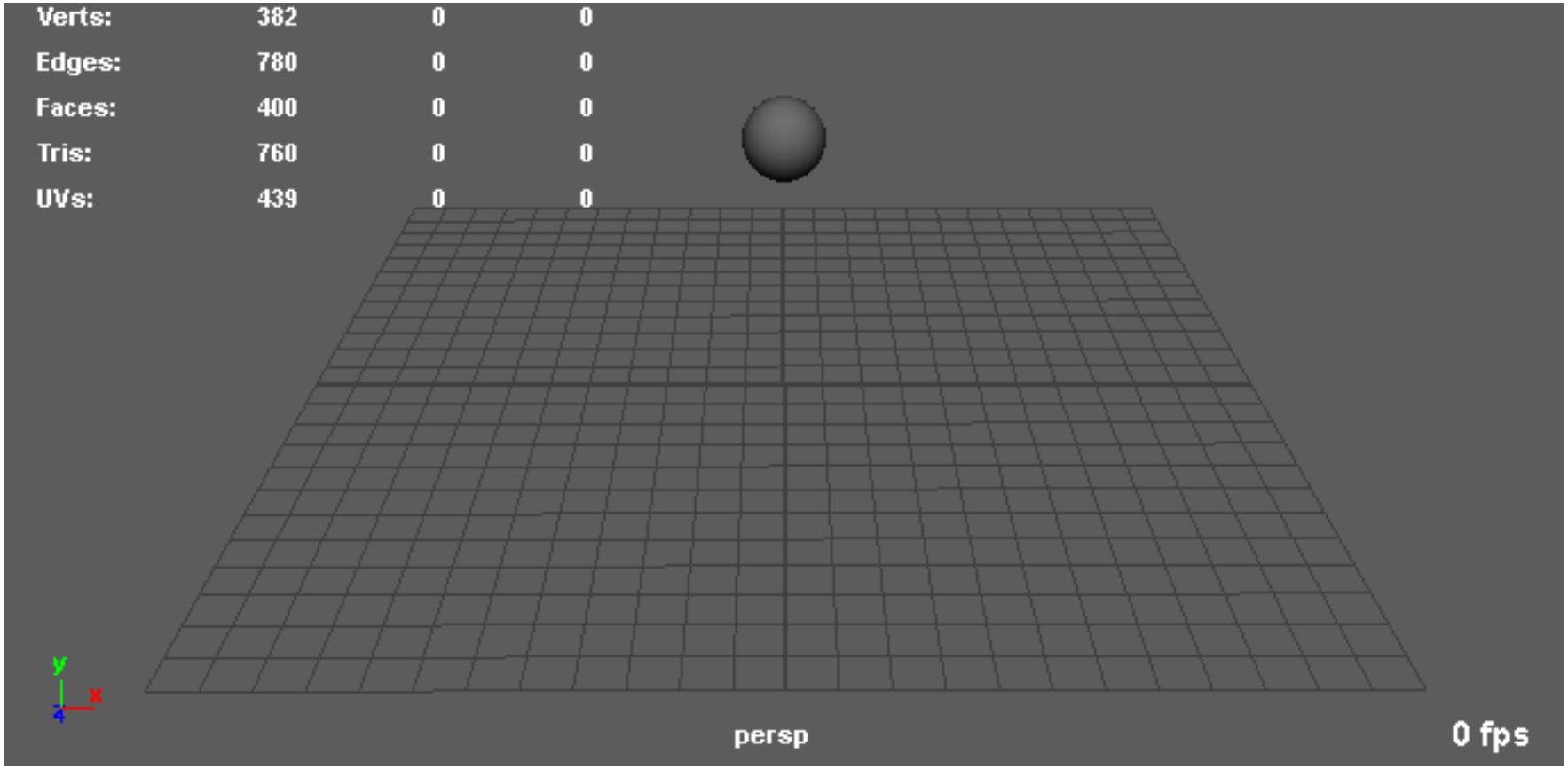
Bouncing Ball Animation Attempt 03 (slide 25)



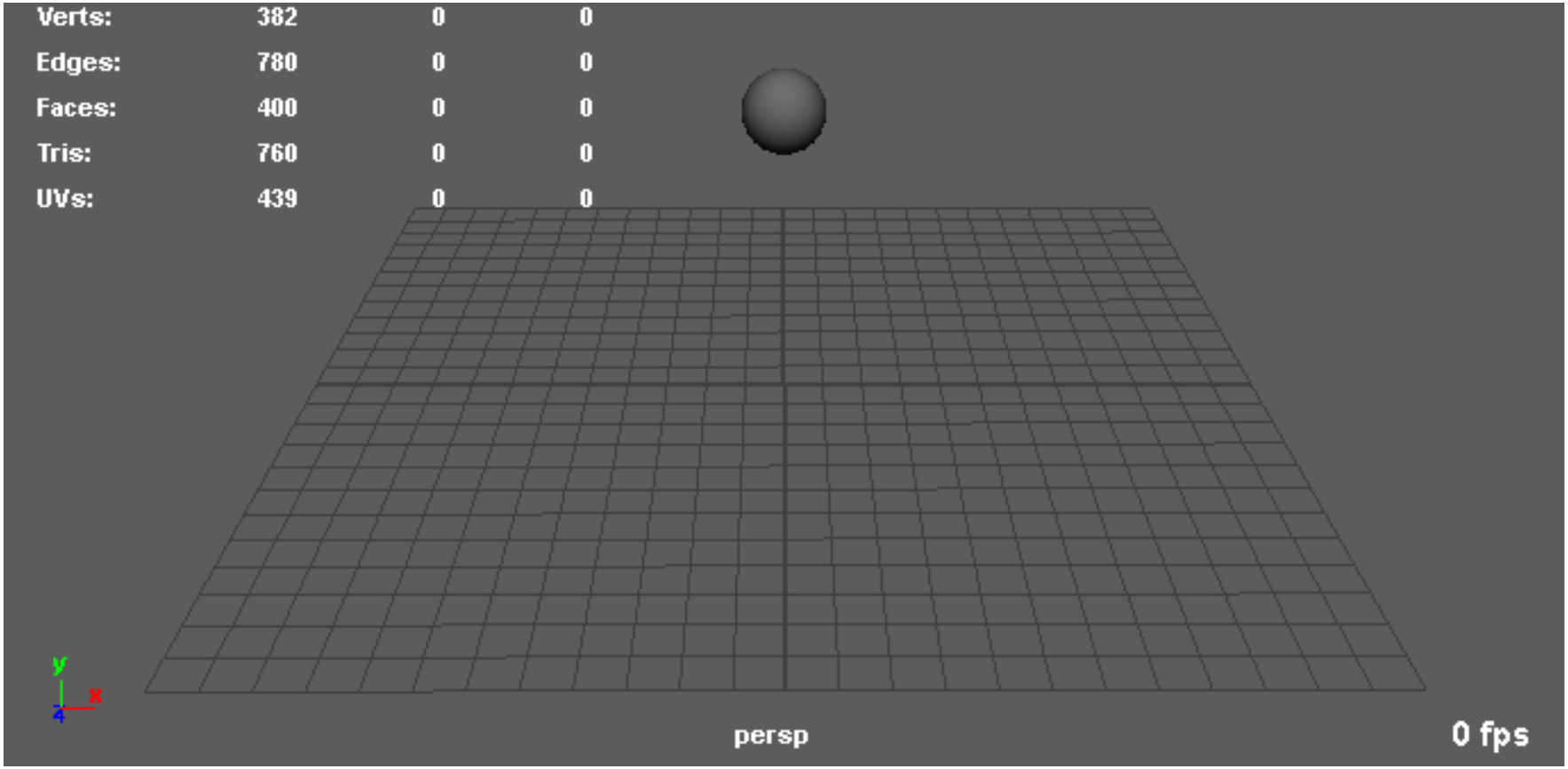
Bouncing Ball Animation Attempt 03 (slide 25)



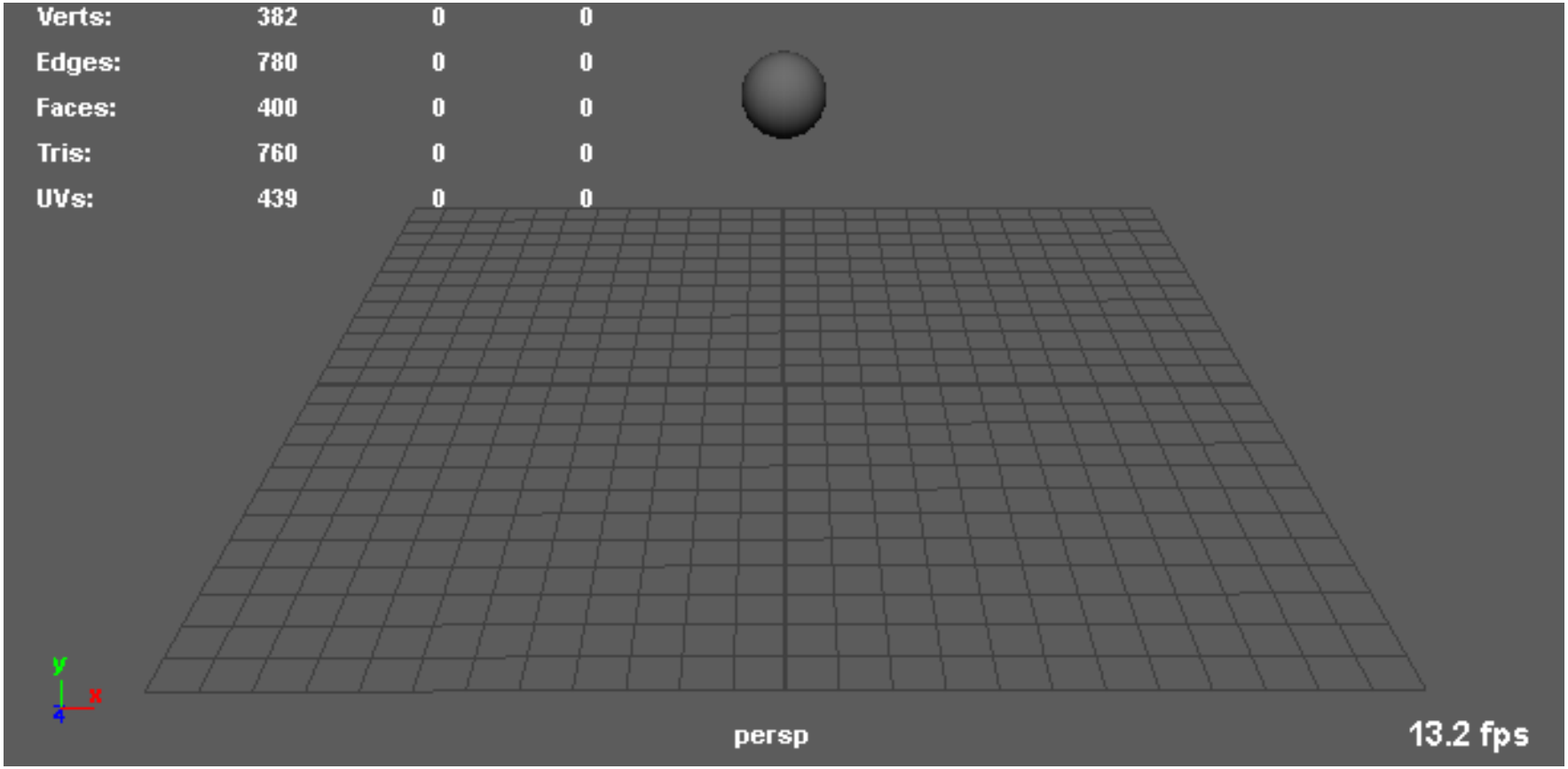
Bouncing Ball Animation Attempt 03 (slide 25)



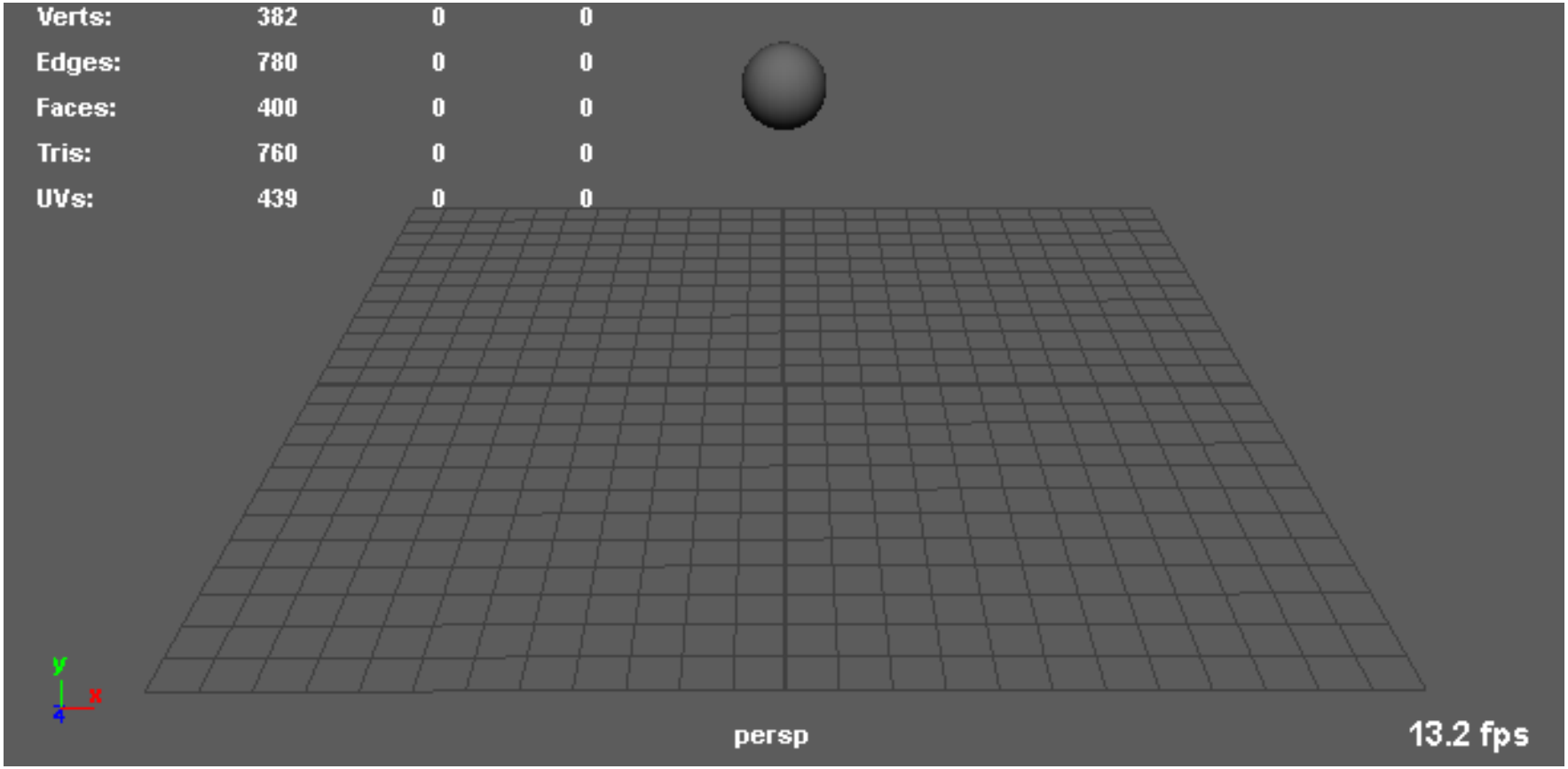
Bouncing Ball Animation Attempt 03 (slide 25)



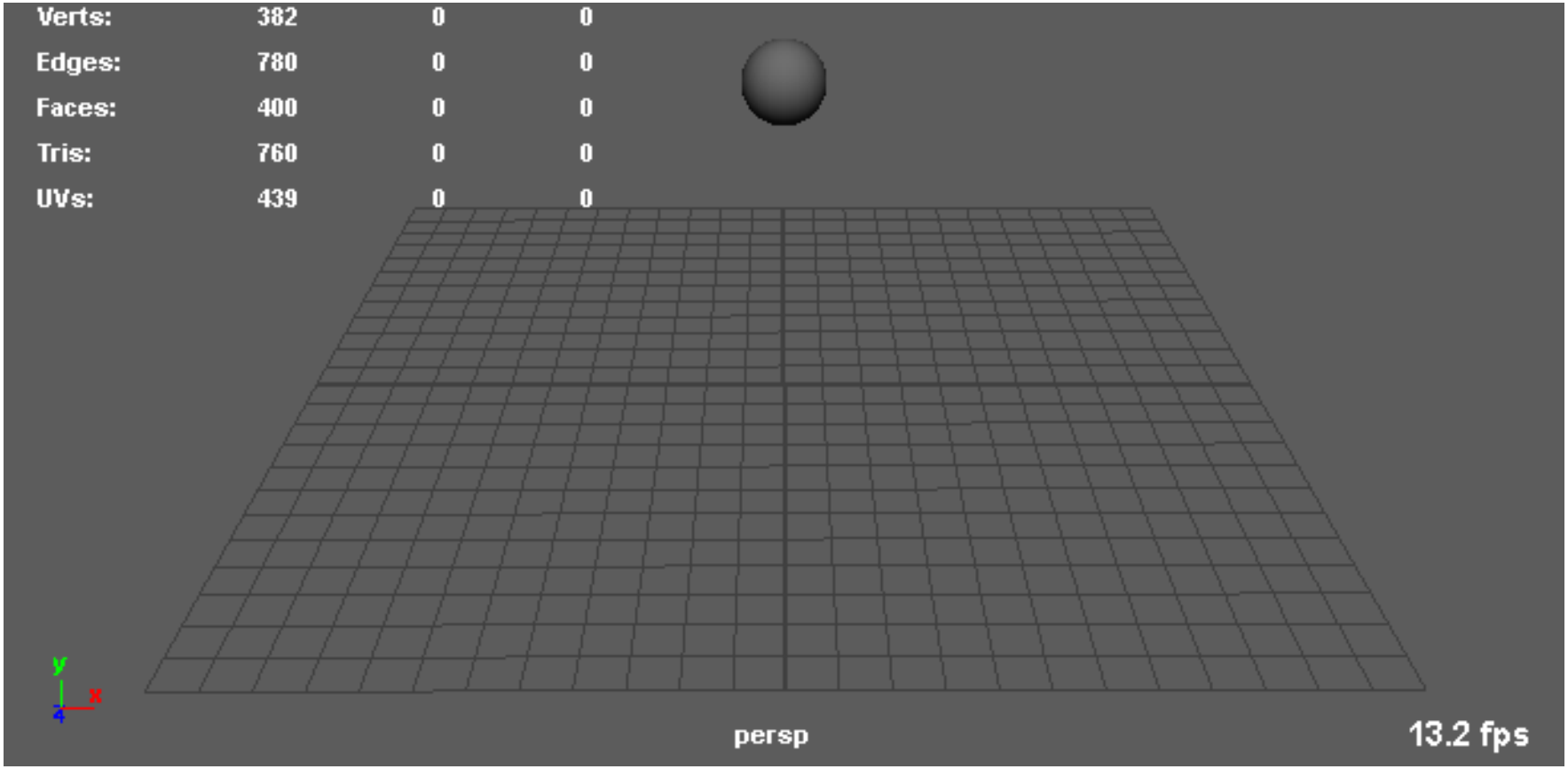
Bouncing Ball Animation Attempt 03 (slide 25)



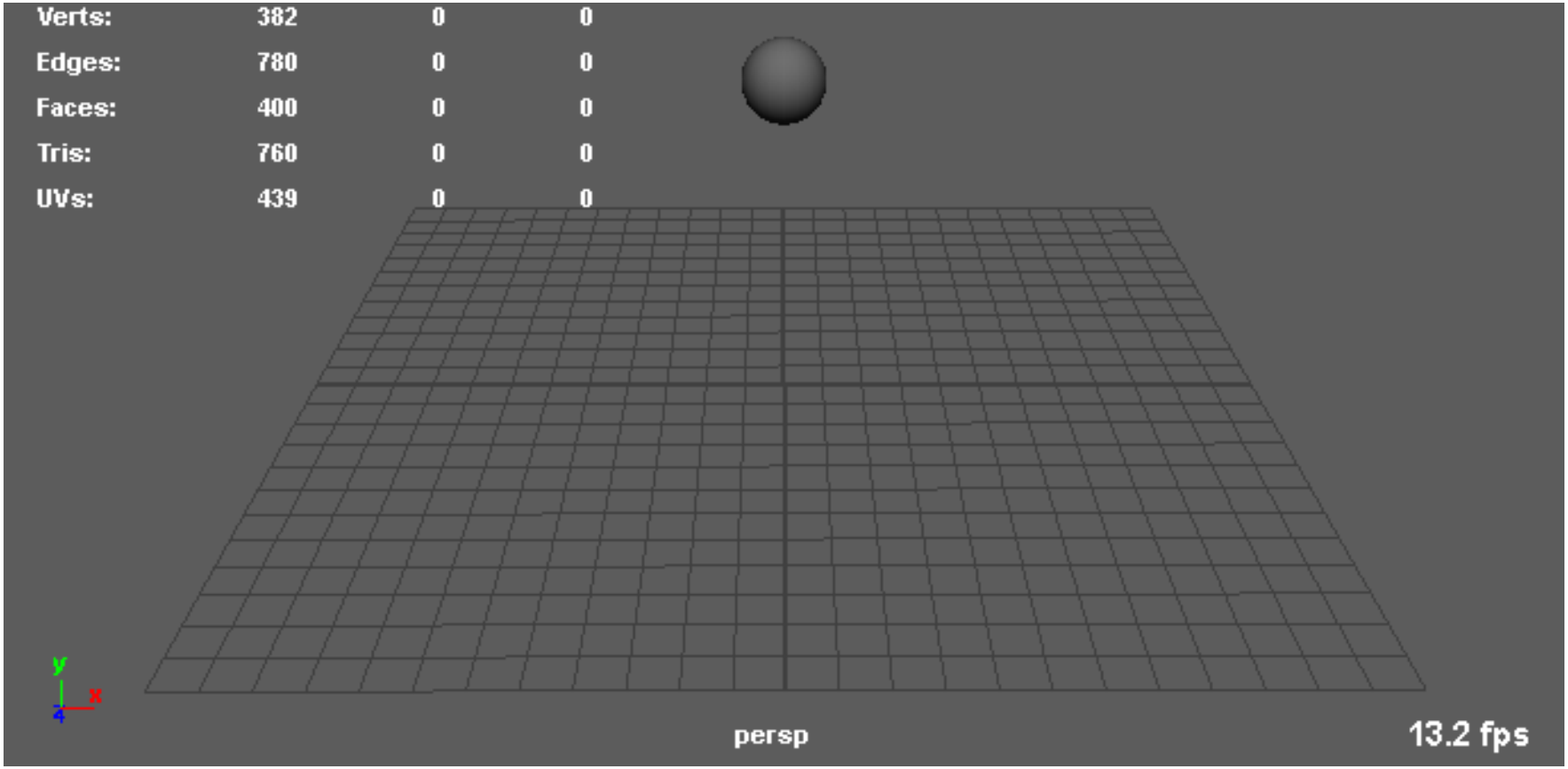
Bouncing Ball Animation Attempt 03 (slide 25)



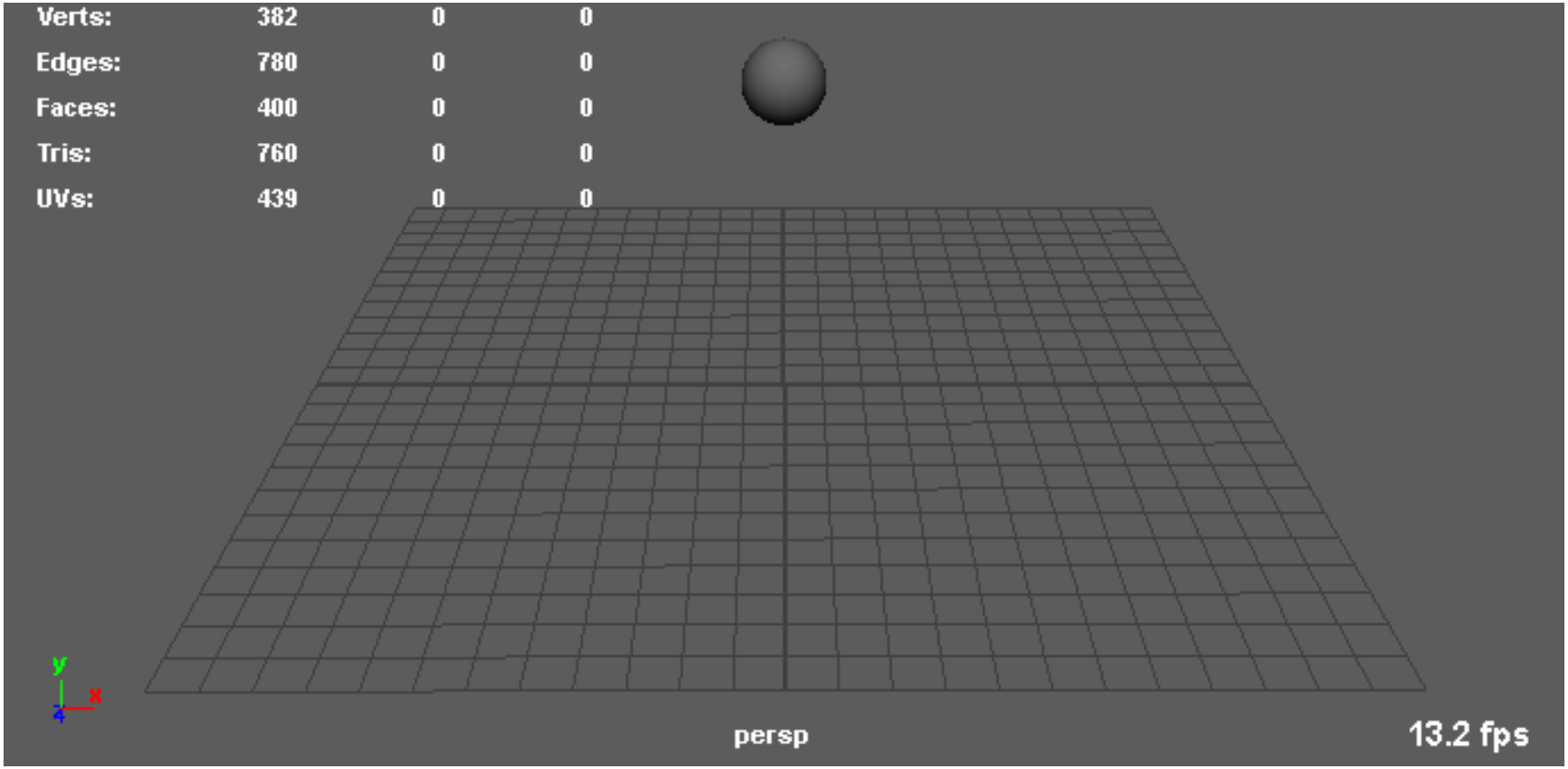
Bouncing Ball Animation Attempt 03 (slide 25)



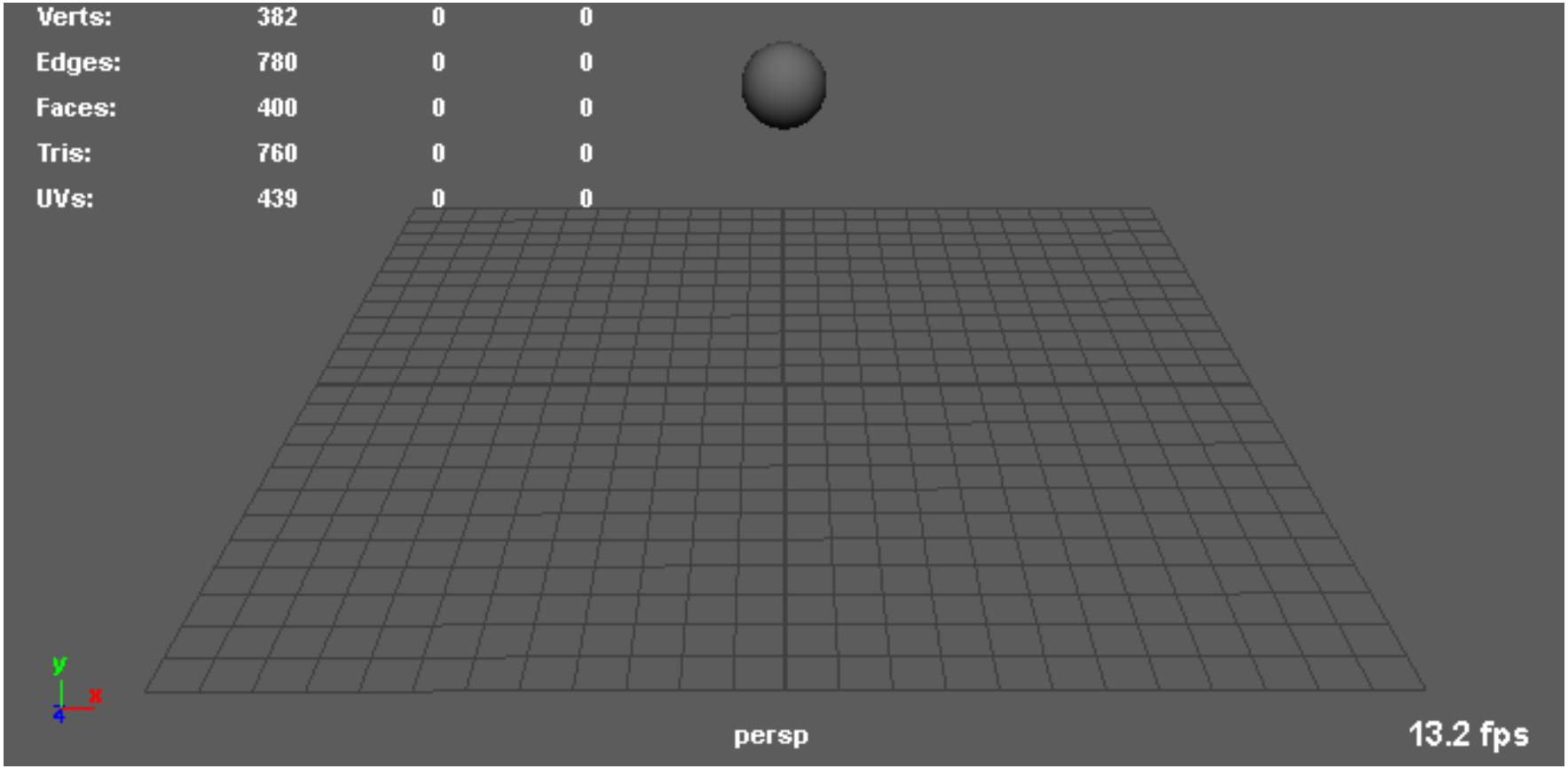
Bouncing Ball Animation Attempt 03 (slide 25)



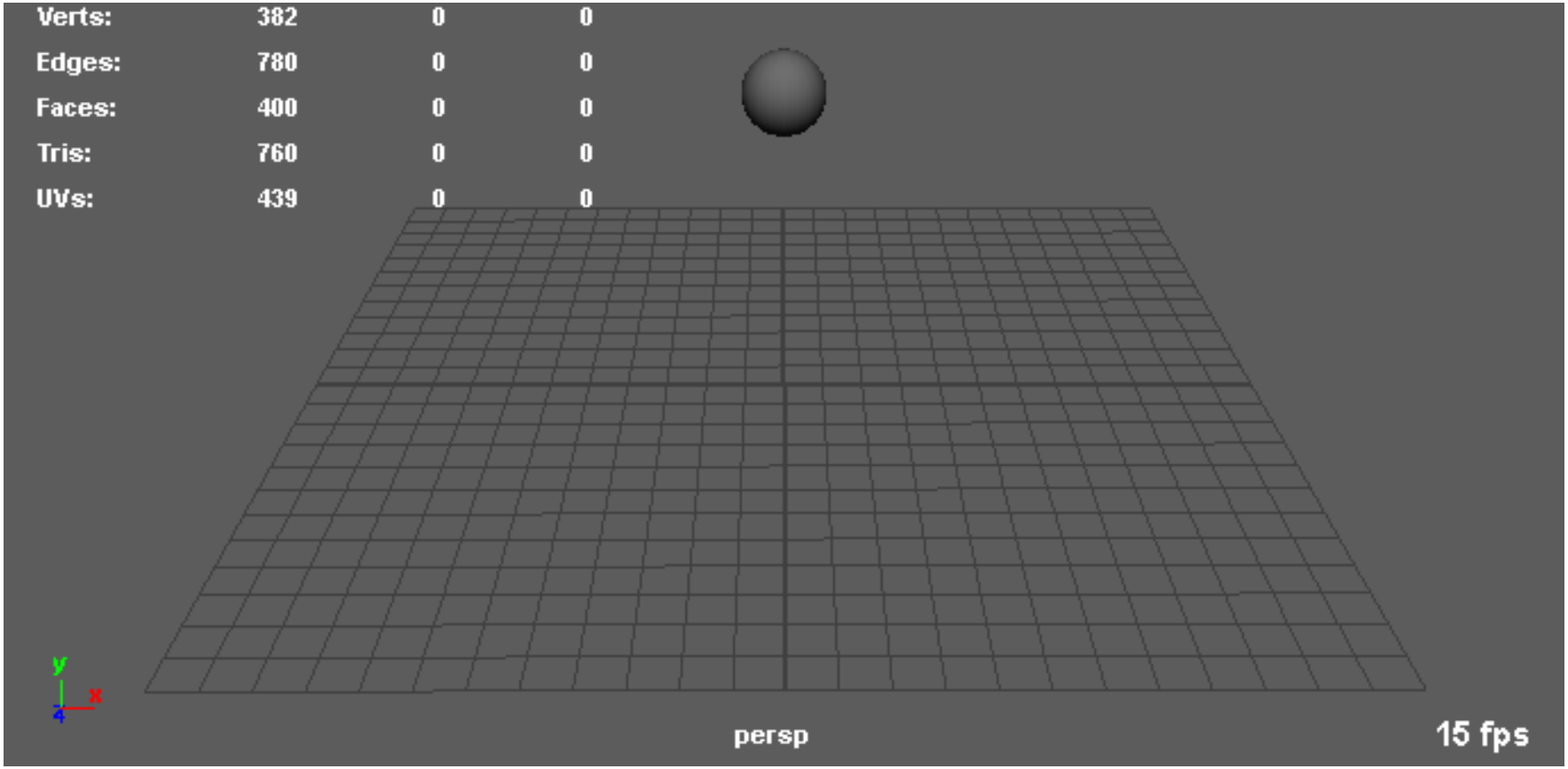
Bouncing Ball Animation Attempt 03 (slide 25)



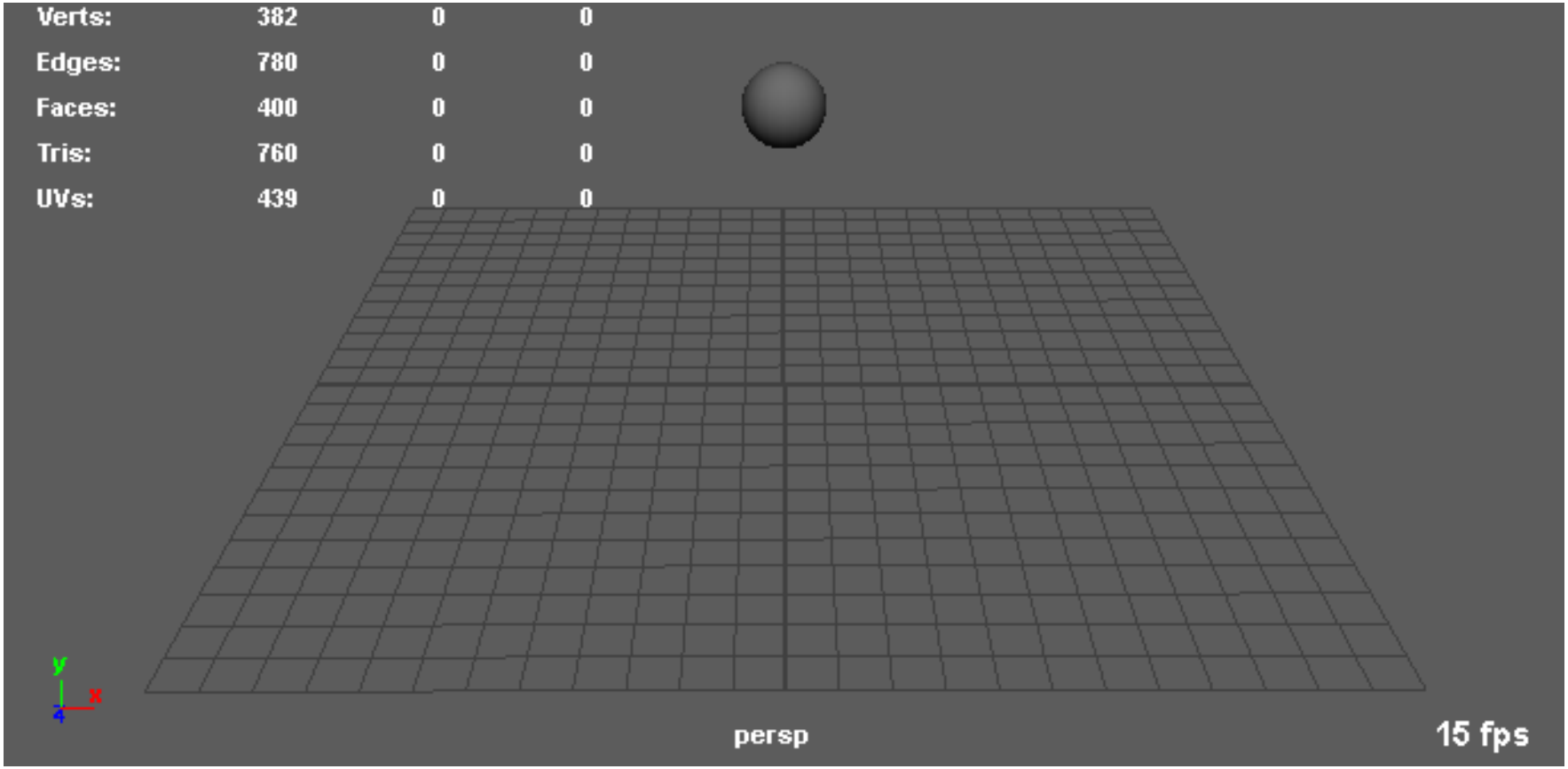
Bouncing Ball Animation Attempt 03 (slide 25)



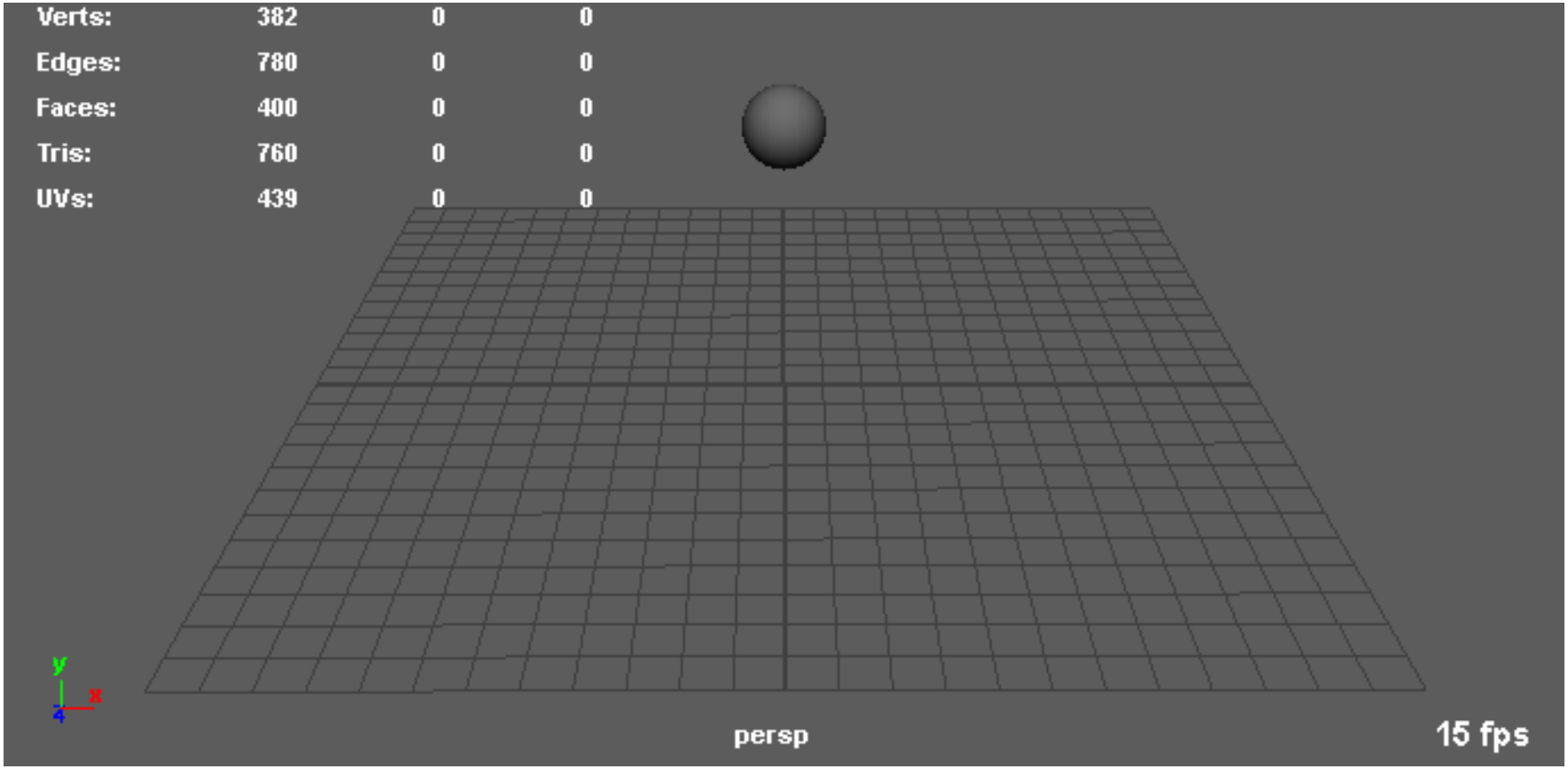
Bouncing Ball Animation Attempt 03 (slide 25)



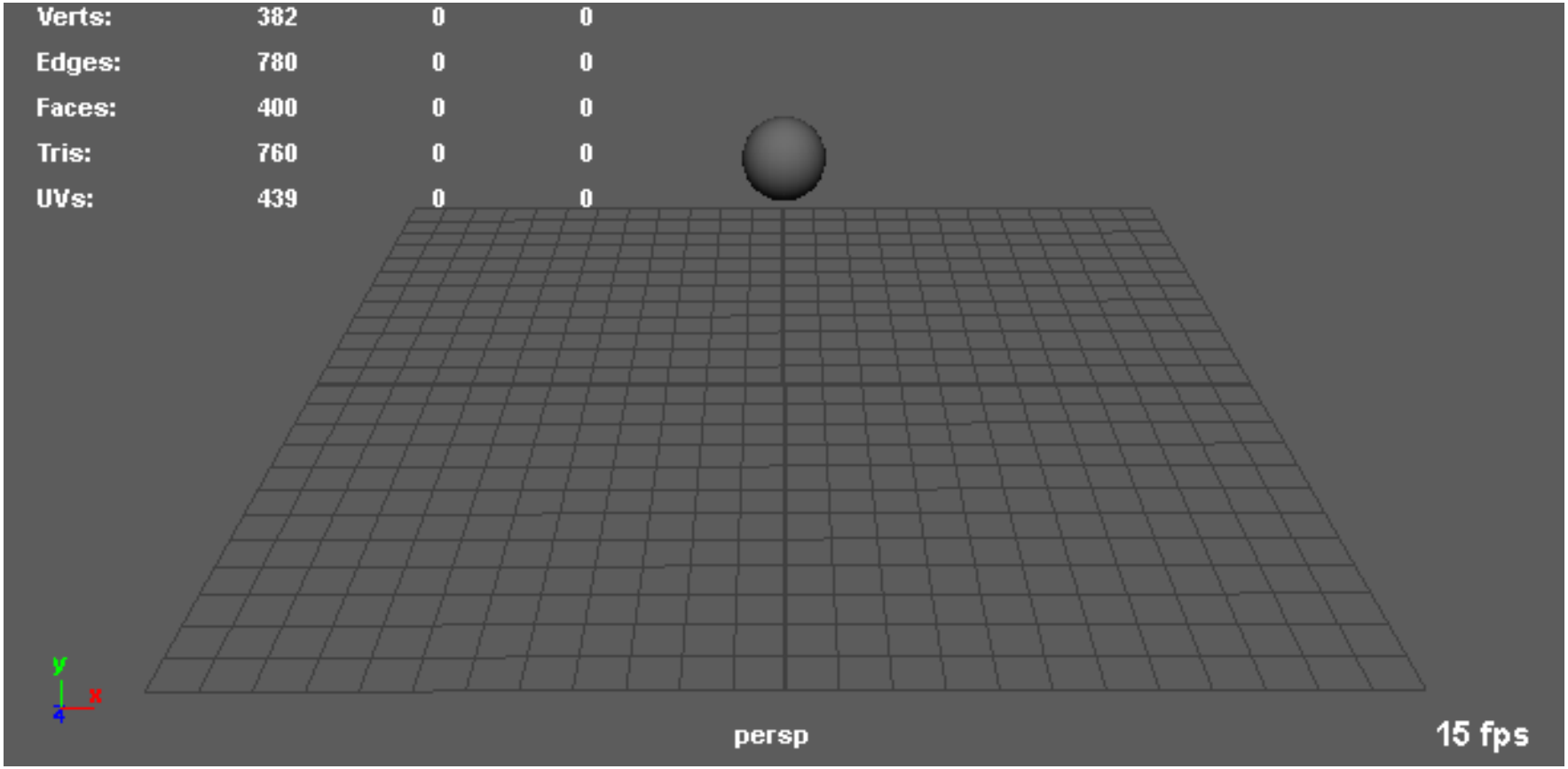
Bouncing Ball Animation Attempt 03 (slide 25)



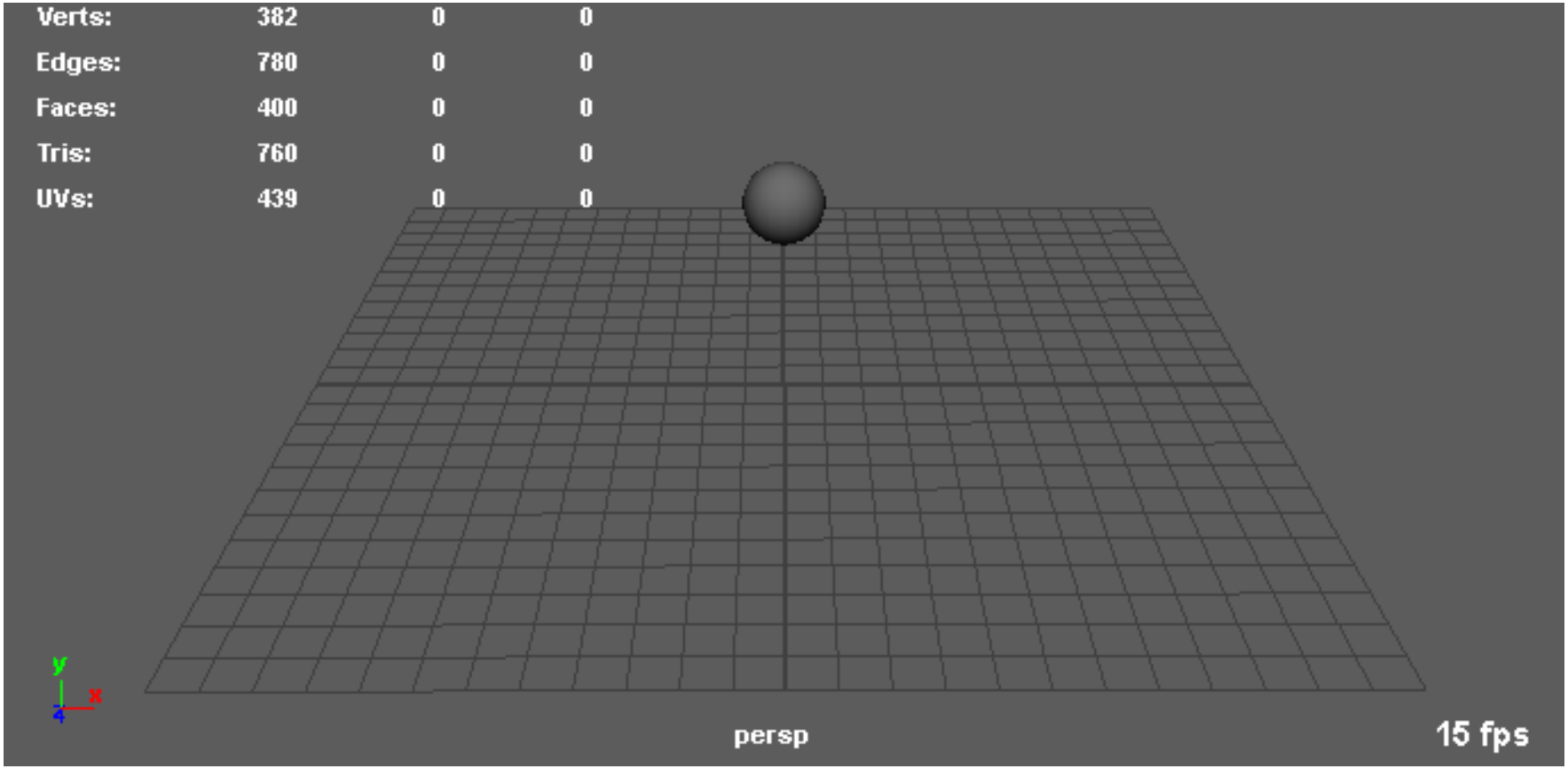
Bouncing Ball Animation Attempt 03 (slide 25)



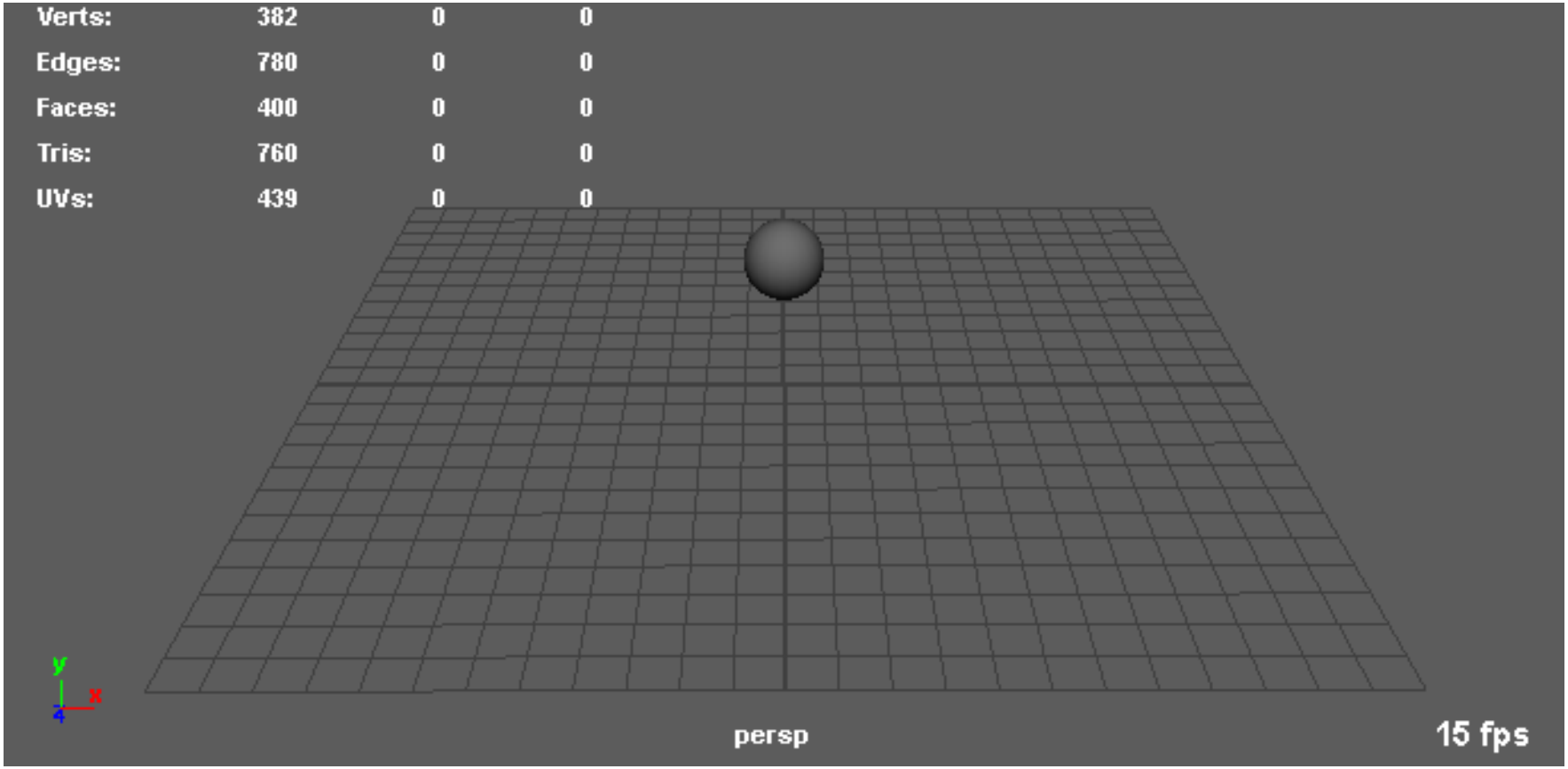
Bouncing Ball Animation Attempt 03 (slide 25)



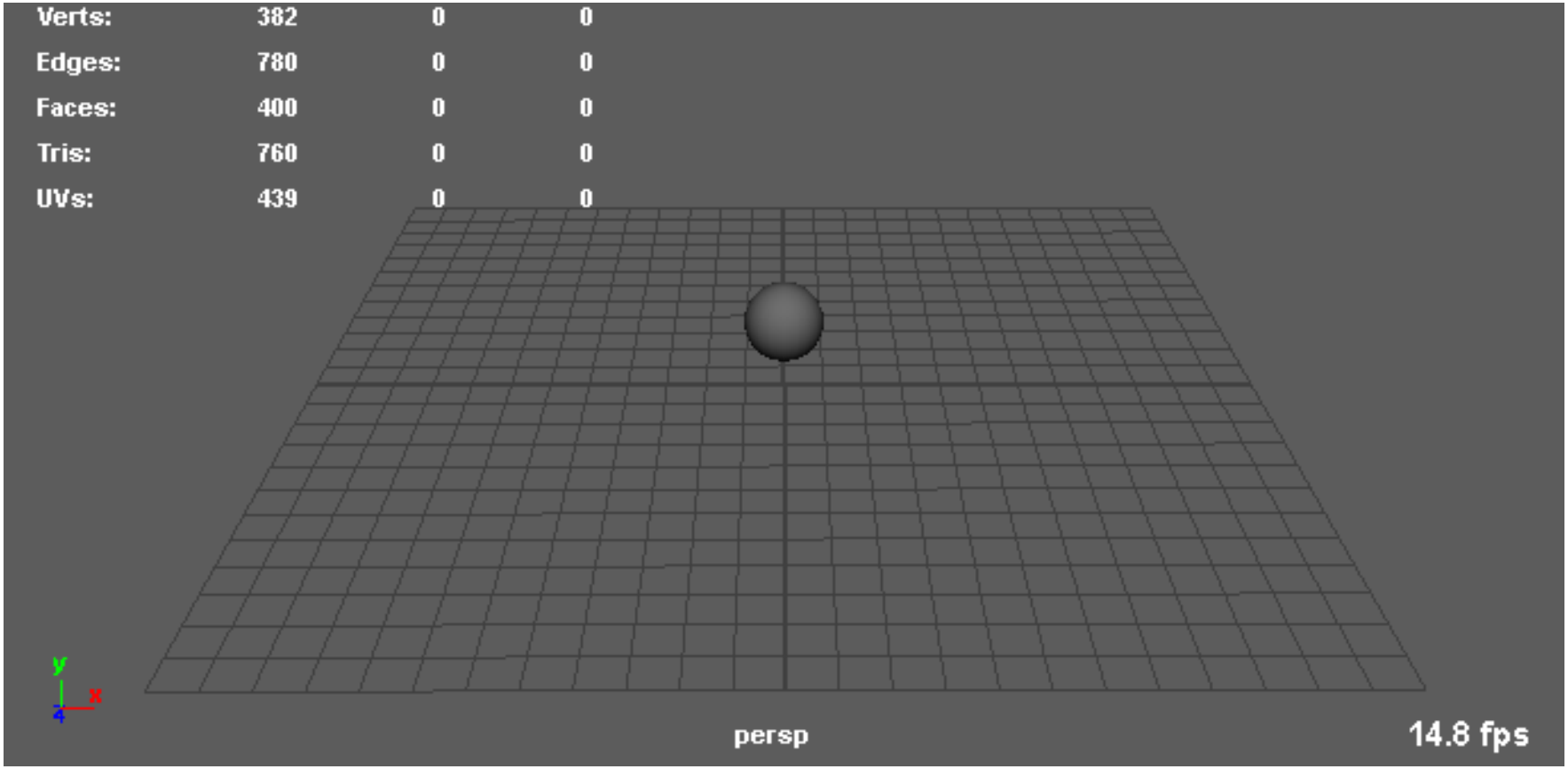
Bouncing Ball Animation Attempt 03 (slide 25)



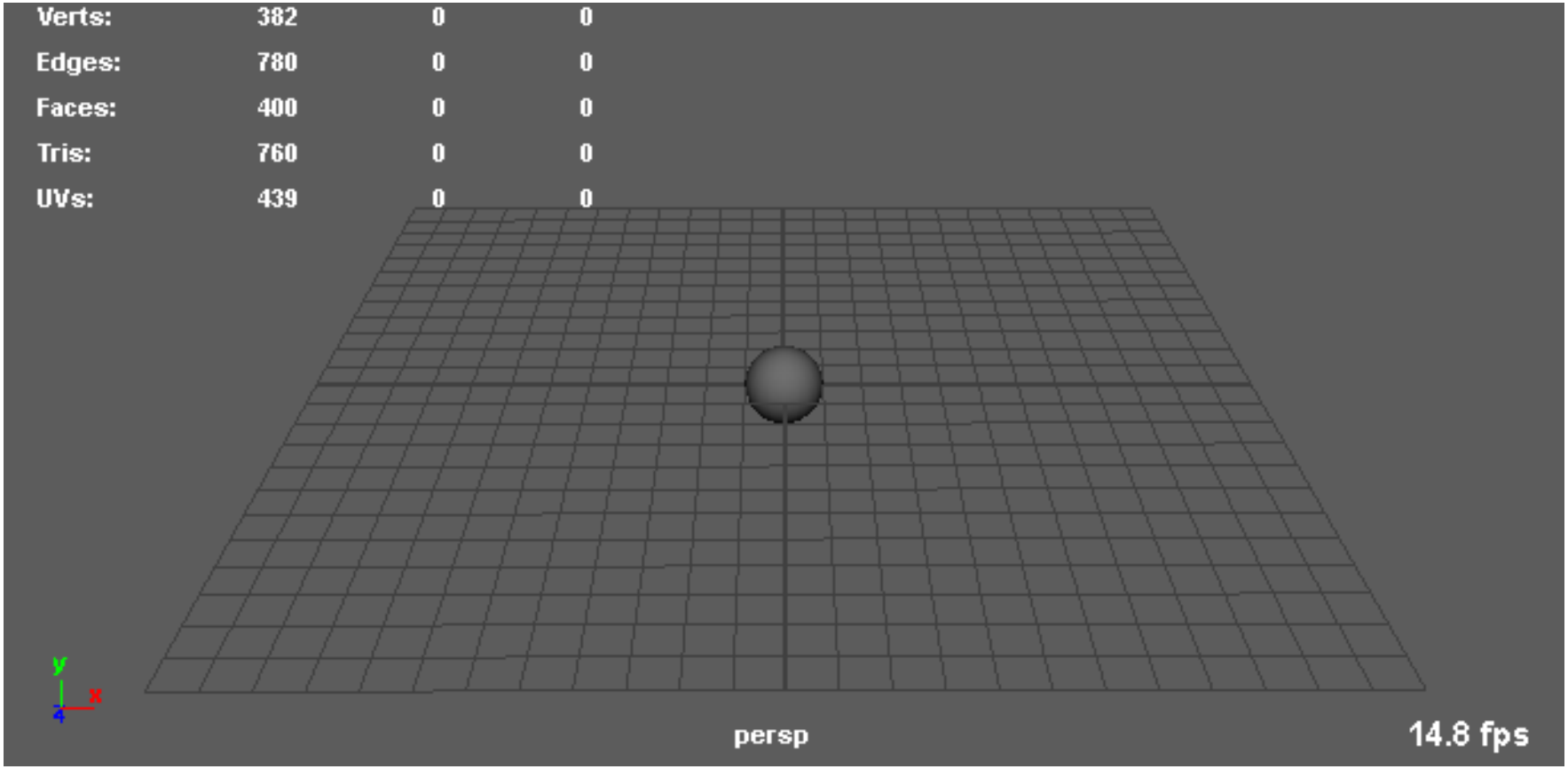
Bouncing Ball Animation Attempt 03 (slide 25)



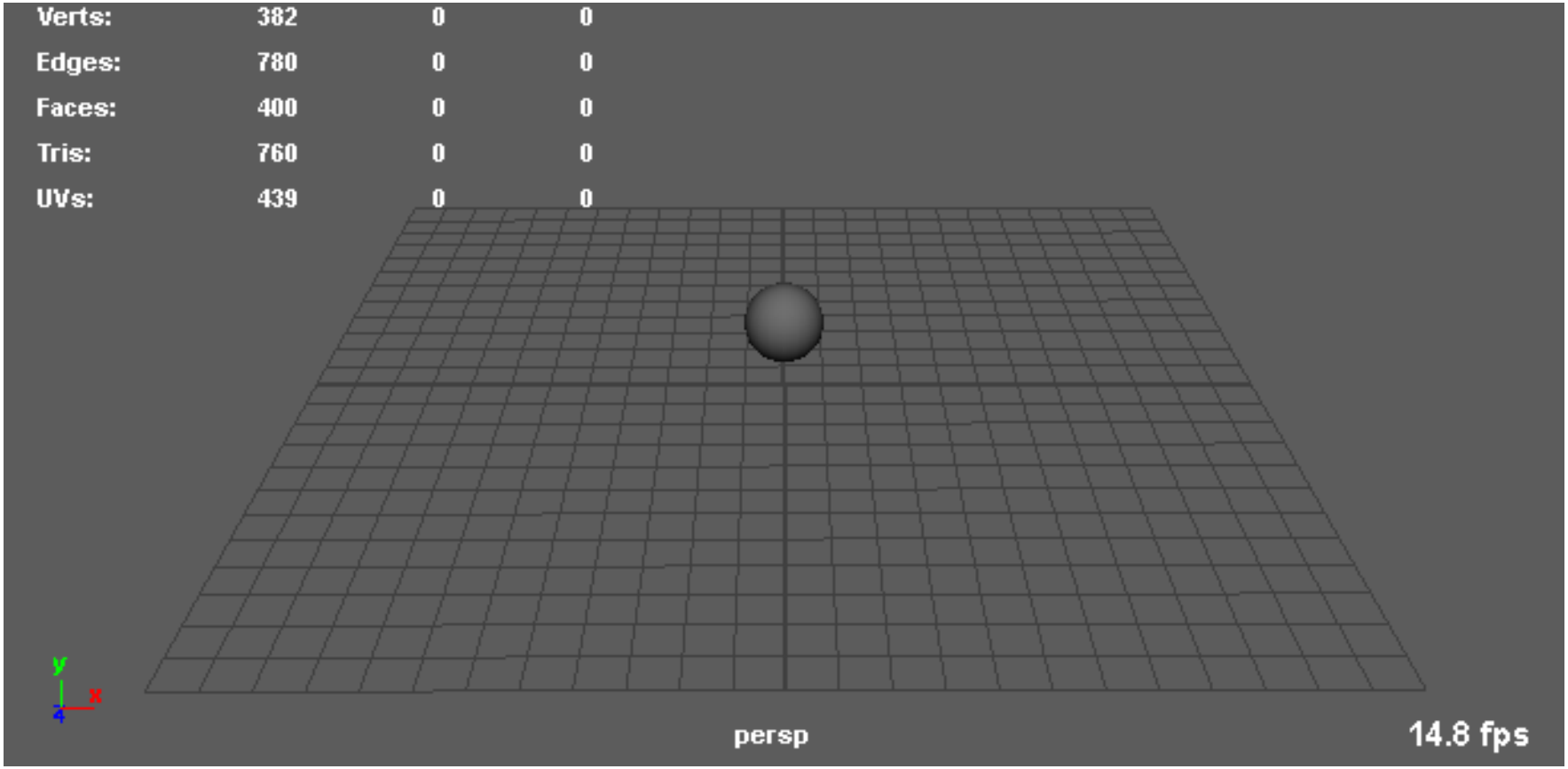
Bouncing Ball Animation Attempt 03 (slide 25)



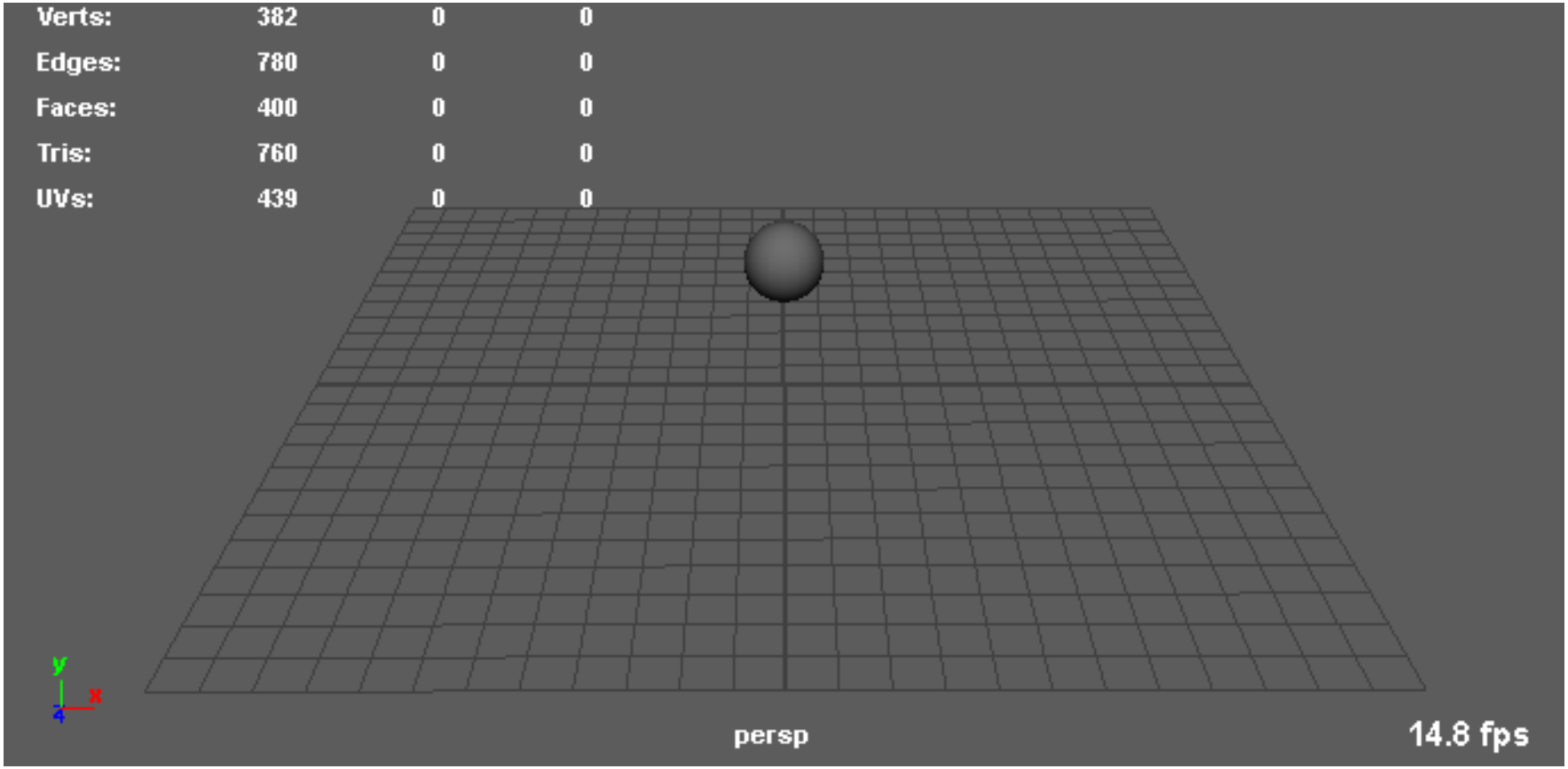
Bouncing Ball Animation Attempt 03 (slide 25)



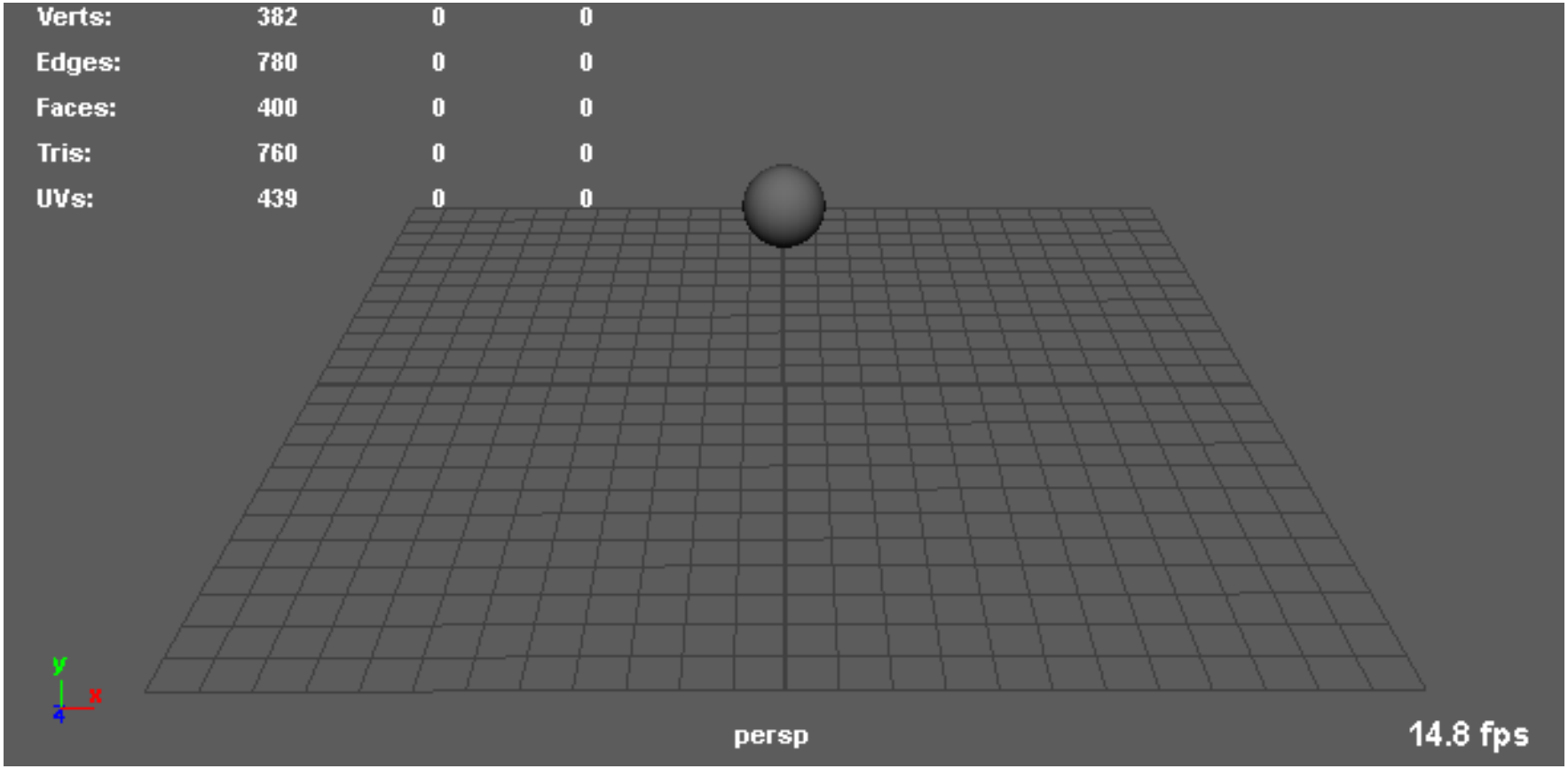
Bouncing Ball Animation Attempt 03 (slide 25)



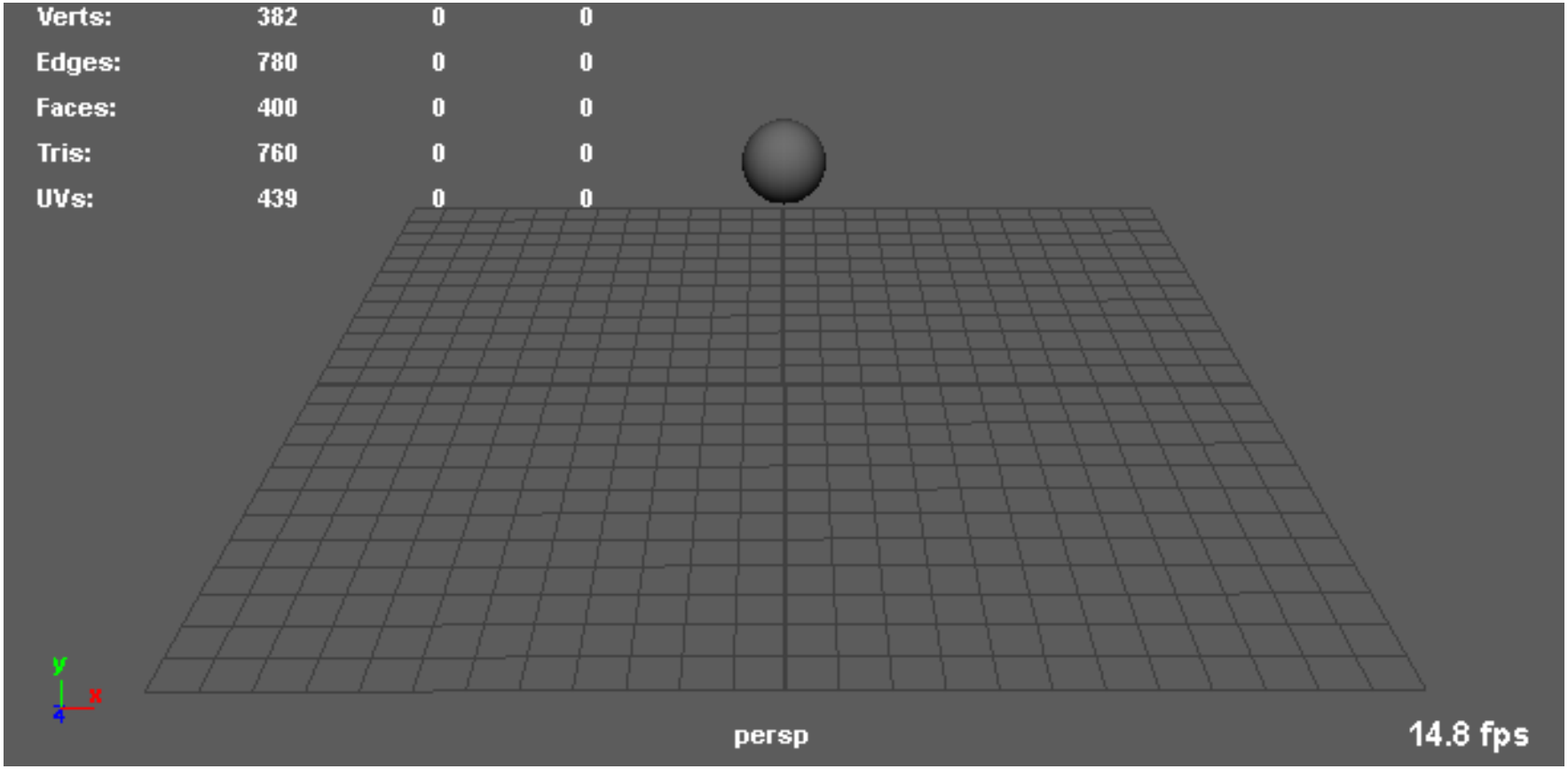
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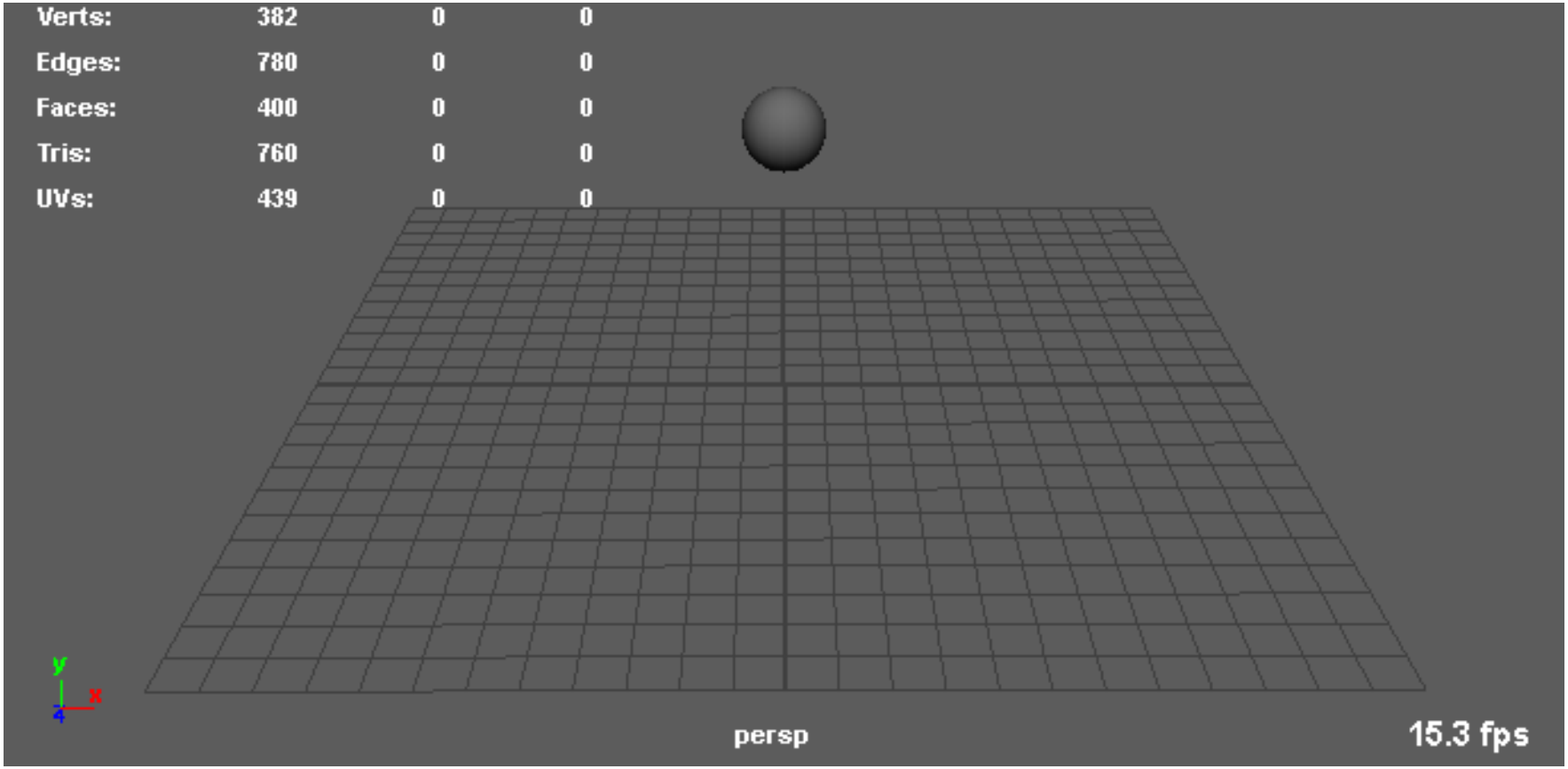
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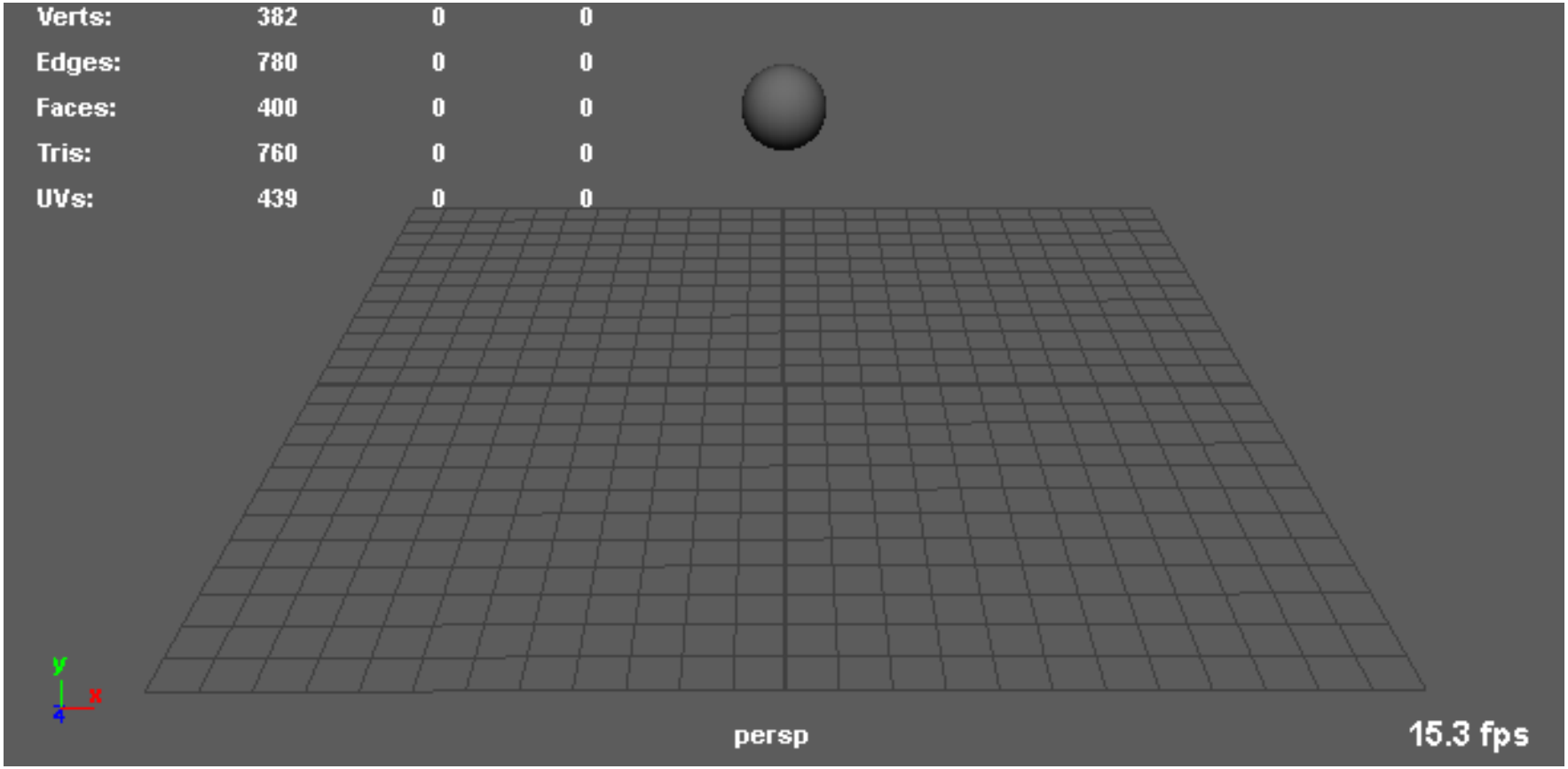
Bouncing Ball Animation Attempt 03 (slide 25)



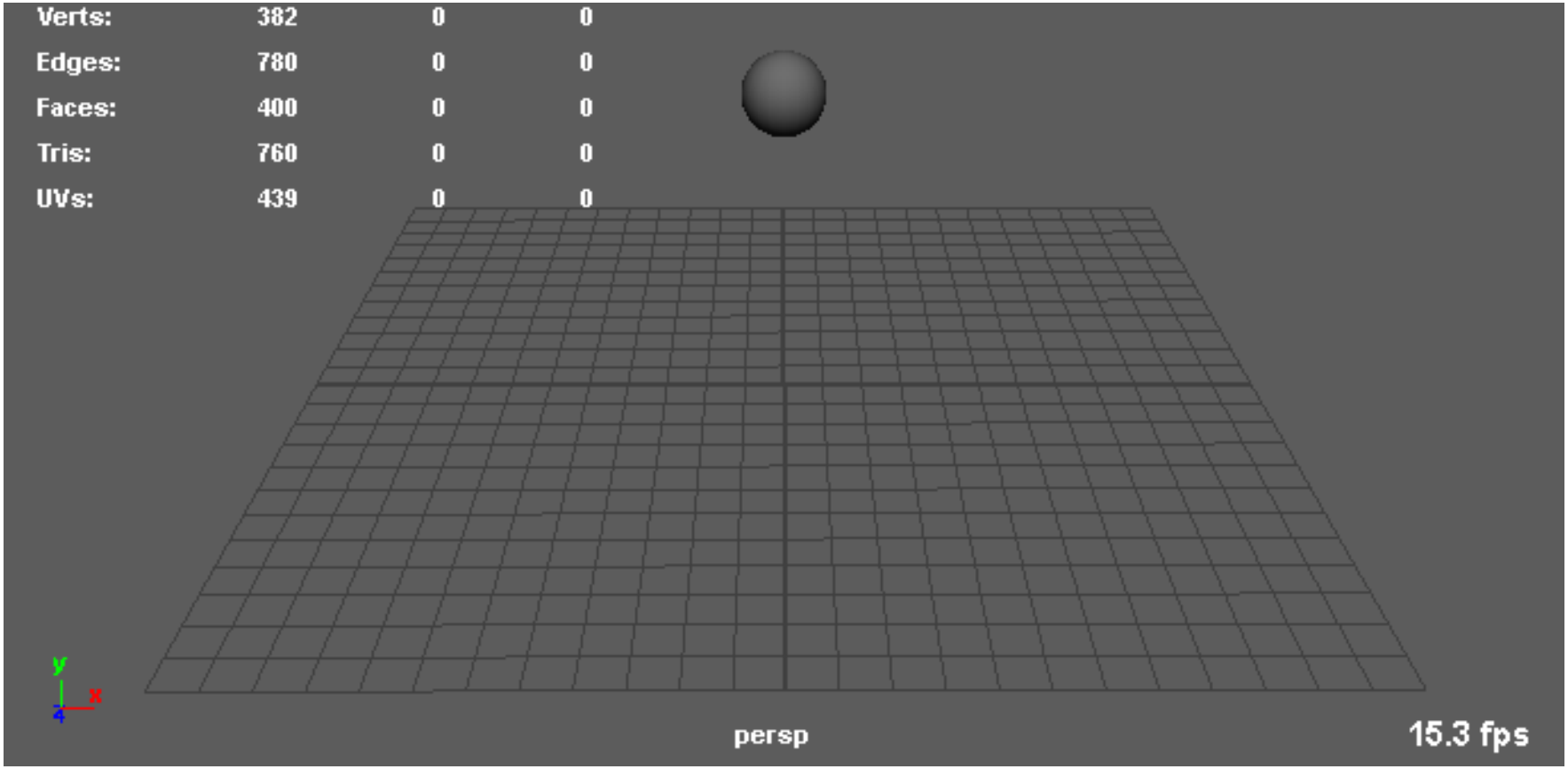
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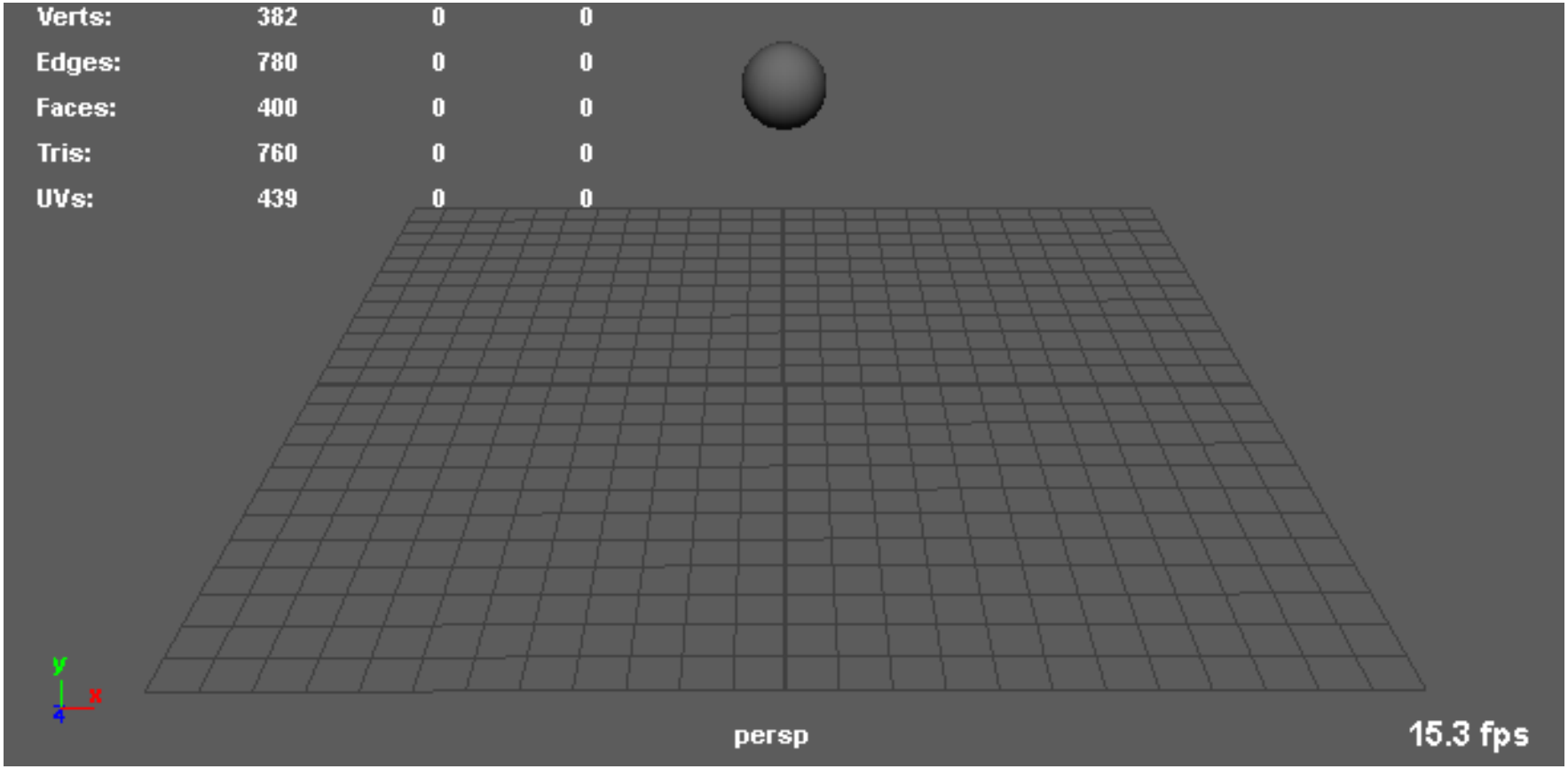
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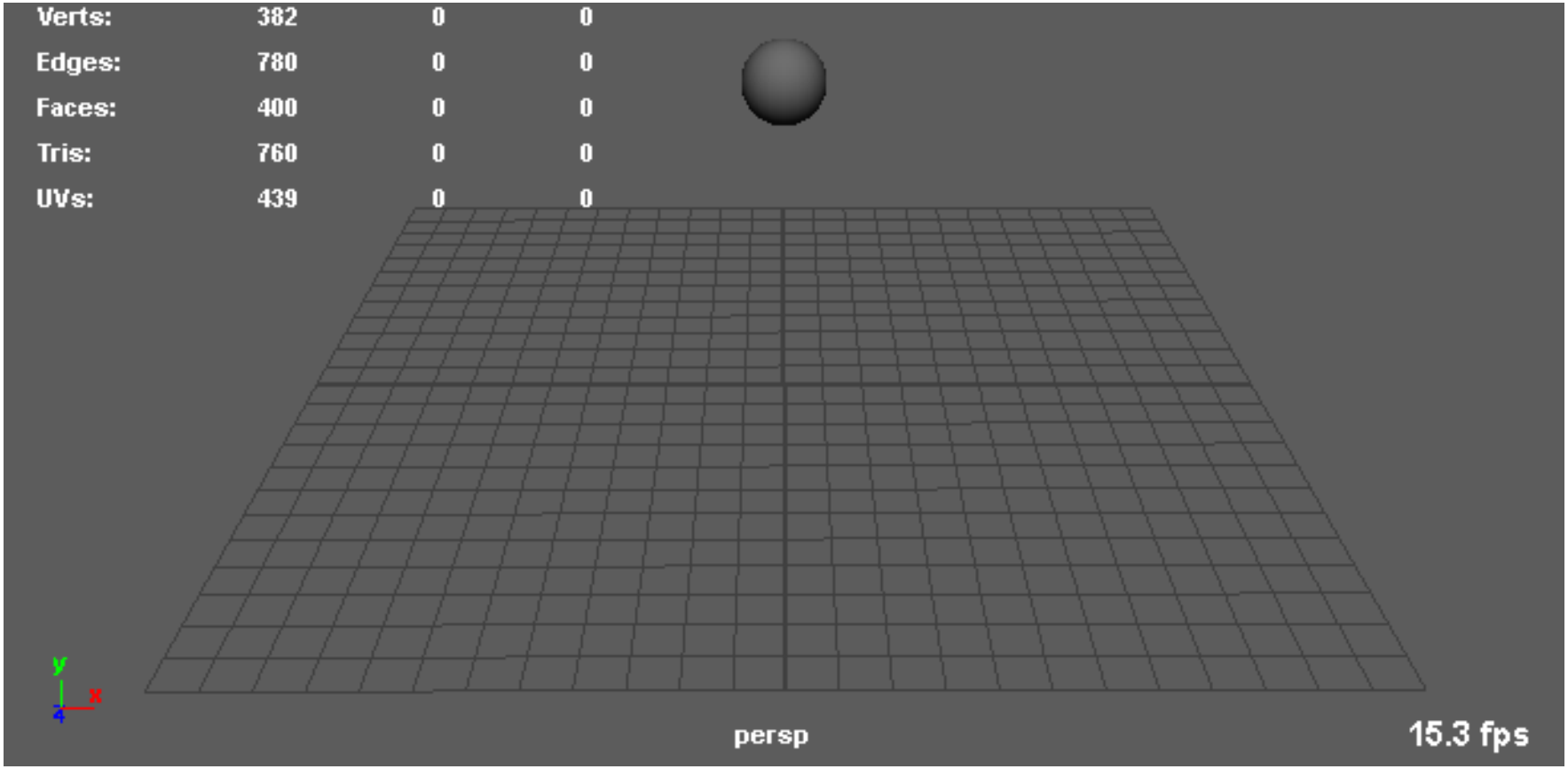
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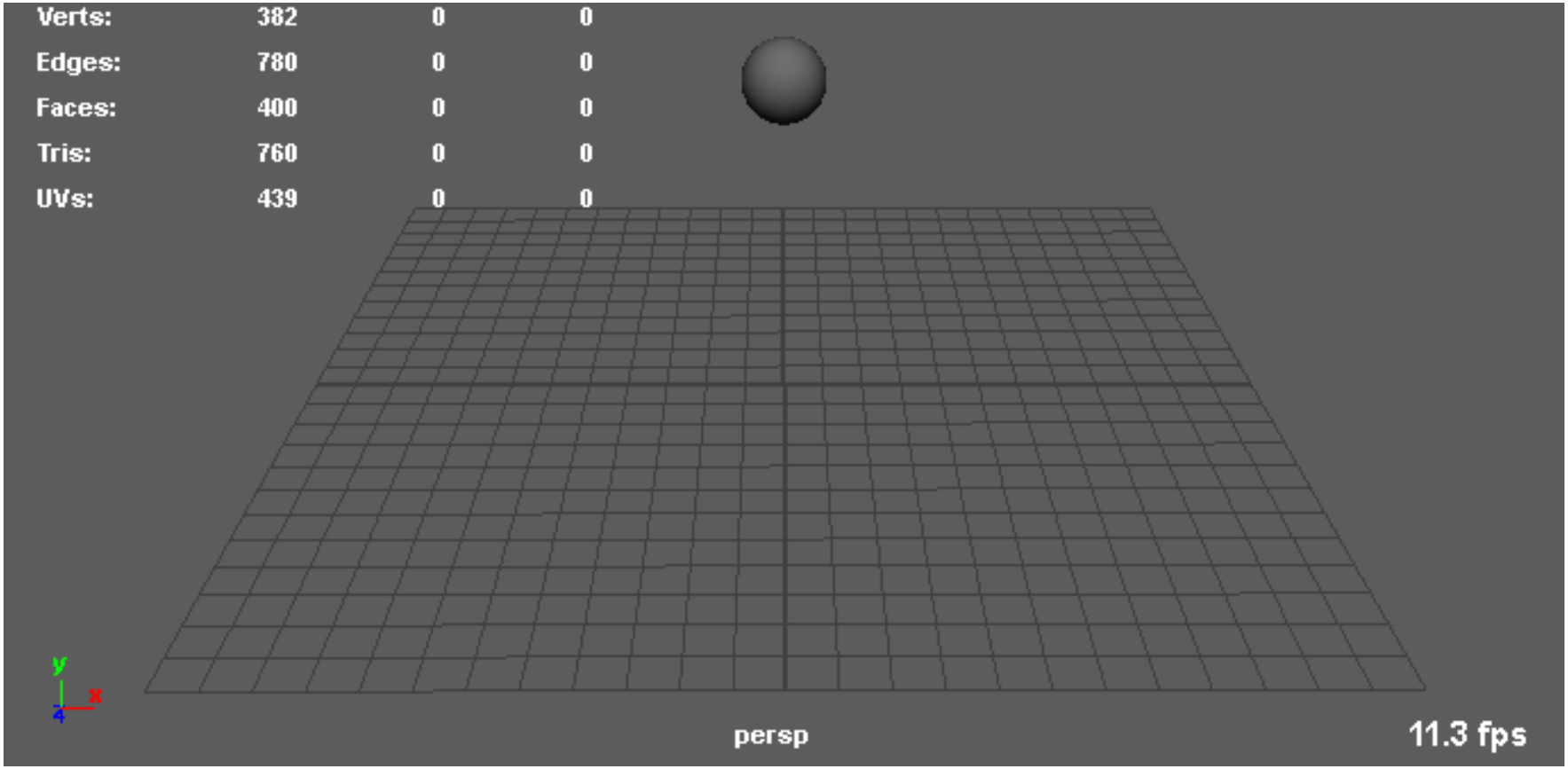
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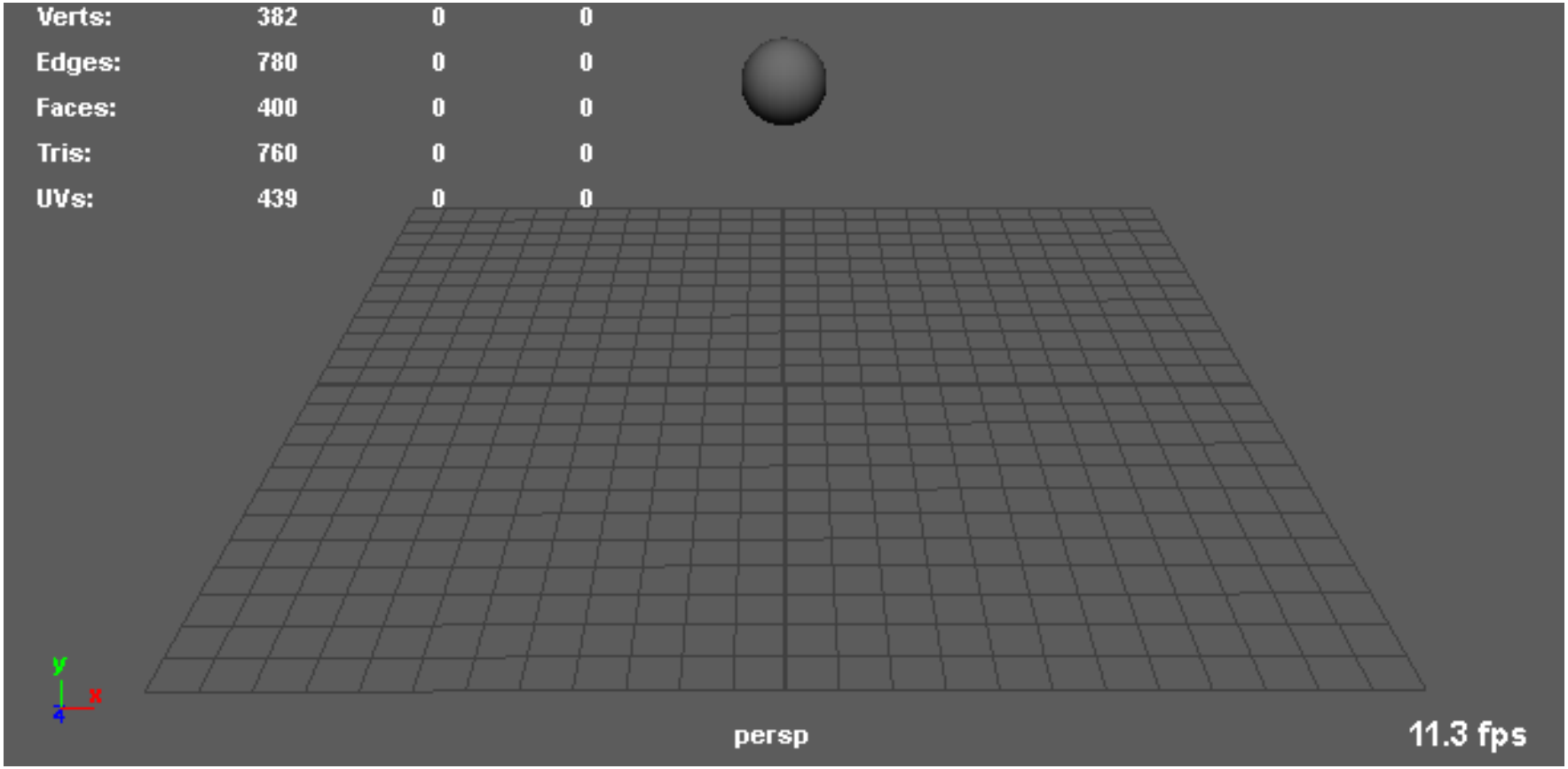
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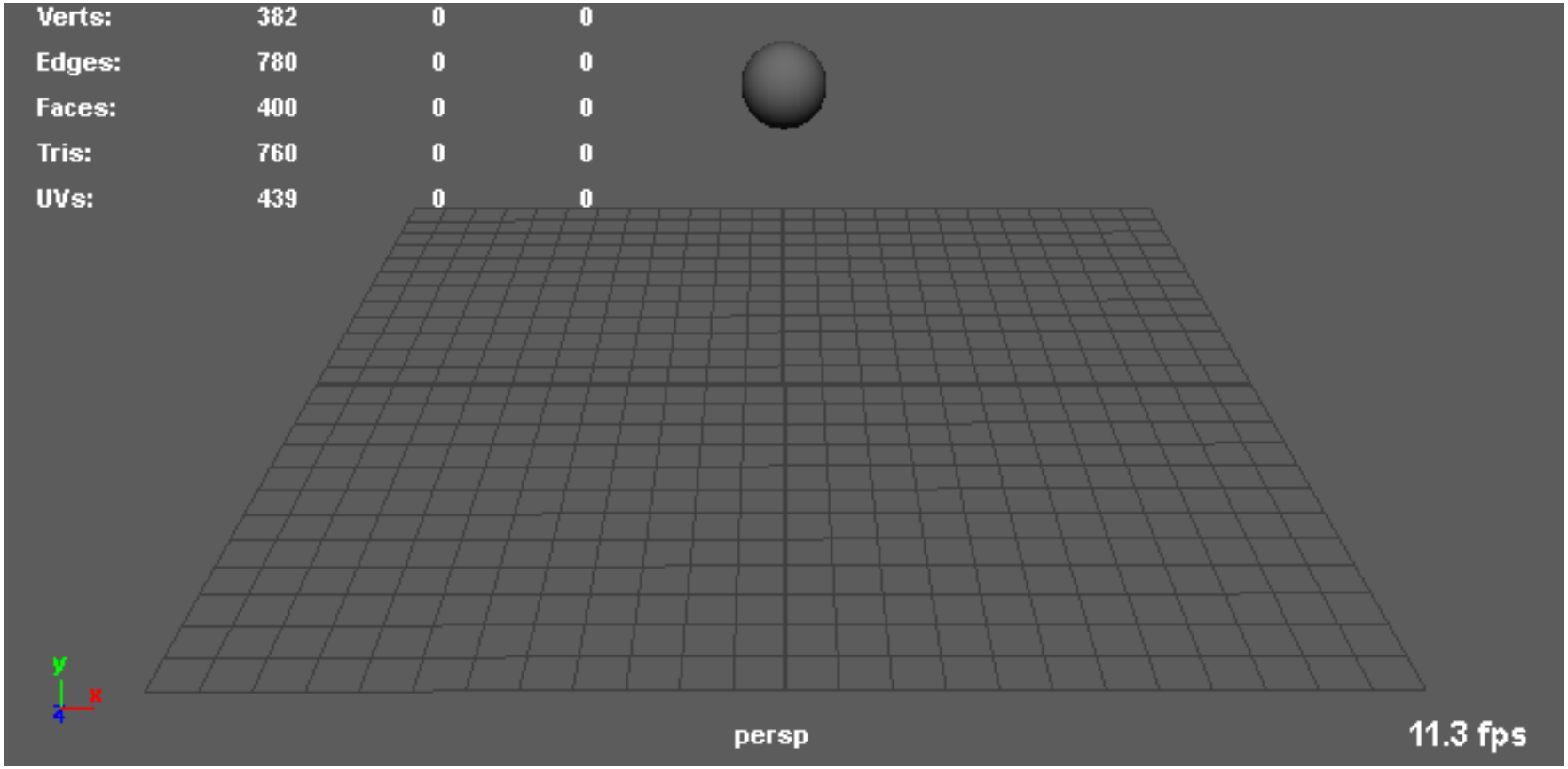
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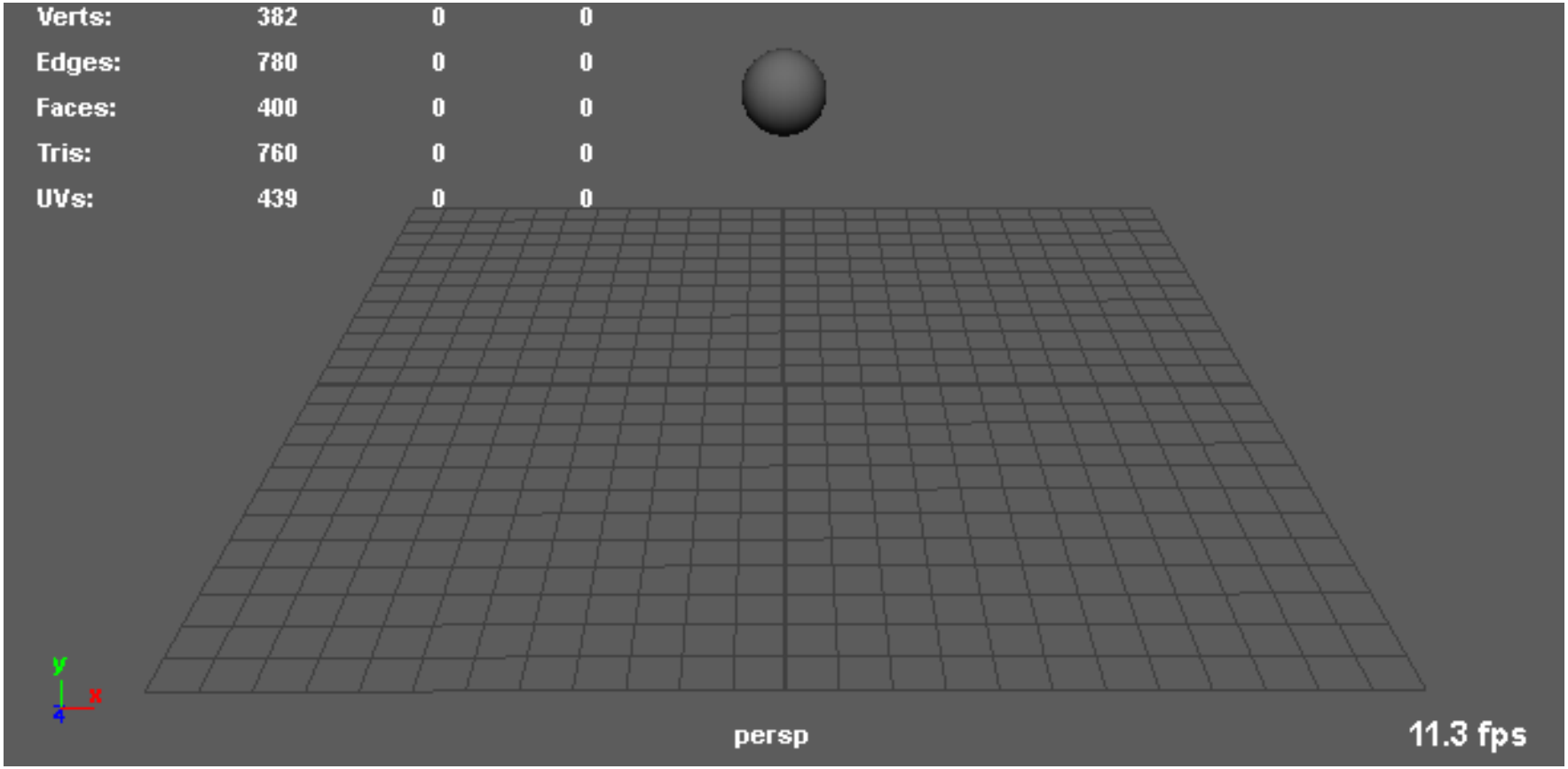
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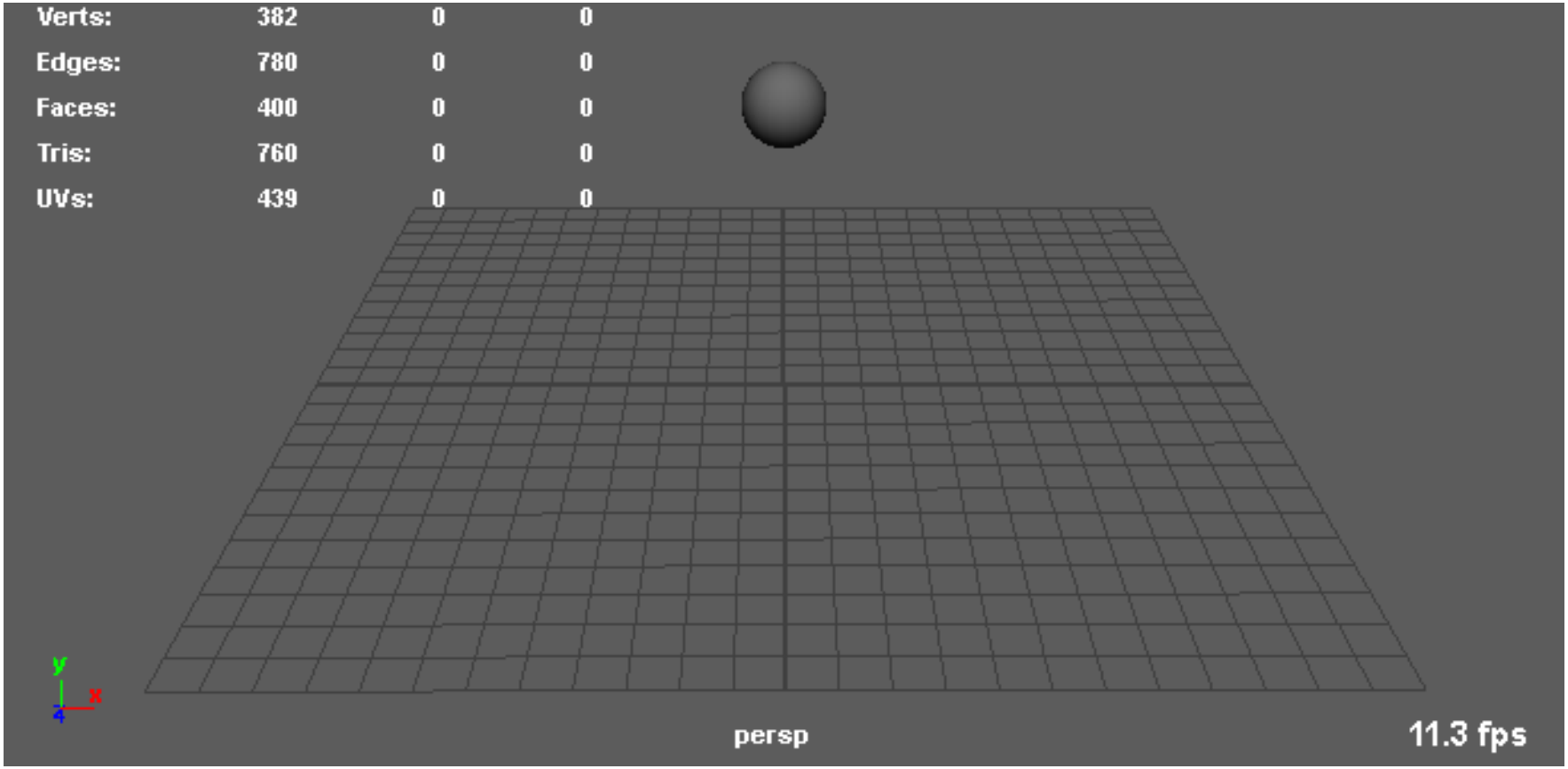
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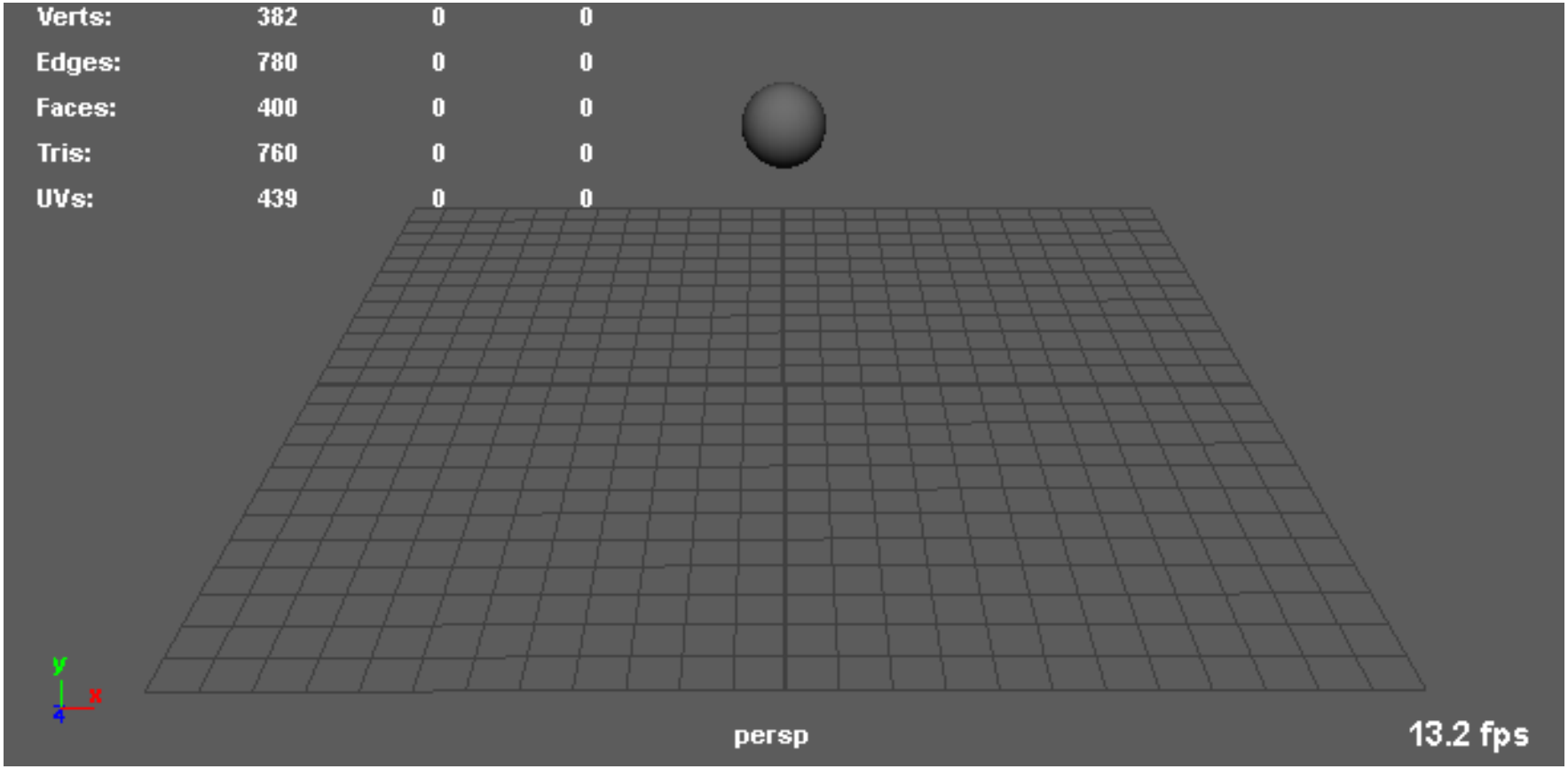
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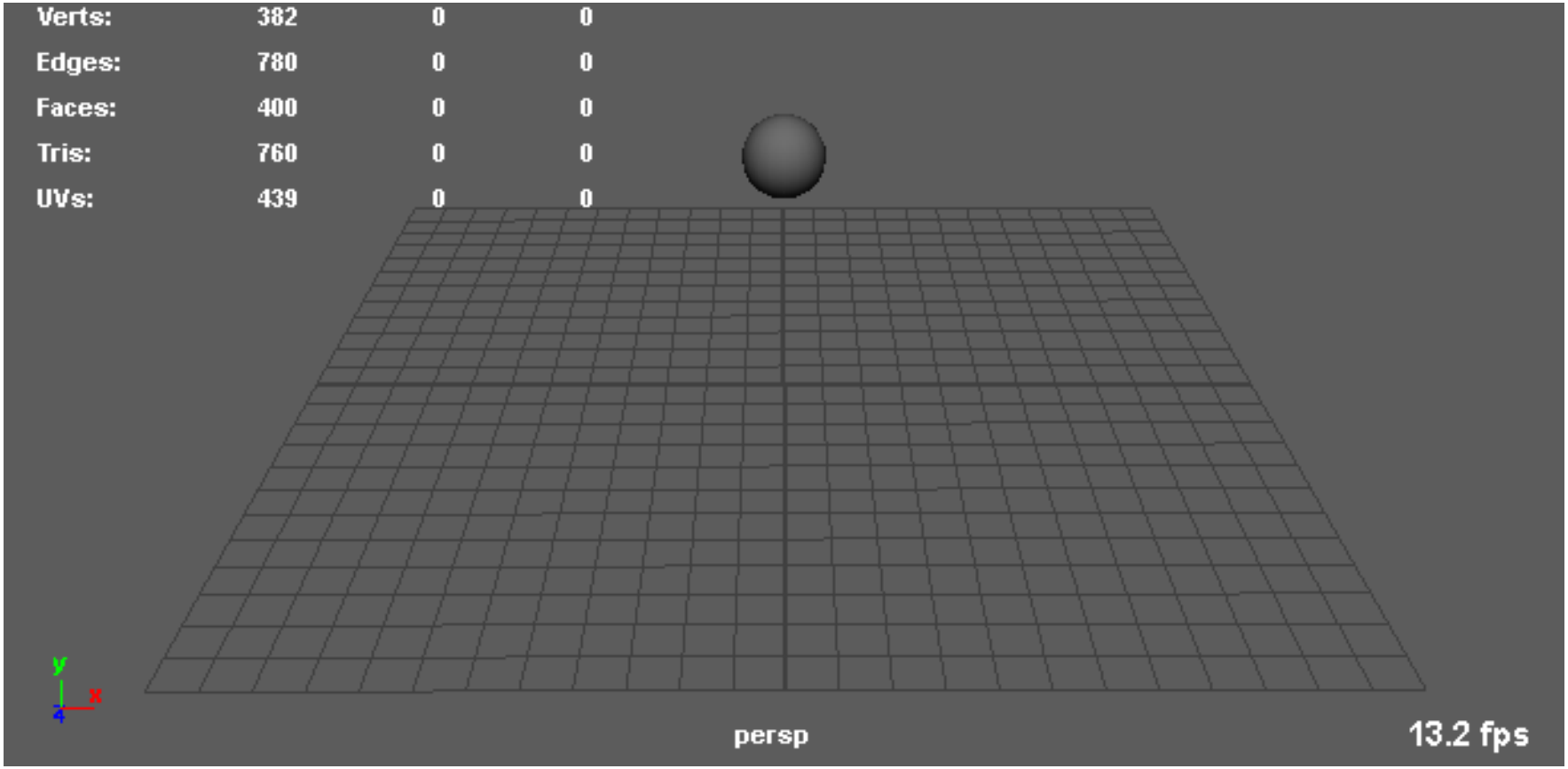
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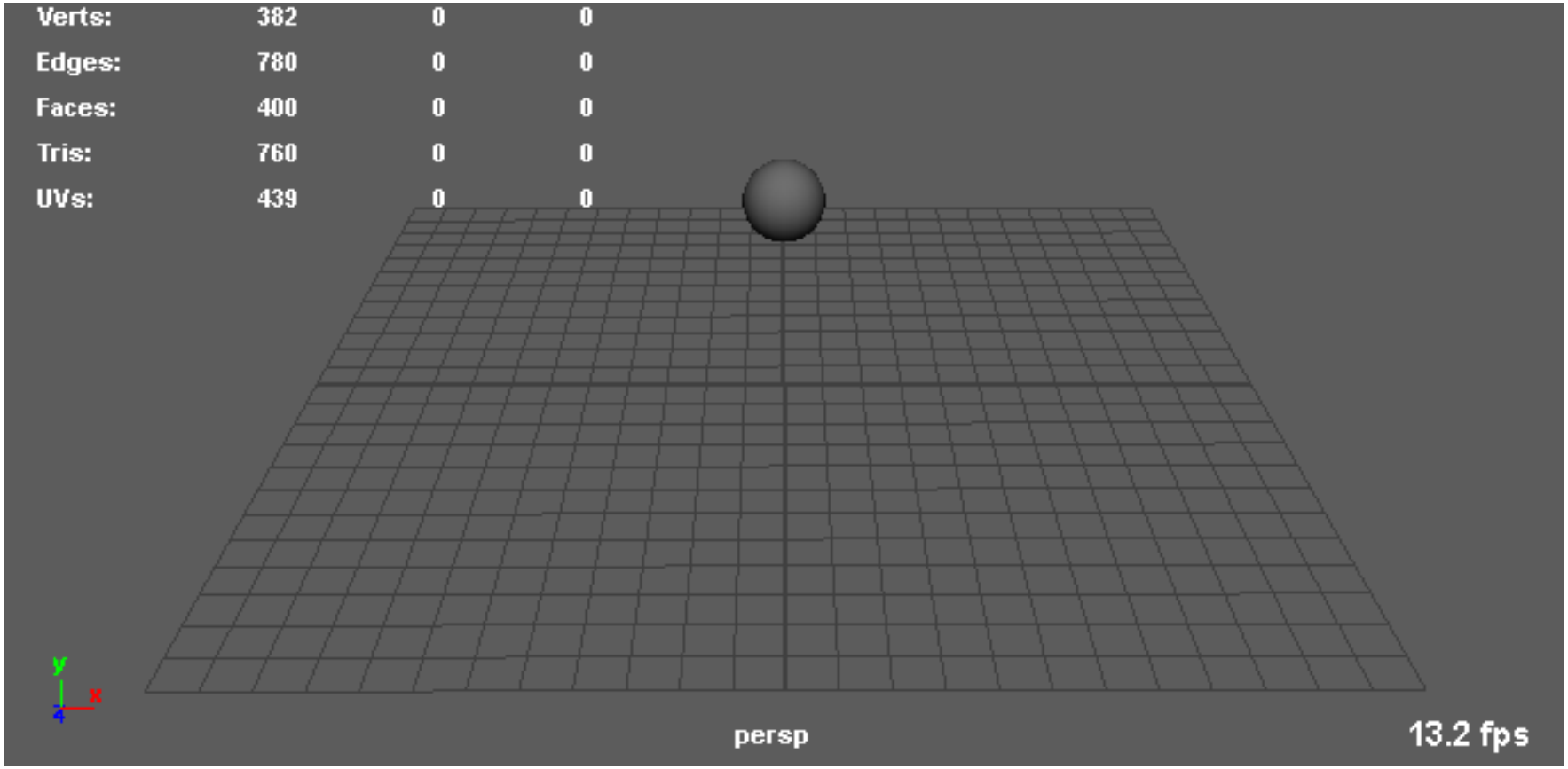
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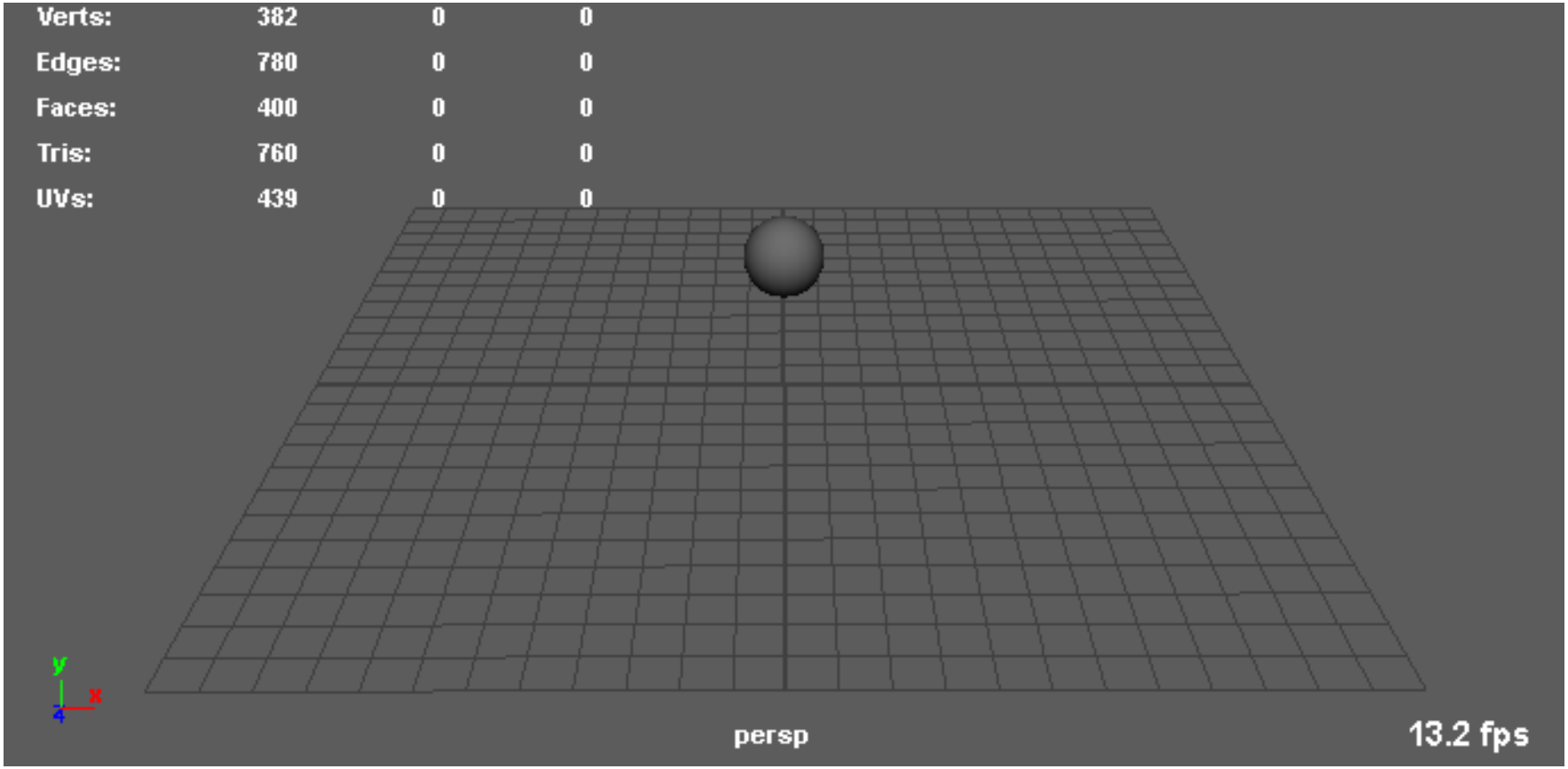
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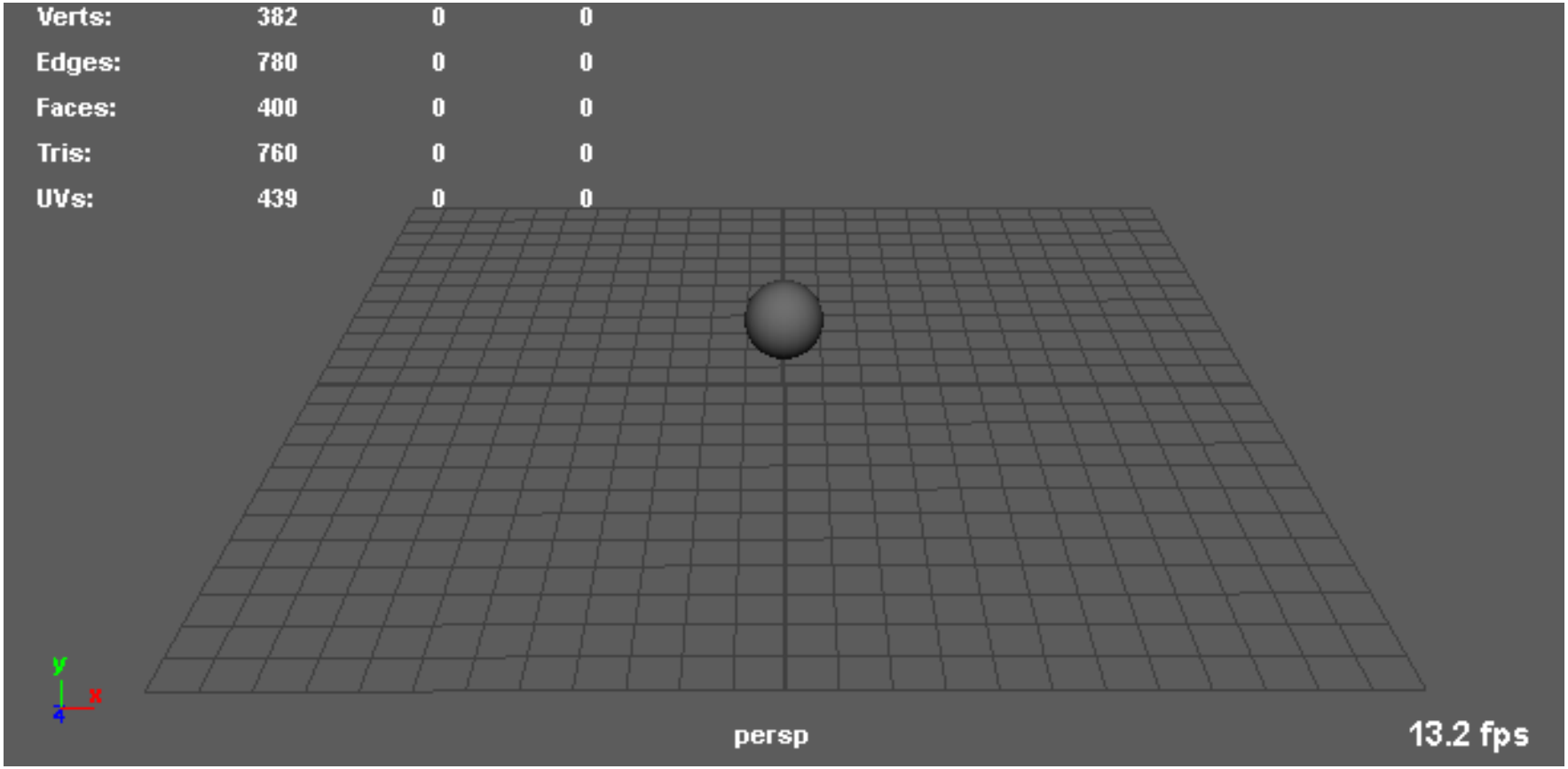
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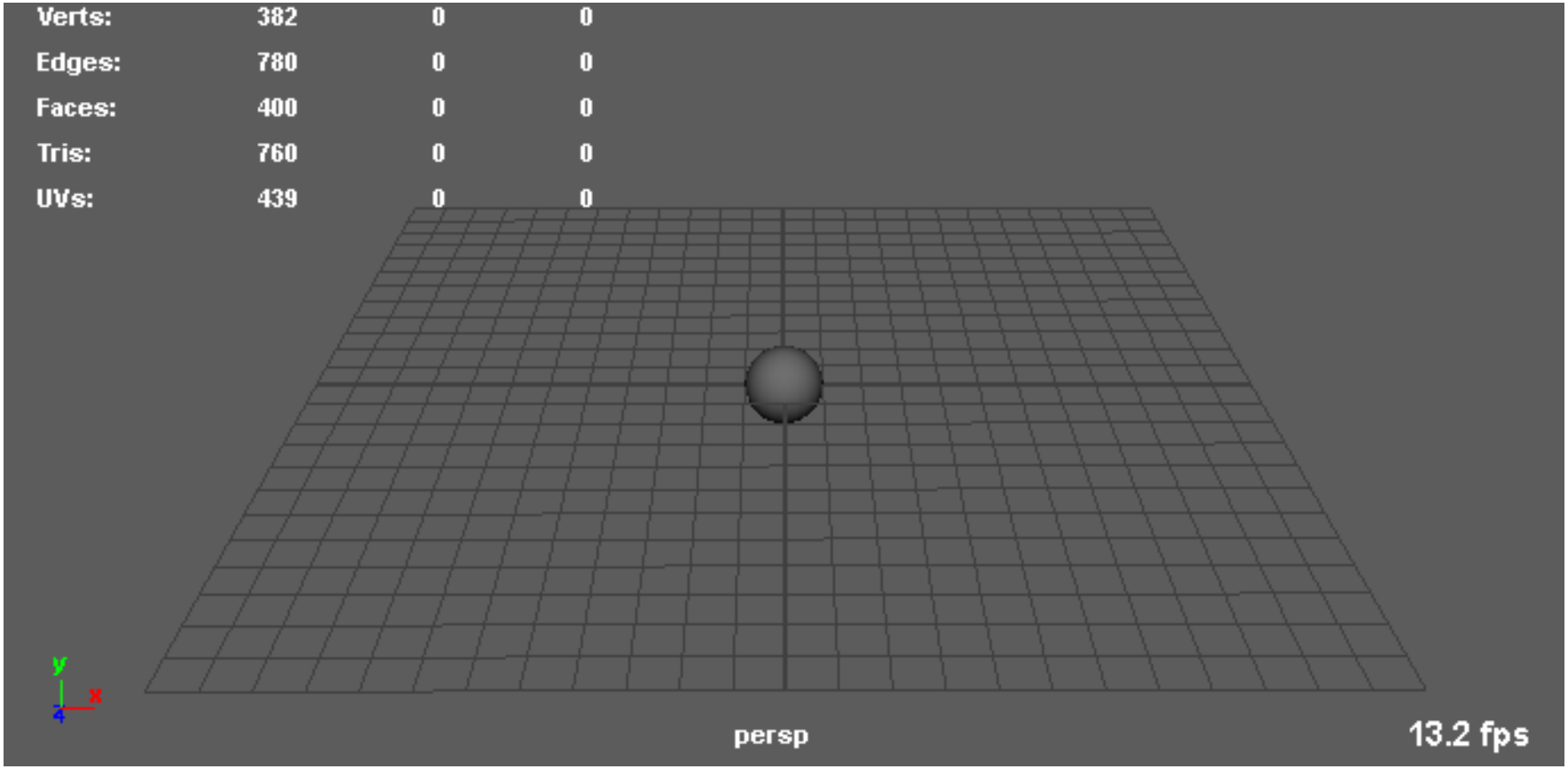
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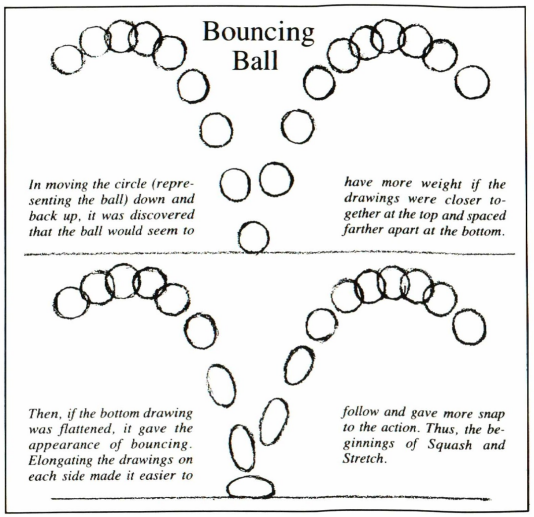
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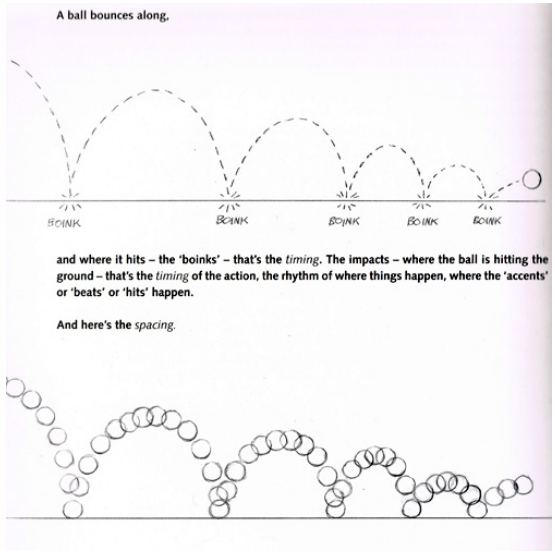


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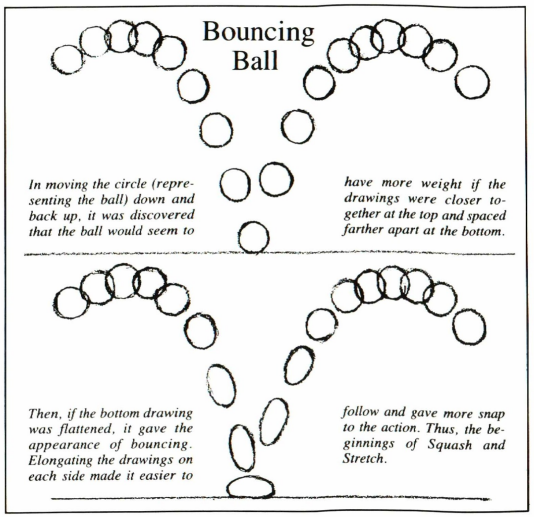
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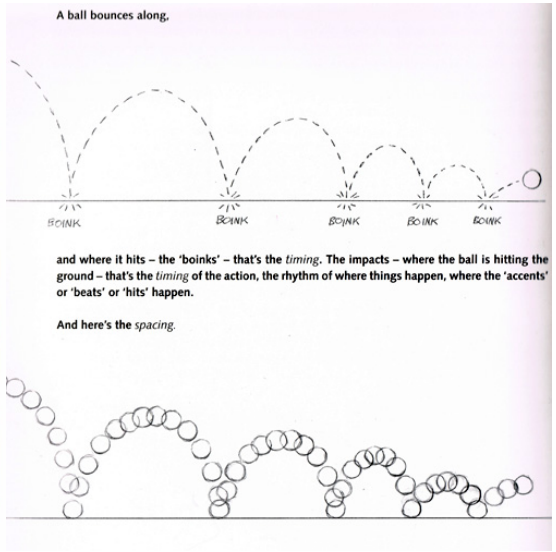




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- Nice bouncing ball examples. Follow Through

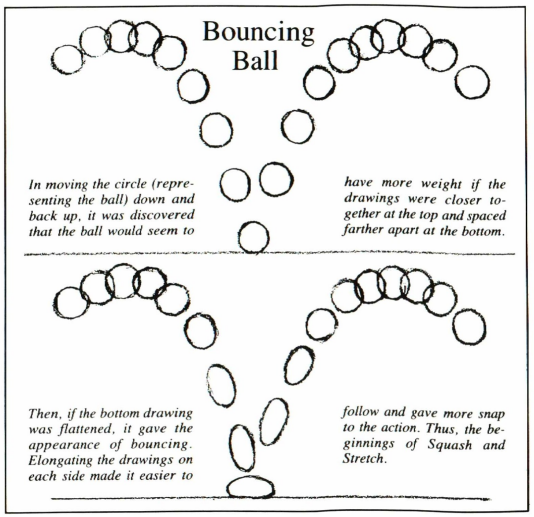
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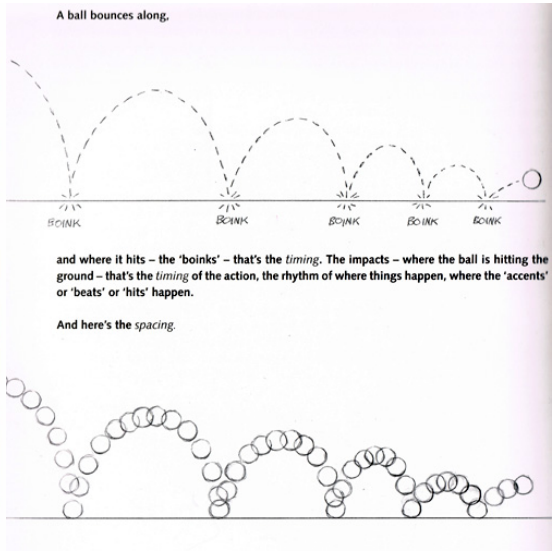




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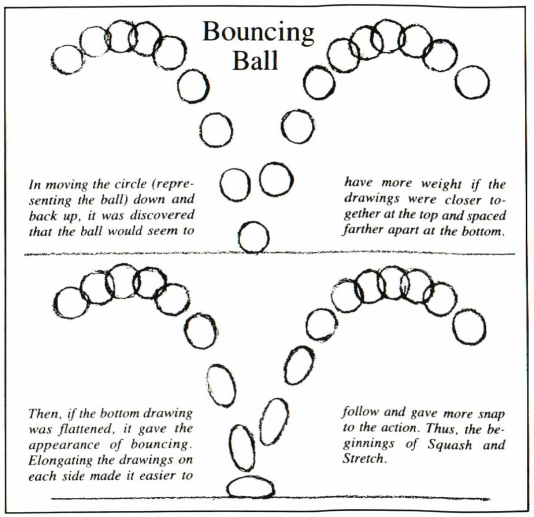
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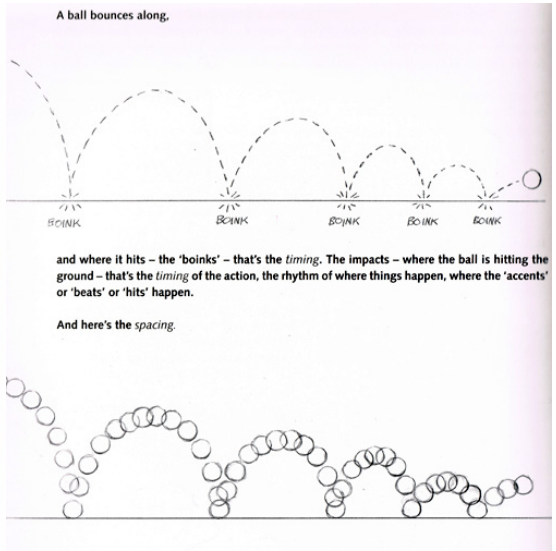




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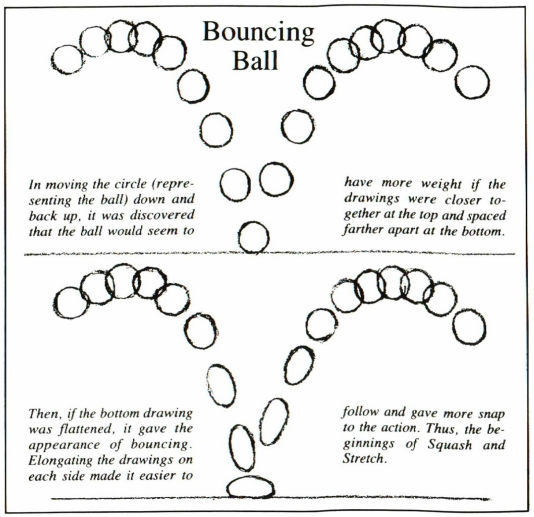
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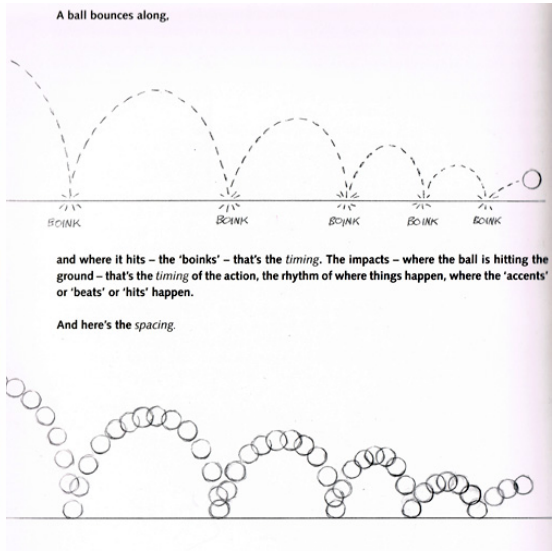




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